

Topological Energy Condensate Theory (TECT): 4

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I. TECT MASTER CONTEXT: MATH 0-7 SUMMARY

A. Topological Energy Condensate Theory

1. Research Goal

The goal of Topological Energy Condensate Theory (TECT) is to derive gravity, gauge fields, and Standard Model fermions from a single underlying energy condensate.

The fundamental hypothesis is that the universe is a condensed energy medium whose equilibrium configuration forms a body-centered cubic (BCC) energy structure.

Particles and gauge fields emerge as collective excitations and topological defects of this condensate.

Energy condensate $\rightarrow$ BCC structure $\rightarrow$ Elastic continuum $\rightarrow$ Gauge fields and fermions

The ultimate target is the derivation of the Standard Model gauge group

$$SU(3) \times SU(2) \times U(1)$$

together with gravity as emergent phenomena.

B. Fundamental Condensate Structure

The condensate is assumed to form a BCC reciprocal star.

The BCC reciprocal shell is

$$S_{BCC} = \frac{1}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}$$

This shell contains

$$|S_{BCC}| = 12$$

wavevectors.

The quartic interaction closure coefficient derived from the BCC amplitude expansion is

$$C_4 = 540$$

which plays a central role in the BCC structure selection theorem.

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C. Emergent Elastic Continuum

At long wavelengths the condensate behaves as an elastic medium.
The displacement field

$$u_i(x)$$

produces strain

$$\epsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$$

and elastic free energy

$$F = \int d^3x (C_{11}\epsilon_{ii}^2 + C_{12}\epsilon_{ii}\epsilon_{jj} + 2C_{44}\epsilon_{ij}^2)$$

The low-energy sector is dominated by transverse modes.
These modes possess an internal rotation symmetry

$$SO(2)$$

which becomes the origin of the emergent electromagnetic gauge symmetry.

D. Emergent Electromagnetism

The transverse polarization rotation

$$SO(2)$$

is promoted to a local symmetry

$$SO(2) \rightarrow U(1)$$

producing a gauge connection

$$A_\mu$$

with curvature

$$F_{\mu\nu}$$

The effective infrared action becomes Maxwell theory

$$S = -\frac{1}{4} \int F_{\mu\nu} F^{\mu\nu}$$

Lorentz invariance emerges from the isotropy of the infrared elastic sector.

E. Internal Geometry

Topological defects of the condensate possess an internal moduli space

$$\mathbb{CP}^2$$

which can be written as the coset

$$\mathbb{CP}^2 \simeq \frac{SU(3)}{SU(2) \times U(1)}$$

This structure provides the geometric origin of the color gauge group.

F. Topological Charge

Defects are classified by

$$\pi_2(\mathbb{CP}^2) = \mathbb{Z}$$

Let

$$B \in \mathbb{Z}$$

be the defect topological charge.

This quantity is interpreted as baryon number.

G. Wess-Zumino-Witten Phase

The defect sector carries a WZW phase

$$\exp(i\pi B)$$

which implies

$$\exp(i\pi B) = (-1)^B$$

Thus defects with odd charge exhibit fermionic statistics.

H. Fermion Zero Modes

Fermions arise as defect-bound zero modes of a twisted Dirac operator.

The internal space is

$$\mathbb{CP}^2$$

which admits spin^c structures.

The twisted chiral Dirac operator is

$$D_+^{(L,E)}$$

The Atiyah-Singer index theorem gives

$$\text{Index}(D_+^{(L,E)}) = \frac{(2m+q)^2 - 1}{8}$$

For the primitive twisting

$$(q, m) = (1, 1)$$

one obtains

$$\text{Index}(D_+) = 1$$

which implies

$$n_L - n_R = 1$$

This establishes topological chirality protection.

I. Mirror Sector

Index theory does not eliminate mirror pairs.
One may still have

$$(n_L, n_R) = (I + r, r)$$

Mirror removal requires a dynamical lifting operator

$$M = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & \mu^\dagger \\ 0 & \mu & 0 \end{pmatrix}$$

If

$$\mu \neq 0$$

all mirror pairs acquire nonzero energy and the spectrum becomes

$$n_L = I, \quad n_R = 0$$

J. Dirac Operator Emergence

Consider the linearized condensate equation

$$i\partial_t \Psi = \mathcal{L}(-i\nabla) \Psi$$

Assume an isolated valley crossing at momentum

$$k_*$$

The projected Hamiltonian near the valley is

$$h_+(p) = \sum_i p_i V_i$$

Since this is a 2×2 Hermitian matrix,

$$h_+(p) = a_0(p) \mathbf{1}_2 + \sum_{a=1}^3 a_a(p) \sigma^a$$

If the traceless coefficient matrix is nondegenerate, the Hamiltonian reduces to

$$h_+(p) \sim p_i \sigma^i$$

which is a Weyl fermion.

K. Dirac Doubling

If an opposite valley exists at

$$-k_*$$

satisfying

$$h_-(p) = -h_+(-p)$$

then the doubled Hamiltonian becomes

$$H = \begin{pmatrix} p_i \sigma^i & 0 \\ 0 & -p_i \sigma^i \end{pmatrix}$$

This is the massless Dirac Hamiltonian.

L. BCC Shell Valley Structure

Choose one reciprocal pair

$$\pm G_*$$

from the BCC shell.

Decompose momentum as

$$p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2$$

The projected Hamiltonian takes the form

$$h_+(p) = \varepsilon_0(p) \mathbf{1} + v_{\parallel} p_{\parallel} \sigma^3 + v_1 p_1 \sigma^1 + v_2 p_2 \sigma^2$$

If

$$v_{\parallel} v_1 v_2 \neq 0$$

this is Weyl.

Opposite valley doubling gives Dirac.

M. Current Proven Results

The following results are rigorously established:

1. BCC condensate selection theorem
2. Elastic continuum emergence
3. $U(1)$ gauge emergence
4. Maxwell dynamics
5. Lorentz symmetry emergence
6. CP^2 internal geometry
7. $SU(3)$ Yang-Mills emergence
8. Chirality index theorem
9. Mirror lifting theorem
10. Weyl valley theorem
11. Dirac emergence from valley doubling

N. Open Problems

Remaining frontier problems are:

1. Derive the full microscopic TECT condensate PDE
2. Prove the existence of BCC shell valleys
3. Compute velocity coefficients

$$v_{\parallel}, v_1, v_2$$

and verify nondegeneracy

4. Derive the $SU(2)$ electroweak sector
5. Verify anomaly cancellation

O. Current Research Frontier

The fermion emergence chain currently reads

$$CP^2 \rightarrow \text{Chiral Index} \rightarrow \text{Mirror Lifting} \rightarrow \text{Weyl Valleys} \rightarrow \text{Dirac Fermion}$$

The remaining goal is the full recovery of

$$SU(3) \times SU(2) \times U(1)$$

from the condensate geometry.

II. EMERGENT $SU(2)$ GAUGE STRUCTURE FROM THE $\mathbb{CP}^2$ INTERNAL GEOMETRY

A. Statement of the result

We show that the internal geometry

$$\mathbb{CP}^2 \simeq \frac{SU(3)}{U(2)}$$

naturally carries a rank-two complex bundle whose local orthonormal frame redundancy is $U(2)$. Consequently, the corresponding bundle connection decomposes canonically into

$$\mathfrak{u}(2) \simeq \mathfrak{su}(2) \oplus \mathfrak{u}(1),$$

which yields an emergent $SU(2)$ gauge connection together with an additional $U(1)$ connection.

Thus the internal $\mathbb{CP}^2$ geometry does not merely encode an $SU(3)$ tangent-algebra structure; it also contains a distinguished local $U(2)$ frame bundle, and therefore an electroweak-type precursor

$$U(2) \cong \frac{SU(2) \times U(1)}{\mathbb{Z}_2}.$$

B. Normalized order parameter and projective reduction

Let

$$z(x) \in \mathbb{C}^3, \quad z^\dagger z = 1.$$

The physical internal state is invariant under the local phase identification

$$z(x) \sim e^{i\alpha(x)} z(x).$$

Hence the physical configuration space is the complex projective space

$$[z(x)] \in \mathbb{CP}^2.$$

Equivalently, define the rank-one Hermitian projector

$$P = zz^\dagger, \quad P^2 = P, \quad P^\dagger = P, \quad \text{Tr } P = 1.$$

Then P is phase-invariant and therefore represents a point of $\mathbb{CP}^2$.

C. The tautological line and the orthogonal rank-two complement

At each point $[z] \in \mathbb{CP}^2$, define the tautological complex line

$$L_{[z]} := \text{span}_{\mathbb{C}}\{z\} \subset \mathbb{C}^3.$$

Its Hermitian orthogonal complement is

$$Q_{[z]} := L_{[z]}^\perp = \{v \in \mathbb{C}^3 \mid z^\dagger v = 0\}.$$

Since $\dim_{\mathbb{C}} \mathbb{C}^3 = 3$ and $\dim_{\mathbb{C}} L = 1$, one has

$$\dim_{\mathbb{C}} Q = 2.$$

Thus $Q \rightarrow \mathbb{CP}^2$ is a rank-two complex vector bundle.

The corresponding orthogonal projector onto Q is

$$\Pi := \mathbf{1}_3 - P = \mathbf{1}_3 - zz^\dagger.$$

Indeed,

$$\Pi^2 = \Pi, \quad \Pi^\dagger = \Pi, \quad \Pi z = 0.$$

D. Local orthonormal frames and the $U(2)$ redundancy

Choose locally an orthonormal frame of Q ,

$$e(x) = (e_1(x), e_2(x)),$$

satisfying

$$z^\dagger e_a = 0, \quad e_a^\dagger e_b = \delta_{ab}, \quad a, b = 1, 2.$$

Then

$$\Pi = \sum_{a=1}^2 e_a e_a^\dagger.$$

Any other local orthonormal frame $\tilde{e}$ of the same bundle Q is related to e by

$$\tilde{e} = e U, \quad U(x) \in U(2).$$

Indeed,

$$\tilde{e}^\dagger \tilde{e} = U^\dagger e^\dagger e U = U^\dagger U = \mathbf{1}_2.$$

Since

$$\sum_{a=1}^2 \tilde{e}_a \tilde{e}_a^\dagger = e U U^\dagger e^\dagger = e e^\dagger = \Pi,$$

the projector Π , and hence the physical point of $\mathbb{CP}^2$, is unchanged.

Therefore the local frame choice on Q has an exact $U(2)$ redundancy. This is the precise geometric origin of the emergent gauge structure.

E. Emergent $U(2)$ connection

Define the local bundle connection by

$$\mathcal{A}_\mu := -i e^\dagger \partial_\mu e.$$

Because $e^\dagger e = \mathbf{1}_2$, differentiation gives

$$(\partial_\mu e^\dagger) e + e^\dagger (\partial_\mu e) = 0,$$

hence

$$\mathcal{A}_\mu^\dagger = (+i \partial_\mu e^\dagger e) = (-i e^\dagger \partial_\mu e) = \mathcal{A}_\mu.$$

So $\mathcal{A}_\mu$ is Hermitian and takes values in $\mathfrak{u}(2)$.

Under a local frame change

$$e \mapsto e U(x), \quad U(x) \in U(2),$$

we obtain

$$\mathcal{A}_\mu \mapsto \mathcal{A}'_\mu = -i(eU)^\dagger \partial_\mu (eU) = U^\dagger \mathcal{A}_\mu U - iU^\dagger \partial_\mu U.$$

Thus $\mathcal{A}_\mu$ transforms exactly as a $U(2)$ gauge connection.

F. Canonical decomposition into $SU(2)$ and $U(1)$

Every $\mathfrak{u}(2)$ -valued Hermitian matrix admits the decomposition

$$\mathcal{A}_\mu = W_\mu + \frac{1}{2}B_\mu \mathbf{1}_2,$$

where

$$W_\mu \in \mathfrak{su}(2), \quad \text{Tr } W_\mu = 0,$$

and

$$B_\mu := \text{Tr } \mathcal{A}_\mu.$$

Expanding the traceless part in Pauli matrices,

$$W_\mu = \frac{1}{2}W_\mu^a \sigma^a, \quad a = 1, 2, 3.$$

Now let

$$U(x) = e^{i\beta(x)/2} S(x), \quad S(x) \in SU(2).$$

Then the transformation law splits into

$$W_\mu \mapsto S^\dagger W_\mu S - iS^\dagger \partial_\mu S,$$

and

$$B_\mu \mapsto B_\mu + \partial_\mu \beta.$$

Hence W_μ is an $SU(2)$ gauge connection and B_μ is a $U(1)$ gauge connection.

Therefore the local frame redundancy of the rank-two bundle Q yields the gauge structure

$$U(2) \cong \frac{SU(2) \times U(1)}{\mathbb{Z}_2}.$$

G. Coset derivation from $SU(3)/U(2)$

The same result may be derived from the coset structure directly. Choose a local $SU(3)$ matrix

$$g(x) = (z(x), e_1(x), e_2(x)) \in SU(3),$$

whose first column is z and whose remaining columns form an orthonormal basis of Q . Then the Maurer–Cartan form is

$$\Omega_\mu := g^\dagger \partial_\mu g \in \mathfrak{su}(3).$$

Block-decompose it as

$$\Omega_\mu = \begin{pmatrix} ia_\mu & -\beta_\mu^\dagger \\ \beta_\mu & i\mathcal{A}_\mu \end{pmatrix},$$

with $a_\mu \in \mathbb{R}$, $\beta_\mu \in \mathbb{C}^2$, and $\mathcal{A}_\mu \in \mathfrak{u}(2)$, subject to

$$a_\mu + \text{Tr } \mathcal{A}_\mu = 0$$

because $\Omega_\mu \in \mathfrak{su}(3)$ is traceless.

The right action of the stabilizer

$$h(x) = \begin{pmatrix} e^{i\alpha(x)} & 0 \\ 0 & U(x) \end{pmatrix} \in U(1) \times U(2)$$

acts as

$$g(x) \mapsto g(x)h(x),$$

and induces

$$\mathcal{A}_\mu \mapsto U^\dagger \mathcal{A}_\mu U - iU^\dagger \partial_\mu U,$$

while

$$\beta_\mu \mapsto U^\dagger \beta_\mu e^{i\alpha}.$$

Thus the composite connection is precisely the stabilizer connection of the coset $SU(3)/U(2)$.

H. Covariant derivative on Q -doublets

Let $\chi(x) \in \mathbb{C}^2$ be a local section of the rank-two bundle Q , expressed in the frame e . Under frame changes,

$$\chi \mapsto U^{-1} \chi.$$

Then the covariant derivative

$$D_\mu \chi := \partial_\mu \chi - i\mathcal{A}_\mu \chi$$

transforms covariantly:

$$D_\mu \chi \mapsto U^{-1} D_\mu \chi.$$

After the decomposition

$$\mathcal{A}_\mu = \frac{1}{2} W_\mu^a \sigma^a + \frac{1}{2} B_\mu \mathbf{1}_2,$$

we obtain

$$D_\mu \chi = \left(\partial_\mu - \frac{i}{2} W_\mu^a \sigma^a - \frac{i}{2} B_\mu \right) \chi.$$

This is exactly the standard form of a doublet-valued covariant derivative under $SU(2) \times U(1)$.

I. Projector form of the sigma model and bundle geometry

The $\mathbb{CP}^2$ nonlinear sigma model may be written as

$$\mathcal{L}_{\mathbb{CP}^2} = \frac{\rho}{2} \text{Tr}(\partial_\mu P \partial^\mu P).$$

Since

$$P = zz^\dagger, \quad \Pi = \mathbf{1} - P,$$

one finds

$$\partial_\mu P = (\Pi \partial_\mu z) z^\dagger + z(\partial_\mu z^\dagger \Pi).$$

Thus only derivatives orthogonal to the tautological line contribute. Equivalently, the dynamics naturally lives on the quotient bundle data

$$L \oplus Q = \underline{\mathbb{C}}^3.$$

This is the geometric reason that the stabilizer connection $\mathcal{A}_\mu \in \mathfrak{u}(2)$ appears canonically in the low-energy description.

J. Curvature and field strengths

The $U(2)$ curvature is

$$\mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - i[\mathcal{A}_\mu, \mathcal{A}_\nu].$$

Decomposing again,

$$\mathcal{F}_{\mu\nu} = F_{\mu\nu}^{(W)} + \frac{1}{2} G_{\mu\nu} \mathbf{1}_2,$$

where

$$F_{\mu\nu}^{(W)} = \frac{1}{2} F_{\mu\nu}^a \sigma^a, \quad F_{\mu\nu}^a = \partial_\mu W_\nu^a - \partial_\nu W_\mu^a + \epsilon^{abc} W_\mu^b W_\nu^c,$$

and

$$G_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu.$$

Therefore the nonabelian $SU(2)$ curvature and the abelian $U(1)$ curvature both arise directly from the same geometric $U(2)$ connection.

K. Infrared gauge dynamics

Assume:

1. the low-energy theory is local,
2. the Q -frame redundancy is exact,
3. a derivative expansion is valid,
4. no explicit $U(2)$ -breaking operator survives in the infrared.

Then the leading gauge-field self-action must be built from the curvatures:

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4g_2^2} F_{\mu\nu}^a F^{a\mu\nu} - \frac{1}{4g_1^2} G_{\mu\nu} G^{\mu\nu} + \dots$$

Mass terms of the form

$$W_\mu^a W^{a\mu}, \quad B_\mu B^\mu$$

are forbidden by exact local frame redundancy.

Thus the $\mathbb{CP}^2$ internal geometry implies, at the structural infrared level, an emergent $SU(2) \times U(1)$ gauge sector.

L. What is proven and what remains open

What is established here is the following theorem:

a. Geometric $SU(2)$ emergence theorem. If the TECT internal low-energy sector is described by a normalized $z \in \mathbb{C}^3$ modulo phase, so that the physical internal manifold is $\mathbb{CP}^2$, then the orthogonal rank-two bundle

$$Q = (\mathbb{C}z)^\perp$$

has local orthonormal frame redundancy $U(2)$. Its bundle connection decomposes canonically into

$$\mathfrak{u}(2) = \mathfrak{su}(2) \oplus \mathfrak{u}(1),$$

thereby producing an emergent $SU(2)$ gauge connection W_μ and an emergent $U(1)$ gauge connection B_μ .

What is not yet established is the full Standard Model identification, namely:

1. the exact hypercharge embedding,
2. chiral fermion representation assignments,
3. electroweak symmetry breaking,
4. anomaly cancellation,
5. the Weinberg-angle prediction.

Therefore the present derivation closes the geometric origin of an electroweak-type $U(2)$ gauge bundle, but not yet the complete Standard Model electroweak sector.

III. FINAL SUMMARY

Starting from the TECT internal order parameter

$$z \in \mathbb{C}^3, \quad z^\dagger z = 1,$$

with the physical phase identification $z \sim e^{i\alpha} z$, the low-energy internal manifold is

$$\mathbb{CP}^2 \simeq SU(3)/U(2).$$

At each point $[z] \in \mathbb{CP}^2$, the orthogonal complement

$$Q_{[z]} = \{v \in \mathbb{C}^3 \mid z^\dagger v = 0\}$$

is a complex rank-two vector space. Hence $Q \rightarrow \mathbb{CP}^2$ is a rank-two Hermitian bundle.

A local orthonormal frame $e = (e_1, e_2)$ of Q is defined only up to

$$e \mapsto e U(x), \quad U(x) \in U(2),$$

so the local frame redundancy is exactly $U(2)$.

The associated bundle connection

$$\mathcal{A}_\mu = -ie^\dagger \partial_\mu e$$

transforms as a $U(2)$ gauge field and admits the canonical decomposition

$$\mathcal{A}_\mu = \frac{1}{2} W_\mu^a \sigma^a + \frac{1}{2} B_\mu \mathbf{1}_2,$$

which yields an emergent $SU(2)$ connection W_μ and an emergent $U(1)$ connection B_μ .

Thus the $\mathbb{CP}^2$ internal geometry contains, in a mathematically precise sense, an electroweak-type gauge precursor

$$U(2) \cong \frac{SU(2) \times U(1)}{\mathbb{Z}_2}.$$

Under locality, exact frame redundancy, and derivative expansion, the leading infrared gauge dynamics is

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4g_2^2} F_{\mu\nu}^a F^{a\mu\nu} - \frac{1}{4g_1^2} G_{\mu\nu} G^{\mu\nu} + \dots$$

Therefore:

$\mathbb{CP}^2 \text{ internal geometry} \implies Q \text{ rank-two bundle} \implies U(2) \text{ frame redundancy} \implies SU(2) \times U(1) \text{ gauge precursor.}$

This closes the geometric emergence of the $SU(2)$ connection at the bundle level, but the full Standard Model electroweak reconstruction remains an open problem.

IV. INDUCED YANG–MILLS DYNAMICS FOR THE COMPOSITE $U(2)$ CONNECTION

Assumption Audit

- A1: The internal manifold is $\mathbb{CP}^2$, and the orthogonal complement

$$Q = (\mathbb{C}z)^\perp$$

is a rank-two Hermitian bundle.

Proved

- A2: A local orthonormal frame $e = (e_1, e_2)$ of Q defines a composite connection

$$\mathcal{A}_\mu = -ie^\dagger \partial_\mu e \in \mathfrak{u}(2),$$

transforming as a $U(2)$ gauge connection.

Proved

- A3: The infrared effective action is local and admits a derivative expansion.

Needs extra assumption

- A4: There exists a dynamical field χ transforming in the fundamental of the local $U(2)$ frame bundle.

Needs extra assumption

- A5: Integrating out χ produces a gauge-invariant effective action for $\mathcal{A}_\mu$.

Provable once A4 is assumed

- A6: The induced kinetic coefficient is nonzero and positive.

Provable in explicit matter models, not yet from full TECT

A. Goal

We distinguish three logically different statements:

1. **Geometric statement:** $\mathcal{A}_\mu$ is a $U(2)$ connection.
2. **Kinematic EFT statement:** if a local gauge-invariant effective action exists, its leading self-interaction is built from the curvature.
3. **Dynamical statement:** a nonzero Yang–Mills kinetic term is actually induced by integrating out charged low-energy degrees of freedom.

Only the first is unconditional. The second is conditional on locality and exact gauge redundancy. The third is conditional on the existence of dynamical Q -charged modes.

B. Composite $U(2)$ connection and curvature

Let $e = (e_1, e_2)$ be a local orthonormal frame of the rank-two bundle Q . Then

$$\mathcal{A}_\mu := -ie^\dagger \partial_\mu e$$

is Hermitian and transforms under a local frame change $e \mapsto eU(x)$, $U(x) \in U(2)$, as

$$\mathcal{A}_\mu \mapsto U^\dagger \mathcal{A}_\mu U - iU^\dagger \partial_\mu U.$$

Its curvature is

$$\mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - i[\mathcal{A}_\mu, \mathcal{A}_\nu].$$

Under the same gauge transformation,

$$\mathcal{F}_{\mu\nu} \mapsto U^\dagger \mathcal{F}_{\mu\nu} U.$$

Decompose

$$\mathcal{A}_\mu = W_\mu + \frac{1}{2} B_\mu \mathbf{1}_2, \quad W_\mu = \frac{1}{2} W_\mu^a \sigma^a \in \mathfrak{su}(2),$$

with

$$B_\mu = \text{Tr } \mathcal{A}_\mu.$$

Then

$$\mathcal{F}_{\mu\nu} = F_{\mu\nu}^{(W)} + \frac{1}{2} G_{\mu\nu} \mathbf{1}_2,$$

where

$$F_{\mu\nu}^{(W)} = \partial_\mu W_\nu - \partial_\nu W_\mu - i[W_\mu, W_\nu],$$

and

$$G_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu.$$

C. What gauge invariance alone implies

Assume only that the infrared effective action for $\mathcal{A}_\mu$ is local and invariant under the exact local $U(2)$ frame redundancy. Then:

a. Claim 1. A mass term

$$\int d^4x \text{Tr}(\mathcal{A}_\mu \mathcal{A}^\mu)$$

is not gauge invariant, hence is forbidden.

b. Proof. Under

$$\mathcal{A}_\mu \mapsto U^\dagger \mathcal{A}_\mu U - iU^\dagger \partial_\mu U,$$

the quantity $\text{Tr}(\mathcal{A}_\mu \mathcal{A}^\mu)$ picks up inhomogeneous derivative terms involving $U^\dagger \partial_\mu U$, and therefore is not invariant. Thus an exact local $U(2)$ redundancy forbids such a term. $\square$

c. Claim 2. The leading local gauge-invariant self-interaction of $\mathcal{A}_\mu$ begins with curvature invariants.

d. Proof. Gauge invariants without derivatives built solely from $\mathcal{A}_\mu$ do not exist because $\mathcal{A}_\mu$ is not tensorial. The first tensorial object is $\mathcal{F}_{\mu\nu}$, which transforms covariantly. Therefore the first local gauge-invariant terms are built from $\mathcal{F}_{\mu\nu}$, e.g.

$$\text{Tr}(\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}), \quad (\text{Tr } \mathcal{F}_{\mu\nu})(\text{Tr } \mathcal{F}^{\mu\nu}).$$

$\square$

Thus, if a local self-action exists, then to leading order it must take the form

$$\mathcal{L}_{\text{gauge}} = -\frac{c_2}{2} \text{Tr}(F_{\mu\nu}^{(W)} F^{(W)\mu\nu}) - \frac{c_1}{4} G_{\mu\nu} G^{\mu\nu} + \dots$$

for some coefficients c_1, c_2 .

This conclusion is kinematic and does not yet show that $c_1, c_2 \neq 0$.

D. Conditional dynamical theorem from integrating out Q -doublets

Assume now the existence of a dynamical field χ transforming in the fundamental representation of the local $U(2)$ frame bundle:

$$\chi(x) \mapsto U^{-1}(x)\chi(x).$$

Its covariant derivative is

$$D_\mu \chi = \partial_\mu \chi - i\mathcal{A}_\mu \chi.$$

We present two explicit model classes.

1. Model A: massive complex scalar doublet

Let χ be a complex scalar doublet with Euclidean action

$$S_\chi^{(E)} = \int d^4x \left[(D_\mu \chi)^\dagger (D_\mu \chi) + M^2 \chi^\dagger \chi \right].$$

Integrating out χ gives

$$Z[\mathcal{A}] = \int \mathcal{D}\chi \mathcal{D}\chi^\dagger e^{-S_\chi^{(E)}} = \left[\det(-D^2 + M^2) \right]^{-1},$$

hence

$$\Gamma_{\text{eff}}^{(E)}[\mathcal{A}] = -\log Z[\mathcal{A}] = \text{Tr} \log(-D^2 + M^2).$$

Using the proper-time representation,

$$\Gamma_{\text{eff}}^{(E)} = - \int_0^\infty \frac{ds}{s} e^{-sM^2} \text{Tr} e^{-s(-D^2)}.$$

For a Laplace-type operator on flat four-dimensional spacetime, the heat-kernel expansion is

$$\text{Tr} e^{-s(-D^2)} \sim \frac{1}{(4\pi s)^2} \int d^4x \text{tr} [a_0 + sa_1 + s^2a_2 + \dots].$$

For a pure gauge connection background,

$$a_2 = \frac{1}{12} \mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}.$$

Therefore,

$$\Gamma_{\text{eff}}^{(E)} \supset -\frac{1}{12(4\pi)^2} \left(\int_0^\infty \frac{ds}{s} e^{-sM^2} \right) \int d^4x \text{tr}(\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}).$$

After renormalization, this yields a local term of the form

$$\Gamma_{\text{eff}}^{(E)} \supset \kappa_s \int d^4x \text{tr}(\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}),$$

with a nonzero coefficient κ_s depending on the matter content and renormalization scale.

Thus a massive scalar Q -doublet induces a Yang–Mills kinetic term.

2. Model B: massive Dirac doublet

Let ψ be a Dirac fermion in the fundamental of $U(2)$, with Euclidean action

$$S_\psi^{(E)} = \int d^4x \bar{\psi}(\gamma^\mu D_\mu + M)\psi.$$

Then

$$Z[\mathcal{A}] = \det(\gamma^\mu D_\mu + M), \quad \Gamma_{\text{eff}}^{(E)} = -\log \det(\gamma^\mu D_\mu + M).$$

Squaring the Dirac operator,

$$(\gamma^\mu D_\mu + M)(-\gamma^\nu D_\nu + M) = -D^2 + M^2 + \frac{1}{2}\sigma^{\mu\nu}\mathcal{F}_{\mu\nu},$$

so the heat-kernel again generates

$$\Gamma_{\text{eff}}^{(E)} \supset \kappa_f \int d^4x \text{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu}),$$

with nonzero κ_f .

Hence massive Q -charged fermions also induce a Yang–Mills kinetic term.

E. Decomposition into $SU(2)$ and $U(1)$

Since

$$\mathcal{F}_{\mu\nu} = F_{\mu\nu}^{(W)} + \frac{1}{2}G_{\mu\nu}\mathbf{1}_2,$$

we have

$$\text{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu}) = \text{tr}(F_{\mu\nu}^{(W)}F^{(W)\mu\nu}) + \frac{1}{2}G_{\mu\nu}G^{\mu\nu}.$$

Indeed, the mixed term vanishes because

$$\text{tr}(F_{\mu\nu}^{(W)}) = 0.$$

Using

$$F_{\mu\nu}^{(W)} = \frac{1}{2}F_{\mu\nu}^a\sigma^a, \quad \text{tr}(\sigma^a\sigma^b) = 2\delta^{ab},$$

we find

$$\text{tr}(F_{\mu\nu}^{(W)}F^{(W)\mu\nu}) = \frac{1}{2}F_{\mu\nu}^aF^{a\mu\nu}.$$

Therefore the induced term splits as

$$\Gamma_{\text{eff}} \supset \int d^4x \left[\frac{\kappa}{2}F_{\mu\nu}^aF^{a\mu\nu} + \frac{\kappa}{2}G_{\mu\nu}G^{\mu\nu} \right]$$

up to normalization conventions and possible representation-dependent coefficients. After canonically normalizing fields, this becomes

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4g_2^2}F_{\mu\nu}^aF^{a\mu\nu} - \frac{1}{4g_1^2}G_{\mu\nu}G^{\mu\nu} + \dots$$

F. Precise logical status of the theorem

We may now state the main result carefully.

a. Conditional induced-gauge theorem. Suppose that:

1. the TECT internal sector contains the rank-two bundle $Q \rightarrow \mathbb{CP}^2$,
2. the corresponding local orthonormal frame redundancy $U(2)$ is exact,
3. there exist dynamical gapped fields transforming in the fundamental of Q ,
4. these fields can be integrated out in a local derivative expansion.

Then the infrared effective action for the composite connection $\mathcal{A}_\mu$ contains a gauge-invariant Yang–Mills term

$$\Gamma_{\text{eff}}[\mathcal{A}] \supset \int d^4x \left[-\frac{1}{4g_2^2} F_{\mu\nu}^a F^{a\mu\nu} - \frac{1}{4g_1^2} G_{\mu\nu} G^{\mu\nu} \right].$$

b. Proof. By assumptions (1)–(2), $\mathcal{A}_\mu$ is a legitimate $U(2)$ gauge connection. By assumption (3), there exist matter operators with covariant derivative D_μ . By assumption (4), integrating them out yields a local gauge-invariant effective action. Gauge invariance forbids mass terms for $\mathcal{A}_\mu$, while locality and derivative expansion imply that the leading self-interaction is built from $\mathcal{F}_{\mu\nu}$. The explicit heat-kernel expansion of the scalar or Dirac determinant produces a nonzero $\text{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu})$ term. Decomposition into traceless and trace pieces yields the $SU(2)$ and $U(1)$ kinetic terms. $\square$

G. What this does *not* prove

This theorem does not yet prove:

1. that the full microscopic TECT condensate PDE necessarily contains such a Q -charged field,
2. that the induced coefficients g_2, g_1 match the Standard Model,
3. that the $U(1)$ factor is precisely hypercharge,
4. that the $SU(2)$ factor acts chirally only on left-handed fermions,
5. that anomaly cancellation holds.

Thus the derivation closes the *structural mechanism* for induced $SU(2) \times U(1)$ gauge dynamics, but not yet the complete electroweak sector.

V. FINAL SUMMARY

The composite connection

$$\mathcal{A}_\mu = -ie^\dagger \partial_\mu e \in \mathfrak{u}(2)$$

arises rigorously from the local orthonormal frame redundancy of the rank-two bundle

$$Q = (\mathbb{C}z)^\perp \rightarrow \mathbb{CP}^2.$$

From gauge invariance alone, one can prove:

1. a local mass term for $\mathcal{A}_\mu$ is forbidden,
2. the leading local self-interaction must be built from the curvature

$$\mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - i[\mathcal{A}_\mu, \mathcal{A}_\nu].$$

Therefore any local infrared gauge-invariant effective action must begin as

$$\mathcal{L}_{\text{eff}} = -\frac{c_2}{2} \text{Tr}(F_{\mu\nu}^{(W)} F^{(W)\mu\nu}) - \frac{c_1}{4} G_{\mu\nu} G^{\mu\nu} + \dots.$$

However, geometry alone does *not* prove that $c_1, c_2 \neq 0$.

A nonzero Yang–Mills kinetic term is obtained conditionally if there exist dynamical fields χ or ψ transforming in the fundamental representation of the Q -bundle and if these fields may be integrated out in a local derivative expansion. In that case the one-loop determinant generates

$$\Gamma_{\text{eff}}[\mathcal{A}] \supset \kappa \int d^4x \text{tr}(\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}),$$

which decomposes into

$$-\frac{1}{4g_2^2} F_{\mu\nu}^a F^{a\mu\nu} - \frac{1}{4g_1^2} G_{\mu\nu} G^{\mu\nu}.$$

Hence the mathematically honest conclusion is:

Geometric $U(2)$ emergence is proved;

Yang–Mills dynamics is conditionally induced once Q -charged matter exists.

What remains open is the full microscopic TECT derivation of the required charged modes, their chirality structure, hypercharge embedding, and anomaly cancellation.

VI. DO Q -CHARGED MATTER MODES EXIST? A PRECISE BUNDLE-THEORETIC AUDIT

Assumption Audit

- A1: The internal order parameter is a normalized complex vector

$$z \in \mathbb{C}^3, \quad z^\dagger z = 1,$$

modulo local phase, so that the physical internal manifold is

$$\mathbb{CP}^2.$$

Proved

- A2: The orthogonal complement

$$Q = (\mathbb{C}z)^\perp$$

is a rank-two Hermitian vector bundle over $\mathbb{CP}^2$, with local orthonormal frame redundancy $U(2)$.

Proved

- A3: Any local section of Q transforms in the fundamental representation of the local $U(2)$ frame bundle.

Provable in current framework

- A4: Physical defect-bound, valley, or collective modes of TECT are realized as sections of Q .

Needs extra assumption

- A5: In adiabatic or restricted defect constructions, the fluctuation space around a defect contains a natural Q -valued sector.

Provable in restricted constructions

- A6: The relevant zero modes exist as kernels of a linearized operator or twisted Dirac-type operator.

Provable in model operators, not yet from full TECT PDE

- A7: These modes are exactly the Standard Model chiral electroweak doublets.

Currently unproved

A. Goal

We aim to separate the following logically distinct statements:

1. **Bundle-theoretic statement:** sections of Q transform as $U(2)$ fundamentals.
2. **Restricted physical statement:** for certain defect or collective-coordinate constructions, a natural fluctuation sector is Q -valued.
3. **Full microscopic TECT statement:** the actual low-energy matter modes of the condensate necessarily include such Q -valued fields.
4. **Standard Model identification:** those Q -valued modes coincide with chiral electroweak doublets.

Only the first is fully unconditional at present.

B. Rank-two bundle Q and local sections

Let

$$z(x) \in \mathbb{C}^3, \quad z^\dagger z = 1, \quad z \sim e^{i\alpha} z.$$

The physical internal configuration is the projective class $[z] \in \mathbb{CP}^2$.

Define the tautological complex line

$$L_{[z]} = \mathbb{C}z$$

and the orthogonal complement

$$Q_{[z]} = (\mathbb{C}z)^\perp = \{v \in \mathbb{C}^3 \mid z^\dagger v = 0\}.$$

Then $Q \rightarrow \mathbb{CP}^2$ is a rank-two complex Hermitian bundle.

Choose a local orthonormal frame

$$e = (e_1, e_2), \quad z^\dagger e_a = 0, \quad e_a^\dagger e_b = \delta_{ab}.$$

Any local section $s(x) \in Q_{[z(x)]}$ can be expanded uniquely as

$$s(x) = e(x) \chi(x), \quad \chi(x) \in \mathbb{C}^2.$$

Under a local frame change

$$e(x) \mapsto e(x)U(x), \quad U(x) \in U(2),$$

the same geometric section s must remain invariant:

$$s = e\chi = (eU)\chi'.$$

Therefore

$$\chi' = U^{-1}\chi.$$

Hence the coefficient field χ transforms in the fundamental representation of the local $U(2)$ frame bundle.

This proves the following unconditional theorem.

a. Theorem 1 (Bundle-fundamental theorem). Let $Q \rightarrow \mathbb{CP}^2$ be the rank-two orthogonal complement bundle of the tautological line. Then any local section of Q , expressed in a local orthonormal frame, transforms in the fundamental representation of the local $U(2)$ frame redundancy.

b. Proof. The proof is exactly the computation above:

$$s = e\chi = (eU)\chi' \implies \chi' = U^{-1}\chi.$$

□

C. Covariant derivative and minimal coupling

Since

$$\mathcal{A}_\mu = -ie^\dagger \partial_\mu e \in \mathfrak{u}(2),$$

the covariant derivative acting on the coefficient field χ is

$$D_\mu \chi = \partial_\mu \chi - i\mathcal{A}_\mu \chi.$$

Indeed, differentiating the section $s = e\chi$,

$$\partial_\mu s = (\partial_\mu e)\chi + e \partial_\mu \chi.$$

Projecting back to Q with $e^\dagger$, we obtain

$$e^\dagger \partial_\mu s = e^\dagger (\partial_\mu e)\chi + \partial_\mu \chi = i\mathcal{A}_\mu \chi + \partial_\mu \chi.$$

Thus the covariant derivative is forced by the bundle geometry:

$$D_\mu \chi = \partial_\mu \chi - i\mathcal{A}_\mu \chi.$$

This establishes a second unconditional fact: if a physical field is realized as a section of Q , then it is necessarily minimally coupled to the composite $U(2)$ connection.

a. Corollary 1. Any physical excitation identified with a local section of Q is automatically a $U(2)$ doublet and couples to the composite connection $\mathcal{A}_\mu$ through the covariant derivative $D_\mu = \partial_\mu - i\mathcal{A}_\mu$.

D. Restricted route I: tangent fluctuations of $\mathbb{CP}^2$

The tangent space of $\mathbb{CP}^2$ at $[z]$ may be described as

$$T_{[z]}\mathbb{CP}^2 \cong \text{Hom}(L_{[z]}, Q_{[z]}).$$

Since $L_{[z]}$ is one-dimensional,

$$\text{Hom}(L, Q) \cong L^* \otimes Q.$$

Thus infinitesimal deformations δz orthogonal to the phase direction obey

$$z^\dagger \delta z = 0,$$

so that

$$\delta z \in Q_{[z]}.$$

In a local gauge where the line L is trivialized, tangent fluctuations may therefore be represented locally by Q -valued fields.

This gives a precise but limited statement:

a. Theorem 2 (Local tangent-mode theorem). Infinitesimal internal fluctuations of the projective order parameter, modulo phase, are locally Q -valued. Hence their local coordinate fields transform as $U(2)$ fundamentals.

b. Proof. A variation δz preserving the normalization and modding out the phase direction satisfies

$$z^\dagger \delta z = 0,$$

which means $\delta z \in Q$. Expressing $\delta z = e\chi$, Theorem 1 gives

$$\chi \mapsto U^{-1}\chi.$$

□

However, this theorem describes *bosonic tangent fluctuations* of the internal order parameter. It does not yet identify them with fermions, defect zero modes, or Standard Model matter.

E. Restricted route II: defect moduli and normal fluctuations

Suppose there exists a defect background

$$z_0(x; a)$$

depending on collective coordinates a^I . Linearizing around it,

$$z(x) = z_0(x; a) + \delta z(x).$$

If the defect preserves the normalization and projective constraints to linear order, then admissible fluctuations satisfy

$$z_0^\dagger \delta z = 0$$

modulo gauge/phase directions. Hence the normal fluctuation sector lies in

$$\delta z(x) \in Q_{z_0(x)}.$$

Then the coefficients of $\delta z = e\chi$ again transform as a local $U(2)$ doublet.

This yields a restricted structural statement.

a. Theorem 3 (Restricted defect-doublet theorem). If a defect background in the TECT internal sector admits linearized normal fluctuations that are orthogonal to the tautological line $L = \mathbb{C}z_0$, then those fluctuation coefficients form a local $U(2)$ doublet.

b. Proof. The orthogonality condition implies $\delta z \in Q$. Expand in a local orthonormal frame

$$\delta z = e\chi.$$

Then Theorem 1 implies

$$\chi \mapsto U^{-1}\chi.$$

□

Again, this is a *conditional* theorem: it applies once such a defect background exists and once the normal fluctuation sector is indeed Q -valued.

F. Restricted route III: twisted Dirac operators and zero modes

A more ambitious route uses a Dirac-type operator on the internal defect geometry. Let

$$D_Q$$

be a twisted Dirac operator whose twisting bundle is built from Q or from an associated bundle derived from the $\mathbb{CP}^2$ geometry. Then any zero mode

$$D_Q \psi = 0$$

takes values in the corresponding twisted spinor bundle.

If the twisting factor contains Q , then locally

$$\psi = e\eta$$

for some coefficient field $\eta \in \mathbb{C}^2$, and under frame changes

$$\eta \mapsto U^{-1}\eta.$$

Therefore the zero mode again transforms as a $U(2)$ fundamental in its bundle index.

This gives the following model theorem.

- a. Theorem 4 (Model twisted-zero-mode theorem).* Let D_Q be a Dirac-type operator twisted by a bundle containing the factor Q . Then any local zero mode carries a $U(2)$ fundamental index inherited from Q .
- b. Proof.* The twisted spinor bundle has the local product structure

$$S \otimes Q$$

or a direct summand containing Q . Choosing a local frame e of Q , any section admits the decomposition

$$\psi = \sum_{a=1}^2 e_a \otimes \eta^a.$$

Under $e \mapsto eU$, the coefficient vector transforms as

$$\eta \mapsto U^{-1}\eta.$$

Hence the Q -bundle index is in the fundamental representation of $U(2)$. $\square$

This is mathematically correct, but it still does *not* prove that the actual TECT defect operator is of this form.

G. What would be sufficient for a full TECT theorem

We now isolate a sufficient criterion for the existence of physical Q -charged matter.

- a. Proposition 5 (Sufficient criterion).* Suppose the microscopic TECT linearized operator around a background z_0 decomposes as

$$\mathcal{L}_{\text{lin}} = \mathcal{L}_{\parallel} \oplus \mathcal{L}_{\perp}$$

with

$$\mathcal{L}_{\perp} : \Gamma(Q) \rightarrow \Gamma(Q),$$

and suppose $\mathcal{L}_{\perp}$ possesses normalizable low-energy eigenmodes

$$\mathcal{L}_{\perp} \chi_n = \lambda_n \chi_n, \quad |\lambda_n| \ll \Lambda.$$

Then the corresponding low-energy excitations are physical Q -valued modes, hence local $U(2)$ doublets.

- b. Proof.* By assumption, the relevant low-energy modes are sections of Q . By Theorem 1 their local coefficient fields transform in the fundamental representation of the local $U(2)$ frame bundle. $\square$

This proposition is important because it shows exactly what remains to be proven microscopically: one must derive a Q -sector in the linearized condensate operator and show that it carries normalizable low-energy modes.

H. What is actually established at present

We may summarize the precise logical status as follows.

- a. Unconditional facts.*

1. $Q \rightarrow \mathbb{CP}^2$ is a rank-two Hermitian bundle.
2. Local sections of Q transform as $U(2)$ fundamentals.
3. If a physical mode is Q -valued, it couples minimally to the composite $U(2)$ connection.

- b. Conditionally established facts.*

1. Tangent fluctuations of the internal projective order parameter are locally Q -valued.
2. Defect normal modes are Q -doublets if the admissible fluctuation sector is orthogonal to the tautological line.
3. Twisted Dirac zero modes carry a $U(2)$ fundamental bundle index if the twisting bundle contains Q .

c. Not yet established.

1. The full microscopic TECT PDE necessarily produces such Q -valued matter modes.
2. Those modes are fermionic electroweak doublets rather than bosonic internal fluctuations.
3. The action is chiral in exactly the Standard Model way.

VII. FINAL SUMMARY

The central question is whether the TECT internal geometry produces physical matter fields transforming in the fundamental representation of the emergent local $U(2)$ bundle structure.

The precise bundle-theoretic result is:

$$Q = (\mathbb{C}z)^\perp \rightarrow \mathbb{CP}^2$$

is a rank-two Hermitian bundle, and every local section

$$s = e \chi$$

of Q , written in a local orthonormal frame e , transforms as

$$\chi \mapsto U^{-1}\chi, \quad U(x) \in U(2).$$

Therefore:

Any physical excitation realized as a section of Q is automatically a local $U(2)$ doublet.

Moreover, such a mode couples minimally to the composite connection

$$\mathcal{A}_\mu = -ie^\dagger \partial_\mu e$$

through

$$D_\mu \chi = \partial_\mu \chi - i\mathcal{A}_\mu \chi.$$

This is fully rigorous.

What is only conditionally true is the physical realization of such sections: tangent fluctuations, defect normal modes, and twisted Dirac zero modes can all produce Q -valued sectors, but only under additional assumptions about the microscopic operator or defect background.

Thus the honest status is:

The $U(2)$ -doublet transformation law is proved.

The existence of actual TECT Q -charged matter modes is not yet proved microscopically.

A sufficient criterion for the full theorem would be to derive, from the linearized TECT condensate dynamics, an invariant sector

$$\mathcal{L}_\perp : \Gamma(Q) \rightarrow \Gamma(Q)$$

with normalizable low-energy eigenmodes. Only then would the existence of physical $U(2)$ doublet matter be fully established.

VIII. BLOCK DECOMPOSITION OF THE LINEARIZED INTERNAL FLUCTUATION OPERATOR

Assumption Audit

- A1: The physical internal field is a projective complex vector

$$z(x) \in \mathbb{C}^3, \quad z^\dagger z = 1, \quad z \sim e^{i\alpha(x)} z,$$

so the target space is $\mathbb{CP}^2$.

Proved

- A2: At each point z , the ambient space decomposes as

$$\mathbb{C}^3 = L_z \oplus Q_z, \quad L_z = \mathbb{C}z, \quad Q_z = (\mathbb{C}z)^\perp.$$

Proved

- A3: Infinitesimal physical fluctuations modulo normalization and phase are represented by Q -valued perturbations.

Provable in current framework

- A4: The linearized operator preserves the Q -sector:

$$\mathcal{L}_{\text{lin}} : \Gamma(Q) \rightarrow \Gamma(Q).$$

Needs extra assumption in general

- A5: For projective, $U(1)$ -redundant sigma-model/projector-type dynamics, the linearized operator restricted to physical fluctuations closes on Q .

Provable in current framework

- A6: The full microscopic TECT condensate PDE has the same block structure.

Currently unproved

A. Goal

We want to determine when the linearized fluctuation operator around a background

$$z_0(x) \in \mathbb{C}^3, \quad z_0^\dagger z_0 = 1,$$

admits a decomposition into longitudinal and transverse/projective sectors, schematically

$$\mathcal{L}_{\text{lin}} = \mathcal{L}_{\parallel} \oplus \mathcal{L}_{\perp},$$

with

$$\mathcal{L}_{\perp} : \Gamma(Q_{z_0}) \rightarrow \Gamma(Q_{z_0}).$$

We distinguish carefully between:

1. the kinematic decomposition of the ambient space $\mathbb{C}^3 = L \oplus Q$,
2. the representation of physical tangent fluctuations by Q -valued fields,
3. the dynamical closure of the linearized PDE on that Q -sector.

Only the first two are unconditional at present.

B. Ambient-space decomposition and projectors

Let $z_0(x)$ be a background section with

$$z_0^\dagger z_0 = 1.$$

Define the rank-one projector

$$P := z_0 z_0^\dagger, \quad P^2 = P, \quad P^\dagger = P, \quad \text{Tr } P = 1,$$

and the orthogonal complement projector

$$\Pi := \mathbf{1}_3 - P.$$

Then

$$\Pi^2 = \Pi, \quad \Pi^\dagger = \Pi, \quad \Pi z_0 = 0.$$

Hence

$$L_{z_0} = \text{Im}(P) = \mathbb{C}z_0, \quad Q_{z_0} = \text{Im}(\Pi) = (\mathbb{C}z_0)^\perp.$$

Any ambient fluctuation $\delta z \in \mathbb{C}^3$ splits uniquely as

$$\delta z = P\delta z + \Pi\delta z.$$

Since $P\delta z = z_0(z_0^\dagger \delta z)$, the longitudinal component lies along z_0 , while $\Pi\delta z \in Q_{z_0}$.

C. Physical infinitesimal fluctuations of $\mathbb{CP}^2$

Because the physical field is projective, not all δz correspond to physical directions. We must quotient out:

1. normalization changes,
2. phase changes $\delta z \sim i\alpha z_0$.

Linearizing the normalization constraint

$$(z_0 + \delta z)^\dagger (z_0 + \delta z) = 1$$

gives

$$z_0^\dagger \delta z + \delta z^\dagger z_0 = 0,$$

equivalently

$$2 \text{Re}(z_0^\dagger \delta z) = 0.$$

Thus $z_0^\dagger \delta z$ is purely imaginary. But an infinitesimal phase transformation

$$\delta z = i\alpha z_0$$

has exactly

$$z_0^\dagger \delta z = i\alpha.$$

Hence, modulo phase, every physical infinitesimal fluctuation can be represented by a gauge-fixed perturbation satisfying

$$z_0^\dagger \delta z = 0.$$

Therefore physical fluctuations may be represented locally as sections of Q_{z_0} .

a. *Theorem 1 (Physical tangent representation).* The tangent space to $\mathbb{CP}^2$ at $[z_0]$ is represented by

$$T_{[z_0]}\mathbb{CP}^2 \cong Q_{z_0}$$

after choosing a local phase gauge.

b. *Proof.* The ambient variation space is $\mathbb{C}^3$. The linearized unit-sphere condition removes one real direction, and quotienting by the phase $U(1)$ removes another real direction. The remaining four real dimensions form the tangent space of $\mathbb{CP}^2$, and are represented by the complex two-dimensional orthogonal complement

$$Q_{z_0} = \{v \in \mathbb{C}^3 \mid z_0^\dagger v = 0\}.$$

□

D. General linearized operator and obstruction to block-diagonalization

Let the microscopic or effective equation of motion be

$$E[z] = 0,$$

with E covariant under phase redundancy. Linearizing around a background z_0 gives

$$\mathcal{L}_{\text{lin}} \delta z = 0, \quad \mathcal{L}_{\text{lin}} := \left. \frac{\delta E}{\delta z} \right|_{z_0}.$$

In the ambient decomposition $\delta z = P\delta z + \Pi\delta z$, one may always write

$$\mathcal{L}_{\text{lin}} = \begin{pmatrix} P\mathcal{L}_{\text{lin}}P & P\mathcal{L}_{\text{lin}}\Pi \\ \Pi\mathcal{L}_{\text{lin}}P & \Pi\mathcal{L}_{\text{lin}}\Pi \end{pmatrix}$$

relative to $L \oplus Q$.

Thus the Q -sector is dynamically closed if and only if

$$P\mathcal{L}_{\text{lin}}\Pi = 0 \quad \text{on admissible fluctuations,}$$

and similarly the longitudinal sector decouples if

$$\Pi\mathcal{L}_{\text{lin}}P = 0.$$

In a completely general complex 3-component PDE, these off-diagonal blocks need not vanish.

Therefore the statement

$$\mathcal{L}_{\text{lin}} : \Gamma(Q) \rightarrow \Gamma(Q)$$

is not automatic from linear algebra alone.

E. Projective sigma-model structure

Consider now the standard projector-based or $\mathbb{CP}^2$ -sigma-type action

$$S[z] = \frac{\rho}{2} \int d^d x (D_\mu z)^\dagger (D^\mu z),$$

where

$$D_\mu z := \Pi \partial_\mu z = \partial_\mu z - z(z^\dagger \partial_\mu z).$$

Equivalently, in projector form,

$$S[P] = \frac{\rho}{4} \int d^d x \text{Tr}(\partial_\mu P \partial^\mu P).$$

The key property is

$$D_\mu z \in Q_z, \quad z^\dagger D_\mu z = 0.$$

Thus the dynamics depends only on derivatives projected orthogonally to the tautological line.

F. Linearization in background gauge

Let

$$z = z_0 + \eta, \quad z_0^\dagger \eta = 0.$$

This is the natural background gauge fixing of the projective redundancy.

We now linearize $D_\mu z$. Since

$$D_\mu z = \Pi(z) \partial_\mu z, \quad \Pi(z) = \mathbf{1} - z z^\dagger,$$

we obtain to first order

$$\delta(D_\mu z) = (\delta\Pi) \partial_\mu z_0 + \Pi_0 \partial_\mu \eta,$$

where

$$\Pi_0 = \mathbf{1} - z_0 z_0^\dagger, \quad \delta\Pi = -\eta z_0^\dagger - z_0 \eta^\dagger.$$

Using $z_0^\dagger \eta = 0$, we find

$$\delta\Pi \partial_\mu z_0 = -\eta(z_0^\dagger \partial_\mu z_0) - z_0(\eta^\dagger \partial_\mu z_0).$$

The second term is longitudinal, proportional to z_0 . It is a gauge artifact associated with the phase bundle, and may be absorbed into the induced $U(1)$ connection of the tautological line. The genuinely physical transverse fluctuation is therefore the Q -projected quantity

$$\Pi_0 \delta(D_\mu z) = -\eta(z_0^\dagger \partial_\mu z_0) + \Pi_0 \partial_\mu \eta.$$

Since both terms lie in Q , the principal part closes on Q .

This suggests introducing the background $U(1)$ covariant derivative

$$\nabla_\mu^{(0)} \eta := \Pi_0 \partial_\mu \eta - (z_0^\dagger \partial_\mu z_0) \eta.$$

Then

$$\nabla_\mu^{(0)} \eta \in Q_{z_0}.$$

a. Theorem 2 (Closure of the principal linearized operator). For projective $\mathbb{CP}^2$ -sigma dynamics written in terms of $D_\mu z = \Pi \partial_\mu z$, the linearized physical fluctuation operator around a background z_0 , in the gauge $z_0^\dagger \eta = 0$, has principal part acting within $\Gamma(Q_{z_0})$.

b. Proof. The linearized derivative is the Q -valued object

$$\Pi_0 \delta(D_\mu z) = \nabla_\mu^{(0)} \eta.$$

Hence the second-order principal term appearing in the Euler–Lagrange operator is of the form

$$-(\nabla_\mu^{(0)} \nabla^{(0)\mu}) \eta,$$

which maps $\Gamma(Q_{z_0})$ to itself. $\square$

G. Explicit Hessian structure

Let us now sketch the quadratic form. Writing

$$z = z_0 + \eta, \quad z_0^\dagger \eta = 0,$$

the action expands as

$$S[z_0 + \eta] = S[z_0] + \delta S[z_0; \eta] + \frac{1}{2} \delta^2 S[z_0; \eta] + \cdots.$$

If z_0 satisfies the background equation of motion, then

$$\delta S[z_0; \eta] = 0.$$

The quadratic term has the general form

$$\delta^2 S = \int d^d x \eta^\dagger \left[-\nabla_\mu^{(0)} \nabla^{(0)\mu} + \mathcal{M}[z_0] \right] \eta,$$

where $\mathcal{M}[z_0]$ is a background-dependent Hermitian endomorphism of Q .

Thus the linearized fluctuation operator takes the form

$$\mathcal{L}_\perp = -\nabla_\mu^{(0)} \nabla^{(0)\mu} + \mathcal{M}[z_0], \quad \mathcal{L}_\perp : \Gamma(Q_{z_0}) \rightarrow \Gamma(Q_{z_0}).$$

a. Theorem 3 (Restricted block decomposition theorem). For any projective sigma-model or projector-type internal dynamics depending only on the Q -projected derivatives of z , the physical Hessian around a background z_0 defines a closed operator

$$\mathcal{L}_\perp : \Gamma(Q_{z_0}) \rightarrow \Gamma(Q_{z_0}).$$

b. Proof. The proof follows from three facts:

1. physical fluctuations are represented by $\eta \in \Gamma(Q_{z_0})$,
2. the projected derivative $\nabla_\mu^{(0)} \eta$ remains Q -valued,
3. the Hessian is built from repeated applications of this covariant derivative and background endomorphisms preserving Q .

Therefore the quadratic operator maps Q -sections to Q -sections. $\square$

H. Why this does not yet prove the full TECT statement

The above theorem is rigorous for sigma-model/projector-type dynamics, but not yet for an arbitrary microscopic condensate PDE. A general TECT microscopic equation might contain:

1. anisotropic couplings mixing longitudinal and transverse sectors,
2. higher-derivative nonprojective terms,
3. explicit dependence on an ambient $\mathbb{C}^3$ representative rather than only on the projective class $[z]$,
4. couplings to additional condensate variables that reintroduce L - Q mixing.

In those cases the off-diagonal blocks

$$P\mathcal{L}_{\text{lin}}\Pi, \quad \Pi\mathcal{L}_{\text{lin}}P$$

need not vanish.

Hence the full microscopic theorem would require proving that the TECT PDE itself is projective in the relevant internal sector, or at least that its linearization around the chosen background admits a gauge-compatible reduction to Q .

I. Sufficient criterion for full microscopic closure

We isolate the exact condition needed.

a. Proposition 4 (Sufficient criterion for Q -sector closure). Let the TECT internal equation $E[z] = 0$ satisfy the following around a background z_0 :

1. $E[z]$ depends on z only through the projective data $[z]$, equivalently through $P = zz^\dagger$ or Q -projected derivatives,
2. the linearized phase gauge is fixed by $z_0^\dagger \eta = 0$,
3. all background-dependent endomorphisms preserve Q_{z_0} .

Then the physical linearized operator closes on Q_{z_0} :

$$\mathcal{L}_{\text{lin}}^{\text{phys}} : \Gamma(Q_{z_0}) \rightarrow \Gamma(Q_{z_0}).$$

b. Proof. Under assumptions (1)–(3), the equation of motion and its linearization are built only from operators that preserve the orthogonal decomposition into the tautological line and its orthogonal complement after gauge fixing. Therefore the projected physical Hessian acts entirely within $\Gamma(Q_{z_0})$. $\square$

J. Consequences for matter-mode emergence

Once a closed operator

$$\mathcal{L}_\perp : \Gamma(Q) \rightarrow \Gamma(Q)$$

exists, its eigenmodes

$$\mathcal{L}_\perp \chi_n = \lambda_n \chi_n$$

are genuine Q -valued physical excitations. By the bundle-fundamental theorem, their local coefficients transform as

$$\chi_n \mapsto U^{-1} \chi_n, \quad U(x) \in U(2).$$

Therefore they are automatically local $U(2)$ doublets.

This does not yet prove that low-lying or zero eigenmodes exist, but it does show that the existence problem has been reduced to an honest spectral question for a closed Q -bundle operator.

IX. FINAL SUMMARY

Let

$$z_0(x) \in \mathbb{C}^3, \quad z_0^\dagger z_0 = 1,$$

and define

$$P = z_0 z_0^\dagger, \quad \Pi = \mathbf{1} - P.$$

Then the ambient fluctuation space decomposes as

$$\mathbb{C}^3 = L_{z_0} \oplus Q_{z_0}, \quad L_{z_0} = \mathbb{C} z_0, \quad Q_{z_0} = (\mathbb{C} z_0)^\perp.$$

After quotienting the normalization and phase redundancy of the projective field $[z] \in \mathbb{CP}^2$, physical infinitesimal fluctuations are represented by

$$\eta \in \Gamma(Q_{z_0}), \quad z_0^\dagger \eta = 0.$$

For a general microscopic complex three-component PDE, the linearized operator has block form

$$\mathcal{L}_{\text{lin}} = \begin{pmatrix} P \mathcal{L}_{\text{lin}} P & P \mathcal{L}_{\text{lin}} \Pi \\ \Pi \mathcal{L}_{\text{lin}} P & \Pi \mathcal{L}_{\text{lin}} \Pi \end{pmatrix},$$

and the off-diagonal blocks need not vanish. Therefore Q -sector closure is not automatic.

However, for projective sigma-model/projector-type dynamics depending only on Q -projected derivatives, the physical Hessian in background gauge $z_0^\dagger \eta = 0$ takes the form

$$\mathcal{L}_\perp = -\nabla_\mu^{(0)} \nabla^{(0)\mu} + \mathcal{M}[z_0], \quad \mathcal{L}_\perp : \Gamma(Q_{z_0}) \rightarrow \Gamma(Q_{z_0}),$$

with $\nabla_\mu^{(0)}$ the induced background covariant derivative on Q .

Thus:

The Q -sector is geometrically natural and dynamically closed for projective sigma-model-type dynamics.

But:

It is not yet proven that the full microscopic TECT condensate PDE has this same block decomposition.

If such closure holds microscopically, then the existence of physical $U(2)$ -doublet matter reduces to the spectral problem

$$\mathcal{L}_\perp \chi_n = \lambda_n \chi_n$$

for a closed operator acting on Q -valued sections.

X. SUFFICIENT CONDITIONS FOR ZERO MODES AND LIGHT MODES IN THE Q -SECTOR

Assumption Audit

- A1: A closed Q -sector operator exists,

$$\mathcal{L}_\perp : \Gamma(Q) \rightarrow \Gamma(Q).$$

Provable in restricted projector/sigma-model frameworks

- A2: The operator is elliptic/self-adjoint or otherwise admits a controlled spectral theory.

Needs extra assumption

- A3: Topology alone forces zero modes for a generic bosonic second-order operator.

False in general

- A4: A Q -twisted Dirac-type operator exists.

Provable in model constructions, not yet from full TECT

- A5: If the Dirac index is nonzero, then zero modes must exist.

Proved

- A6: Defect backgrounds may create localized bound states through mass inversion or trapping.

Provable under explicit spectral assumptions

- A7: The resulting modes coincide exactly with Standard Model chiral fermions.

Currently unproved

A. Goal

We separate three distinct mechanisms for low-energy modes in the Q -sector:

1. **Index-forced zero modes** for Dirac-type operators,
2. **Defect-trapped bound states** for Schrödinger/Dirac operators in nontrivial backgrounds,
3. **Accidental or symmetry-protected light modes** from small eigenvalues of a self-adjoint operator.

Only the first gives a robust topological theorem. The second gives useful sufficient criteria. The third is the weakest and most model-dependent.

B. Why generic bosonic second-order operators do not admit topological zero-mode theorems

Let

$$\mathcal{L}_\perp = -\nabla^\mu \nabla_\mu + \mathcal{M}(x)$$

be a second-order Hermitian operator acting on Q -valued sections over some internal or effective defect geometry. Even if Q is topologically nontrivial, the existence of a zero mode

$$\mathcal{L}_\perp \chi = 0$$

is not guaranteed by topology alone. Indeed, the spectrum depends on the endomorphism

$$\mathcal{M}(x),$$

and a generic positive shift

$$\mathcal{M}(x) \mapsto \mathcal{M}(x) + m_0^2 \mathbf{1}$$

can lift all eigenvalues away from zero.

Therefore:

a. Proposition 1. For a generic bosonic second-order operator on Q -valued sections, topology of the bundle Q alone does not force the existence of a zero mode.

b. Proof. Take any elliptic self-adjoint operator $\mathcal{L}_\perp$ and add a sufficiently large positive constant $m_0^2 \mathbf{1}$. The spectrum shifts by m_0^2 , eliminating any zero eigenvalue. Hence zero modes are not topologically protected for generic second-order bosonic operators. $\square$

This is a crucial honesty point: we must not overclaim topological protection unless the operator has first-order chiral structure.

C. Dirac-type operators and index protection

Now consider instead a Dirac-type operator twisted by the Q -bundle:

$$D_Q : \Gamma(S^+ \otimes Q \otimes \mathcal{E}) \rightarrow \Gamma(S^- \otimes Q \otimes \mathcal{E}),$$

where $S^\pm$ are chiral spinor bundles and $\mathcal{E}$ denotes any additional twisting.

Its chiral index is

$$\text{Index}(D_Q) = \dim \ker D_Q^+ - \dim \ker D_Q^-.$$

By the Atiyah–Singer index theorem,

$$\text{Index}(D_Q) = \int_M \widehat{A}(M) \text{ch}(Q \otimes \mathcal{E}) \Big|_{\dim M}.$$

Therefore:

a. Theorem 2 (Index-forced zero-mode theorem). If D_Q is a Fredholm Dirac-type operator and

$$\text{Index}(D_Q) \neq 0,$$

then D_Q has at least one zero mode.

b. Proof. If both kernels were empty, then

$$\dim \ker D_Q^+ = \dim \ker D_Q^- = 0,$$

hence $\text{Index}(D_Q) = 0$, contradiction. More generally, nonzero index implies that at least one of $\ker D_Q^\pm$ is nontrivial. $\square$

This is the cleanest route to protected chiral zero modes.

D. Connection with the previously derived $\mathbb{CP}^2$ index

In the model constructions already discussed, the internal defect geometry was associated with $\mathbb{CP}^2$ and a twisted spin^c Dirac operator with index

$$\text{Index}(D_+) = \frac{(2m+q)^2 - 1}{8}.$$

For primitive twisting

$$(q, m) = (1, 1),$$

this gives

$$\text{Index}(D_+) = 1.$$

Hence one obtains

$$n_L - n_R = 1,$$

which forces a chiral zero mode.

This remains mathematically correct for the model Dirac operator, but it does not yet prove that the same operator is derived from the full microscopic TECT condensate PDE.

Thus we have:

a. Corollary 3. If the effective low-energy defect operator in TECT is indeed the previously identified twisted Dirac operator on the relevant internal geometry, then a nonzero chiral zero mode is forced.

E. Defect-localized bound states without index theory

A second mechanism for light or zero modes is trapping by a defect background.

Consider a self-adjoint operator of Schrödinger type

$$\mathcal{L}_\perp = -\partial_\xi^2 + V(\xi)$$

in the coordinate ξ transverse to a codimension-one defect, or more generally

$$\mathcal{L}_\perp = -\nabla^2 + V(x)$$

on the defect background.

If $V(x)$ develops a sufficiently deep localized well, then the discrete spectrum contains a bound state

$$\mathcal{L}_\perp \chi_n = \lambda_n \chi_n, \quad \lambda_n < \lambda_{\text{cont}},$$

where λ_{cont} is the continuum threshold.

This gives:

a. Proposition 4 (Bound-state sufficient criterion). Suppose $\mathcal{L}_\perp$ is self-adjoint and its effective potential $V(x)$ contains a localized attractive region strong enough to create a discrete eigenvalue below the continuum. Then $\mathcal{L}_\perp$ has a normalizable localized light mode.

b. Proof. This is a standard consequence of spectral theory for Schrödinger-type operators: a sufficiently attractive localized potential well creates discrete spectrum below the essential spectrum. $\square$

This proves only a *sufficient* criterion, not a universal theorem.

F. Jackiw–Rebbi-type mechanism for first-order operators

For a one-dimensional transverse profile, consider a Dirac-type operator

$$\mathcal{D} = i\Gamma^\xi \partial_\xi + m(\xi),$$

with asymptotic mass profile satisfying

$$m(+\infty)m(-\infty) < 0.$$

Then the sign change in the mass can support a localized zero mode.

a. Theorem 5 (Jackiw–Rebbi-type sufficient criterion). If the effective first-order defect operator has a mass term whose sign flips across the defect, then there exists a normalizable localized zero mode.

b. Sketch of proof. The zero-mode equation

$$(i\Gamma^\xi \partial_\xi + m(\xi))\psi = 0$$

reduces, after choosing a definite Γ^ξ -eigencomponent, to

$$\partial_\xi \psi = \mp m(\xi)\psi.$$

Hence

$$\psi(\xi) \propto \exp\left(\mp \int^\xi m(\xi') d\xi'\right).$$

When $m(\xi)$ changes sign, one choice of sign yields exponential decay at both ends, producing a normalizable zero mode. $\square$

Again, this is powerful but model-dependent: one needs an explicit effective first-order operator with such a mass profile.

G. Light modes from approximate symmetry or near-BPS structure

A third route is not exact zero modes but parametrically small eigenvalues. Suppose

$$\mathcal{L}_\perp = \mathcal{L}_0 + \varepsilon \Delta\mathcal{L}, \quad 0 < \varepsilon \ll 1,$$

and $\mathcal{L}_0$ has an exact zero mode

$$\mathcal{L}_0 \chi_0 = 0.$$

Then by perturbation theory,

$$\lambda(\varepsilon) = \varepsilon \langle \chi_0, \Delta\mathcal{L} \chi_0 \rangle + O(\varepsilon^2).$$

Hence a small deformation of a symmetry-protected or BPS zero mode produces a light mode.

a. Proposition 6 (Near-zero-mode criterion). If an operator is a small deformation of one admitting an exact zero mode, then the zero mode persists as a light mode with eigenvalue $O(\varepsilon)$, provided no level crossing removes it.

b. Proof. This is standard analytic perturbation theory for isolated eigenvalues of self-adjoint operators. $\square$

This may be relevant if the TECT defect sector is approximately BPS, approximately chiral, or approximately factorized.

H. What is sufficient for a TECT theorem

We can now formulate the exact conditions under which a genuine TECT theorem would follow.

a. Theorem 7 (Sufficient criteria for physical Q -sector low-energy modes). Assume:

1. a closed Q -sector operator exists,

$$\mathcal{L}_\perp : \Gamma(Q) \rightarrow \Gamma(Q),$$

2. one of the following three conditions holds:

(I) $\mathcal{L}_\perp$ is derived from a Fredholm Dirac-type operator D_Q with

$$\text{Index}(D_Q) \neq 0,$$

or

(II) $\mathcal{L}_\perp$ is a self-adjoint defect operator whose effective potential has a localized attractive well generating discrete spectrum below the continuum, or

(III) $\mathcal{L}_\perp$ is a small deformation of an operator with an exact isolated zero mode.

Then $\mathcal{L}_\perp$ possesses either an exact zero mode or a parametrically light normalizable mode.

b. Proof. Case (I) follows from Theorem 2. Case (II) follows from Proposition 4. Case (III) follows from Proposition 6. $\square$

I. What remains missing for the full TECT program

Even if one proves the existence of such a mode, further steps are still needed:

1. identification of the actual microscopic operator producing it,
2. proof that the mode is fermionic rather than merely bosonic,
3. proof of chirality selection consistent with $SU(2)_L$,
4. hypercharge embedding,
5. anomaly cancellation.

Thus the low-energy-mode theorem is necessary, but not sufficient, for full Standard Model recovery.

XI. FINAL SUMMARY

Let

$$\mathcal{L}_\perp : \Gamma(Q) \rightarrow \Gamma(Q)$$

be the closed operator acting on the Q -sector.

There are three logically distinct mechanisms for low-energy modes:

(1) Index-forced zero modes

If $\mathcal{L}_\perp$ arises from a Fredholm Dirac-type operator D_Q with

$$\text{Index}(D_Q) \neq 0,$$

then a zero mode is forced:

$$\ker D_Q \neq 0.$$

This is the strongest and most rigorous mechanism.

(2) Defect-trapped bound states

If $\mathcal{L}_\perp$ is a self-adjoint defect operator with a sufficiently attractive localized effective potential, then it possesses a normalizable bound state

$$\mathcal{L}_\perp \chi = \lambda \chi, \quad \lambda < \lambda_{\text{cont}}.$$

This provides a light localized mode.

(3) Near-zero modes from deformation

If

$$\mathcal{L}_\perp = \mathcal{L}_0 + \varepsilon \Delta \mathcal{L}, \quad 0 < \varepsilon \ll 1,$$

and $\mathcal{L}_0$ has an exact isolated zero mode, then perturbation theory yields

$$\lambda(\varepsilon) = O(\varepsilon),$$

so the zero mode persists as a light mode.

What is not true

For a generic bosonic second-order operator,

$$\mathcal{L}_\perp = -\nabla^2 + \mathcal{M}(x),$$

the topology of Q alone does *not* force zero modes. Thus one must not claim topological protection without a Dirac/chiral structure.

Honest conclusion

The strongest rigorous route to protected Q -sector zero modes is:

derive a Q -twisted Dirac-type operator with nonzero index.

Without this, one may still obtain localized light modes from defect trapping or approximate symmetry, but these are more model-dependent and require explicit spectral analysis.

Therefore the next decisive mathematical task is not merely to postulate a Q -sector, but to show that the effective TECT defect operator in that sector is genuinely of Dirac type, or else to prove a concrete trapping theorem for the microscopic defect background.

XII. EMERGENCE OF AN EFFECTIVE FIRST-ORDER WEYL/DIRAC OPERATOR

Assumption Audit

- A1: The microscopic or effective linearized operator admits a momentum-space or Bloch-type description.

Needs extra assumption

- A2: Around a periodic BCC background, a Bloch decomposition exists.

Provable in periodic background models

- A3: There exists a momentum k_* with an isolated two-dimensional low-energy subspace.

Needs extra assumption

- A4: The projected operator admits a smooth expansion in $p = k - k_*$.

Provable in current framework

- A5: The linear term has full rank and produces a nondegenerate Weyl crossing.

Provable once the crossing assumptions are imposed

- A6: A symmetry relates k_* and $-k_*$, producing valley doubling.

Provable under symmetry assumptions

- A7: Defect backgrounds admit an envelope-mode reduction to an effective first-order operator.

Provable in restricted constructions

- A8: The full TECT BCC condensate realizes all of the above assumptions.

Currently unproved

A. Goal

We aim to prove the following conditional statement:

If a linearized TECT sector has an isolated two-band linear crossing, then its low-energy effective operator is of Weyl type.

A further symmetry pairing of opposite valleys then yields a Dirac operator.

The proof is structural and does not yet establish that the TECT microscopic PDE indeed possesses the required valleys.

B. Abstract linearized operator and spectral projection

Let $\mathcal{L}(k)$ be a Hermitian family of operators depending smoothly on momentum k near a point $k_* \in \mathbb{R}^3$. Assume that at k_* there exists an isolated two-dimensional eigenspace

$$\mathcal{H}_{\text{low}}(k_*) = \text{span}\{u_1, u_2\}, \quad \dim \mathcal{H}_{\text{low}}(k_*) = 2,$$

separated by a spectral gap from the rest of the spectrum.

Let

$$\Pi_* : \mathcal{H} \rightarrow \mathcal{H}_{\text{low}}(k_*)$$

be the corresponding spectral projector. Then for $p = k - k_*$, degenerate perturbation theory gives the effective Hamiltonian

$$H_{\text{eff}}(p) = \Pi_* \mathcal{L}(k_* + p) \Pi_*|_{\mathcal{H}_{\text{low}}} = H_0 + p_i V_i + O(|p|^2),$$

where

$$H_0 := \Pi_* \mathcal{L}(k_*) \Pi_*, \quad V_i := \Pi_* (\partial_{k_i} \mathcal{L})_{k_*} \Pi_*.$$

If the two low-energy levels are degenerate at k_* , then

$$H_0 = E_* \mathbf{1}_2.$$

Subtracting the irrelevant scalar $E_* \mathbf{1}_2$, we obtain

$$\tilde{H}_{\text{eff}}(p) = p_i V_i + O(|p|^2).$$

C. Hermitian 2×2 decomposition

Every Hermitian 2×2 matrix may be decomposed uniquely as

$$M = a_0 \mathbf{1}_2 + a_a \sigma^a, \quad a_0, a_a \in \mathbb{R}, \quad a = 1, 2, 3.$$

Hence each velocity matrix V_i may be written as

$$V_i = v_i^{(0)} \mathbf{1}_2 + v_i^{(a)} \sigma^a.$$

Therefore

$$\tilde{H}_{\text{eff}}(p) = \varepsilon_0(p) \mathbf{1}_2 + d_a(p) \sigma^a + O(|p|^2),$$

with

$$\varepsilon_0(p) = p_i v_i^{(0)}, \quad d_a(p) = p_i v_i^{(a)}.$$

The scalar term $\varepsilon_0(p) \mathbf{1}_2$ merely tilts or shifts the cone and is irrelevant for chirality. The essential part is the traceless term

$$H_W(p) = d_a(p) \sigma^a.$$

D. Weyl criterion

Define the 3×3 real matrix

$$M_{ai} := v_i^{(a)}.$$

Then

$$d_a(p) = M_{ai} p_i.$$

If

$$\det M \neq 0,$$

then $d(p) \neq 0$ for all sufficiently small $p \neq 0$, so the dispersion is conical:

$$E_{\pm}(p) = \varepsilon_0(p) \pm |M p| + O(|p|^2).$$

a. Theorem 1 (Two-band Weyl reduction). Assume:

1. $\mathcal{L}(k)$ is smooth near k_* ,
2. there is an isolated two-dimensional degenerate low-energy subspace at k_* ,
3. the linearized traceless velocity matrix M_{ai} is invertible.

Then the projected low-energy operator is, up to linear changes of momentum coordinates and unitary basis changes in the two-dimensional eigenspace, of Weyl form:

$$H_{\text{eff}}(p) = \varepsilon_0(p) \mathbf{1}_2 + \sum_{a=1}^3 \tilde{v}_a p_a \sigma^a + O(|p|^2),$$

with all $\tilde{v}_a \neq 0$.

b. Proof. By Hermitian 2×2 decomposition,

$$H_{\text{eff}}(p) = \varepsilon_0(p)\mathbf{1}_2 + (Mp) \cdot \sigma + O(|p|^2).$$

If $\det M \neq 0$, then after an invertible linear transformation of momenta

$$p \mapsto p' = Mp,$$

the traceless part becomes

$$p'_a \sigma^a.$$

A further unitary basis change rotates the Pauli matrices by an $SO(3)$ transformation, so the operator is equivalent to the standard Weyl form. $\square$

E. Topological chirality of the crossing

For a Weyl point, the map

$$\hat{d}(p) = \frac{d(p)}{|d(p)|}$$

defines

$$\hat{d} : S_p^2 \rightarrow S^2,$$

whose degree is the chirality of the node:

$$\chi = \text{sgn}(\det M) \in \{\pm 1\}.$$

Thus the nondegenerate two-band crossing carries a topological charge.

a. Corollary 2. Under the hypotheses of Theorem 1, the isolated crossing at k_* is a Weyl point with chirality

$$\chi = \text{sgn}(\det M).$$

F. BCC-shell-adapted coordinates

Suppose now that the crossing occurs near a BCC reciprocal-shell momentum G_* , and let

$$\hat{n} := \frac{G_*}{|G_*|}.$$

Choose orthonormal coordinates

$$p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2.$$

Then the effective Hamiltonian can always be written as

$$H_{\text{eff}}(p) = \varepsilon_0(p)\mathbf{1}_2 + v_{\parallel} p_{\parallel} \sigma^3 + v_1 p_1 \sigma^1 + v_2 p_2 \sigma^2 + O(|p|^2)$$

after suitable basis choices, provided the linear crossing is nondegenerate. The Weyl criterion is then simply

$$v_{\parallel} v_1 v_2 \neq 0.$$

a. Corollary 3 (BCC-shell Weyl form). If a BCC-shell valley G_* supports an isolated nondegenerate two-band crossing, then the projected low-energy Hamiltonian takes the shell-adapted Weyl form above.

G. Valley doubling and Dirac emergence

Assume now that there exists a symmetry Θ relating the two valleys k_* and $-k_*$, for instance inversion or a generalized time-reversal/valley symmetry, such that

$$\Theta \mathcal{L}(k) \Theta^{-1} = \mathcal{L}(-k).$$

Suppose further that the chiralities of the two valleys are opposite, as required for a paired Dirac structure. Let

$$\psi_+(p) \in \mathbb{C}^2$$

be the Weyl spinor near k_* , and

$$\psi_-(p) \in \mathbb{C}^2$$

the partner near $-k_*$.

Define the four-component spinor

$$\Psi(p) = \begin{pmatrix} \psi_+(p) \\ \psi_-(p) \end{pmatrix}.$$

Then the block Hamiltonian takes the form

$$H_D(p) = \begin{pmatrix} H_+(p) & 0 \\ 0 & H_-(p) \end{pmatrix},$$

with

$$H_+(p) = v_i p_i \sigma^i, \quad H_-(p) = -v_i p_i \sigma^i$$

after suitable conventions. This is equivalent to a massless Dirac Hamiltonian

$$H_D(p) = \alpha^i p_i.$$

a. Theorem 4 (Valley-doubling Dirac theorem). If two opposite valleys k_* and $-k_*$ carry Weyl Hamiltonians of opposite chirality and are related by a symmetry pairing, then the combined low-energy theory is a massless Dirac theory.

b. Proof. Stacking the two Weyl Hamiltonians in block form gives

$$H_D(p) = \begin{pmatrix} v_i p_i \sigma^i & 0 \\ 0 & -v_i p_i \sigma^i \end{pmatrix}.$$

Defining

$$\alpha^i = \begin{pmatrix} \sigma^i & 0 \\ 0 & -\sigma^i \end{pmatrix},$$

one obtains

$$H_D(p) = \alpha^i p_i.$$

This is the standard massless Dirac form. $\square$

H. Defect-envelope derivation

We now give the restricted defect version. Let a background defect preserve translational symmetry along coordinates y^μ ($\mu = 0, 1, \dots, d-1$) and vary slowly or nontrivially in transverse coordinates ξ . Consider a linearized fluctuation equation

$$\mathcal{L}_{\text{lin}} \Phi = 0.$$

Assume there exists a localized transverse mode $u_0(\xi)$ satisfying

$$\mathcal{L}_\perp u_0 = 0,$$

or more generally a low-lying subspace spanned by $\{u_a(\xi)\}_{a=1,2}$. Use the ansatz

$$\Phi(y, \xi) = \sum_{a=1}^2 \psi_a(y) u_a(\xi) + \text{higher modes}.$$

Projecting the linearized operator onto this low-energy subspace yields an effective operator on the envelope field $\psi(y) \in \mathbb{C}^2$:

$$H_{\text{env}}(-i\partial_y) = \langle u_a, \mathcal{L}_{\text{lin}} u_b \rangle_\xi.$$

If the leading derivative couplings among the low-energy modes are linear in ∂_{y^μ} and produce a full-rank Pauli structure, then

$$H_{\text{env}}(p_y) = \varepsilon_0(p_y) \mathbf{1}_2 + c_\mu p_y^\mu \sigma^\mu + O(p_y^2),$$

which is again Weyl/Dirac type after valley doubling if a partner sector exists.

a. Theorem 5 (Restricted defect-envelope Weyl theorem). Suppose a defect background admits a two-dimensional localized low-energy subspace and that the projected envelope operator is linear to first order in worldvolume momentum with full-rank Pauli-matrix coefficients. Then the defect worldvolume theory is Weyl type, and Dirac type if an opposite-chirality partner sector is present.

b. Proof. This is the same projected two-band argument as in Theorem 1, now applied to the effective worldvolume operator obtained by transverse-mode projection. $\square$

I. What is actually proven and what is not

We emphasize the exact logical status.

a. Proved structurally.

1. Any isolated two-band linear crossing yields a Weyl Hamiltonian after projection.
2. Any symmetry-paired opposite-chirality valley double yields a Dirac Hamiltonian.
3. The same applies to defect-envelope reductions if a two-dimensional localized low-energy subspace exists.

b. Not yet proved for full TECT.

1. That the microscopic BCC condensate operator actually has isolated valleys at $\pm G_*$.
2. That the low-energy subspace is exactly two-dimensional.
3. That the linear velocity matrix is nondegenerate:

$$v_\parallel v_1 v_2 \neq 0.$$

4. That the defect-background operator produces the required localized doublet.

Thus the Weyl/Dirac emergence theorem is mathematically correct as a conditional theorem, but its full TECT realization remains an open spectral problem.

XIII. FINAL SUMMARY

The emergence of a first-order Weyl or Dirac operator can be formulated as a precise conditional spectral theorem. Let $\mathcal{L}(k)$ be a smooth Hermitian family near k_* , and assume that:

1. there exists an isolated two-dimensional degenerate low-energy subspace at k_* ,

2. the projected linear velocity matrices generate a full-rank traceless part.

Then the projected low-energy operator is

$$H_{\text{eff}}(p) = \varepsilon_0(p)\mathbf{1}_2 + d_a(p)\sigma^a + O(|p|^2), \quad p = k - k_*,$$

with

$$d_a(p) = M_{ai}p_i.$$

If

$$\det M \neq 0,$$

this is equivalent to a Weyl Hamiltonian:

$$H_W(p) \sim v_i p_i \sigma^i.$$

For a BCC-shell-adapted basis,

$$H_{\text{eff}}(p) = \varepsilon_0(p)\mathbf{1}_2 + v_{\parallel}p_{\parallel}\sigma^3 + v_1p_1\sigma^1 + v_2p_2\sigma^2 + O(|p|^2),$$

and the Weyl criterion is

$$v_{\parallel}v_1v_2 \neq 0.$$

If a symmetry relates k_* and $-k_*$ and the two valleys have opposite chirality, then their combined four-component theory is Dirac:

$$H_D(p) = \alpha^i p_i.$$

The same logic applies to defect-envelope reductions: a two-dimensional localized low-energy subspace plus a full-rank linear projected operator implies a Weyl worldvolume theory, and a paired opposite-chirality sector gives Dirac.

Therefore the mathematically honest conclusion is:

Isolated two-band linear crossing $\implies$ Weyl

Opposite-chirality valley pairing $\implies$ Dirac

What remains open in TECT is not the projection theorem itself, but the microscopic proof that the BCC condensate or defect spectrum actually satisfies the hypotheses required by the theorem.

XIV. STRENGTHENED DERIVATION OF THE EMERGENT DIRAC EQUATION: FROM DEFECT/VALLEY PROJECTION TO WEYL AND DIRAC STRUCTURE

Assumption Audit

- A1: The internal orientational sector is described by

$$P = zz^\dagger \in \mathbb{CP}^2, \quad z^\dagger z = 1, \quad z \sim e^{i\alpha} z.$$

Established

- A2: Around a defect background P_B , the physical linearized fluctuation operator closes on the orthogonal rank-two bundle sector

$$Q = (\mathbb{C}z_B)^\perp, \quad \mathcal{L}_\perp : \Gamma(Q) \rightarrow \Gamma(Q).$$

Provable in projector/sigma-model realizations

- A3: The closed operator $\mathcal{L}_\perp$ possesses an isolated two-dimensional low-energy subspace

$$\mathcal{K} = \text{span}\{u_1, u_2\}.$$

Needs spectral assumption

- A4: The projected low-energy operator is smooth in momentum and has nondegenerate linear traceless part.

Provable once A3 is assumed

- A5: An opposite-chirality partner sector exists, either from valley doubling or from a symmetry-related partner localized sector.

Provable under symmetry assumptions

- A6: The odd-level WZW/Berry term fermionizes the one-defect state.

Conditional but structurally established

- A7: The emergent $SU(3)$ connection acts on the low-energy mode bundle by minimal coupling.

Established once the low-energy bundle is identified

- A8: All of A2–A7 hold in the full microscopic TECT condensate PDE without further assumption.

Currently unproved

A. Why the previous square-root argument is insufficient

The chain

$$\text{fermionic statistics} + \text{relativistic dispersion} \implies \text{Dirac equation}$$

is not a mathematical derivation by itself. A relativistic quadratic dispersion relation

$$E^2 = c_*^2 p^2 + M_*^2 c_*^4$$

does not uniquely imply the Dirac equation, because Klein–Gordon, Proca, Majorana, and Dirac theories may all share the same quadratic mass shell.

Therefore the correct strategy is to derive a *first-order projected low-energy operator* from the defect or valley spectrum, and only then identify its statistics via the WZW/Berry quantization.

B. Defect background and Q -sector Hessian

Let

$$P_B(x; X, \xi)$$

be a static or slowly moving charge- B defect background in the $\mathbb{CP}^2$ sector. Choose a representative

$$z_B(x), \quad P_B = z_B z_B^\dagger, \quad z_B^\dagger z_B = 1.$$

Define

$$Q_B := (\mathbb{C}z_B)^\perp, \quad \Pi_B := \mathbf{1}_3 - z_B z_B^\dagger.$$

A physical fluctuation η in background gauge satisfies

$$z_B^\dagger \eta = 0, \quad \eta \in \Gamma(Q_B).$$

For projector/sigma-model dynamics, the quadratic fluctuation operator takes the form

$$\mathcal{L}_\perp = -\nabla_\mu^{(B)} \nabla^{(B)\mu} + \mathcal{M}_B, \quad \mathcal{L}_\perp : \Gamma(Q_B) \rightarrow \Gamma(Q_B),$$

where $\nabla_\mu^{(B)}$ is the induced background covariant derivative on Q_B , and $\mathcal{M}_B$ is a Hermitian endomorphism of Q_B .

This defines the defect-localized spectral problem

$$\mathcal{L}_\perp u_n = \lambda_n u_n.$$

C. Isolated two-mode low-energy sector

Assume now that $\mathcal{L}_\perp$ admits an isolated two-dimensional low-energy sector

$$\mathcal{K} = \text{span}\{u_1, u_2\}, \quad \dim \mathcal{K} = 2,$$

separated by a gap from higher modes. This may arise either:

1. from a localized defect doublet,
2. from a BCC-shell valley pair at momentum k_* ,
3. or from a two-dimensional collective low-energy sector after symmetry reduction.

Let $\Pi_{\mathcal{K}}$ denote the projector onto $\mathcal{K}$.

We expand the full fluctuation field as

$$\eta(x, y) = \sum_{a=1}^2 \psi_a(y) u_a(x) + \eta_{\text{high}}(x, y),$$

where x denotes transverse/internal coordinates and y denotes the slowly varying worldvolume or envelope coordinates. Integrating out the high modes yields an effective operator for

$$\psi(y) = \begin{pmatrix} \psi_1(y) \\ \psi_2(y) \end{pmatrix} \in \mathbb{C}^2.$$

D. Projected effective Hamiltonian

Let the exact linearized generator of low-energy evolution be denoted abstractly by

$$\mathcal{H}_{\text{lin}}(p),$$

where p stands for small momentum around the relevant defect worldvolume or around a valley momentum k_* . Then the projected operator is

$$H_{\text{eff}}(p) = \Pi_{\mathcal{K}} \mathcal{H}_{\text{lin}}(p) \Pi_{\mathcal{K}}|_{\mathcal{K}}.$$

Assuming smoothness in p , degenerate perturbation theory yields

$$H_{\text{eff}}(p) = E_* \mathbf{1}_2 + p_i V_i + O(|p|^2),$$

with Hermitian 2×2 matrices V_i .

Every Hermitian 2×2 matrix decomposes as

$$V_i = v_i^{(0)} \mathbf{1}_2 + v_i^{(a)} \sigma^a, \quad a = 1, 2, 3.$$

Hence

$$H_{\text{eff}}(p) = \varepsilon_0(p) \mathbf{1}_2 + d_a(p) \sigma^a + O(|p|^2),$$

where

$$\varepsilon_0(p) = E_* + p_i v_i^{(0)}, \quad d_a(p) = M_{ai} p_i, \quad M_{ai} := v_i^{(a)}.$$

E. Weyl reduction theorem

a. *Theorem 1 (Projected two-mode Weyl theorem).* Suppose that the 3×3 matrix

$$M_{ai} = v_i^{(a)}$$

has full rank:

$$\det M \neq 0.$$

Then, after an invertible linear change of momentum variables and a unitary basis change in $\mathcal{K}$, the effective operator is of Weyl form:

$$H_{\text{eff}}(p) = \varepsilon_0(p)\mathbf{1}_2 + \tilde{v}_1 p_1 \sigma^1 + \tilde{v}_2 p_2 \sigma^2 + \tilde{v}_3 p_3 \sigma^3 + O(|p|^2),$$

with all $\tilde{v}_i \neq 0$.

b. *Proof.* The traceless part of $H_{\text{eff}}(p)$ is

$$H_{\text{tr}}(p) = (Mp) \cdot \sigma.$$

If $\det M \neq 0$, then M is invertible, so by changing momentum coordinates

$$p' = Mp,$$

the traceless part becomes

$$H_{\text{tr}}(p') = p'_a \sigma^a.$$

A further $SU(2)$ basis rotation acts on the Pauli matrices by $SO(3)$, giving the stated canonical Weyl form. $\square$

Thus an isolated nondegenerate two-mode sector does not merely suggest a first-order operator: it mathematically forces a Weyl operator.

F. BCC-shell form

If the low-energy sector is centered near a BCC reciprocal-shell momentum G_* , define

$$\hat{n} = \frac{G_*}{|G_*|}, \quad p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2.$$

Then the Weyl Hamiltonian may be written as

$$H_+(p) = \varepsilon_0(p)\mathbf{1}_2 + v_{\parallel} p_{\parallel} \sigma^3 + v_1 p_1 \sigma^1 + v_2 p_2 \sigma^2 + O(|p|^2),$$

with the nondegeneracy criterion

$$v_{\parallel} v_1 v_2 \neq 0.$$

G. Partner sector and Dirac doubling

The Weyl operator above gives only a two-component theory. To obtain a four-component Dirac field, one needs an opposite-chirality partner sector.

This may arise in either of two ways:

1. **Valley doubling:** there exists a symmetry-related valley at $-G_*$ with

$$H_-(p) = -H_+(-p),$$

up to unitary equivalence.

2. Defect partner doubling: the defect spectrum contains a second isolated two-mode sector $\mathcal{K}_-$ of opposite chirality, related to $\mathcal{K}_+$ by an antiunitary or discrete symmetry.

Define

$$\Psi(p) = \begin{pmatrix} \psi_+(p) \\ \psi_-(p) \end{pmatrix}.$$

Then the combined projected Hamiltonian is

$$H_D(p) = \begin{pmatrix} H_+(p) & 0 \\ 0 & H_-(p) \end{pmatrix}.$$

When the two sectors have opposite chirality, this is equivalent to the Dirac Hamiltonian

$$H_D(p) = \alpha^i p_i + \beta M_*,$$

after the standard identification of gamma matrices.

a. Theorem 2 (Partner-doubling Dirac theorem). If a projected Weyl sector $H_+(p)$ admits an opposite-chirality partner $H_-(p)$ related by a symmetry pairing, then the combined low-energy operator is unitarily equivalent to a Dirac operator:

$$(i\gamma^\mu \partial_\mu - M_*)\Psi = 0$$

in covariant form, and

$$H_D(p) = \alpha^i p_i + \beta M_*$$

in Hamiltonian form.

b. Proof. Stack the two opposite-chirality Weyl sectors into a four-component field. In a basis adapted to the chirality pairing, the block Hamiltonian becomes

$$H_D(p) = \begin{pmatrix} v_i p_i \sigma^i & 0 \\ 0 & -v_i p_i \sigma^i \end{pmatrix} + M_* \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix},$$

which is the standard Dirac form after identifying the gamma matrices. $\square$

H. Role of the WZW/Berry term: statistics, not operator algebra

The odd-level WZW/Berry term does not by itself determine the Dirac operator. Instead, it fixes the statistics of the quantized low-energy sector.

Let the one-defect low-energy Hilbert space be

$$\mathcal{H}_{\text{low}} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^4 \otimes \mathbb{C}^3,$$

where $\mathbb{C}^4$ is the Dirac spinor module obtained from partner doubling, and $\mathbb{C}^3$ is the internal $SU(3)$ sector.

If the moduli-space quantization yields the phase

$$e^{i\pi B} = (-1)^B$$

for the unit defect, then the second-quantized field obeys fermionic anticommutation relations:

$$\{\hat{\Psi}_\alpha^m(x), \hat{\Psi}_\beta^n(y)\} = 0, \quad \{\hat{\Psi}_\alpha^m(x), \hat{\Psi}_\beta^{n\dagger}(y)\} = \delta_{\alpha\beta} \delta^{mn} \delta^{(3)}(x - y).$$

Thus the WZW/Berry term supplies *fermionic statistics* for the already-derived Dirac sector.

I. Minimal coupling to the emergent $SU(3)$ connection

Let the emergent internal connection act on the low-energy mode bundle in the fundamental representation:

$$D_\mu = \partial_\mu - iA_\mu^a T^a.$$

Projecting the full gauge-covariant fluctuation operator onto the low-energy sector preserves minimal coupling:

$$p_\mu \mapsto p_\mu - A_\mu^a T^a.$$

Therefore the final effective covariant Dirac equation is

$$(i\gamma^\mu D_\mu - M_*)\Psi = 0, \quad D_\mu = \partial_\mu - iA_\mu^a T^a.$$

J. Chirality protection and mirror lifting

If, in addition, the low-energy partner sectors arise from a twisted Dirac operator on the $\mathbb{CP}^2$ defect core with nonzero index

$$\text{Index}(D_+) \neq 0,$$

then one chirality is topologically protected. If extra mirror pairs are present, the previously established mirror-lifting mechanism removes vectorlike pairs, leaving the net chiral sector equal to the index.

Thus the logic is:

$$\text{two-mode projection} \implies \text{Weyl},$$

$$\text{partner doubling} \implies \text{Dirac},$$

$$\text{nonzero twisted index} \implies \text{chiral protection},$$

$$\text{mirror lifting} \implies \text{net physical chirality}.$$

K. Strengthened structural theorem

a. Theorem 3 (Strengthened emergent-Dirac theorem for TECT). Assume:

1. the BCC-condensed TECT background admits the already-established $\mathbb{CP}^2$ orientational sector and emergent $SU(3)$ connection;
2. around a charge- B defect background or a BCC-shell valley background, the physical linearized operator closes on a Q -sector and admits an isolated two-dimensional low-energy subspace $\mathcal{K}_+$;
3. the projected linearized operator on $\mathcal{K}_+$ has nondegenerate linear traceless part, equivalently a full-rank velocity matrix;
4. there exists an opposite-chirality partner sector $\mathcal{K}_-$ related by valley pairing or defect-sector symmetry;
5. the WZW/Berry quantization of the defect sector is odd-level, so that the quantized low-energy field is fermionic;
6. the emergent $SU(3)$ connection acts by minimal coupling on the low-energy mode bundle.

Then the infrared one-defect field is a fermionic Dirac field

$$\Psi \in S_{\text{Dirac}} \otimes \mathbb{C}^3$$

satisfying

$$(i\gamma^\mu D_\mu - M_*) \Psi = 0, \quad D_\mu = \partial_\mu - iA_\mu^a T^a.$$

b. Proof. By assumptions (2)–(3), the projected low-energy operator on $\mathcal{K}_+$ is Weyl by Theorem 1. By assumption (4), doubling with the opposite-chirality partner yields the Dirac operator by Theorem 2. By assumption (5), second quantization gives fermionic statistics. By assumption (6), the operator is gauge-covariant under the already-emergent $SU(3)$ connection. Hence the infrared effective equation is precisely the covariant Dirac equation displayed above. $\square$

L. What is genuinely proved and what is still open

What is genuinely proved by the theorem above is not merely a consistency argument from relativistic dispersion, but a proper projected low-energy derivation:

$$\text{isolated two-mode sector} \implies \text{Weyl operator} \implies \text{Dirac operator after partner doubling.}$$

What remains open is the purely microscopic spectral problem of proving that the full TECT condensate PDE actually satisfies the hypotheses of the theorem.

Specifically, it remains to prove directly from the microscopic BCC condensate operator:

1. the existence of the relevant isolated two-mode sector,
2. the full-rank linear velocity matrix,
3. the opposite-chirality partner pairing,
4. and the exact defect/valley spectral realization of the low-energy bundle.

XV. FINAL SUMMARY OF THE STRENGTHENED DIRAC DERIVATION

The previous argument

$$\text{fermionic statistics} + \text{relativistic dispersion} \implies \text{Dirac equation}$$

is not mathematically sufficient, because a quadratic relativistic mass shell does not uniquely determine a Dirac operator.

The strengthened derivation replaces that step with the following chain:

$$\text{TECT defect or valley background} \implies \text{closed } Q\text{-sector} \implies \text{isolated 2-mode low-energy subspace}$$

$$\implies \text{projected } 2 \times 2 \text{ Hermitian operator} \implies \text{Weyl Hamiltonian} \implies \text{opposite-chirality partner doubling} \implies \text{Dirac Hamiltonian.}$$

More precisely, if the projected low-energy operator has the form

$$H_{\text{eff}}(p) = E_* \mathbf{1}_2 + p_i V_i + O(p^2),$$

then Hermitian 2×2 decomposition gives

$$H_{\text{eff}}(p) = \varepsilon_0(p) \mathbf{1}_2 + d_a(p) \sigma^a + O(p^2), \quad d_a(p) = M_{ai} p_i.$$

If

$$\det M \neq 0,$$

the operator is Weyl. If there is an opposite-chirality partner sector, the doubled four-component theory is Dirac:

$$(i\gamma^\mu D_\mu - M_*) \Psi = 0.$$

The odd WZW/Berry term does not generate the Dirac algebra; instead, it ensures that the resulting low-energy field is fermionic upon second quantization.

Therefore the mathematically strongest honest conclusion is:

Dirac emergence in TECT is rigorously reduced to a projected two-mode spectral theorem.

What still remains open is the microscopic proof that the full BCC condensate or defect operator indeed realizes the required isolated two-mode sector and its opposite-chirality partner.

XVI. ASSUMPTION-REDUCTION AUDIT FOR THE EMERGENT DIRAC THEOREM

Goal

We revisit the strengthened conditional Dirac theorem and classify each assumption into one of the following categories:

Eliminable	Replaceable by a theorem	Weakable	Currently indispensable
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The objective is to reduce the theorem to its minimal logically necessary assumptions.

Initial conditional theorem

The strengthened Dirac theorem previously used the following assumptions:

(A1) The internal orientational sector is described by

$$P = zz^\dagger \in \mathbb{CP}^2.$$

(A2) Around the relevant background, the physical linearized fluctuation operator closes on the orthogonal bundle sector

$$Q = (\mathbb{C}z_B)^\perp, \quad \mathcal{L}_\perp : \Gamma(Q) \rightarrow \Gamma(Q).$$

(A3) The operator $\mathcal{L}_\perp$ possesses an isolated two-dimensional low-energy subspace

$$\mathcal{K} = \text{span}\{u_1, u_2\}.$$

(A4) The projected operator has nondegenerate linear traceless part, i.e.

$$H_{\text{eff}}(p) = E_* \mathbf{1}_2 + p_i V_i + O(p^2), \quad \det M \neq 0.$$

(A5) There exists an opposite-chirality partner sector.

(A6) The WZW/Berry term yields fermionic statistics.

(A7) The emergent $SU(3)$ connection acts by minimal coupling on the low-energy mode bundle.

(A8) All of the above follow from the full microscopic TECT PDE.

We now reduce this list.

A. Reduction of A1

a. Claim. Assumption (A1) is not an independent assumption for the Dirac step if the prior $\mathbb{CP}^2$ -sector construction has already been established.

b. Reason. The Dirac theorem is not responsible for proving the existence of the $\mathbb{CP}^2$ orientational sector. It uses that sector as already-established input. Therefore A1 should be treated as prior proven structure, not as an additional assumption of the Dirac step.

A1 is removable from the Dirac theorem as an independent hypothesis.

B. Reduction of A7

a. Claim. Assumption (A7) is replaceable by a theorem once the low-energy subspace is identified as a subbundle of a gauge-covariant fluctuation bundle.

b. Proof. Suppose the full fluctuation operator is gauge-covariant under the emergent connection:

$$\mathcal{D}_\mu = \partial_\mu - iA_\mu^a T^a.$$

Let $\Pi_{\mathcal{K}}$ be the spectral projector onto an isolated low-energy subspace $\mathcal{K}$. If $\mathcal{K}$ is preserved by the gauge-covariant structure, then the projected operator is

$$H_{\text{eff}} = \Pi_{\mathcal{K}} \mathcal{H}(\mathcal{D}) \Pi_{\mathcal{K}}.$$

Hence every appearance of ∂_μ in the low-energy theory is inherited as

$$\partial_\mu \mapsto \partial_\mu - iA_\mu^a T^a.$$

Thus minimal coupling is not an extra postulate but a consequence of projecting a gauge-covariant operator onto a gauge-stable low-energy subspace. $\square$

Therefore:

A7 can be replaced by a projection-covariance theorem.

C. Reduction of A6

a. Claim. Assumption (A6) is not needed to derive the Dirac operator itself; it is needed only for fermionic statistics of the quantized field.

b. Proof. The derivation of the first-order operator proceeds through spectral projection onto a two-mode sector and decomposition of Hermitian 2×2 matrices:

$$H_{\text{eff}}(p) = \varepsilon_0(p) \mathbf{1}_2 + d_a(p) \sigma^a + O(p^2).$$

No use of the WZW/Berry term is made in this derivation. The WZW/Berry term enters only at the stage of quantizing the one-particle sector and assigning exchange statistics. $\square$

Hence:

A6 is removable from the operator-level Dirac theorem.

It remains necessary only for the stronger statement:

“the emergent Dirac field is fermionic.”

D. Reduction of A8

a. Claim. Assumption (A8) is too strong and should be weakened.

b. Reason. The operator-level Dirac theorem need not assume derivation from the full microscopic TECT PDE. It suffices that the relevant effective low-energy fluctuation operator exists in the projector or defect sector and is gauge-covariant. Therefore the theorem should first be stated at the level of the effective sector, not the full microscopic PDE.

Thus:

A8 can be weakened to an effective-sector hypothesis.

This does not solve the microscopic problem, but it avoids overloading the Dirac theorem with a claim it does not need.

E. Reduction of A2

a. Claim. Assumption (A2) is partly replaceable by prior geometric theorems, but not fully removable.

b. Reason. The existence of the bundle

$$Q = (\mathbb{C}z_B)^\perp$$

and the representation of physical tangent fluctuations as Q -valued fields are already established geometrically. Also, for projector/sigma-model-type dynamics, the physical Hessian closes on Q :

$$\mathcal{L}_\perp : \Gamma(Q) \rightarrow \Gamma(Q).$$

Thus the statement is already a theorem in that class of models.

However, for a completely general microscopic operator, closure of the Q -sector is not automatic: the off-diagonal blocks

$$P\mathcal{L}_{\text{lin}}\Pi, \quad \Pi\mathcal{L}_{\text{lin}}P$$

need not vanish.

Therefore:

A2 is replaceable by a theorem in the projector/sigma-model sector,

but

not removable in full generality.

F. Analysis of A4

a. Claim. Assumption (A4) is not fully independent once A3 is given, but one nontrivial residue remains.

b. Reason. If an isolated two-dimensional low-energy subspace $\mathcal{K}$ exists and the projected operator is smooth in momentum, then the expansion

$$H_{\text{eff}}(p) = E_* \mathbf{1}_2 + p_i V_i + O(p^2)$$

is automatic by degenerate perturbation theory.

Hence the only genuinely nontrivial part of A4 is not the expansion itself, but the nondegeneracy:

$$\det M \neq 0.$$

Thus:

The Taylor expansion part of A4 is theorem-level once A3 holds,

while

$\det M \neq 0$ remains an independent spectral condition.

G. Analysis of A3

a. Claim. Assumption (A3) is currently indispensable.

b. Reason. The Weyl reduction theorem fundamentally depends on the existence of an isolated two-dimensional low-energy subspace. Without such a sector, the projected operator is not a 2×2 Hermitian matrix, and the Pauli-matrix reduction cannot even begin.

Could A3 be replaced by topology alone? In general, no. Could it be replaced by the existence of a nonzero index? Only if a first-order Dirac-type operator has already been identified, which is precisely what one is trying to derive.

Thus A3 is not presently derivable from the already-established ingredients alone.

A3 is currently indispensable.

H. Analysis of A5

a. Claim. Assumption (A5) is currently indispensable for the Dirac, as opposed to Weyl, conclusion.
b. Reason. A single isolated two-mode sector yields a Weyl operator, not a four-component Dirac operator. To obtain Dirac one needs either:

1. a symmetry-related opposite valley,
2. or a second localized partner sector of opposite chirality.

Without such a partner, the logically strongest conclusion is only

Weyl emergence.

Therefore:

A5 is currently indispensable for Dirac doubling.

I. Minimal reduced theorem

After removing or weakening all reducible assumptions, the operator-level theorem becomes:

a. Minimal reduced theorem. Suppose:

- (R1) the relevant physical fluctuation sector closes on a gauge-covariant bundle sector (Q -sector in the projector/sigma-model realization);
- (R2) the closed operator possesses an isolated two-dimensional low-energy subspace

$$\mathcal{K} = \text{span}\{u_1, u_2\};$$

- (R3) the linear traceless velocity matrix on $\mathcal{K}$ is nondegenerate:

$$\det M \neq 0;$$

- (R4) there exists an opposite-chirality partner sector related by symmetry.

Then the infrared effective operator is Dirac.

b. Proof. By (R2), projection onto $\mathcal{K}$ yields a 2×2 Hermitian low-energy operator. By smoothness, it expands linearly near the crossing point. By (R3), the traceless linear part is Weyl. By (R4), partner doubling yields the Dirac operator. Gauge covariance is inherited by projection from (R1). $\square$

J. Sharp conclusion of the reduction audit

Therefore the assumptions split as follows:

Original assumption	Status after audit
A1	remove (already prior structure)
A2	replace by theorem in projector sector; keep in full generality
A3	currently indispensable
A4	reduce to $\det M \neq 0$
A5	currently indispensable for Dirac
A6	remove from operator theorem; keep for fermionic statistics
A7	replace by gauge-covariant projection theorem
A8	weaken to effective-sector existence

Hence the true core of the unresolved problem is not the whole previous theorem, but the pair:

(i) isolated two-mode low-energy sector

(ii) opposite-chirality partner sector

together with the residual nondegeneracy condition

$\det M \neq 0.$

XVII. FINAL SUMMARY

The previous strengthened Dirac theorem contained several assumptions that should not all be kept as independent hypotheses.

After a rigorous reduction audit:

1. The CP^2 orientational sector is prior structure, not a new assumption of the Dirac step.
2. The WZW/Berry term is not needed to derive the Dirac operator itself; it is needed only to fix fermionic statistics after quantization.
3. Minimal coupling to the emergent $SU(3)$ connection follows from gauge-covariant projection, once the low-energy subspace is identified.
4. The full microscopic TECT-PDE hypothesis is stronger than necessary for the operator-level Dirac theorem and should be weakened to an effective-sector hypothesis.

Therefore the reduced operator-level theorem depends essentially on only three unresolved ingredients:

isolated two-dimensional low-energy subspace

nondegenerate linear traceless velocity matrix ($\det M \neq 0$)

opposite-chirality partner sector

Among these, the first and third are presently the most fundamental and have not yet been derived from the already-established geometric structure alone.

Thus the honest status is:

many auxiliary assumptions can be removed or weakened,

but

the 2-mode hypothesis and partner-doubling hypothesis remain genuinely open.

XVIII. CAN THE TWO-MODE HYPOTHESIS BE REDUCED BY SYMMETRY AND INDEX?

Assumption Audit

- A1: The orthogonal complement

$$Q = (\mathbb{C}z)^\perp$$

is a rank-two complex Hermitian bundle.

Proved

- A2: Physical fluctuations in the projective sector are locally Q -valued and transform as $U(2)$ fundamentals.

Proved

- A3: The relevant little-group representation at the candidate valley/defect sector is two-dimensional irreducible.

Provable once the symmetry is specified

- A4: That two-dimensional irrep is spectrally isolated in the infrared.

Needs extra spectral assumption

- A5: A Dirac-type operator with nonzero index constrains the internal chiral multiplicity.

Provable

- A6: Nonzero index alone forces a two-component Weyl sector.

False in general

- A7: A symmetry pairs opposite-chirality sectors or opposite valleys.

Provable under symmetry assumptions

A. Goal

We want to reduce the unresolved assumption

$$\dim \mathcal{K} = 2$$

as much as possible.

We distinguish three notions that are often conflated:

1. **Fiber dimension:** the bundle Q has complex rank 2.
2. **Internal zero-mode multiplicity:** the number of independent zero modes of an internal Dirac-type operator.
3. **Low-energy projected Hilbert-space dimension:** the dimension of the isolated spectral subspace $\mathcal{K}$ carrying the Weyl reduction.

Only the first is already unconditional. The second can sometimes be controlled by index theory. The third is the one needed for the Weyl projection theorem.

B. Why rank-two of Q does not by itself prove $\dim \mathcal{K} = 2$

Since $Q \rightarrow \mathbb{CP}^2$ is a rank-two complex bundle, any local section may be written as

$$s = e \chi, \quad \chi \in \mathbb{C}^2.$$

This shows that the *fiber* carries a two-dimensional complex representation of the local frame redundancy $U(2)$.

However, the Hilbert space of fluctuations is not the fiber itself but a function space:

$$\Gamma(Q), \quad L^2(Q), \quad \ker \mathcal{L}_\perp, \quad \text{etc.}$$

Its dimension depends on the spectrum of the relevant operator, not just on the fiber rank.

a. Proposition 1. The statement

$$\text{rank}_{\mathbb{C}}(Q) = 2$$

does not imply

$$\dim \mathcal{K} = 2$$

for a low-energy eigenspace $\mathcal{K} \subset \Gamma(Q)$.

b. Proof. Take any Hermitian operator acting on sections of a rank-two bundle over a compact or noncompact base. Its eigenspaces may have dimension $0, 1, 2, 3, \dots$, depending on degeneracies and the geometry of the base. Bundle rank determines the local coefficient space, not the global spectral multiplicity. $\square$

Thus the already-proved Q -bundle theorem is necessary, but not sufficient, for two-mode Weyl reduction.

C. Representation-theoretic forcing of a two-dimensional sector

Now suppose the relevant background admits a symmetry group G , and let G_{k_*} be the little group of the candidate valley momentum k_* , or the symmetry group preserving the defect sector under consideration.

Let the low-energy subspace $\mathcal{K}$ carry a representation

$$\rho : G_{k_*} \rightarrow U(\mathcal{K}).$$

If $\mathcal{K}$ is an irreducible G_{k_*} -module and ρ is a two-dimensional irrep, then automatically

$$\dim \mathcal{K} = 2.$$

a. Theorem 2 (Two-dimensional irrep forcing theorem). Let G_{k_*} be the relevant little group of a candidate valley or defect sector. Assume:

1. the low-energy sector is invariant under G_{k_*} ,
2. it transforms irreducibly under G_{k_*} ,
3. the corresponding irreducible representation is two-dimensional.

Then

$$\dim \mathcal{K} = 2.$$

b. Proof. This is immediate from irreducibility and the definition of representation dimension: if $\mathcal{K}$ carries a two-dimensional irreducible representation, then its vector-space dimension is exactly two. $\square$

So the problem of proving $\dim \mathcal{K} = 2$ may be reduced to the problem of identifying a relevant symmetry and showing that the infrared sector is carried by a two-dimensional irrep.

D. What representation theory cannot do by itself

Even if one identifies a two-dimensional irrep, this alone does not prove that the corresponding sector is the actual low-energy one, nor that it is spectrally isolated.

a. Proposition 3. Representation theory alone does not imply that a given two-dimensional irrep is:

1. the lowest-energy sector,
2. spectrally isolated from other sectors,
3. associated with a nondegenerate linear crossing.

b. Proof. The symmetry decomposition classifies invariant subspaces but does not determine their eigenvalues or relative spectral ordering. Two different irreps may occur at lower energy, several copies of the same irrep may appear, or the chosen irrep may mix through accidental degeneracies if not spectrally isolated. $\square$

Hence symmetry can reduce the unresolved assumption from

“there exists an isolated two-dimensional low-energy sector”

to

“a specific two-dimensional irrep is realized and spectrally isolated.”

This is a real reduction, but not a complete elimination.

E. Index theory: what it can force

Now consider a Dirac-type operator

$$D_Q : \Gamma(S^+ \otimes E) \rightarrow \Gamma(S^- \otimes E),$$

where the twisting bundle E may involve Q or an associated bundle.

Its index is

$$\text{Index}(D_Q) = \dim \ker D_Q^+ - \dim \ker D_Q^-.$$

If

$$\text{Index}(D_Q) = 1,$$

then at least one chiral zero mode exists and the net internal chiral multiplicity is one.

a. Theorem 4 (Index reduction of internal multiplicity). If D_Q is Fredholm and

$$\text{Index}(D_Q) = 1,$$

and if mirror/vectorlike partners are absent or lifted, then the net physical internal chiral zero-mode multiplicity is exactly one.

b. Proof. By index theory,

$$\dim \ker D_Q^+ - \dim \ker D_Q^- = 1.$$

If all paired mirror modes are removed, the remaining unpaired chiral zero-mode sector has multiplicity one. $\square$

This is very useful, but note carefully what it proves: it constrains the number of independent *internal chiral zero-mode families*. It does not, by itself, prove that the propagating worldvolume or valley Hamiltonian is a 2×2 Weyl operator.

F. Why index alone does not prove the two-mode Weyl sector

A common but incorrect inference is:

$$\text{Index}(D_Q) = 1 \implies \dim \mathcal{K} = 2.$$

This is false in general.

a. Proposition 5. A nonzero Dirac index does not by itself imply that the effective low-energy propagating operator is a two-component Weyl Hamiltonian.

b. Proof. The index counts the difference of chiral internal zero modes, not the dimension of a projected momentum-space crossing subspace. To obtain a Weyl Hamiltonian one also needs:

1. a first-order effective kinetic operator along the propagating directions,
2. an isolated two-state sector on which that operator acts,
3. a full-rank linear Pauli structure.

Index theory alone supplies none of these three statements. $\square$

Therefore index reduces multiplicity, but it does not replace the two-mode spectral hypothesis.

G. A useful combined theorem: symmetry + index

The strongest honest reduction available at present is to combine symmetry and index.

a. Theorem 6 (Symmetry-index reduction theorem). Suppose:

1. the relevant low-energy sector decomposes into irreducible representations of a symmetry group G ,
2. one candidate infrared sector carries a two-dimensional irrep ρ_2 ,
3. the internal Dirac-type operator has index 1, and mirror modes are lifted,
4. the sector carrying ρ_2 is the unique surviving chiral low-energy family.

Then the unresolved multiplicity problem reduces to a purely spectral isolation statement for that two-dimensional irrep.

b. Proof. Condition (2) fixes the representation dimension to two. Condition (3) fixes the surviving internal family count to one. Therefore all multiplicity beyond the desired two-component sector has been removed except for the possibility that additional spectrally nearby states remain. Hence the only remaining issue is whether the ρ_2 sector is actually isolated in the infrared. $\square$

This is the sharpest honest reduction we can currently make without performing the explicit spectral calculation.

H. Partner-sector hypothesis

Even after the two-mode problem is reduced, one still needs partner doubling to obtain Dirac rather than merely Weyl.

Suppose there is a discrete or antiunitary symmetry Θ such that

$$\Theta \mathcal{H}(k) \Theta^{-1} = \mathcal{H}(\sigma(k)),$$

where $\sigma(k)$ maps the relevant valley/sector to its partner and reverses chirality.

a. Theorem 7 (Symmetry-forced partner theorem). If the symmetry Θ maps a Weyl sector to an inequivalent sector of opposite chirality, and if the partner sector is spectrally realized, then the doubled low-energy theory is Dirac.

b. Proof. The original sector gives a Weyl Hamiltonian

$$H_+(p) = v_i p_i \sigma^i.$$

The symmetry-related sector has opposite chirality,

$$H_-(p) \simeq -v_i p_i \sigma^i.$$

Stacking the two sectors yields the standard Dirac block form. $\square$

Again, symmetry reduces the issue, but does not by itself prove that the partner is actually present and isolated.

I. Minimal remaining assumptions after reduction

After the symmetry/index audit, the original two-mode hypothesis is reduced but not eliminated. The remaining irreducible assumptions are:

R1: a specific two-dimensional irrep is realized in the infrared

R2: that irrep is spectrally isolated

R3: the projected linear traceless velocity matrix has full rank

R4: an opposite-chirality partner sector is spectrally realized

Everything else has either already been proved, or reduced to these residual spectral statements.

J. Sharp logical conclusion

We therefore have the following hierarchy:

$$\text{CP}^2 \text{ geometry} \implies \text{rank}(Q) = 2$$

does *not* imply

$$\dim \mathcal{K} = 2.$$

However,

$$\text{symmetry with a 2D irrep} \implies \dim \mathcal{K} = 2$$

for that irrep sector.

Also,

$$\text{Index}(D_Q) = 1$$

does *not* imply Weyl by itself, but it does imply a single surviving internal chiral family after mirror lifting.

Hence the strongest honest reduction is:

symmetry fixes the two-dimensional representation,

index fixes the surviving internal family count,

and only spectral isolation/full-rank crossing remain to be checked.

XIX. FINAL SUMMARY

The unresolved hypothesis

$$\dim \mathcal{K} = 2$$

cannot be eliminated purely from the already-established CP^2 bundle geometry.

Indeed:

$$\text{rank}(Q) = 2$$

is a statement about local fiber dimension, whereas

$$\dim \mathcal{K}$$

is a statement about the spectral multiplicity of a low-energy eigenspace. These are not the same.

What can be proved is the following:

1. If the relevant valley or defect sector transforms in a two-dimensional irreducible representation of the appropriate little symmetry group, then that representation sector has dimension 2.
2. If the relevant internal Dirac-type operator has

$$\text{Index}(D_Q) = 1,$$

and mirror modes are removed, then the net surviving internal chiral family count is 1.

3. Symmetry plus index therefore reduces the multiplicity problem dramatically: the remaining unresolved issue is not arbitrary large degeneracy, but whether the distinguished two-dimensional irrep is actually realized as an isolated low-energy sector with full-rank linear crossing.

Thus the strongest honest conclusion is:

Symmetry can force the 2-dimensional representation structure;

index can force a single surviving internal chiral family;

but

spectral isolation and nondegenerate linear crossing still require actual operator analysis.

XX. LITTLE-GROUP ANALYSIS FOR THE BCC SHELL DIRECTION AND THE LIMITS OF SYMMETRY FORCING

Assumption Audit

- A1: The relevant BCC shell direction may be represented by

$$\hat{n} = \frac{1}{\sqrt{2}}(1, 1, 0).$$

Valid group-theoretic setup

- A2: The stabilizer of the oriented direction $+\hat{n}$ inside the cubic point group is C_{2v} .

Provable

- A3: The stabilizer of the unoriented pair $\{\pm\hat{n}\}$ is D_{2h} .

Provable

- A4: All ordinary complex irreducible representations of C_{2v} and D_{2h} are one-dimensional.

Proved

- A5: Therefore ordinary little-group symmetry alone cannot force a two-dimensional irreducible infrared sector.

Proved

- A6: A spinorial/double-group route to 2-dimensional irreps exists only if a genuine spinorial structure has already been independently established.

Conditionally true

- A7: The valley-pair sector $\text{span}\{\phi_+, \phi_-\}$ is a natural two-dimensional candidate space.

Proved

- A8: Symmetry of the pair alone protects an exact two-fold degeneracy.

False in general

A. Goal

We analyze the shell direction

$$\hat{n} = \frac{1}{\sqrt{2}}(1, 1, 0)$$

inside the cubic symmetry group, determine its little group, and decide whether ordinary symmetry can force the desired

$$\dim \mathcal{K} = 2$$

required for Weyl reduction.

B. The BCC shell and the [110] direction

The BCC shell used in the TECT construction is

$$S_{\text{BCC}} = \frac{1}{\sqrt{2}}\{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}.$$

Choose the representative direction

$$\hat{n} = \frac{1}{\sqrt{2}}(1, 1, 0).$$

Complete it to an orthonormal frame by

$$\hat{m} = \frac{1}{\sqrt{2}}(1, -1, 0), \quad \hat{\ell} = (0, 0, 1).$$

Then

$$\{\hat{n}, \hat{m}, \hat{\ell}\}$$

is an orthonormal basis adapted to the shell direction.

C. Oriented stabilizer of $\hat{n}$

Let $G_{\hat{n}}^+$ denote the subgroup of the cubic point group that preserves the oriented direction $\hat{n}$:

$$G_{\hat{n}}^+ = \{R \in O_h \mid R\hat{n} = \hat{n}\}.$$

In the adapted basis $(\hat{n}, \hat{m}, \hat{\ell})$, any such R must act trivially on the $\hat{n}$ -axis and orthogonally on the perpendicular plane

$$\hat{n}^\perp = \text{span}\{\hat{m}, \hat{\ell}\}.$$

Inside the cubic group, the allowed actions on this plane are precisely the sign flips

$$(\hat{m}, \hat{\ell}) \mapsto (\pm\hat{m}, \pm\hat{\ell})$$

with determinant ± 1 , subject to crystallographic compatibility.

A convenient generating set is:

$$r_{\hat{n}}(\pi) : (\hat{n}, \hat{m}, \hat{\ell}) \mapsto (\hat{n}, -\hat{m}, -\hat{\ell}),$$

$$\sigma_{n\ell} : (\hat{n}, \hat{m}, \hat{\ell}) \mapsto (\hat{n}, -\hat{m}, \hat{\ell}),$$

$$\sigma_{nm} : (\hat{n}, \hat{m}, \hat{\ell}) \mapsto (\hat{n}, \hat{m}, -\hat{\ell}).$$

These satisfy

$$r_{\hat{n}}(\pi) = \sigma_{n\ell}\sigma_{nm}, \quad \sigma_{n\ell}^2 = \sigma_{nm}^2 = e, \quad \sigma_{n\ell}\sigma_{nm} = \sigma_{nm}\sigma_{n\ell}.$$

Hence

$$G_{\hat{n}}^+ \cong C_{2v} \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

a. Theorem 1. The stabilizer of the oriented BCC shell direction

$$\hat{n} = \frac{1}{\sqrt{2}}(1, 1, 0)$$

inside the cubic point group is

$$G_{\hat{n}}^+ \cong C_{2v}.$$

b. Proof. Any symmetry preserving $\hat{n}$ must preserve the orthogonal decomposition

$$\mathbb{R}^3 = \mathbb{R}\hat{n} \oplus \hat{n}^\perp.$$

Within the cubic group, only the identity, the π -rotation about $\hat{n}$, and the two mirror reflections containing the $\hat{n}$ -axis preserve the $[110]$ direction with orientation. These four commuting elements form

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \cong C_{2v}.$$

□

D. Pair stabilizer of $\{\pm\hat{n}\}$

Now let

$$G_{\hat{n}}^\pm = \{R \in O_h \mid R\hat{n} = \pm\hat{n}\}$$

be the stabilizer of the unoriented pair $\{\pm\hat{n}\}$.

Relative to the oriented stabilizer, this enlarged group contains one additional operation that reverses $\hat{n}$, for instance inversion I , or equivalently a suitable twofold rotation about an axis perpendicular to $\hat{n}$. Therefore

$$G_{\hat{n}}^\pm = G_{\hat{n}}^+ \times \{e, \iota\}, \quad \iota^2 = e,$$

so that

$$G_{\hat{n}}^\pm \cong C_{2v} \times \mathbb{Z}_2 \cong D_{2h} \cong \mathbb{Z}_2^3.$$

a. *Theorem 2.* The stabilizer of the shell pair $\{\pm\hat{n}\}$ is

$$G_{\hat{n}}^{\pm} \cong D_{2h}.$$

b. *Proof.* The oriented stabilizer is C_{2v} . Adjoining any symmetry that maps $\hat{n} \mapsto -\hat{n}$ while preserving the line spanned by $\hat{n}$ doubles the group. Since all generators square to the identity and commute in this line-preserving subgroup, the resulting group is

$$\mathbb{Z}_2^3 \cong D_{2h}.$$

□

E. Representation-theoretic consequence

Both C_{2v} and D_{2h} are abelian. Therefore every ordinary complex irreducible representation is one-dimensional.

a. *Theorem 3 (No ordinary 2D irrep theorem).* All ordinary complex irreducible representations of

$$C_{2v} \quad \text{and} \quad D_{2h}$$

are one-dimensional.

b. *Proof.* Every irreducible representation of a finite abelian group over $\mathbb{C}$ is one-dimensional. Since

$$C_{2v} \cong \mathbb{Z}_2 \times \mathbb{Z}_2, \quad D_{2h} \cong \mathbb{Z}_2^3$$

are both abelian, the claim follows. □

c. *Corollary 4.* Ordinary little-group symmetry of a generic BCC shell direction cannot by itself force an irreducible two-dimensional low-energy sector:

$$\dim \mathcal{K} = 2$$

cannot be deduced from ordinary C_{2v} or D_{2h} representation theory alone.

This closes one possible route completely.

F. Why this matters

The previous hope was:

$$\text{little group has a 2D irrep} \implies \dim \mathcal{K} = 2.$$

For the BCC shell direction $[110]$, this route fails at the ordinary point-group level because the relevant little groups are abelian.

Thus the unresolved 2-mode hypothesis cannot be reduced through ordinary little-group irreducibility at a generic shell direction.

G. The valley-pair sector as a two-dimensional candidate space

Even though the ordinary irreps are one-dimensional, there is still a natural two-dimensional candidate space coming from the valley pair itself:

$$\mathcal{V}_{\pm} = \text{span}\{\phi_+, \phi_-\},$$

where ϕ_+ is localized near $+G_*$ and ϕ_- near $-G_*$.

This space is indeed two-dimensional as a vector space. However, under the pair stabilizer, it is generally a reducible representation. Indeed, in the basis

$$\phi_{\text{even}} = \frac{1}{\sqrt{2}}(\phi_+ + \phi_-), \quad \phi_{\text{odd}} = \frac{1}{\sqrt{2}}(\phi_+ - \phi_-),$$

the pair-exchange symmetry is diagonal.

a. Proposition 5. The valley-pair space

$$\mathcal{V}_\pm = \text{span}\{\phi_+, \phi_-\}$$

provides a two-dimensional candidate sector, but it is generically reducible under the ordinary pair-stabilizer symmetry.

b. Proof. The symmetry exchanging $+G_* \leftrightarrow -G_*$ acts by

$$\phi_+ \leftrightarrow \phi_-.$$

Passing to the even/odd basis diagonalizes this action. Hence the two-dimensional pair space decomposes into one-dimensional invariant subspaces. $\square$

Therefore a valley pair gives a 2-dimensional *candidate space*, but not a symmetry-protected irreducible doublet.

H. Generic intervalley splitting

Because the pair space is reducible, a symmetry-allowed effective Hamiltonian on it can contain a mixing term. In the (ϕ_+, ϕ_-) basis, the most general Hermitian pair Hamiltonian is

$$H_\pm = E_0 \mathbf{1}_2 + a \sigma^3 + b \sigma^1 + c \sigma^2.$$

If pair-exchange symmetry is present, it may constrain some coefficients, but in general an intervalley mixing term of the form

$$m \sigma^1$$

or its equivalent is not forbidden. Then the two energies split:

$$E_\pm = E_0 \pm |m|.$$

a. Proposition 6. The existence of the pair sector $\mathcal{V}_\pm$ does not by itself protect an exact two-fold degeneracy.

b. Proof. Because the representation is reducible, symmetry-allowed off-diagonal mixing generically diagonalizes into nondegenerate even/odd combinations. Hence degeneracy is not protected unless an additional symmetry forbids the mixing term. $\square$

This is another crucial closure:

$$\boxed{\pm G_* \text{ pair} \neq \text{automatically protected 2-mode Weyl sector.}}$$

I. Double-group / spinorial route

There is, however, one important caveat. If the relevant low-energy object already carries a spinorial or projective representation of the little group, then one must pass to the double group. The double cover of the rotational D_2 subgroup is the quaternion group Q_8 , which does admit a genuine 2-dimensional irreducible representation. More generally, spinorial double groups of C_{2v} or D_{2h} can support 2-dimensional irreps.

But this route is logically delicate.

a. Proposition 7. A double-group 2-dimensional irrep may provide the desired 2-mode structure only after an independent spinorial/fermionic structure has already been established.

b. Reason. If one uses the spinorial double group to *derive* the two-component sector while the spinorial nature itself is still being derived from that same sector, the reasoning becomes circular.

Therefore the double-group route is legitimate only in the following order:

$$\text{independent fermionic/spinorial structure} \implies \text{double-group representation analysis} \implies \text{2D irrep.}$$

It cannot be used as the first, unconditional proof of the two-mode hypothesis.

J. Net result of the little-group audit

We can now state the exact conclusion.

a. Theorem 8 (Sharp little-group conclusion for the BCC shell). For the BCC shell direction

$$G_* \parallel (1, 1, 0),$$

the ordinary oriented little group is

$$C_{2v},$$

the ordinary pair stabilizer is

$$D_{2h},$$

and both have only one-dimensional ordinary complex irreducible representations.

Hence the unresolved 2-mode hypothesis cannot be reduced to ordinary little-group irreducibility.

The only remaining symmetry-based possibilities are:

1. a reducible 2-dimensional valley-pair space $\text{span}\{\phi_+, \phi_-\}$, which still requires an extra symmetry to forbid splitting;
2. a genuine 2-dimensional spinorial irrep of the double group, which is legitimate only after an independently established spinorial/fermionic structure.

XXI. FINAL SUMMARY

Let

$$G_* \parallel (1, 1, 0)$$

be a representative BCC shell direction.

Then:

$$G_{\hat{n}}^+ = \{R \in O_h \mid R\hat{n} = \hat{n}\} \cong C_{2v},$$

and

$$G_{\hat{n}}^\pm = \{R \in O_h \mid R\hat{n} = \pm\hat{n}\} \cong D_{2h}.$$

Since both groups are abelian, all ordinary complex irreducible representations are one-dimensional:

ordinary little-group representation theory does not force $\dim \mathcal{K} = 2$.

Therefore the generic BCC shell direction does *not* support the hoped-for reduction

$$\text{“2D irrep} \Rightarrow \dim \mathcal{K} = 2\text{”}$$

at the ordinary cubic-symmetry level.

What remains are only two symmetry-based candidate mechanisms:

1. the valley-pair space

$$\mathcal{V}_\pm = \text{span}\{\phi_+, \phi_-\},$$

which is two-dimensional but generically reducible and not automatically degenerate;

2. the spinorial/double-group route, which may provide a genuine 2-dimensional irrep, but only after an independently justified spinorial structure is already available.

Thus the strongest honest conclusion is:

The ordinary BCC shell little group does not solve the 2-mode problem.

The unresolved issue is now sharply reduced to one of the following:

either an additional symmetry forbids intervalley splitting,

or

a genuine spinorial/double-group structure must be established independently.

XXII. SYMMETRY CLASSIFICATION OF ALLOWED AND FORBIDDEN MASS TERMS IN THE VALLEY-PAIR SECTOR

Assumption Audit

- A1: The low-energy valley-pair space is

$$\mathcal{V}_{\pm} = \text{span}\{\phi_+, \phi_-\},$$

where ϕ_+ and ϕ_- are modes localized near $+G_*$ and $-G_*$, respectively.

Valid definition

- A2: Any Hermitian effective Hamiltonian on $\mathcal{V}_{\pm}$ is of the form

$$H_{\text{val}}(p) = h_0(p)\mathbf{1} + h_a(p)\sigma^a.$$

Proved

- A3: Discrete symmetries can be represented on $\mathcal{V}_{\pm}$ by unitary or antiunitary operators after a basis choice.

Provable

- A4: The symmetry constraints on $h_a(p)$ can be classified algebraically.

Proved

- A5: The corresponding symmetries are actually realized in the full microscopic TECT BCC background.

Currently unproved

- A6: Forbidding a mass term automatically yields a full Weyl/Dirac cone.

False in general

A. General valley-pair Hamiltonian

Let

$$\mathcal{V}_{\pm} = \text{span}\{\phi_+, \phi_-\}$$

with ordered basis

$$\Phi = \begin{pmatrix} \phi_+ \\ \phi_- \end{pmatrix}.$$

A general Hermitian effective Hamiltonian depending on small momentum p around the pair sector is

$$H_{\text{val}}(p) = h_0(p)\mathbf{1} + h_1(p)\sigma^1 + h_2(p)\sigma^2 + h_3(p)\sigma^3,$$

with real-valued coefficients $h_{\mu}(p)$.

Interpretationally:

- $h_0\mathbf{1}$ is an overall shift/tilt term,
- $h_3\sigma^3$ measures valley imbalance,
- $h_1\sigma^1 + h_2\sigma^2$ mixes the two valleys.

At $p = 0$, a “mass/splitting term” means any nonzero traceless constant term

$$m_a\sigma^a.$$

Such a term generically gaps or splits the pair sector.

B. Valley exchange (inversion-like) symmetry

Suppose there is a unitary symmetry P exchanging the two valleys:

$$P\phi_+ = \phi_-, \quad P\phi_- = \phi_+.$$

In the chosen basis,

$$P = \sigma^1.$$

If the symmetry acts on momentum by

$$p \mapsto -p,$$

then invariance means

$$P H_{\text{val}}(p) P^{-1} = H_{\text{val}}(-p).$$

Using

$$\sigma^1 \sigma^1 \sigma^1 = \sigma^1, \quad \sigma^1 \sigma^2 \sigma^1 = -\sigma^2, \quad \sigma^1 \sigma^3 \sigma^1 = -\sigma^3,$$

we obtain

$$h_0(p) = h_0(-p),$$

$$h_1(p) = h_1(-p), \quad h_2(p) = -h_2(-p), \quad h_3(p) = -h_3(-p).$$

Therefore, at $p = 0$,

$$h_2(0) = h_3(0) = 0,$$

but

$$h_1(0)$$

is allowed.

a. Theorem 1 (Valley-exchange constraint). If the pair-exchange symmetry is represented by

$$P = \sigma^1$$

and sends $p \rightarrow -p$, then the allowed constant terms at $p = 0$ are

$$h_0(0)\mathbf{1} + h_1(0)\sigma^1,$$

while the terms

$$\sigma^2, \quad \sigma^3$$

are forbidden at $p = 0$.

b. Proof. Immediate from the parity constraints above. $\square$

c. Consequence. Valley exchange symmetry alone does *not* protect the pair degeneracy, because the mixing term

$$m \sigma^1$$

is symmetry-allowed.

Thus:

inversion-like valley exchange alone does not forbid intervalley splitting.

C. Time-reversal symmetry in the spinless valley basis

Assume a spinless antiunitary time-reversal symmetry

$$T = K,$$

where K is complex conjugation, acting by

$$T H_{\text{val}}(p) T^{-1} = H_{\text{val}}(-p).$$

Since

$$K \sigma^1 K^{-1} = \sigma^1, \quad K \sigma^2 K^{-1} = -\sigma^2, \quad K \sigma^3 K^{-1} = \sigma^3,$$

we get

$$h_0(p) = h_0(-p), \quad h_1(p) = h_1(-p), \quad h_2(p) = -h_2(-p), \quad h_3(p) = h_3(-p).$$

Hence at $p = 0$,

$$h_2(0) = 0,$$

but

$$h_1(0), h_3(0)$$

are allowed.

a. Theorem 2 (Spinless time-reversal constraint). If

$$T = K$$

acts on the valley-pair space, then at $p = 0$ the only forbidden traceless constant term is

$$\sigma^2,$$

while

$$\sigma^1, \quad \sigma^3$$

remain allowed.

b. Proof. Immediate from antiunitary conjugation by K . $\square$

Therefore:

spinless time-reversal alone is also insufficient to protect the valley degeneracy.

D. Combined PT symmetry

Combine the two previous symmetries:

$$PT = \sigma^1 K.$$

Then

$$(PT) H_{\text{val}}(p) (PT)^{-1} = H_{\text{val}}(p)$$

if both P and T reverse momentum, so the combination leaves p unchanged.

Using

$$\sigma^1(\sigma^1)^* \sigma^1 = \sigma^1, \quad \sigma^1(\sigma^2)^* \sigma^1 = \sigma^2, \quad \sigma^1(\sigma^3)^* \sigma^1 = -\sigma^3,$$

one obtains

$$h_0(p) \text{ arbitrary}, \quad h_1(p) \text{ arbitrary}, \quad h_2(p) \text{ arbitrary}, \quad h_3(p) = 0.$$

So PT kills only the σ^3 term.

a. Theorem 3 (PT-constraint). If

$$PT = \sigma^1 K$$

acts on the valley-pair space and leaves p invariant, then the term

$$h_3(p)\sigma^3$$

is forbidden, but

$$h_1(p)\sigma^1, \quad h_2(p)\sigma^2$$

are allowed.

b. Proof. Direct computation. $\square$

Hence:

PT removes valley imbalance σ^3 but still allows intervalley mixing.

E. Chiral / sublattice-like symmetry

Now consider a unitary chiral symmetry Γ satisfying

$$\Gamma^2 = \mathbf{1}, \quad \Gamma H_{\text{val}}(p) \Gamma^{-1} = -H_{\text{val}}(p).$$

A natural valley-basis choice is

$$\Gamma = \sigma^3.$$

Then

$$\sigma^3 \mathbf{1} \sigma^3 = \mathbf{1}, \quad \sigma^3 \sigma^1 \sigma^3 = -\sigma^1, \quad \sigma^3 \sigma^2 \sigma^3 = -\sigma^2, \quad \sigma^3 \sigma^3 \sigma^3 = \sigma^3.$$

The anticommutation condition implies

$$h_0(p) = 0, \quad h_3(p) = 0,$$

while

$$h_1(p), \quad h_2(p)$$

are allowed.

a. Theorem 4 (Chiral symmetry constraint). If

$$\Gamma = \sigma^3, \quad \Gamma H \Gamma^{-1} = -H,$$

then the valley-pair Hamiltonian is constrained to the form

$$H_{\text{val}}(p) = h_1(p)\sigma^1 + h_2(p)\sigma^2.$$

In particular, both

$$\mathbf{1} \quad \text{and} \quad \sigma^3$$

are forbidden.

b. Proof. Immediate from the anticommutation relations above. $\square$

This is the first symmetry that genuinely forbids the usual constant splitting term

$$m \sigma^3.$$

However, it still allows

$$m_1 \sigma^1 + m_2 \sigma^2$$

unless further momentum/parity conditions force these coefficients to vanish at the expansion point.

F. Particle-hole-like antiunitary symmetry

Consider an antiunitary particle-hole-like symmetry C satisfying

$$C H_{\text{val}}(p) C^{-1} = -H_{\text{val}}(-p).$$

Take for instance

$$C = \sigma^1 K.$$

Using the same algebra as above, we find

$$h_0(p) = -h_0(-p), \quad h_1(p) = -h_1(-p), \quad h_2(p) = -h_2(-p), \quad h_3(p) = +h_3(-p).$$

Thus at $p = 0$,

$$h_0(0) = h_1(0) = h_2(0) = 0,$$

but

$$h_3(0)$$

is allowed.

a. Theorem 5 (Particle-hole-like constraint). For

$$C = \sigma^1 K$$

with

$$CH(p)C^{-1} = -H(-p),$$

the only allowed constant traceless term at $p = 0$ is

$$m \sigma^3.$$

b. Proof. Direct computation. $\square$

So a particle-hole-like symmetry by itself does not protect the valley pair either.

G. Combined chiral and valley-exchange symmetry

Now combine:

1. valley exchange:

$$P = \sigma^1, \quad PH(p)P^{-1} = H(-p),$$

2. chiral symmetry:

$$\Gamma = \sigma^3, \quad \Gamma H(p)\Gamma^{-1} = -H(p).$$

From chiral symmetry alone:

$$H(p) = h_1(p)\sigma^1 + h_2(p)\sigma^2.$$

Applying P -symmetry then gives

$$h_1(p) = h_1(-p), \quad h_2(p) = -h_2(-p).$$

Hence at $p = 0$,

$$h_2(0) = 0,$$

but

$$h_1(0)$$

is still allowed.

a. Corollary 6. The combination of valley exchange and chiral symmetry still allows a constant term

$$m\sigma^1.$$

Therefore it does not completely protect the two-fold degeneracy.

So even this combination is not enough.

H. When is the pair degeneracy actually protected?

To forbid *all* traceless constant terms

$$m_a\sigma^a,$$

one needs symmetry constraints strong enough to eliminate $\sigma^1, \sigma^2, \sigma^3$ simultaneously.

But in a pure 2×2 valley-pair space, this is extremely restrictive. In fact, if all traceless constant terms are forbidden, the Hamiltonian at $p = 0$ reduces to

$$H(0) = E_0 \mathbf{1},$$

so degeneracy is protected there. However, the linearized momentum dependence must still produce a full-rank map to the Pauli matrices in order to yield a Weyl cone.

Thus:

a. Proposition 7. Symmetry-based protection of the valley-pair degeneracy at $p = 0$ is a necessary condition for a Weyl/Dirac reduction from the pair sector, but it is not sufficient.

b. Proof. A protected degeneracy guarantees only that $H(0)$ is proportional to the identity. A Weyl point requires additionally that the linearized traceless coefficients span the full Pauli algebra:

$$H(p) - H(0) = M_{ai}p_i\sigma^a + O(p^2), \quad \det M \neq 0.$$

Without this, one may obtain only a line node, quadratic touching, or no conical structure at all. $\square$

I. Sharp classification table

At $p = 0$, the symmetry constraints may be summarized as follows:

Symmetry	Forbidden traceless constant terms
$P = \sigma^1$	σ^2, σ^3
$T = K$	σ^2
$PT = \sigma^1 K$	σ^3
$\Gamma = \sigma^3$	σ^3 and $\mathbf{1}$ (for all p)
$C = \sigma^1 K$	σ^1, σ^2 and $\mathbf{1}$ at $p = 0$

But no single one of these standard symmetries forbids all three Pauli masses simultaneously in the 2×2 valley-pair sector.

J. Main conclusion

We may now state the precise theorem.

a. Theorem 8 (No standard-symmetry protection theorem for pure valley pair). Consider the pure two-dimensional valley-pair sector

$$\mathcal{V}_\pm = \text{span}\{\phi_+, \phi_-\}$$

with general Hermitian Hamiltonian

$$H_{\text{val}}(p) = h_0(p)\mathbf{1} + h_a(p)\sigma^a.$$

Then none of the standard discrete symmetries

$$P, \quad T, \quad PT, \quad \Gamma, \quad C$$

acting in the natural 2×2 valley basis is, by itself, sufficient to forbid all traceless constant mass/splitting terms.

Therefore a pure valley-pair sector does not generically provide a symmetry-protected Weyl node.

b. Proof. The claim follows from the explicit algebraic classifications above: in every case at least one traceless Pauli term remains symmetry-allowed. $\square$

Hence the valley-pair space is a useful candidate sector, but not by itself the final solution of the 2-mode problem.

K. What this means for TECT

The sharp conclusion is:

ordinary valley-pair symmetry is too weak to guarantee an exact protected two-mode Weyl sector.

Therefore, if TECT is to obtain a robust Dirac sector, one of the following stronger mechanisms is needed:

1. an additional symmetry stronger than the standard P, T, PT, Γ, C actions on the pure pair basis;
2. an enlarged low-energy space in which the valley label combines with an independent internal two-mode structure;
3. a genuine first-order defect/valley operator whose zero-mode structure is protected by index theory before projection to the pair basis.

This is a real reduction of the problem: the obstacle is no longer vague, but identified precisely as the insufficiency of standard 2×2 valley-pair symmetry constraints.

XXIII. FINAL SUMMARY

On the two-dimensional valley-pair space

$$\mathcal{V}_{\pm} = \text{span}\{\phi_+, \phi_-\},$$

the most general Hermitian effective Hamiltonian is

$$H_{\text{val}}(p) = h_0(p)\mathbf{1} + h_a(p)\sigma^a.$$

A symmetry can protect a degeneracy only if it forbids the constant traceless Pauli terms

$$m_a\sigma^a.$$

The explicit algebraic classification shows:

1. Valley-exchange/inversion-like symmetry $P = \sigma^1$ forbids σ^2, σ^3 , but still allows σ^1 .
2. Spinless time-reversal $T = K$ forbids only σ^2 , leaving σ^1, σ^3 allowed.
3. $PT = \sigma^1 K$ forbids σ^3 , but still allows σ^1, σ^2 .
4. Chiral symmetry $\Gamma = \sigma^3$ enforces

$$H = h_1\sigma^1 + h_2\sigma^2,$$

thus forbidding $\mathbf{1}$ and σ^3 , but not σ^1, σ^2 .

5. Particle-hole-like symmetry $C = \sigma^1 K$ forbids σ^1, σ^2 at $p = 0$, but still allows σ^3 .

Therefore:

No standard single symmetry on the pure 2×2 valley-pair space forbids all mass/splitting terms.

Hence a pure valley-pair sector is not generically a symmetry-protected Weyl sector.

The honest implication for TECT is that a robust Dirac/Weyl emergence must rely on something stronger than ordinary valley-pair symmetry alone, such as:

an additional internal two-mode structure,

or

a genuine first-order defect/valley operator with protected zero modes.

Thus the remaining problem is now sharply localized: it is not merely to find a two-dimensional pair space, but to identify the stronger structure that forbids its symmetry-allowed mass terms and generates a full-rank first-order crossing.

XXIV. THE 4-DIMENSIONAL INTERNAL $\otimes$ VALLEY SPACE AS THE NATURAL CARRIER OF THE DIRAC ALGEBRA

Assumption Audit

- A1: The internal low-energy sector carries a complex 2-dimensional space

$$\mathbb{C}_{\text{int}}^2,$$

coming from the rank-two bundle $Q \rightarrow \mathbb{CP}^2$.

Proved

- A2: The valley pair $\{\pm G_*\}$ defines a candidate complex 2-dimensional space

$$\mathbb{C}_{\text{val}}^2 = \text{span}\{\phi_+, \phi_-\}.$$

Valid as a candidate sector

- A3: Therefore the natural candidate low-energy space is

$$\mathcal{H}_{\text{low}} = \mathbb{C}_{\text{int}}^2 \otimes \mathbb{C}_{\text{val}}^2.$$

Provable as a construction

- A4: The 4-dimensional space above admits a representation of the Dirac/Clifford algebra.

Proved

- A5: The actual projected TECT low-energy Hamiltonian closes on that Clifford subalgebra.

Needs extra spectral/dynamical input

- A6: Discrete symmetries on the tensor-product space constrain the allowed mass terms.

Provable algebraically

- A7: Those symmetries are realized by the full microscopic TECT background.

Currently unproved

A. Why the pure valley-pair space is insufficient

The pure valley-pair space

$$\mathbb{C}_{\text{val}}^2$$

admits only the Pauli basis

$$\mathbf{1}, \sigma^1, \sigma^2, \sigma^3.$$

As shown previously, standard discrete symmetries acting on this 2×2 space do not forbid all traceless constant terms simultaneously. Thus the pure pair sector does not generically provide a symmetry-protected Weyl or Dirac structure.

This motivates enlarging the low-energy candidate space by an independent internal doublet:

$$\mathcal{H}_{\text{low}} = \mathbb{C}_{\text{int}}^2 \otimes \mathbb{C}_{\text{val}}^2.$$

B. Canonical operator basis on the tensor-product space

Let

$$\tau^a, \quad a = 1, 2, 3,$$

denote Pauli matrices acting on $\mathbb{C}_{\text{int}}^2$, and let

$$\sigma^a, \quad a = 1, 2, 3,$$

act on $\mathbb{C}_{\text{val}}^2$.

Then every Hermitian operator on

$$\mathcal{H}_{\text{low}} = \mathbb{C}_{\text{int}}^2 \otimes \mathbb{C}_{\text{val}}^2$$

can be expanded uniquely as

$$H = c_{00} \mathbf{1} \otimes \mathbf{1} + c_{a0} \tau^a \otimes \mathbf{1} + c_{0b} \mathbf{1} \otimes \sigma^b + c_{ab} \tau^a \otimes \sigma^b,$$

with all coefficients real.

This gives the full Hermitian basis of dimension

$$1 + 3 + 3 + 9 = 16.$$

C. Clifford generators inside the 4×4 operator algebra

We now show that this 4-dimensional space naturally supports a Dirac algebra.

Define

$$\alpha^1 = \tau^1 \otimes \sigma^1, \quad \alpha^2 = \tau^1 \otimes \sigma^2, \quad \alpha^3 = \tau^1 \otimes \sigma^3,$$

and

$$\beta = \tau^3 \otimes \mathbf{1}.$$

Also define

$$\gamma^5 := -i\alpha^1\alpha^2\alpha^3\beta = \tau^2 \otimes \mathbf{1}.$$

a. Theorem 1 (Clifford realization theorem). The matrices defined above satisfy

$$\{\alpha^i, \alpha^j\} = 2\delta^{ij} \mathbf{1}_4, \quad \{\alpha^i, \beta\} = 0, \quad \beta^2 = \mathbf{1}_4.$$

b. Proof. First,

$$\alpha^i \alpha^j = (\tau^1 \tau^1) \otimes (\sigma^i \sigma^j) = \mathbf{1} \otimes (\sigma^i \sigma^j).$$

Hence

$$\alpha^i \alpha^j + \alpha^j \alpha^i = \mathbf{1} \otimes (\sigma^i \sigma^j + \sigma^j \sigma^i) = 2\delta^{ij} \mathbf{1}_4.$$

Next,

$$\alpha^i \beta + \beta \alpha^i = (\tau^1 \tau^3 + \tau^3 \tau^1) \otimes \sigma^i = 0,$$

because Pauli matrices anticommute for distinct indices. Finally,

$$\beta^2 = (\tau^3)^2 \otimes \mathbf{1} = \mathbf{1}_4.$$

□

Thus the tensor-product space is the minimal natural carrier of a 3 + 1-dimensional Dirac Hamiltonian.

D. Interpretation of the tensor factors

The above realization is not unique, but it has a suggestive interpretation:

- the internal Q -doublet sector supplies the τ -structure that can anticommute with the valley sector;
- the valley pair supplies the σ -structure associated with the momentum-dependent crossing.

In this sense, the internal doublet and the valley doublet together provide exactly enough algebra to support a Dirac operator, whereas either factor alone is too small or too weakly constrained.

E. General projected Hamiltonian on $\mathcal{H}_{\text{low}}$

A general Hermitian effective Hamiltonian on the 4-dimensional candidate space is

$$H_{\text{eff}}(p) = h_{00}(p) \mathbf{1} \otimes \mathbf{1} + h_{a0}(p) \tau^a \otimes \mathbf{1} + h_{0b}(p) \mathbf{1} \otimes \sigma^b + h_{ab}(p) \tau^a \otimes \sigma^b.$$

To obtain a Dirac form, one needs a restriction to the Clifford subalgebra generated by

$$\alpha^i = \tau^1 \otimes \sigma^i, \quad \beta = \tau^3 \otimes \mathbf{1}.$$

That is, one wants

$$H_{\text{eff}}(p) = \varepsilon_0(p) \mathbf{1}_4 + v_i p_i (\tau^1 \otimes \sigma^i) + m (\tau^3 \otimes \mathbf{1}) + \dots$$

a. Proposition 2. The tensor-product space $\mathbb{C}_{\text{int}}^2 \otimes \mathbb{C}_{\text{val}}^2$ is sufficient to support a Dirac Hamiltonian, but a generic Hermitian operator on that space is not of Dirac form.

b. Proof. Sufficiency follows from Theorem 1. Non-genericity follows because the full Hermitian algebra has dimension 16, while the Dirac Hamiltonian occupies only a small subspace spanned by

$$\mathbf{1}_4, \quad \tau^1 \otimes \sigma^1, \quad \tau^1 \otimes \sigma^2, \quad \tau^1 \otimes \sigma^3, \quad \tau^3 \otimes \mathbf{1}.$$

A generic Hermitian operator contains many additional allowed terms. □

Therefore the remaining problem is not algebraic possibility, but dynamical projection: why should the microscopic or effective TECT Hamiltonian select precisely this Clifford sector?

F. Mass-term classification in the 4×4 space

Relative to the chosen kinetic generators

$$\alpha^i = \tau^1 \otimes \sigma^i,$$

a mass term is any Hermitian matrix M satisfying

$$\{M, \alpha^i\} = 0 \quad \text{for all } i = 1, 2, 3.$$

a. Theorem 3 (Mass classification theorem). Within the 4×4 tensor-product algebra, the Hermitian matrices that anticommute with all three $\alpha^i = \tau^1 \otimes \sigma^i$ are precisely

$$M = a(\tau^2 \otimes \mathbf{1}) + b(\tau^3 \otimes \mathbf{1}),$$

with $a, b \in \mathbb{R}$.

b. Proof. Write the most general Hermitian matrix as

$$M = c_{00} \mathbf{1} \otimes \mathbf{1} + c_{a0} \tau^a \otimes \mathbf{1} + c_{0b} \mathbf{1} \otimes \sigma^b + c_{ab} \tau^a \otimes \sigma^b.$$

We require

$$\{M, \tau^1 \otimes \sigma^i\} = 0 \quad \forall i.$$

A direct basis computation shows:

- any term containing $\mathbf{1} \otimes \mathbf{1}$ or $\tau^1 \otimes \mathbf{1}$ commutes, not anticommutes;
- any term containing $\mathbf{1} \otimes \sigma^b$ or $\tau^1 \otimes \sigma^b$ fails to anticommute for all i ;
- terms $\tau^2 \otimes \mathbf{1}$ and $\tau^3 \otimes \mathbf{1}$ anticommute with $\tau^1 \otimes \sigma^i$ because

$$\{\tau^2, \tau^1\} = 0, \quad \{\tau^3, \tau^1\} = 0,$$

while $\mathbf{1}$ commutes with all σ^i .

No other basis elements satisfy the full set of anticommutation conditions simultaneously. $\square$

Thus the 4×4 space already has a two-dimensional mass space. This is very different from the pure 2×2 valley pair, where all Pauli terms compete on equal footing.

G. Chiral symmetry in the 4×4 space

Take

$$\Gamma = \tau^2 \otimes \mathbf{1} = \gamma^5.$$

Then

$$\{\Gamma, \alpha^i\} = 0, \quad [\Gamma, \tau^2 \otimes \mathbf{1}] = 0, \quad \{\Gamma, \tau^3 \otimes \mathbf{1}\} = 0.$$

If one imposes the chiral condition

$$\Gamma H \Gamma^{-1} = -H,$$

then any term commuting with Γ is forbidden, while anticommuting terms are allowed.

a. Corollary 4. If

$$\Gamma = \tau^2 \otimes \mathbf{1}$$

is an exact chiral symmetry, then the mass term

$$\tau^2 \otimes \mathbf{1}$$

is forbidden, while

$$\tau^3 \otimes \mathbf{1}$$

remains allowed.

Thus chiral symmetry removes one of the two possible Dirac masses, but not both.

H. Valley-exchange symmetry in the 4×4 space

Let valley exchange be represented by

$$P_v = \mathbf{1} \otimes \sigma^1.$$

Then

$$P_v(\tau^1 \otimes \sigma^1)P_v^{-1} = \tau^1 \otimes \sigma^1,$$

$$P_v(\tau^1 \otimes \sigma^2)P_v^{-1} = -\tau^1 \otimes \sigma^2,$$

$$P_v(\tau^1 \otimes \sigma^3)P_v^{-1} = -\tau^1 \otimes \sigma^3.$$

The mass terms

$$\tau^2 \otimes \mathbf{1}, \quad \tau^3 \otimes \mathbf{1}$$

are both invariant under P_v .

a. Corollary 5. Valley exchange alone does not forbid either of the Dirac masses in the 4×4 tensor-product space.

So valley exchange helps organize the kinetic structure, but does not solve the mass problem.

I. Internal parity between the two Q -modes

Suppose now there is an internal symmetry acting as

$$P_{\text{int}} = \tau^1 \otimes \mathbf{1}$$

on the Q -doublet factor. Then

$$P_{\text{int}}(\tau^3 \otimes \mathbf{1})P_{\text{int}}^{-1} = -\tau^3 \otimes \mathbf{1},$$

$$P_{\text{int}}(\tau^2 \otimes \mathbf{1})P_{\text{int}}^{-1} = -\tau^2 \otimes \mathbf{1}.$$

Hence both Dirac masses are odd under this symmetry.

a. Theorem 6 (Internal-exchange protection theorem). If the low-energy theory has an exact internal exchange symmetry

$$P_{\text{int}} = \tau^1 \otimes \mathbf{1}$$

and the Hamiltonian is invariant under it, then both Dirac masses

$$\tau^2 \otimes \mathbf{1}, \quad \tau^3 \otimes \mathbf{1}$$

are forbidden.

b. Proof. Both masses change sign under conjugation by τ^1 , hence cannot appear in an invariant Hamiltonian. $\square$

This is the first strong algebraic result of the 4×4 analysis: an exact symmetry exchanging the two internal Q -modes would forbid the full Dirac mass space.

However, whether such an internal exchange symmetry exists in TECT is not yet established.

J. Sharp conclusion of the 4×4 audit

We may now state the central result.

a. Theorem 7 (Why the 4-dimensional space is the right target). The tensor-product space

$$\mathbb{C}_{\text{int}}^2 \otimes \mathbb{C}_{\text{val}}^2$$

is the minimal natural low-energy candidate space that:

1. admits a representation of the 3 + 1-dimensional Dirac/Clifford algebra;
2. contains a genuine Dirac kinetic sector

$$v_i p_i (\tau^1 \otimes \sigma^i);$$

3. allows a nontrivial mass classification;
4. can, under additional internal symmetry, forbid the entire Dirac mass space.

By contrast, the pure valley-pair 2×2 space is too small and too weakly constrained to do so.

b. Proof. Items (1)–(3) follow from Theorems 1 and 3. Item (4) follows from Theorem 6. The insufficiency of the pure valley-pair space was proved earlier. $\square$

K. What is proved and what remains open

What is proved:

1. the 4-dimensional internal $\otimes$ valley space is algebraically sufficient for Dirac;
2. the pure valley-pair space is not;
3. the full Dirac mass space in this 4×4 setting is exactly two-dimensional;
4. an internal exchange symmetry would forbid all Dirac masses.

What remains open:

1. whether the actual TECT low-energy projected operator closes on this tensor-product space;
2. whether its kinetic term really takes the form

$$v_i p_i (\tau^1 \otimes \sigma^i);$$

3. whether an exact internal symmetry exchanging the two Q -modes exists;
4. whether the necessary low-energy sector is spectrally isolated in the microscopic theory.

XXV. FINAL SUMMARY

The pure valley-pair space

$$\mathbb{C}_{\text{val}}^2$$

is insufficient to produce a robust symmetry-protected Weyl/Dirac sector.

The natural next candidate is the tensor-product space

$$\mathcal{H}_{\text{low}} = \mathbb{C}_{\text{int}}^2 \otimes \mathbb{C}_{\text{val}}^2,$$

where $\mathbb{C}_{\text{int}}^2$ comes from the rank-two Q -bundle and $\mathbb{C}_{\text{val}}^2$ from the valley pair.

On this 4-dimensional space one can define

$$\alpha^i = \tau^1 \otimes \sigma^i, \quad \beta = \tau^3 \otimes \mathbf{1},$$

which satisfy the Dirac/Clifford relations

$$\{\alpha^i, \alpha^j\} = 2\delta^{ij}, \quad \{\alpha^i, \beta\} = 0, \quad \beta^2 = 1.$$

Thus this space is the minimal natural carrier of a $3 + 1$ -dimensional Dirac operator. The complete mass classification relative to these kinetic generators is:

$$M \text{ anticommutes with all } \alpha^i \iff M = a(\tau^2 \otimes \mathbf{1}) + b(\tau^3 \otimes \mathbf{1}).$$

Hence the Dirac mass space is exactly two-dimensional.

Moreover, if an exact internal exchange symmetry

$$P_{\text{int}} = \tau^1 \otimes \mathbf{1}$$

is present, then both masses are forbidden, since both are odd under this symmetry.

Therefore the sharp honest conclusion is:

The 4-dimensional internal $\otimes$ valley space is the first setting in which a genuine Dirac algebra and a nontrivial full mass-protect

What remains open is not the algebraic possibility, but the dynamical realization: whether the actual TECT low-energy operator projects onto this space, selects the Dirac kinetic sector, and realizes the required internal exchange symmetry.

XXVI. COMPLETE SYMMETRY AUDIT ON THE 4×4 INTERNAL $\otimes$ VALLEY SPACE: WHY STANDARD FACTORIZED SYMMETRIES DO NOT ISOLATE THE PURE DIRAC KINETIC SECTOR

Assumption Audit

- A1: The candidate infrared space is

$$\mathcal{H}_{\text{low}} = \mathbb{C}_{\text{int}}^2 \otimes \mathbb{C}_{\text{val}}^2.$$

Valid candidate construction

- A2: A general Hermitian operator on $\mathcal{H}_{\text{low}}$ admits the expansion

$$H = c_{00} \mathbf{1} \otimes \mathbf{1} + c_{a0} \tau^a \otimes \mathbf{1} + c_{0b} \mathbf{1} \otimes \sigma^b + c_{ab} \tau^a \otimes \sigma^b.$$

Proved

- A3: Internal exchange and chiral symmetry reduce H to

$$H = \tau^1 \otimes (b_0 \mathbf{1} + b_i \sigma^i).$$

Proved

- A4: Standard factorized valley/discrete symmetries can eliminate the scalar $\tau^1 \otimes \mathbf{1}$ while preserving all three kinetic generators $\tau^1 \otimes \sigma^i$.

False in general

- A5: A non-factorized symmetry mixing internal and valley factors may do better.

Conditionally true

- A6: A microscopic first-order projection may dynamically select the Dirac sector without symmetry isolation.

Conditionally true

A. Goal

We want to answer the following precise question:

Which minimal exact symmetry set, if any, forces $H_{\text{eff}}(p) = \varepsilon_0(p)\mathbf{1}_4 + v_i p_i (\tau^1 \otimes \sigma^i)$

up to higher-order corrections?

We show that standard *factorized* symmetries do not suffice.

B. General Hermitian Hamiltonian

Let

$$\mathcal{H}_{\text{low}} = \mathbb{C}_{\text{int}}^2 \otimes \mathbb{C}_{\text{val}}^2,$$

with Pauli matrices

$$\tau^a \text{ acting on } \mathbb{C}_{\text{int}}^2, \quad \sigma^b \text{ acting on } \mathbb{C}_{\text{val}}^2.$$

The most general Hermitian Hamiltonian is

$$H(p) = c_{00}(p) \mathbf{1} \otimes \mathbf{1} + c_{a0}(p) \tau^a \otimes \mathbf{1} + c_{0b}(p) \mathbf{1} \otimes \sigma^b + c_{ab}(p) \tau^a \otimes \sigma^b.$$

C. Step 1: internal exchange symmetry

Impose an exact internal exchange symmetry

$$X = \tau^1 \otimes \mathbf{1}, \quad XH(p)X^{-1} = H(p).$$

Using

$$\tau^1 \tau^1 \tau^1 = \tau^1, \quad \tau^1 \tau^2 \tau^1 = -\tau^2, \quad \tau^1 \tau^3 \tau^1 = -\tau^3,$$

we see that all terms containing τ^2 or τ^3 are forbidden, while terms containing $\mathbf{1}$ or τ^1 survive. Thus

$$H(p) = a_0(p) \mathbf{1} \otimes \mathbf{1} + a_b(p) \mathbf{1} \otimes \sigma^b + b_0(p) \tau^1 \otimes \mathbf{1} + b_b(p) \tau^1 \otimes \sigma^b.$$

a. Theorem 1. Internal exchange invariance under $X = \tau^1 \otimes \mathbf{1}$ reduces the 16-dimensional Hermitian operator space to the 8-dimensional subspace

$$\text{span}\{\mathbf{1} \otimes \mathbf{1}, \mathbf{1} \otimes \sigma^b, \tau^1 \otimes \mathbf{1}, \tau^1 \otimes \sigma^b\}.$$

D. Step 2: chiral symmetry

Now impose chiral symmetry

$$\Gamma = \tau^2 \otimes \mathbf{1}, \quad \Gamma H(p) \Gamma^{-1} = -H(p).$$

Since

$$\tau^2 \mathbf{1} \tau^2 = \mathbf{1}, \quad \tau^2 \tau^1 \tau^2 = -\tau^1,$$

the anticommutation condition kills all terms with internal factor $\mathbf{1}$, and retains terms with internal factor τ^1 . Therefore

$$H(p) = \tau^1 \otimes (b_0(p) \mathbf{1} + b_i(p) \sigma^i).$$

a. Theorem 2. The simultaneous imposition of internal exchange

$$X = \tau^1 \otimes \mathbf{1}$$

and chiral symmetry

$$\Gamma = \tau^2 \otimes \mathbf{1}$$

reduces the effective Hamiltonian to

$$H(p) = \tau^1 \otimes (b_0(p)\mathbf{1} + b_i(p)\sigma^i).$$

This is the maximal reduction obtainable from these two exact symmetries alone.

E. Interpretation

The surviving terms are exactly:

$$\tau^1 \otimes \mathbf{1} \quad (\text{scalar in valley space}),$$

and

$$\tau^1 \otimes \sigma^1, \quad \tau^1 \otimes \sigma^2, \quad \tau^1 \otimes \sigma^3 \quad (\text{valley-vector sector}).$$

Hence the problem is reduced to a 4-parameter family. If one could eliminate the scalar b_0 while keeping all three b_i , then one would indeed arrive at the desired Dirac kinetic form.

F. Step 3: standard factorized valley symmetry

Consider a unitary valley exchange

$$P_v = \mathbf{1} \otimes \sigma^1, \quad P_v H(p) P_v^{-1} = H(-p).$$

Using

$$\sigma^1 \sigma^1 \sigma^1 = \sigma^1, \quad \sigma^1 \sigma^2 \sigma^1 = -\sigma^2, \quad \sigma^1 \sigma^3 \sigma^1 = -\sigma^3,$$

we obtain

$$b_0(p) = b_0(-p), \quad b_1(p) = b_1(-p), \quad b_2(p) = -b_2(-p), \quad b_3(p) = -b_3(-p).$$

Thus P_v alone leaves

$$\tau^1 \otimes \mathbf{1}, \quad \tau^1 \otimes \sigma^1$$

even, and

$$\tau^1 \otimes \sigma^2, \quad \tau^1 \otimes \sigma^3$$

odd.

So the scalar term survives.

G. Step 4: standard factorized antiunitary symmetries

We now test the standard antiunitary factorized possibilities.

Case A: spinless time reversal

Let

$$\Theta_0 = \mathbf{1} \otimes \mathbf{1} K.$$

Then

$$\Theta_0(\tau^1 \otimes \sigma^1)\Theta_0^{-1} = +\tau^1 \otimes \sigma^1,$$

$$\Theta_0(\tau^1 \otimes \sigma^2)\Theta_0^{-1} = -\tau^1 \otimes \sigma^2,$$

$$\Theta_0(\tau^1 \otimes \sigma^3)\Theta_0^{-1} = +\tau^1 \otimes \sigma^3,$$

and

$$\Theta_0(\tau^1 \otimes \mathbf{1})\Theta_0^{-1} = +\tau^1 \otimes \mathbf{1}.$$

Thus

$$b_0, b_1, b_3 \text{ are even, } \quad b_2 \text{ is odd.}$$

Combining with valley exchange P_v , we get

$$b_0 \text{ even, } \quad b_1 \text{ even, } \quad b_2 \text{ odd, } \quad b_3 \equiv 0.$$

Hence

$$H(p) = \tau^1 \otimes (b_0(p)\mathbf{1} + b_1(p)\sigma^1 + b_2(p)\sigma^2).$$

This is not the pure Dirac kinetic sector.

Case B: Kramers-like valley antiunitary symmetry

Let

$$\Theta_v = \mathbf{1} \otimes i\sigma^2 K.$$

Since

$$(i\sigma^2)\sigma^{i*}(-i\sigma^2) = -\sigma^i,$$

we have

$$\Theta_v(\tau^1 \otimes \mathbf{1})\Theta_v^{-1} = +\tau^1 \otimes \mathbf{1}, \quad \Theta_v(\tau^1 \otimes \sigma^i)\Theta_v^{-1} = -\tau^1 \otimes \sigma^i.$$

Therefore

$$b_0 \text{ is even, } \quad b_i \text{ are odd.}$$

Combining with P_v , we obtain

$$b_0 \text{ even, } \quad b_1 \equiv 0, \quad b_2, b_3 \text{ odd.}$$

Hence

$$H(p) = \tau^1 \otimes (b_0(p)\mathbf{1} + b_2(p)\sigma^2 + b_3(p)\sigma^3).$$

Again, the scalar survives and only two Pauli directions remain.

Case C: mixed internal-valley antiunitary symmetry

Let

$$\tilde{\Theta} = \tau^3 \otimes i\sigma^2 K.$$

Then

$$\tau^3 \tau^1 \tau^3 = -\tau^1,$$

so

$$\tilde{\Theta}(\tau^1 \otimes \mathbf{1})\tilde{\Theta}^{-1} = -\tau^1 \otimes \mathbf{1},$$

while

$$\tilde{\Theta}(\tau^1 \otimes \sigma^i)\tilde{\Theta}^{-1} = +\tau^1 \otimes \sigma^i.$$

Thus

$$b_0 \text{ is odd, } \quad b_i \text{ are even.}$$

Combining with P_v , we obtain

$$b_0 \equiv 0, \quad b_1 \text{ even,} \quad b_2 \equiv 0, \quad b_3 \equiv 0.$$

So

$$H(p) = \tau^1 \otimes b_1(p)\sigma^1.$$

This kills the scalar, but collapses the kinetic sector to only one Pauli direction.

a. Theorem 3. With X and Γ already imposed, none of the standard factorized antiunitary choices

$$\Theta_0, \quad \Theta_v, \quad \tilde{\Theta}$$

together with valley exchange P_v isolates the full pure Dirac kinetic sector

$$\tau^1 \otimes \sigma^1, \quad \tau^1 \otimes \sigma^2, \quad \tau^1 \otimes \sigma^3$$

while simultaneously removing the scalar

$$\tau^1 \otimes \mathbf{1}.$$

b. Proof. Direct from the three cases above. $\square$

H. A more general no-go statement for factorized symmetries

The previous computation suggests a stronger structural statement.

Let $\mathcal{S}$ be any additional symmetry that acts *factorizably* on the reduced subspace:

$$\mathcal{S} = U_{\text{int}} \otimes U_{\text{val}}$$

or

$$\mathcal{S} = U_{\text{int}} \otimes U_{\text{val}} K.$$

After imposing X and Γ , the Hamiltonian has the form

$$H = \tau^1 \otimes (b_0 \mathbf{1} + b_i \sigma^i).$$

Suppose the symmetry preserves the full valley-vector triplet, meaning that under conjugation it acts as

$$\sigma^i \mapsto R_{ij} \sigma^j, \quad R \in O(3),$$

while

$$\mathbf{1} \mapsto \mathbf{1}.$$

The internal factor contributes only an overall sign $s = \pm 1$ on τ^1 .

Then the scalar term and the vector triplet transform with the same internal sign s ; the only difference is that the vector rotates by R , while the scalar does not rotate. Therefore a single such factorized symmetry cannot impose a parity rule that eliminates

$$\tau^1 \otimes \mathbf{1}$$

while preserving all three independent components of

$$\tau^1 \otimes \sigma^i$$

as a full triplet.

a. Theorem 4 (Factorized-symmetry no-go theorem). After reduction by internal exchange X and chiral symmetry Γ , no additional single factorized symmetry that preserves the valley Pauli triplet as a whole can kill the scalar

$$\tau^1 \otimes \mathbf{1}$$

while retaining all three kinetic generators

$$\tau^1 \otimes \sigma^1, \quad \tau^1 \otimes \sigma^2, \quad \tau^1 \otimes \sigma^3.$$

b. Proof. In the reduced sector, every term has the common internal factor τ^1 . A factorized symmetry acts on that internal factor by a single sign $s = \pm 1$, common to both the scalar and vector terms. On the valley factor, the identity is fixed, while the vector $(\sigma^1, \sigma^2, \sigma^3)$ transforms as an orthogonal triplet. Therefore the symmetry can at most distinguish the scalar from the vector by rotation structure, but not by an independent sign rule that removes the scalar while leaving the entire vector triplet intact. Any additional constraint strong enough to kill the scalar either also kills some vector components, or requires non-factorized action mixing the two tensor factors. $\square$

This is the cleanest mathematical statement of the obstruction.

I. Consequences

At this point the logic is sharp:

1. Internal exchange X and chiral symmetry Γ reduce the problem to the 4-term family

$$\tau^1 \otimes (b_0 \mathbf{1} + b_i \sigma^i).$$

2. Standard factorized valley/discrete symmetries can reduce this further, but always leave one of the following problems:

- (a) the scalar b_0 survives, or
- (b) one or two of the three kinetic directions are killed.

3. Therefore the exact pure Dirac kinetic sector

$$H_{\text{Dirac}}(p) = v_i p_i (\tau^1 \otimes \sigma^i)$$

cannot be isolated by standard factorized symmetry alone.

J. What can still work?

There remain only two logically viable routes.

a. Route I: non-factorized symmetry. A symmetry that mixes internal and valley indices nontrivially may distinguish

$$\tau^1 \otimes \mathbf{1}$$

from

$$\tau^1 \otimes \sigma^i$$

more strongly than any factorized symmetry can. Such a symmetry could, in principle, kill the scalar while keeping a full Clifford triplet.

b. Route II: dynamical first-order projection. Even without symmetry isolation, a microscopic first-order operator or an explicit projected defect/valley spectral calculation may show that the actual low-energy Hamiltonian closes on the Dirac sector

$$v_i p_i (\tau^1 \otimes \sigma^i)$$

because all other reduced-sector operators have vanishing matrix elements in the relevant low-energy subspace.

c. Theorem 5 (Sharp remaining alternatives). After the no-go result above, the pure Dirac kinetic sector can be justified only by one of the following:

1. an exact non-factorized symmetry acting on

$$\mathbb{C}_{\text{int}}^2 \otimes \mathbb{C}_{\text{val}}^2,$$

or

2. an explicit dynamical projection theorem for the microscopic or defect linearized operator.

d. Proof. The factorized route has been exhausted by Theorem 4. Therefore any successful remaining mechanism must lie outside that class. $\square$

XXVII. FINAL SUMMARY

On the 4-dimensional candidate low-energy space

$$\mathcal{H}_{\text{low}} = \mathbb{C}_{\text{int}}^2 \otimes \mathbb{C}_{\text{val}}^2,$$

the exact imposition of internal exchange

$$X = \tau^1 \otimes \mathbf{1}$$

and chiral symmetry

$$\Gamma = \tau^2 \otimes \mathbf{1}$$

reduces the most general Hermitian effective Hamiltonian to

$$H(p) = \tau^1 \otimes (b_0(p)\mathbf{1} + b_i(p)\sigma^i).$$

This is the maximal universal algebraic reduction obtainable from those two symmetries.

The desired pure Dirac kinetic sector would be

$$H_{\text{Dirac}}(p) = v_i p_i (\tau^1 \otimes \sigma^i),$$

which requires:

1. elimination of the scalar term

$$\tau^1 \otimes \mathbf{1},$$

2. retention of all three kinetic generators

$$\tau^1 \otimes \sigma^1, \quad \tau^1 \otimes \sigma^2, \quad \tau^1 \otimes \sigma^3.$$

Explicit classification shows that standard factorized valley/discrete symmetries (valley exchange, spinless time reversal, Kramers-like antiunitary valley symmetry, and mixed factorized antiunitary symmetries) cannot achieve both goals simultaneously:

either the scalar survives, or one/two of the three kinetic directions are lost.

More generally:

no standard factorized symmetry preserving the valley Pauli triplet as a whole can remove $\tau^1 \otimes \mathbf{1}$ while preserving all $\tau^1 \otimes \sigma^i$.

Therefore the problem is now sharply localized. The remaining viable mechanisms are only:

(i) an exact non-factorized symmetry mixing internal and valley sectors,

or

(ii) an explicit dynamical first-order projection of the microscopic/defect operator.

This is the strongest honest conclusion available at the present stage.

XXVIII. OPERATOR-STRUCTURE THEOREM: BLOCK-OFF-DIAGONAL FIRST-ORDER FORM IMPLIES PROJECTED DIRAC KINETICS

Assumption Audit

- A1: The relevant infrared sector is represented on a complex 4-dimensional space

$$\mathcal{H}_{\text{low}} \cong \mathbb{C}_{\text{int}}^2 \otimes \mathbb{C}_{\text{val}}^2.$$

Valid candidate low-energy construction

- A2: There exists a chiral grading

$$\Gamma^2 = \mathbf{1}, \quad \Gamma^\dagger = \Gamma, \quad \{\Gamma, \mathcal{D}(p)\} = 0$$

for the low-energy operator.

Needs extra structural assumption

- A3: In the chiral decomposition, the off-diagonal block is first-order in momentum:

$$A(p) = p_i M_i + O(|p|^2).$$

Provable once a first-order structure is assumed

- A4: The principal symbol is nondegenerate.

Residual nondegeneracy assumption

- A5: A 2×2 complex linear symbol can be reduced to Pauli form by basis changes.

Proved

- A6: Therefore the full 4×4 operator is reducible to Dirac kinetic form.

Provable from A2–A5

- A7: The full microscopic TECT defect/valley operator actually realizes A2–A4.

Currently unproved

A. Goal

We seek a theorem of the following type:

If the projected low-energy operator is block-off-diagonal and first-order, then Dirac kinetics follows automatically.

This avoids relying on an insufficient symmetry-isolation argument.

B. Chiral grading and block decomposition

Let $\mathcal{H}_{\text{low}}$ be a complex 4-dimensional Hilbert space with a grading operator

$$\Gamma = \Gamma^\dagger, \quad \Gamma^2 = \mathbf{1}.$$

Then $\mathcal{H}_{\text{low}}$ splits as

$$\mathcal{H}_{\text{low}} = \mathcal{H}_+ \oplus \mathcal{H}_-, \quad \Gamma|_{\mathcal{H}_\pm} = \pm 1,$$

with

$$\dim_{\mathbb{C}} \mathcal{H}_+ = \dim_{\mathbb{C}} \mathcal{H}_- = 2.$$

If the effective operator $\mathcal{D}(p)$ anticommutes with Γ ,

$$\{\Gamma, \mathcal{D}(p)\} = 0,$$

then in the decomposition $\mathcal{H}_+ \oplus \mathcal{H}_-$ it has the form

$$\mathcal{D}(p) = \begin{pmatrix} 0 & A(p) \\ A(p)^\dagger & 0 \end{pmatrix},$$

where

$$A(p) : \mathcal{H}_- \rightarrow \mathcal{H}_+$$

is a 2×2 complex matrix.

a. Theorem 1 (Off-diagonalization theorem). If a Hermitian operator $\mathcal{D}$ anticommutes with a grading Γ , then in the Γ -eigenbasis it is block-off-diagonal.

b. Proof. Write

$$\mathcal{D} = \begin{pmatrix} D_{++} & D_{+-} \\ D_{-+} & D_{--} \end{pmatrix}, \quad \Gamma = \begin{pmatrix} \mathbf{1}_2 & 0 \\ 0 & -\mathbf{1}_2 \end{pmatrix}.$$

Then

$$\Gamma \mathcal{D} + \mathcal{D} \Gamma = \begin{pmatrix} 2D_{++} & 0 \\ 0 & -2D_{--} \end{pmatrix}.$$

Thus $\{\Gamma, \mathcal{D}\} = 0$ implies

$$D_{++} = D_{--} = 0.$$

Hermiticity gives $D_{-+} = D_{+-}^\dagger$. $\square$

C. First-order principal symbol

Assume now that

$$A(p) = p_i M_i + O(|p|^2), \quad i = 1, 2, 3,$$

for some complex 2×2 matrices M_i . Then the principal part of the low-energy operator is

$$\mathcal{D}_1(p) = \begin{pmatrix} 0 & p_i M_i \\ p_i M_i^\dagger & 0 \end{pmatrix}.$$

The spectrum of $\mathcal{D}_1(p)$ is determined by

$$\mathcal{D}_1(p)^2 = \begin{pmatrix} (p_i M_i)(p_j M_j)^\dagger & 0 \\ 0 & (p_i M_i)^\dagger(p_j M_j) \end{pmatrix}.$$

Hence the low-energy dispersion is governed by the singular values of the 2×2 matrix

$$A_1(p) := p_i M_i.$$

D. Classification of 2×2 complex linear symbols

Every complex 2×2 matrix admits the decomposition

$$A_1(p) = a_0(p)\mathbf{1}_2 + a_i(p)\sigma^i,$$

with complex coefficients a_0, a_i linear in p .

We now distinguish the traceless and scalar parts. The scalar term $a_0(p)\mathbf{1}_2$ tilts or rescales, but does not obstruct the Weyl structure of the traceless part. The essential point is that, after invertible basis changes on $\mathcal{H}_+$ and $\mathcal{H}_-$, a nondegenerate linear symbol may be brought to canonical form.

a. Lemma 2 (Biunitary reduction of the principal symbol). Let

$$A_1(p) = p_i M_i$$

be a 2×2 complex linear symbol. Suppose its determinant is not identically zero and that its zero at $p = 0$ is isolated and linear. Then there exist invertible changes of basis

$$U_+ \in U(2), \quad U_- \in U(2),$$

and an invertible linear change of momentum variables $p \mapsto p'$, such that

$$U_+ A_1(p) U_-^\dagger = p'_1 \sigma^1 + p'_2 \sigma^2 + p'_3 \sigma^3$$

up to an irrelevant scalar linear term proportional to $\mathbf{1}_2$.

b. Sketch of proof. The space of traceless 2×2 complex matrices is spanned by the Pauli matrices. The linear map

$$p \mapsto A_1(p) - \frac{1}{2} \text{tr}(A_1(p)) \mathbf{1}_2$$

defines a linear map $\mathbb{R}^3 \rightarrow \mathfrak{sl}(2, \mathbb{C})$. Under left-right unitary basis changes and momentum redefinition, any full-rank such map may be put into canonical Pauli form. $\square$

E. Canonical Dirac form

After the reduction above, the principal operator becomes

$$\mathcal{D}_1(p) \sim \begin{pmatrix} 0 & p_i \sigma^i \\ p_i \sigma^i & 0 \end{pmatrix}$$

in an appropriate basis, where we have absorbed velocities into the momentum coordinates.

Now define

$$\alpha^i = \begin{pmatrix} 0 & \sigma^i \\ \sigma^i & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} \mathbf{1}_2 & 0 \\ 0 & -\mathbf{1}_2 \end{pmatrix}.$$

Then

$$\{\alpha^i, \alpha^j\} = 2\delta^{ij}\mathbf{1}_4, \quad \{\alpha^i, \beta\} = 0, \quad \beta^2 = \mathbf{1}_4.$$

Hence

$$\mathcal{D}_1(p) = p_i \alpha^i$$

is exactly the massless Dirac kinetic operator.

a. Theorem 3 (Block-off-diagonal Dirac theorem). Suppose a Hermitian low-energy operator on a 4-dimensional complex space satisfies:

1. there exists a grading Γ with $\{\Gamma, \mathcal{D}(p)\} = 0$;
2. in the Γ -decomposition,

$$\mathcal{D}(p) = \begin{pmatrix} 0 & A(p) \\ A(p)^\dagger & 0 \end{pmatrix};$$

3. the principal part of $A(p)$ is linear,

$$A(p) = p_i M_i + O(|p|^2);$$

4. the linear symbol is nondegenerate.

Then, after basis changes and linear momentum redefinition, the principal part of $\mathcal{D}$ is the massless Dirac operator

$$\mathcal{D}_1(p) = p_i \alpha^i.$$

b. Proof. By Theorem 1, the operator is off-diagonal. By Lemma 2, the principal symbol of the off-diagonal block can be reduced to Pauli form. Substituting into the full 4×4 completion yields

$$\mathcal{D}_1(p) = \begin{pmatrix} 0 & p_i \sigma^i \\ p_i \sigma^i & 0 \end{pmatrix} = p_i \alpha^i.$$

□

F. Relation to the internal $\otimes$ valley basis

The canonical Dirac basis above is abstract. In the concrete tensor-product basis

$$\mathcal{H}_{\text{low}} = \mathbb{C}_{\text{int}}^2 \otimes \mathbb{C}_{\text{val}}^2,$$

one convenient choice is

$$\Gamma = \tau^3 \otimes \mathbf{1},$$

so that the grading is carried by the internal factor. Then the off-diagonal generators are naturally

$$\alpha^i = \tau^1 \otimes \sigma^i$$

or an equivalent Clifford triple obtained by unitary conjugation.

Thus the earlier algebraic construction of the Dirac generators is recovered as a concrete realization of the abstract block-off-diagonal theorem.

G. Mass terms

A mass term $m\beta$ is compatible with the grading if β anticommutes with all α^i and commutes with Γ . In the canonical basis one has

$$\beta = \begin{pmatrix} \mathbf{1}_2 & 0 \\ 0 & -\mathbf{1}_2 \end{pmatrix}.$$

Hence the massive Dirac operator is

$$\mathcal{D}(p) = p_i \alpha^i + m\beta + O(|p|^2).$$

The important point here is that the kinetic sector no longer needs to be extracted by forbidding all possible terms via symmetry. Once the low-energy operator is known to be first-order and block-off-diagonal, the Dirac kinetic structure is automatic.

H. What remains to be proved in TECT

The block-off-diagonal Dirac theorem shows that the genuinely load-bearing unresolved ingredients are now reduced to:

1. existence of a 4-dimensional isolated infrared sector;
2. existence of a grading Γ for the projected operator;
3. derivation of a first-order off-diagonal principal symbol from the TECT defect/valley linearization;
4. nondegeneracy of that symbol.

This is much sharper than the previous symmetry-only formulations.

a. Corollary 4 (Sharp remaining TECT problem). To derive the Dirac kinetic term in TECT, it is sufficient to show that the projected microscopic or defect low-energy operator has a chiral block-off-diagonal first-order principal symbol with nondegenerate 2×2 off-diagonal block.

No further symmetry-isolation theorem is required.

I. Why this is stronger than the previous route

Previously the strategy was:

$$\text{symmetry constraints} \implies \text{few allowed terms} \implies \text{hopefully Dirac.}$$

Now the structure theorem is:

$$\text{chiral grading} + \text{first-order off-diagonal symbol} \implies \text{Dirac kinetic automatically.}$$

This is mathematically stronger because it derives the operator form directly rather than trying to eliminate all competitors one by one by symmetry.

XXIX. FINAL SUMMARY

The previous symmetry-based route was too weak to isolate the pure Dirac kinetic sector on the 4×4 internal $\otimes$ valley space.

A stronger operator-theoretic route is available.

Let the projected infrared operator $\mathcal{D}(p)$ act on a 4-dimensional complex space and suppose:

1. there exists a grading operator

$$\Gamma^2 = \mathbf{1}, \quad \{\Gamma, \mathcal{D}(p)\} = 0;$$

2. in the Γ -eigenbasis, $\mathcal{D}(p)$ is block-off-diagonal:

$$\mathcal{D}(p) = \begin{pmatrix} 0 & A(p) \\ A(p)^\dagger & 0 \end{pmatrix};$$

3. the off-diagonal block has first-order principal symbol

$$A(p) = p_i M_i + O(|p|^2);$$

4. the symbol is nondegenerate.

Then, after basis changes and linear momentum redefinition, the principal part is exactly the massless Dirac kinetic operator:

$$\mathcal{D}_1(p) = p_i \alpha^i.$$

Equivalently, in a concrete internal $\otimes$ valley realization one may take

$$\Gamma = \tau^3 \otimes \mathbf{1}, \quad \alpha^i = \tau^1 \otimes \sigma^i,$$

up to unitary equivalence.

Therefore the mathematically decisive unresolved problem is no longer a symmetry-selection problem, but the following operator-structure problem:

Does the projected TECT defect/valley operator possess a chiral block-off-diagonal first-order nondegenerate symbol?

If yes, Dirac kinetics follows automatically.

XXX. WHEN DOES A CHIRAL BLOCK-OFF-DIAGONAL FIRST-ORDER STRUCTURE ACTUALLY EMERGE FROM DEFECT/VALLEY LINEARIZATION?

Assumption Audit

- A1: The relevant infrared sector is a spectrally isolated complex 4-dimensional space

$$\mathcal{H}_{\text{low}} \cong \mathbb{C}^4.$$

Needs spectral input

- A2: This space admits a $2 + 2$ decomposition

$$\mathcal{H}_{\text{low}} = \mathcal{H}_+ \oplus \mathcal{H}_-, \quad \dim \mathcal{H}_\pm = 2.$$

Needs structural input

- A3: The leading reduced operator is odd with respect to this grading:

$$\Pi_\pm \mathcal{D}_1 \Pi_\pm = 0 \quad \Leftrightarrow \quad \{\Gamma, \mathcal{D}_1\} = 0.$$

Provable once one of several sufficient mechanisms holds

- A4: The off-diagonal block is first-order in the propagating momentum:

$$A(p) = p_i M_i.$$

Provable in first-order/envelope reductions

- A5: The principal symbol is nondegenerate.

Residual nondegeneracy condition

- A6: A generic second-order bosonic Hessian automatically has the above properties.

False in general

A. Goal

We want to isolate minimal sufficient conditions under which the reduced defect/valley operator takes the form

$$\mathcal{D}_1(p) = \begin{pmatrix} 0 & p_i M_i \\ p_i M_i^\dagger & 0 \end{pmatrix},$$

so that the previous block-off-diagonal Dirac theorem applies.

The logical point is crucial:

$$\text{CP}^2 \text{ geometry alone} \not\Rightarrow \text{Dirac operator},$$

but

$$\text{graded first-order reduced operator} \Rightarrow \text{Dirac kinetic}.$$

Thus the issue is to identify the origin of the grading and the vanishing of the diagonal blocks.

B. General reduced low-energy operator

Let $\mathcal{L}$ denote the linearized microscopic or effective operator around a defect or valley background. Let

$$\Pi_{\text{low}}$$

be the projector onto a 4-dimensional isolated low-energy sector

$$\mathcal{H}_{\text{low}} = \Pi_{\text{low}} \mathcal{H}.$$

The reduced operator is

$$\mathcal{D}(p) := \Pi_{\text{low}} \mathcal{L}(p) \Pi_{\text{low}}.$$

Assume now that $\mathcal{H}_{\text{low}}$ is decomposed as

$$\mathcal{H}_{\text{low}} = \mathcal{H}_+ \oplus \mathcal{H}_-, \quad \dim \mathcal{H}_\pm = 2,$$

with projectors $\Pi_\pm$. Then the reduced operator splits into blocks:

$$\mathcal{D}(p) = \begin{pmatrix} \Pi_+ \mathcal{D}(p) \Pi_+ & \Pi_+ \mathcal{D}(p) \Pi_- \\ \Pi_- \mathcal{D}(p) \Pi_+ & \Pi_- \mathcal{D}(p) \Pi_- \end{pmatrix}.$$

The Dirac route becomes available precisely when the diagonal blocks vanish to leading order.

C. Abstract grading criterion

Define the grading operator

$$\Gamma := \Pi_+ - \Pi_-.$$

Then

$$\Gamma^\dagger = \Gamma, \quad \Gamma^2 = \mathbf{1}.$$

A direct computation gives:

a. Proposition 1. The following are equivalent:

1.

$$\Pi_+ \mathcal{D}_1 \Pi_+ = 0 \quad \text{and} \quad \Pi_- \mathcal{D}_1 \Pi_- = 0;$$

2.

$$\{\Gamma, \mathcal{D}_1\} = 0;$$

3. $\mathcal{D}_1$ is block-off-diagonal in the decomposition

$$\mathcal{H}_{\text{low}} = \mathcal{H}_+ \oplus \mathcal{H}_-.$$

b. Proof. Write

$$\Gamma = \begin{pmatrix} \mathbf{1}_2 & 0 \\ 0 & -\mathbf{1}_2 \end{pmatrix}, \quad \mathcal{D}_1 = \begin{pmatrix} D_{++} & D_{+-} \\ D_{-+} & D_{--} \end{pmatrix}.$$

Then

$$\Gamma \mathcal{D}_1 + \mathcal{D}_1 \Gamma = \begin{pmatrix} 2D_{++} & 0 \\ 0 & -2D_{--} \end{pmatrix}.$$

Hence the anticommutation condition is equivalent to $D_{++} = D_{--} = 0$, which is exactly the block-off-diagonal statement. $\square$

Thus everything reduces to proving or explaining the oddness of the leading operator.

D. Why generic second-order Hessians do not suffice

Consider a generic bosonic second-order reduced operator

$$\mathcal{H}_{\text{bos}}(p) = K_{ij} p_i p_j + B_i p_i + M,$$

acting on a finite-dimensional internal space. Even if a grading exists on that space, there is no general reason for either M or the second-order symbol $K_{ij} p_i p_j$ to anticommute with the grading. In fact, diagonal blocks are generically present.

a. Proposition 2. A generic second-order bosonic Hessian does not automatically produce a chiral block-off-diagonal first-order reduced operator.

b. Proof. The operator contains diagonal mass/endomorphism terms and second-order kinetic terms with no generic parity under a grading. Without a further structural constraint, there is no reason for

$$\Pi_\pm \mathcal{H}_{\text{bos}} \Pi_\pm$$

to vanish. $\square$

This is an important no-go statement: the Dirac structure cannot be read off from arbitrary quadratic fluctuation theory.

E. Sufficient mechanism I: exact grading symmetry

The cleanest route is an exact involution J on the full or reduced theory satisfying:

$$J^2 = \mathbf{1}, \quad J^\dagger = J,$$

and for the leading reduced operator,

$$J\mathcal{D}_1 J^{-1} = -\mathcal{D}_1.$$

If the low-energy sector is J -invariant, then letting

$$\mathcal{H}_\pm := \ker(J \mp \mathbf{1})$$

gives a grading and the diagonal blocks vanish.

a. Theorem 3 (Exact-grading sufficient criterion). Suppose the leading reduced operator $\mathcal{D}_1$ admits an exact involutive symmetry J with

$$J\mathcal{D}_1 J^{-1} = -\mathcal{D}_1.$$

Then $\mathcal{D}_1$ is odd with respect to the J -grading and hence block-off-diagonal.

b. Proof. By decomposing the low-energy space into the ± 1 eigenspaces of J , one obtains a grading operator $\Gamma = J$. The anticommutation relation is exactly

$$\{\Gamma, \mathcal{D}_1\} = 0.$$

Proposition 1 then implies block-off-diagonality. $\square$

This is the most rigorous operator route.

F. Sufficient mechanism II: constrained mode reduction with opposite parity sectors

A second route is more concrete and often more physical. Suppose the low-energy sector is spanned by modes

$$u_a^+, \quad a = 1, 2,$$

and

$$u_b^-, \quad b = 1, 2,$$

which are eigenmodes of some internal parity/exchange operator J :

$$Ju_a^\pm = \pm u_a^\pm.$$

Suppose further that the leading perturbation operator $V_1(p)$ is odd under J :

$$JV_1(p)J^{-1} = -V_1(p).$$

Then the matrix elements within the same parity sector vanish:

$$\langle u_a^\pm, V_1(p)u_b^\pm \rangle = 0.$$

a. Theorem 4 (Selection-rule criterion). Let J be an involution on the low-energy sector and let $V_1(p)$ be odd under J . Then the projected leading matrix elements satisfy

$$\Pi_\pm V_1(p)\Pi_\pm = 0.$$

Hence the reduced leading operator is block-off-diagonal.

b. Proof. For $u, v \in \mathcal{H}_+$,

$$\langle u, V_1 v \rangle = \langle Ju, V_1 Jv \rangle = \langle u, JV_1 Jv \rangle = \langle u, -V_1 v \rangle = -\langle u, V_1 v \rangle,$$

so the matrix element vanishes. The same argument applies on $\mathcal{H}_-$. $\square$

This is especially useful because it does not require a full anticommuting operator identity on the whole space; it suffices to know the parity of the leading reduced perturbation.

G. Sufficient mechanism III: envelope/worldvolume reduction of first-order transport

Now consider a defect background depending on transverse coordinates ξ , with slow worldvolume coordinates y^μ . Suppose there are four low-energy modes grouped into two doublets $u_a^\pm(\xi)$, and use the ansatz

$$\Phi(y, \xi) = \sum_{a=1}^2 \psi_a^+(y) u_a^+(\xi) + \sum_{b=1}^2 \psi_b^-(y) u_b^-(\xi) + \dots$$

Assume the projected worldvolume operator is first-order:

$$\langle u_a^\pm, \mathcal{L} u_b^\pm \rangle_\xi = O(p^2), \quad \langle u_a^+, \mathcal{L} u_b^- \rangle_\xi = p_i (M_i)_{ab} + O(p^2),$$

where p is worldvolume momentum.

Then the reduced operator has the form

$$\mathcal{D}_1(p) = \begin{pmatrix} 0 & p_i M_i \\ p_i M_i^\dagger & 0 \end{pmatrix}.$$

a. Theorem 5 (Envelope-reduction sufficient criterion). If the low-energy defect/worldvolume reduction yields two J -even and two J -odd modes, and the leading projected transport operator is J -odd and first-order in worldvolume momentum, then the reduced principal operator is block-off-diagonal first-order.

b. Proof. Block-off-diagonality follows from Theorem 4. First-order structure follows from the assumed worldvolume expansion of the mixed matrix elements. $\square$

This theorem is close to what one actually wants for defect-bound Dirac fermions.

H. Sufficient mechanism IV: valley linearization at a symmetry-related crossing pair

Now consider a valley-based construction. Suppose at $+G_*$ there is a two-dimensional internal doublet, and similarly at $-G_*$. Let the total low-energy space be

$$\mathcal{H}_{\text{low}} = \mathcal{H}_{+G_*} \oplus \mathcal{H}_{-G_*}, \quad \dim \mathcal{H}_{\pm G_*} = 2.$$

Assume there exists an involution J distinguishing the two valley sectors,

$$J|_{\mathcal{H}_{\pm G_*}} = \pm 1.$$

Let the linearized perturbation in momentum about the valley pair be odd under J , so that it maps $+G_*$ modes to $-G_*$ modes at leading order while same-valley leading diagonal terms vanish after subtraction of the common energy.

Then again

$$\mathcal{D}_1(p) = \begin{pmatrix} 0 & p_i M_i \\ p_i M_i^\dagger & 0 \end{pmatrix}.$$

a. Theorem 6 (Valley-pair odd-coupling criterion). If a valley-pair low-energy sector admits a grading by valley label and the leading reduced perturbation is odd with respect to that grading, then the principal reduced operator is block-off-diagonal. If the off-diagonal block is a nondegenerate linear 2×2 symbol, the principal reduced operator is Dirac.

b. Proof. The first statement is Theorem 4 with J taken as the valley grading. The second follows from the previously established block-off-diagonal Dirac theorem. $\square$

This is a clean statement, but whether the odd-coupling hypothesis is actually satisfied in the TECT BCC valley problem remains open.

I. Nondegeneracy and Dirac completion

Once

$$\mathcal{D}_1(p) = \begin{pmatrix} 0 & p_i M_i \\ p_i M_i^\dagger & 0 \end{pmatrix}$$

is established, the only remaining issue is symbol nondegeneracy.

Assume:

$$\det(p_i M_i) \neq 0 \quad \text{for generic } p \neq 0 \text{ near } 0.$$

Then, by the earlier 2×2 symbol classification, one may bring $p_i M_i$ to Pauli form, and thus the full principal operator becomes

$$\mathcal{D}_1(p) \sim p_i \alpha^i.$$

a. Corollary 7. Under any of Theorems 3–6 plus symbol nondegeneracy, the reduced principal operator is the massless Dirac operator.

J. What is actually proved and what is not

We have proved several *sufficient criteria* for the desired structure. But we have not yet shown that the full microscopic TECT operator necessarily satisfies them.

What is proved:

1. exact oddness under a grading implies block-off-diagonalization;
2. a selection rule from parity/exchange-odd leading perturbations implies vanishing diagonal matrix elements;
3. envelope/worldvolume reduction with J -odd first-order transport yields the desired reduced structure;
4. valley-pair grading plus odd leading coupling also yields the desired structure.

What is not proved:

1. that the actual TECT defect Hessian possesses such a grading J ;
2. that the actual reduced leading perturbation is J -odd;
3. that the diagonal blocks vanish in the microscopic defect or valley calculation;
4. that the off-diagonal 2×2 symbol is nondegenerate.

Thus the theorem has reduced the problem to a very specific operator-analysis task.

XXXI. FINAL SUMMARY

To obtain Dirac kinetics from the TECT defect/valley sector, one does *not* need to isolate the Dirac form purely by discrete symmetry elimination.

A stronger operator route is available.

Let the reduced infrared sector be a 4-dimensional space

$$\mathcal{H}_{\text{low}} = \mathcal{H}_+ \oplus \mathcal{H}_-, \quad \dim \mathcal{H}_\pm = 2.$$

If the leading reduced operator $\mathcal{D}_1$ is odd with respect to the grading

$$\Gamma = \Pi_+ - \Pi_-, \quad \{\Gamma, \mathcal{D}_1\} = 0,$$

then

$$\mathcal{D}_1 = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}.$$

If, moreover,

$$A_1(p) = p_i M_i$$

is first-order and nondegenerate, then the principal reduced operator is unitarily equivalent to the massless Dirac operator:

$$\mathcal{D}_1(p) \sim p_i \alpha^i.$$

Several sufficient mechanisms can produce the oddness condition:

1. an exact involutive grading symmetry J with

$$J\mathcal{D}_1 J^{-1} = -\mathcal{D}_1;$$

2. a parity/exchange selection rule forcing same-sector matrix elements to vanish;
3. a defect envelope reduction with two even and two odd low-energy modes and a J -odd first-order transport operator;
4. a valley-pair reduction in which the leading coupling is odd under the valley grading.

Thus the decisive unresolved TECT problem is now sharply localized:

Does the actual defect/valley reduced operator admit such a grading and odd first-order principal coupling?

If yes, Dirac kinetics follows automatically.

XXXII. PROJECTED MATRIX-ELEMENT THEOREM FOR DEFECT ENVELOPE REDUCTION

Assumption Audit

- A1: The linearized operator around a defect background admits an expansion

$$\mathcal{L}(p) = \mathcal{L}_0 + \mathcal{L}_1(p) + \mathcal{L}_{\geq 2}(p),$$

where $\mathcal{L}_1(p)$ is linear in the propagating momentum/worldvolume derivatives, and $\mathcal{L}_{\geq 2}(p) = O(|p|^2)$.

Needs structural input

- A2: The low-energy subspace $\mathcal{K} = \ker \mathcal{L}_0$ (or the isolated near-kernel of $\mathcal{L}_0$) is four-dimensional and decomposes as

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \dim \mathcal{K}_\pm = 2.$$

Needs spectral input

- A3: There exists an involution J such that

$$J^2 = \mathbf{1}, \quad J^\dagger = J, \quad [J, \mathcal{L}_0] = 0,$$

and

$$\mathcal{K}_\pm = \ker(J \mp \mathbf{1}) \cap \mathcal{K}.$$

Needs structural identification of J

- A4: The leading perturbation is odd under J :

$$J\mathcal{L}_1(p)J^{-1} = -\mathcal{L}_1(p).$$

Provable once a parity/exchange rule is established

- A5: The complement of $\mathcal{K}$ is spectrally gapped for $\mathcal{L}_0$, so that Schur-complement corrections from high modes begin at $O(|p|^2)$.

Provable once a gap exists

- A6: $\mathcal{L}_1(p)$ is first-order in the propagating derivatives.

Provable in envelope/transport reductions

A. Setup

Let P_B be a static defect background, and let $\mathcal{L}(p)$ denote the linearized operator acting on fluctuations around P_B , where p stands for the slow worldvolume momentum (or envelope momentum near a valley pair).

Assume

$$\mathcal{L}(p) = \mathcal{L}_0 + \mathcal{L}_1(p) + \mathcal{L}_{\geq 2}(p), \quad \mathcal{L}_{\geq 2}(p) = O(|p|^2).$$

Let $\mathcal{K}$ be the isolated low-energy space of $\mathcal{L}_0$, and choose orthonormal bases

$$\{u_a^+\}_{a=1}^2 \subset \mathcal{K}_+, \quad \{u_b^-\}_{b=1}^2 \subset \mathcal{K}_-.$$

Define the low-energy projector

$$\Pi = \Pi_+ + \Pi_-,$$

with

$$\Pi_{\pm} = \sum_{a=1}^2 |u_a^{\pm}\rangle \langle u_a^{\pm}|.$$

Our goal is to compute the effective projected operator

$$\mathcal{D}_{\text{eff}}(p) = \Pi_{\text{eff}}(p) \mathcal{L}(p) \Pi_{\text{eff}}(p)$$

to leading order, where $\Pi_{\text{eff}}(p)$ is the exact low-energy spectral projector.

B. Matrix elements of the leading perturbation

Define the matrix elements

$$(\mathcal{M}_{++}(p))_{ab} := \langle u_a^+, \mathcal{L}_1(p) u_b^+ \rangle,$$

$$(\mathcal{M}_{--}(p))_{ab} := \langle u_a^-, \mathcal{L}_1(p) u_b^- \rangle,$$

$$(\mathcal{M}_{+-}(p))_{ab} := \langle u_a^+, \mathcal{L}_1(p) u_b^- \rangle, \quad \mathcal{M}_{-+}(p) = \mathcal{M}_{+-}(p)^{\dagger}$$

if $\mathcal{L}_1(p)$ is Hermitian.

We first prove that the diagonal blocks vanish under J -oddness.

a. *Theorem 1 (Diagonal-block vanishing theorem).* Assume

$$J\mathcal{L}_1(p)J^{-1} = -\mathcal{L}_1(p), \quad Ju_a^\pm = \pm u_a^\pm.$$

Then

$$\mathcal{M}_{++}(p) = 0, \quad \mathcal{M}_{--}(p) = 0.$$

b. *Proof.* For $u_a^+, u_b^+ \in \mathcal{K}_+$,

$$\langle u_a^+, \mathcal{L}_1(p)u_b^+ \rangle = \langle Ju_a^+, \mathcal{L}_1(p)Ju_b^+ \rangle = \langle u_a^+, J\mathcal{L}_1(p)Ju_b^+ \rangle.$$

Since J is Hermitian and involutive and $\mathcal{L}_1$ is J -odd,

$$J\mathcal{L}_1(p)J = -\mathcal{L}_1(p),$$

so

$$\langle u_a^+, \mathcal{L}_1(p)u_b^+ \rangle = -\langle u_a^+, \mathcal{L}_1(p)u_b^+ \rangle,$$

hence it vanishes. The same argument applies on $\mathcal{K}_-$. $\square$

Therefore the leading projected operator on $\mathcal{K}$ is purely off-diagonal:

$$\Pi\mathcal{L}_1(p)\Pi = \begin{pmatrix} 0 & \mathcal{M}_{+-}(p) \\ \mathcal{M}_{+-}(p)^\dagger & 0 \end{pmatrix}.$$

C. First-order structure of the off-diagonal block

Assume now that $\mathcal{L}_1(p)$ is linear in the propagating momentum:

$$\mathcal{L}_1(p) = p_\mu \mathcal{V}^\mu$$

for some operators $\mathcal{V}^\mu$. Then

$$(\mathcal{M}_{+-}(p))_{ab} = p_\mu \langle u_a^+, \mathcal{V}^\mu u_b^- \rangle.$$

Define the 2×2 coefficient matrices

$$(M^\mu)_{ab} := \langle u_a^+, \mathcal{V}^\mu u_b^- \rangle.$$

Then

$$\mathcal{M}_{+-}(p) = p_\mu M^\mu.$$

a. *Theorem 2 (Off-diagonal symbol theorem).* If

$$\mathcal{L}_1(p) = p_\mu \mathcal{V}^\mu$$

and the diagonal blocks vanish by Theorem 1, then the projected first-order operator on $\mathcal{K}$ has the form

$$\Pi\mathcal{L}_1(p)\Pi = \begin{pmatrix} 0 & p_\mu M^\mu \\ p_\mu (M^\mu)^\dagger & 0 \end{pmatrix}.$$

b. *Proof.* Immediate from the definitions of the projected matrix elements. $\square$

This is the exact matrix-element origin of the 2×2 off-diagonal symbol.

D. Inclusion of high-mode corrections: Feshbach reduction

Let

$$Q := \mathbf{1} - \Pi$$

denote the high-energy projector. Then, formally, the Schur complement / Feshbach reduced operator on the low-energy subspace is

$$\mathcal{D}_{\text{eff}}(p) = \Pi\mathcal{L}(p)\Pi - \Pi\mathcal{L}(p)Q(Q\mathcal{L}(p)Q)^{-1}Q\mathcal{L}(p)\Pi.$$

Assume that $Q\mathcal{L}_0Q$ is invertible with spectral gap $\Delta > 0$, so the inverse exists and is bounded. Since $\mathcal{L}_1(p) = O(|p|)$, the correction term is

$$O(|p|^2).$$

a. Theorem 3 (Leading-order stability under Feshbach reduction). Suppose $Q\mathcal{L}_0Q$ is invertible on the high-energy sector. Then

$$\mathcal{D}_{\text{eff}}(p) = \Pi\mathcal{L}_1(p)\Pi + O(|p|^2).$$

Hence the leading first-order effective operator is exactly the projected matrix-element operator of Theorem 2.

b. Proof. Expand the Schur complement in powers of p . Since

$$\Pi\mathcal{L}_0\Pi = 0$$

on the low-energy kernel (or after subtraction of the common low-energy eigenvalue), and

$$\Pi\mathcal{L}_1Q = O(|p|), \quad Q\mathcal{L}_1\Pi = O(|p|),$$

the correction

$$\Pi\mathcal{L}Q(Q\mathcal{L}Q)^{-1}Q\mathcal{L}\Pi$$

is quadratic in p . $\square$

This is crucial: once the symmetry/selection rule kills the diagonal first-order blocks, high-mode elimination does not reintroduce them at $O(p)$.

E. Projected Dirac theorem in matrix-element form

Combining the previous results gives the main theorem of this step.

a. Theorem 4 (Projected matrix-element Dirac theorem). Assume:

1. $\mathcal{L}(p) = \mathcal{L}_0 + \mathcal{L}_1(p) + O(|p|^2)$;
2. the low-energy space $\mathcal{K}$ is four-dimensional and splits as

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \dim \mathcal{K}_\pm = 2;$$

3. there exists an involution J such that

$$[J, \mathcal{L}_0] = 0, \quad Ju_a^\pm = \pm u_a^\pm;$$

4. the leading perturbation is J -odd:

$$J\mathcal{L}_1(p)J^{-1} = -\mathcal{L}_1(p);$$

5. $\mathcal{L}_1(p) = p_\mu \mathcal{V}^\mu$ is first-order in the propagating momentum;
6. the high-energy sector is spectrally gapped;
7. the 2×2 symbol

$$A_1(p) := p_\mu M^\mu, \quad (M^\mu)_{ab} = \langle u_a^+, \mathcal{V}^\mu u_b^- \rangle$$

is nondegenerate.

Then the effective reduced operator satisfies

$$\mathcal{D}_{\text{eff}}(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix} + O(|p|^2),$$

and, after basis changes and linear momentum redefinition, its principal part is the massless Dirac operator

$$\mathcal{D}_1(p) \sim p_i \alpha^i.$$

b. Proof. Theorem 1 kills the diagonal projected blocks at $O(p)$. Theorem 2 identifies the off-diagonal projected block as the linear symbol $A_1(p)$. Theorem 3 shows that high-mode elimination changes the effective operator only at $O(p^2)$. Finally, the previously proved block-off-diagonal Dirac theorem converts a nondegenerate 2×2 first-order symbol into the massless Dirac principal operator. $\square$

F. Concrete defect-envelope realization

Now let us rewrite the theorem in the envelope language most relevant for defect-bound modes.

Take coordinates $x = (y, \xi)$, where y are slow worldvolume coordinates and ξ are transverse coordinates. Let

$$\Phi(y, \xi) = \sum_{a=1}^2 \psi_a^+(y) u_a^+(\xi) + \sum_{b=1}^2 \psi_b^-(y) u_b^-(\xi) + \Phi_{\text{high}}.$$

Assume the projected transport operator is of the form

$$\langle u_a^\pm, \mathcal{L} u_b^\pm \rangle_\xi = O(\partial_y^2),$$

$$\langle u_a^+, \mathcal{L} u_b^- \rangle_\xi = C_{ab}^\mu (-i\partial_{y^\mu}) + O(\partial_y^2),$$

with

$$\det(C^\mu p_\mu) \neq 0$$

for generic small $p \neq 0$. Then the worldvolume effective equation is

$$\begin{pmatrix} 0 & C^\mu (-i\partial_{y^\mu}) \\ C^{\mu\dagger} (-i\partial_{y^\mu}) & 0 \end{pmatrix} \Psi(y) = 0 + O(\partial_y^2),$$

where

$$\Psi(y) = \begin{pmatrix} \psi^+(y) \\ \psi^-(y) \end{pmatrix}.$$

By Theorem 4, this is Dirac at principal order.

a. Corollary 5 (Defect-envelope Dirac corollary). If the defect low-mode basis splits into two J -even and two J -odd modes, and the leading worldvolume transport operator is J -odd and first-order, then the defect worldvolume theory is Dirac at principal order, provided the 2×2 mixed symbol is nondegenerate.

G. What this does not prove

This theorem is strong, but not magic. It does *not* prove:

1. that the actual TECT defect background has the required 4-dimensional isolated low-mode sector;
2. that the actual defect linearization admits the involution J ;
3. that the leading transport operator is J -odd;
4. that the mixed symbol is nondegenerate.

These are now the true remaining microscopic questions.

XXXIII. FINAL SUMMARY

Let $\mathcal{L}(p)$ be the linearized operator around a defect background, expanded as

$$\mathcal{L}(p) = \mathcal{L}_0 + \mathcal{L}_1(p) + O(|p|^2).$$

Assume the low-energy subspace $\mathcal{K}$ is four-dimensional and decomposes as

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \dim \mathcal{K}_\pm = 2,$$

with respect to an involution J .

If the leading perturbation is odd under J ,

$$J\mathcal{L}_1(p)J^{-1} = -\mathcal{L}_1(p),$$

then the projected same-sector matrix elements vanish:

$$\langle u_a^\pm, \mathcal{L}_1(p)u_b^\pm \rangle = 0.$$

If, furthermore,

$$\mathcal{L}_1(p) = p_\mu \mathcal{V}^\mu,$$

then the mixed matrix elements define a 2×2 first-order symbol

$$A_1(p) = p_\mu M^\mu, \quad (M^\mu)_{ab} = \langle u_a^+, \mathcal{V}^\mu u_b^- \rangle.$$

After Feshbach reduction, the effective operator is

$$\mathcal{D}_{\text{eff}}(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix} + O(|p|^2).$$

If $A_1(p)$ is nondegenerate, the principal reduced operator is unitarily equivalent to the massless Dirac operator.

Thus the decisive reduced microscopic problem is now:

Find a defect low-mode basis $u_a^\pm$ and an involution J such that the leading projected transport operator is J -odd.

If that is achieved, Dirac kinetics follows automatically from the projected matrix elements.

XXXIV. CONCRETE CP^2 DEFECT LOW-MODE ANSATZ AND THE PROJECTED OVERLAP THEOREM

Assumption Audit

- A1: Near the defect core, the coordinates split as

$$x = (y, \xi),$$

where y^μ are worldvolume coordinates and ξ is the transverse coordinate.

Local geometric setup

- A2: The defect background is reflection-symmetric across the core:

$$P_B(-\xi) = P_B(\xi)$$

or gauge-equivalent to this form.

Needs structural input

- A3: The orthogonal bundle

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp$$

admits a local orthonormal frame $e_a(\xi)$, $a = 1, 2$, chosen in a reflection-even gauge:

$$e_a(-\xi) = e_a(\xi).$$

Locally achievable once A2 holds

- A4: There exist two localized scalar profiles

$$f_+(\xi), \quad f_-(\xi),$$

with parities

$$f_+(-\xi) = f_+(\xi), \quad f_-(-\xi) = -f_-(\xi).$$

Needs spectral input

- A5: The leading worldvolume transport operator is first-order,

$$\mathcal{L}_1 = -i \mathcal{V}^\mu(\xi) \partial_{y^\mu},$$

and is odd under reflection:

$$\mathcal{V}^\mu(-\xi) = -\mathcal{V}^\mu(\xi).$$

Needs operator-structure input

- A6: The induced mixed symbol is nondegenerate.

Residual nondegeneracy condition

A. Defect background and Q -frame

Let the defect background be represented by

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \quad z_B^\dagger z_B = 1.$$

Assume that the defect is centered at $\xi = 0$ and reflection-symmetric:

$$P_B(-\xi) = P_B(\xi).$$

Then the rank-two orthogonal complement bundle

$$Q_B(\xi) := (\mathbb{C}z_B(\xi))^\perp$$

is well-defined along the transverse line.

Choose a local orthonormal frame

$$e(\xi) = (e_1(\xi), e_2(\xi)), \quad z_B^\dagger e_a = 0, \quad e_a^\dagger e_b = \delta_{ab},$$

in a reflection-even gauge:

$$e_a(-\xi) = e_a(\xi).$$

B. Concrete low-mode ansatz

Let

$$f_+(\xi), \quad f_-(\xi)$$

be two real localized scalar profiles satisfying

$$\int d\xi |f_\pm(\xi)|^2 = 1, \quad \int d\xi f_+(\xi)f_-(\xi) = 0,$$

and parity conditions

$$f_+(-\xi) = f_+(\xi), \quad f_-(-\xi) = -f_-(\xi).$$

Define four low modes by

$$u_a^+(\xi) := f_+(\xi)e_a(\xi), \quad u_a^-(\xi) := f_-(\xi)e_a(\xi), \quad a = 1, 2.$$

These satisfy orthonormality

$$\langle u_a^\pm, u_b^\pm \rangle = \delta_{ab}, \quad \langle u_a^+, u_b^- \rangle = 0,$$

with inner product

$$\langle u, v \rangle = \int d\xi u(\xi)^\dagger v(\xi).$$

Define reflection

$$(Ju)(\xi) := u(-\xi).$$

Then, using the parities of $f_\pm$ and the even gauge for e_a ,

$$Ju_a^\pm = \pm u_a^\pm.$$

Hence

$$\mathcal{K}_+ = \text{span}\{u_1^+, u_2^+\}, \quad \mathcal{K}_- = \text{span}\{u_1^-, u_2^-\},$$

gives a concrete 2 + 2 graded low-energy ansatz.

a. Theorem 1 (Concrete 2 + 2 ansatz theorem). Under the assumptions above, the four low modes

$$u_a^\pm(\xi) = f_\pm(\xi)e_a(\xi)$$

provide a concrete realization of a 4-dimensional low-energy candidate space with reflection grading

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad Ju_a^\pm = \pm u_a^\pm.$$

b. Proof. Immediate from orthonormality of e_a , normalization/orthogonality of $f_\pm$, and parity of the profiles. $\square$

C. Leading projected transport operator

Assume the linearized operator around the defect has the worldvolume expansion

$$\mathcal{L} = \mathcal{L}_0 - i\mathcal{V}^\mu(\xi)\partial_{y^\mu} + O(\partial_y^2).$$

We define the leading transport piece

$$\mathcal{L}_1 := -i\mathcal{V}^\mu(\xi)\partial_{y^\mu}.$$

Assume now that $\mathcal{V}^\mu$ is reflection-odd:

$$\mathcal{V}^\mu(-\xi) = -\mathcal{V}^\mu(\xi).$$

We compute the projected matrix elements.

D. Diagonal block vanishing

For the $+$ sector:

$$\langle u_a^+, \mathcal{L}_1 u_b^+ \rangle = -i \partial_{y^\mu} \int d\xi f_+(\xi)^2 e_a(\xi)^\dagger \mathcal{V}^\mu(\xi) e_b(\xi).$$

Under $\xi \mapsto -\xi$:

- $f_+(\xi)^2$ is even,
- $e_a^\dagger \mathcal{V}^\mu e_b$ is odd because e_a, e_b are even and $\mathcal{V}^\mu$ is odd.

Hence the integrand is odd, so

$$\langle u_a^+, \mathcal{L}_1 u_b^+ \rangle = 0.$$

Similarly, for the $-$ sector:

$$\langle u_a^-, \mathcal{L}_1 u_b^- \rangle = -i \partial_{y^\mu} \int d\xi f_-(\xi)^2 e_a(\xi)^\dagger \mathcal{V}^\mu(\xi) e_b(\xi) = 0,$$

because $f_-(\xi)^2$ is also even.

a. Theorem 2 (Diagonal overlap vanishing theorem). If the low-mode ansatz is

$$u_a^\pm = f_\pm e_a$$

with f_+ even, f_- odd, e_a even, and $\mathcal{V}^\mu$ odd, then

$$\langle u_a^\pm, \mathcal{L}_1 u_b^\pm \rangle = 0$$

for all a, b .

b. Proof. The diagonal integrands are products of an even scalar density and an odd operator kernel, hence odd under $\xi \mapsto -\xi$. Their integrals vanish. $\square$

This is the concrete matrix-element realization of the abstract J -odd theorem.

E. Mixed block

Now compute the mixed overlap:

$$\langle u_a^+, \mathcal{L}_1 u_b^- \rangle = -i \partial_{y^\mu} \int d\xi f_+(\xi) f_-(\xi) e_a(\xi)^\dagger \mathcal{V}^\mu(\xi) e_b(\xi).$$

Under reflection:

- $f_+ f_-$ is odd,
- $e_a^\dagger \mathcal{V}^\mu e_b$ is odd.

Therefore the integrand is even, so the integral may be nonzero.

Define

$$(M^\mu)_{ab} := \int d\xi f_+(\xi) f_-(\xi) e_a(\xi)^\dagger \mathcal{V}^\mu(\xi) e_b(\xi).$$

Then

$$\langle u_a^+, \mathcal{L}_1 u_b^- \rangle = -i (M^\mu)_{ab} \partial_{y^\mu}.$$

Similarly,

$$\langle u_a^-, \mathcal{L}_1 u_b^+ \rangle = -i (M^{\mu\dagger})_{ab} \partial_{y^\mu}$$

if $\mathcal{V}^\mu$ is Hermitian in the internal indices.

a. Theorem 3 (Mixed overlap theorem). Under the same assumptions,

$$\langle u_a^+, \mathcal{L}_1 u_b^- \rangle = -i (M^\mu)_{ab} \partial_{y^\mu},$$

with

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a(\xi)^\dagger \mathcal{V}^\mu(\xi) e_b(\xi).$$

No parity argument forces M^μ to vanish.

b. Proof. The mixed integrand is reflection-even, so it is generically nonzero. The first-order derivative factor is inherited directly from $\mathcal{L}_1$. $\square$

F. Effective projected operator

Let

$$\Psi(y) = \begin{pmatrix} \psi^+(y) \\ \psi^-(y) \end{pmatrix}, \quad \psi^\pm(y) \in \mathbb{C}^2,$$

and use the low-mode ansatz

$$\Phi(y, \xi) = \sum_{a=1}^2 \psi_a^+(y) u_a^+(\xi) + \sum_{b=1}^2 \psi_b^-(y) u_b^-(\xi) + \Phi_{\text{high}}.$$

The projected leading operator on $\mathcal{K}$ is therefore

$$\mathcal{D}_{\text{proj}}^{(1)} = \begin{pmatrix} 0 & -i M^\mu \partial_{y^\mu} \\ -i M^{\mu\dagger} \partial_{y^\mu} & 0 \end{pmatrix}.$$

a. Theorem 4 (Concrete projected overlap theorem). Under assumptions A1–A6, the leading projected defect-envelope operator on the low-mode ansatz

$$u_a^\pm = f_\pm e_a$$

takes the block-off-diagonal first-order form

$$\boxed{\mathcal{D}_{\text{proj}}^{(1)} = \begin{pmatrix} 0 & -i M^\mu \partial_{y^\mu} \\ -i M^{\mu\dagger} \partial_{y^\mu} & 0 \end{pmatrix}.}$$

b. Proof. The diagonal blocks vanish by Theorem 2. The mixed blocks are given by Theorem 3. $\square$

G. Inclusion of high modes

If the complement of $\mathcal{K}$ is gapped for $\mathcal{L}_0$, then Feshbach/Schur-complement reduction gives

$$\mathcal{D}_{\text{eff}} = \mathcal{D}_{\text{proj}}^{(1)} + O(\partial_y^2).$$

Thus the leading $O(\partial_y)$ structure is not spoiled by high-mode elimination.

a. Corollary 5. If the high-energy complement is spectrally gapped, then the principal worldvolume operator equals the projected overlap operator of Theorem 4 up to $O(\partial_y^2)$.

H. Dirac completion

If the symbol

$$A_1(p) = p_\mu M^\mu$$

is nondegenerate for generic small $p \neq 0$, then by the previously established block-off-diagonal symbol theorem, there exist basis changes and linear momentum redefinitions such that

$$\mathcal{D}_{\text{proj}}^{(1)}(p) \sim p_i \alpha^i.$$

a. Corollary 6 (Concrete defect-envelope Dirac corollary). If the mixed symbol

$$A_1(p) = p_\mu M^\mu$$

constructed from the overlap integrals is nondegenerate, then the principal defect-envelope effective operator is the massless Dirac operator.

I. What is genuinely achieved

This construction does not merely postulate an abstract grading. It produces an explicit J -graded low-mode ansatz

$$u_a^\pm = f_\pm e_a$$

and reduces the entire Dirac question to concrete overlap data:

$$\int d\xi f_+ f_- e_a^\dagger \mathcal{V}^\mu e_b.$$

Thus the previously abstract conditions become explicit computable objects.

J. What is still not proved

What remains open is:

1. the existence of an actual TECT defect background with the assumed reflection structure;
2. the existence of a localized even/odd low-mode pair $f_\pm$;
3. the oddness of the true projected transport kernel $\mathcal{V}^\mu$;
4. the nondegeneracy of the overlap symbol M^μ .

These are now sharply posed concrete questions.

XXXV. FINAL SUMMARY

A concrete low-mode ansatz for a reflection-symmetric CP^2 defect is

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi), \quad a = 1, 2,$$

where:

- $e_a(\xi)$ is an even local orthonormal frame of the rank-two bundle

$$Q_B(\xi) = (\mathbb{C} z_B(\xi))^\perp;$$

- f_+ is an even localized scalar profile;
- f_- is an odd localized scalar profile.

For a leading first-order transport operator

$$\mathcal{L}_1 = -i \mathcal{V}^\mu(\xi) \partial_{y^\mu},$$

with $\mathcal{V}^\mu(\xi)$ odd under reflection, the projected same-sector overlaps vanish:

$$\langle u_a^\pm, \mathcal{L}_1 u_b^\pm \rangle = 0.$$

The mixed overlaps survive and define a 2×2 first-order symbol:

$$\langle u_a^+, \mathcal{L}_1 u_b^- \rangle = -i(M^\mu)_{ab} \partial_{y^\mu},$$

with

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a(\xi)^\dagger \mathcal{V}^\mu(\xi) e_b(\xi).$$

Therefore the principal projected operator is

$$\mathcal{D}_{\text{proj}}^{(1)} = \begin{pmatrix} 0 & -iM^\mu \partial_{y^\mu} \\ -iM^{\mu\dagger} \partial_{y^\mu} & 0 \end{pmatrix}.$$

If the mixed symbol

$$A_1(p) = p_\mu M^\mu$$

is nondegenerate, then the principal effective operator is Dirac.

Thus the Dirac problem for the defect sector is reduced to a concrete overlap computation:

Find $f_\pm$, e_a , $\mathcal{V}^\mu$ such that the mixed overlap symbol M^μ is nondegenerate.

XXXVI. A MINIMAL REFLECTION-SYMMETRIC CP^2 DEFECT ANSATZ AND EXPLICIT EVALUATION OF THE MIXED SYMBOL

Assumption Audit

- A1: In a tubular neighborhood of the defect core, one may use local coordinates

$$x = (y, \xi),$$

where y^μ are worldvolume coordinates and $\xi \in \mathbb{R}$ is the transverse coordinate.

Local setup

- A2: The defect admits a reflection-symmetric local profile in the transverse coordinate.

Model assumption

- A3: The rank-two orthogonal bundle

$$Q_B(\xi) = (\mathbb{C} z_B(\xi))^\perp$$

is locally trivial along the transverse line and admits a smooth orthonormal frame.

Locally proved

- A4: The low-energy scalar profiles may be modeled by one even and one odd localized function.

Model ansatz

- A5: The projected transport kernel can be approximated in factorized form

$$e_a^\dagger(\xi) \mathcal{V}^\mu(\xi) e_b(\xi) = g(\xi) \Sigma_{ab}^\mu.$$

Model assumption

- A6: The matrices Σ^μ are chosen so that the induced symbol is nondegenerate.

Proved once chosen

A. Minimal local defect profile

We choose the simplest reflection-symmetric representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi),$$

with

$$z_B^\dagger z_B = 1.$$

Then the associated projector is

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger = \begin{pmatrix} \cos^2 \theta(\xi) & 0 & \cos \theta(\xi) \sin \theta(\xi) \\ 0 & 0 & 0 \\ \cos \theta(\xi) \sin \theta(\xi) & 0 & \sin^2 \theta(\xi) \end{pmatrix}.$$

Because θ is even, $P_B(-\xi) = P_B(\xi)$.

B. Explicit Q_B -frame

The orthogonal complement bundle

$$Q_B(\xi) = (\mathbb{C} z_B(\xi))^\perp$$

admits the explicit orthonormal frame

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Indeed,

$$z_B^\dagger e_1 = -\cos \theta \sin \theta + \sin \theta \cos \theta = 0, \quad z_B^\dagger e_2 = 0,$$

and

$$e_a^\dagger e_b = \delta_{ab}.$$

Moreover, since $\theta(\xi)$ is even,

$$e_1(-\xi) = e_1(\xi), \quad e_2(-\xi) = e_2(\xi).$$

Thus this is a concrete reflection-even gauge.

- a. *Proposition 1.* The pair (e_1, e_2) above is a reflection-even orthonormal frame of $Q_B(\xi)$.
- b. *Proof.* Direct computation. $\square$

C. Minimal even/odd localized scalar profiles

We choose the standard localized even/odd pair

$$f_+(\xi) = N_+ \operatorname{sech}(\kappa \xi), \quad f_-(\xi) = N_- \tanh(\kappa \xi) \operatorname{sech}(\kappa \xi),$$

with $\kappa > 0$.

Normalize them by

$$\int_{-\infty}^{\infty} |f_+(\xi)|^2 d\xi = 1, \quad \int_{-\infty}^{\infty} |f_-(\xi)|^2 d\xi = 1.$$

Using

$$\int_{-\infty}^{\infty} \text{sech}^2(\kappa\xi) d\xi = \frac{2}{\kappa},$$

we obtain

$$N_+ = \sqrt{\frac{\kappa}{2}}.$$

Similarly,

$$\int_{-\infty}^{\infty} \tanh^2(\kappa\xi) \text{sech}^2(\kappa\xi) d\xi = \frac{2}{3\kappa},$$

so

$$N_- = \sqrt{\frac{3\kappa}{2}}.$$

Therefore

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}} \text{sech}(\kappa\xi), \quad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \text{sech}(\kappa\xi).$$

These satisfy

$$f_+(-\xi) = f_+(\xi), \quad f_-(-\xi) = -f_-(\xi),$$

and

$$\int d\xi f_+(\xi) f_-(\xi) = 0$$

by parity.

D. Concrete low-mode ansatz

Define

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi), \quad a = 1, 2.$$

Then

$$Ju_a^\pm = \pm u_a^\pm, \quad (Ju)(\xi) := u(-\xi),$$

and

$$\mathcal{K}_+ = \text{span}\{u_1^+, u_2^+\}, \quad \mathcal{K}_- = \text{span}\{u_1^-, u_2^-\}.$$

E. Factorized odd transport kernel

We now impose the simplest concrete transport ansatz:

$$\mathcal{L}_1 = -i \mathcal{V}^\mu(\xi) \partial_{y^\mu},$$

with

$$e_a^\dagger(\xi) \mathcal{V}^\mu(\xi) e_b(\xi) = g(\xi) \Sigma_{ab}^\mu,$$

where:

1. $g(\xi)$ is a real odd scalar profile,

$$g(-\xi) = -g(\xi);$$

2. Σ^μ is a fixed 2×2 Hermitian matrix acting on the internal doublet.

Then the mixed symbol becomes

$$M^\mu = \int d\xi f_+(\xi) f_-(\xi) g(\xi) \Sigma^\mu = I_g \Sigma^\mu,$$

where

$$I_g := \int_{-\infty}^{\infty} d\xi f_+(\xi) f_-(\xi) g(\xi).$$

- a. Theorem 2.* Under the factorized ansatz above, the projected mixed symbol factorizes as

$$M^\mu = I_g \Sigma^\mu.$$

- b. Proof.* Immediate from the factorized kernel assumption. $\square$
Thus the entire question reduces to the scalar overlap I_g .

F. Explicit evaluation of the scalar overlap

Choose the simplest odd profile

$$g(\xi) = g_0 \tanh(\kappa \xi), \quad g_0 \in \mathbb{R}.$$

Then

$$I_g = g_0 \int_{-\infty}^{\infty} d\xi f_+(\xi) f_-(\xi) \tanh(\kappa \xi).$$

Substituting the explicit $f_\pm$,

$$f_+(\xi) f_-(\xi) = \sqrt{\frac{\kappa}{2}} \sqrt{\frac{3\kappa}{2}} \tanh(\kappa \xi) \operatorname{sech}^2(\kappa \xi) = \frac{\sqrt{3}\kappa}{2} \tanh(\kappa \xi) \operatorname{sech}^2(\kappa \xi).$$

Hence

$$I_g = \frac{\sqrt{3}\kappa g_0}{2} \int_{-\infty}^{\infty} d\xi \tanh^2(\kappa \xi) \operatorname{sech}^2(\kappa \xi).$$

Using

$$\int_{-\infty}^{\infty} \tanh^2(\kappa \xi) \operatorname{sech}^2(\kappa \xi) d\xi = \frac{2}{3\kappa},$$

we obtain

$$I_g = \frac{\sqrt{3}\kappa g_0}{2} \cdot \frac{2}{3\kappa} = \frac{g_0}{\sqrt{3}}.$$

Therefore

$$\boxed{M^\mu = \frac{g_0}{\sqrt{3}} \Sigma^\mu.}$$

In particular, $M^\mu \neq 0$ whenever $g_0 \neq 0$ and $\Sigma^\mu \neq 0$.

a. *Corollary 3.* For the explicit choice

$$g(\xi) = g_0 \tanh(\kappa \xi),$$

the mixed overlap is manifestly nonzero:

$$I_g = \frac{g_0}{\sqrt{3}} \neq 0 \quad (g_0 \neq 0).$$

G. Nondegenerate symbol condition

Now suppose the propagating directions are labeled by $i = 1, 2, 3$, and choose

$$\Sigma^i = v_i \sigma^i, \quad v_i \in \mathbb{R} \setminus \{0\}.$$

Then

$$A_1(p) := p_i M^i = \frac{g_0}{\sqrt{3}} \sum_{i=1}^3 v_i p_i \sigma^i.$$

Therefore

$$\det A_1(p) = -\frac{g_0^2}{3} (v_1^2 p_1^2 + v_2^2 p_2^2 + v_3^2 p_3^2).$$

Hence

$$\det A_1(p) \neq 0 \quad \text{for generic } p \neq 0$$

provided

$$v_1 v_2 v_3 \neq 0.$$

a. *Theorem 4 (Explicit nondegeneracy theorem).* Under the explicit ansatz

$$M^i = \frac{g_0}{\sqrt{3}} v_i \sigma^i, \quad v_i \neq 0,$$

the mixed first-order symbol is nondegenerate and therefore yields the Dirac principal operator.

b. *Proof.* The determinant formula above shows nondegeneracy away from $p = 0$. The earlier block-off-diagonal Dirac theorem then applies. $\square$

H. What has and has not been shown

What has been shown is an explicit existence model:

1. a concrete reflection-symmetric CP^2 -type background;
2. a concrete even Q -frame;
3. a concrete even/odd localized low-mode pair;
4. a concrete odd transport kernel leading to

$$M^\mu \neq 0;$$

5. a concrete choice of Σ^μ producing a nondegenerate Dirac symbol.

What has *not* been shown is that this specific ansatz is the unique or exact microscopic TECT defect profile.

Thus the present section proves the *existence and consistency* of a fully explicit model realizing the desired projected Dirac mechanism, but not yet its necessity in the full TECT PDE.

I. Dimension caveat

One should also be explicit about dimensional interpretation:

- if the propagating coordinates y^μ provide only two spatial directions, the result is a 2 + 1-dimensional Dirac operator;
- to obtain a full 3 + 1-dimensional Dirac kinetic term from a defect-envelope picture, one needs three independent propagating spatial directions in the reduced effective symbol.

Therefore the defect-envelope construction is certainly sufficient for lower-dimensional Dirac structure, and for full 3 + 1-dimensional Dirac one must verify that the relevant reduced sector indeed carries three independent momentum directions.

XXXVII. FINAL SUMMARY

A concrete minimal reflection-symmetric local CP^2 defect ansatz is

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi),$$

with an explicit even orthonormal frame of

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp$$

given by

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

A concrete even/odd low-mode pair is

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}} \operatorname{sech}(\kappa\xi), \quad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \operatorname{sech}(\kappa\xi).$$

Thus

$$u_a^\pm(\xi) = f_\pm(\xi)e_a(\xi), \quad a = 1, 2,$$

gives a concrete 2 + 2 graded low-energy ansatz.

If the leading transport kernel is factorized as

$$e_a^\dagger(\xi)\mathcal{V}^\mu(\xi)e_b(\xi) = g(\xi)\Sigma_{ab}^\mu, \quad g(-\xi) = -g(\xi),$$

then the mixed symbol is

$$M^\mu = \left(\int d\xi f_+(\xi)f_-(\xi)g(\xi) \right) \Sigma^\mu.$$

For the explicit odd profile

$$g(\xi) = g_0 \tanh(\kappa\xi),$$

one finds

$$\int d\xi f_+(\xi)f_-(\xi)g(\xi) = \frac{g_0}{\sqrt{3}}.$$

Hence

$$M^\mu = \frac{g_0}{\sqrt{3}} \Sigma^\mu.$$

In particular, $M^\mu \neq 0$ whenever $g_0 \neq 0$.

If moreover

$$\Sigma^i = v_i \sigma^i, \quad v_i \neq 0,$$

then

$$A_1(p) = p_i M^i = \frac{g_0}{\sqrt{3}} \sum_i v_i p_i \sigma^i$$

is nondegenerate, and the principal projected operator is Dirac.

Thus the explicit model demonstrates:

the projected Dirac mechanism is concretely realizable in a minimal reflection-symmetric CP^2 -defect ansatz.

What remains open is whether the actual microscopic TECT defect linearization realizes this same structure without further assumptions.

XXXVIII. REVERSE ENGINEERING A MINIMAL DEFECT OPERATOR MODEL FOR THE EXPLICIT LOW-MODE ANSATZ

Assumption Audit

- A1: The defect infrared sector may be modeled as a graded 2-channel operator acting on

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \dim \mathcal{K}_\pm = 2.$$

Valid as a minimal effective model

- A2: The low modes may be written as

$$u_a^\pm(\xi) = f_\pm(\xi) e_a(\xi), \quad a = 1, 2,$$

with a common local Q_B -frame $e_a(\xi)$.

Locally valid

- A3: Given explicit profiles $f_\pm$, one may reconstruct scalar channel operators

$$H_\pm = -\partial_\xi^2 + U_\pm(\xi)$$

for which $f_\pm$ are exact zero modes.

Proved

- A4: The leading transport operator may be modeled by an odd kernel derived from the transverse derivative of an even defect profile.

Provable in the model

- A5: This model reproduces the nonzero mixed overlap symbol M^μ .

Proved in the model

- A6: This already proves the full microscopic TECT operator.

False

A. Step 1: explicit defect profile and Q_B -frame

We retain the minimal reflection-symmetric defect profile

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi),$$

with associated orthogonal frame

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

This gives a reflection-even local orthonormal frame of

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp.$$

B. Step 2: chosen explicit low-mode profiles

We keep the previously chosen normalized even/odd profiles

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}} \operatorname{sech}(\kappa\xi), \quad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \operatorname{sech}(\kappa\xi).$$

Then the concrete low-mode ansatz is

$$u_a^+(\xi) = f_+(\xi)e_a(\xi), \quad u_a^-(\xi) = f_-(\xi)e_a(\xi), \quad a = 1, 2.$$

C. Step 3: reverse-engineered scalar channel operators

We now ask for scalar one-dimensional operators

$$H_\pm = -\partial_\xi^2 + U_\pm(\xi)$$

such that

$$H_+f_+ = 0, \quad H_-f_- = 0.$$

This requirement determines the potentials uniquely:

$$U_+(\xi) = \frac{f_+''(\xi)}{f_+(\xi)}, \quad U_-(\xi) = \frac{f_-''(\xi)}{f_-(\xi)}.$$

For

$$f_+(\xi) \propto \operatorname{sech}(\kappa\xi),$$

one computes

$$\frac{f_+''(\xi)}{f_+(\xi)} = \kappa^2 - 2\kappa^2 \operatorname{sech}^2(\kappa\xi).$$

Hence

$$\boxed{H_+ = -\partial_\xi^2 + \kappa^2 - 2\kappa^2 \operatorname{sech}^2(\kappa\xi)}.$$

For

$$f_-(\xi) \propto \tanh(\kappa\xi) \operatorname{sech}(\kappa\xi),$$

one computes

$$\frac{f_-''(\xi)}{f_-(\xi)} = \kappa^2 - 6\kappa^2 \operatorname{sech}^2(\kappa\xi).$$

Hence

$$H_- = -\partial_\xi^2 + \kappa^2 - 6\kappa^2 \operatorname{sech}^2(\kappa\xi).$$

a. Theorem 1 (Reverse-engineered zero-mode theorem). The chosen profiles $f_\pm$ are exact zero modes of the scalar operators $H_\pm$ above:

$$H_+ f_+ = 0, \quad H_- f_- = 0.$$

b. Proof. This is immediate from the construction $U_\pm = f_\pm''/f_\pm$. $\square$

D. Step 4: graded minimal defect operator

Define the reflection grading

$$J = \begin{pmatrix} +1 & 0 \\ 0 & -1 \end{pmatrix}$$

on the $(+, -)$ channel space. The minimal static defect operator is

$$\mathcal{L}_0 = \begin{pmatrix} H_+ & 0 \\ 0 & H_- \end{pmatrix} \otimes \mathbf{1}_2^{(\text{int})}.$$

Then

$$[J, \mathcal{L}_0] = 0,$$

and the four modes

$$u_a^\pm(\xi) = f_\pm(\xi) e_a(\xi)$$

belong to $\ker \mathcal{L}_0$.

a. Corollary 2. The explicit ansatz $u_a^\pm = f_\pm e_a$ is not merely ad hoc: it is the exact zero-mode basis of the minimal graded defect operator $\mathcal{L}_0$.

E. Step 5: origin of the odd transport kernel

We now model the leading worldvolume transport operator as arising from the transverse variation of the defect background. The most natural local source is

$$\partial_\xi P_B(\xi) = z_B' z_B^\dagger + z_B z_B'^\dagger.$$

Since $P_B(\xi)$ is even under reflection,

$$P_B(-\xi) = P_B(\xi),$$

it follows that

$$\partial_\xi P_B(-\xi) = -\partial_\xi P_B(\xi).$$

Thus any leading projected transport coefficient constructed linearly from $\partial_\xi P_B$ is automatically odd.

Equivalently, because

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi),$$

we have

$$\theta'(-\xi) = -\theta'(\xi).$$

Hence a transport kernel of the form

$$\mathcal{V}^\mu(\xi) = c^\mu \theta'(\xi) \Sigma^\mu$$

is automatically reflection-odd.

a. Theorem 3 (Oddness-from-defect-gradient theorem). If the leading transport kernel is linear in the transverse gradient of an even defect profile, e.g.

$$\mathcal{V}^\mu(\xi) \propto \partial_\xi P_B(\xi) \quad \text{or} \quad \theta'(\xi),$$

then $\mathcal{V}^\mu$ is odd under reflection:

$$\mathcal{V}^\mu(-\xi) = -\mathcal{V}^\mu(\xi).$$

b. Proof. Derivative of an even function is odd. $\square$

This gives the physical origin of the J -oddness assumption used previously.

F. Step 6: explicit mixed overlap in the reverse-engineered model

Assume the factorized transport ansatz

$$e_a^\dagger(\xi) \mathcal{V}^\mu(\xi) e_b(\xi) = g(\xi) \Sigma_{ab}^\mu, \quad g(\xi) = g_0 \tanh(\kappa \xi).$$

Then

$$M_{ab}^\mu = \int d\xi f_+(\xi) f_-(\xi) g(\xi) \Sigma_{ab}^\mu.$$

Using the explicit profiles,

$$f_+(\xi) f_-(\xi) = \frac{\sqrt{3}\kappa}{2} \tanh(\kappa \xi) \operatorname{sech}^2(\kappa \xi),$$

hence

$$M^\mu = \frac{\sqrt{3}\kappa g_0}{2} \left(\int_{-\infty}^{\infty} d\xi \tanh^2(\kappa \xi) \operatorname{sech}^2(\kappa \xi) \right) \Sigma^\mu.$$

Since

$$\int_{-\infty}^{\infty} d\xi \tanh^2(\kappa \xi) \operatorname{sech}^2(\kappa \xi) = \frac{2}{3\kappa},$$

we obtain

$$M^\mu = \frac{g_0}{\sqrt{3}} \Sigma^\mu.$$

a. Corollary 4. The reverse-engineered operator model reproduces the previously derived nonzero mixed overlap symbol exactly.

G. Step 7: nondegenerate Dirac symbol

Choose

$$\Sigma^i = v_i \sigma^i, \quad v_i \neq 0.$$

Then

$$A_1(p) = p_i M^i = \frac{g_0}{\sqrt{3}} \sum_{i=1}^3 v_i p_i \sigma^i,$$

so

$$\det A_1(p) = -\frac{g_0^2}{3} \sum_{i=1}^3 v_i^2 p_i^2.$$

Therefore $A_1(p)$ is nondegenerate for generic $p \neq 0$, and the projected principal operator is Dirac.

a. Theorem 5 (Explicit minimal effective Dirac realization). The minimal graded defect operator

$$\mathcal{L}_0 = \begin{pmatrix} -\partial_\xi^2 + \kappa^2 - 2\kappa^2 \operatorname{sech}^2(\kappa\xi) & 0 \\ 0 & -\partial_\xi^2 + \kappa^2 - 6\kappa^2 \operatorname{sech}^2(\kappa\xi) \end{pmatrix} \otimes \mathbf{1}_2$$

together with the odd first-order transport kernel

$$\mathcal{L}_1 = -i g_0 \tanh(\kappa\xi) \Sigma^\mu \partial_{y^\mu}$$

admits the exact low-mode basis

$$u_a^\pm(\xi) = f_\pm(\xi) e_a(\xi),$$

and induces the projected principal symbol

$$A_1(p) = \frac{g_0}{\sqrt{3}} p_i \Sigma^i.$$

If

$$\Sigma^i = v_i \sigma^i, \quad v_i \neq 0,$$

then the principal reduced operator is Dirac.

b. Proof. Combine Theorem 1, Theorem 3, Corollary 4, and the nondegeneracy of the Pauli symbol. $\square$

H. What this achieves and what it does not

This construction proves that the previously chosen explicit ansatz is realizable as the exact low-energy sector of a concrete minimal graded defect operator model.

However, it does not prove that the full microscopic TECT defect linearization is literally equal to this model. At present, the correct honest statement is:

the explicit ansatz is derivable from a consistent minimal effective defect operator model.

This is stronger than a mere trial function argument, but weaker than a full microscopic TECT derivation.

I. Why this still matters

The present step closes an important logical gap. Previously one might object that the explicit profiles $f_\pm$ and the odd overlap kernel were simply guessed. Now we have shown:

1. the profiles arise as exact zero modes of explicit channel operators;
2. the odd transport kernel arises naturally from the transverse derivative of an even defect profile;
3. the nonzero mixed overlap and Dirac symbol follow by direct computation.

Hence the entire mechanism is not a formal possibility only, but an explicitly realizable operator model.

XXXIX. FINAL SUMMARY

The explicit defect-envelope ansatz

$$u_a^\pm(\xi) = f_\pm(\xi)e_a(\xi)$$

can be reverse-engineered from a concrete minimal graded defect operator.

For the chosen profiles

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}} \operatorname{sech}(\kappa\xi), \quad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \operatorname{sech}(\kappa\xi),$$

the scalar channel operators

$$H_+ = -\partial_\xi^2 + \kappa^2 - 2\kappa^2 \operatorname{sech}^2(\kappa\xi),$$

$$H_- = -\partial_\xi^2 + \kappa^2 - 6\kappa^2 \operatorname{sech}^2(\kappa\xi)$$

satisfy

$$H_+ f_+ = 0, \quad H_- f_- = 0.$$

Thus the full graded static defect operator

$$\mathcal{L}_0 = \begin{pmatrix} H_+ & 0 \\ 0 & H_- \end{pmatrix} \otimes \mathbf{1}_2$$

has the four exact zero modes $u_a^\pm = f_\pm e_a$.

The leading transport kernel becomes naturally odd if it is built from the transverse gradient of an even defect profile:

$$\mathcal{V}^\mu(\xi) \propto \partial_\xi P_B(\xi) \quad \text{or} \quad \theta'(\xi).$$

With the concrete odd model

$$g(\xi) = g_0 \tanh(\kappa\xi),$$

the mixed symbol evaluates explicitly to

$$M^\mu = \frac{g_0}{\sqrt{3}} \Sigma^\mu.$$

Choosing

$$\Sigma^i = v_i \sigma^i, \quad v_i \neq 0,$$

gives

$$A_1(p) = \frac{g_0}{\sqrt{3}} \sum_i v_i p_i \sigma^i,$$

which is nondegenerate and therefore yields the Dirac principal operator.

Hence the honest conclusion is:

the explicit low-mode ansatz and nonzero Dirac symbol are realized by a concrete minimal effective defect operator model.

What remains open is whether the full microscopic TECT defect linearization reduces to this same model or to a sufficiently close effective approximation.

XL. MATCHING THEOREM BETWEEN THE MINIMAL EFFECTIVE DEFECT OPERATOR AND THE CP^2 PROJECTOR HESSIAN

Assumption Audit

- A1: The relevant internal effective sector is described by a projector action

$$S[P] = \int d^{d+1}x \left[\frac{\rho}{4} \text{Tr}(\partial_\mu P \partial^\mu P) - U(P, \partial P, \dots) \right], \quad P^2 = P, \quad P^\dagger = P, \quad \text{Tr } P = 1.$$

Effective-sector assumption

- A2: There exists a static defect background depending only on one transverse coordinate:

$$P_B = P_B(\xi).$$

Local defect setup

- A3: Physical fluctuations are tangent to the projector manifold and may be represented by Q_B -valued fields.

Proved in the projector framework

- A4: The defect background is reflection-symmetric:

$$P_B(-\xi) = P_B(\xi)$$

up to gauge.

Structural assumption

- A5: The Hessian may be expanded into static channel operators plus worldvolume transport terms.

Provable

- A6: Exact equality with the previously chosen minimal Pöschl–Teller channel potentials is not automatic.

Not proved generally

A. Projector fluctuation geometry

Let $P_B(\xi)$ be a static defect background, and let the full projector be

$$P(x) = P_B(\xi) + \delta P(y, \xi), \quad x = (y, \xi),$$

where y^μ are worldvolume coordinates and ξ is the transverse coordinate.

Because $P^2 = P$, linearized fluctuations obey

$$P_B \delta P + \delta P P_B = \delta P.$$

Equivalently,

$$\delta P = P_B \delta P Q_B + Q_B \delta P P_B, \quad Q_B := \mathbf{1} - P_B.$$

Thus the fluctuation is tangent to the projector manifold.

Using a local normalized representative

$$P_B = z_B z_B^\dagger, \quad z_B^\dagger z_B = 1,$$

a convenient tangent parametrization is

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0.$$

Hence

$$\eta(\xi, y) \in Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp.$$

- a. Proposition 1.* The physical fluctuation problem reduces to a rank-two complex field η valued in the bundle Q_B .
- b. Proof.* This is the standard tangent parametrization of rank-one Hermitian projectors. $\square$

B. Quadratic Hessian around a static background

Expand the action to quadratic order:

$$S[P_B + \delta P] = S[P_B] + \delta S[P_B; \delta P] + \frac{1}{2} \delta^2 S[P_B; \delta P] + \dots$$

Since P_B solves the background Euler–Lagrange equation,

$$\delta S[P_B; \delta P] = 0.$$

The quadratic fluctuation action takes the generic form

$$\delta^2 S = \int d^{d+1}x \eta^\dagger \left[-\rho D_\mu^{(B)} D^{(B)\mu} + \mathcal{M}_B(\xi) \right] \eta,$$

where:

1. $D_\mu^{(B)}$ is the induced connection on Q_B ,
2. $\mathcal{M}_B(\xi)$ is a Hermitian 2×2 endomorphism built from the defect background $P_B(\xi)$ and its derivatives.

Since the background depends only on ξ , we may split

$$D_\mu^{(B)} D^{(B)\mu} = D_y^{(B)2} + D_\xi^{(B)2}.$$

Thus the fluctuation operator is of the schematic form

$$\mathcal{H}_B = -\rho D_\xi^{(B)2} - \rho D_y^{(B)2} + \mathcal{M}_B(\xi).$$

a. Theorem 2 (Projector Hessian channel theorem). The quadratic fluctuation operator of a static CP^2 projector defect is a matrix Schrödinger operator in the transverse coordinate ξ , with a background-dependent endomorphism potential on the rank-two bundle Q_B .

- b. Proof.* Directly from the second variation of the projector sigma-model action. $\square$
This already matches the structural form of the minimal effective operator:

$$-\partial_\xi^2 + U(\xi).$$

C. Choice of a local Q_B -frame and channel decomposition

Choose a local orthonormal frame

$$e(\xi) = (e_1(\xi), e_2(\xi))$$

of Q_B . Expanding

$$\eta(\xi, y) = e_a(\xi) \chi^a(\xi, y),$$

the quadratic operator becomes an operator on the $\mathbb{C}^2$ -valued coefficient field

$$\chi = (\chi^1, \chi^2)^T.$$

In general,

$$\mathcal{H}_B = -\rho \partial_\xi^2 + \mathcal{A}_\xi(\xi) \partial_\xi + \mathcal{U}_B(\xi) - \rho \partial_y^2 + \dots,$$

where $\mathcal{A}_\xi$ arises from the frame connection and $\mathcal{U}_B$ is a Hermitian 2×2 potential matrix.

By a local unitary gauge choice along the one-dimensional transverse line, one may set the connection to zero locally:

$$\mathcal{A}_\xi = 0.$$

Thus, locally,

$$\mathcal{H}_B = -\rho \partial_\xi^2 + \mathcal{U}_B(\xi) - \rho \partial_y^2 + \dots$$

a. Corollary 3. Locally along the transverse line, the projector Hessian is gauge-equivalent to a matrix Schrödinger operator

$$-\partial_\xi^2 + \mathcal{U}_B(\xi)$$

on the internal doublet.

This is the first precise matching with the minimal effective defect operator.

D. Reflection symmetry and parity decomposition

Assume now that the defect is reflection-symmetric:

$$P_B(-\xi) = P_B(\xi).$$

Then, in a reflection-even frame $e_a(-\xi) = e_a(\xi)$, the coefficient-space potential obeys

$$\mathcal{U}_B(-\xi) = \mathcal{U}_B(\xi).$$

Hence the static channel operator commutes with reflection

$$(J\chi)(\xi) := \chi(-\xi).$$

Therefore the transverse spectrum splits into even and odd channels:

$$\chi_+(-\xi) = \chi_+(\xi), \quad \chi_-(-\xi) = -\chi_-(\xi).$$

a. Theorem 4 (Parity channel decomposition theorem). For a reflection-symmetric static defect background, the static projector Hessian decomposes into even and odd transverse parity sectors.

b. Proof. The operator has even coefficients in ξ , hence commutes with reflection. Standard spectral decomposition then gives parity eigenchannels. $\square$

This is the operator origin of the previously postulated f_+ and f_- channels.

E. Matching to the minimal scalar channels

The minimal effective defect model used

$$H_+ = -\partial_\xi^2 + U_+(\xi), \quad H_- = -\partial_\xi^2 + U_-(\xi),$$

with

$$U_+(\xi) = \kappa^2 - 2\kappa^2 \operatorname{sech}^2(\kappa\xi), \quad U_-(\xi) = \kappa^2 - 6\kappa^2 \operatorname{sech}^2(\kappa\xi).$$

The exact claim that the projector Hessian gives precisely these two Pöschl–Teller wells is too strong in general and cannot be proved without the explicit microscopic/effective potential $U(P, \partial P, \dots)$.

However, a weaker and correct matching statement is available:

a. Theorem 5 (Effective channel matching theorem). If the reflection-symmetric projector Hessian has one localized even low mode and one localized odd low mode in each internal channel, then its low-energy parity-reduced transverse spectrum is effectively modeled by a pair of scalar channel operators

$$H_\pm = -\partial_\xi^2 + U_\pm^{\text{eff}}(\xi),$$

for suitable even potentials $U_\pm^{\text{eff}}(\xi)$. The previously used Pöschl–Teller choice is an explicit solvable representative of this effective class.

b. Proof. Given any normalized isolated even/odd low modes $f_\pm$, one may reconstruct effective scalar potentials by

$$U_\pm^{\text{eff}}(\xi) = \frac{f_\pm''(\xi)}{f_\pm(\xi)}.$$

The Pöschl–Teller wells are a solvable representative realizing exactly the chosen explicit profiles. $\square$

Thus the relation between the projector Hessian and the minimal model is:

$$\text{projector Hessian} \implies \text{parity channels} \implies \text{effective scalar wells} \implies \text{explicit Pöschl–Teller representative}.$$

F. Origin of the odd transport kernel

We now turn on slow worldvolume dependence y^μ . The full Hessian contains mixed terms in which worldvolume derivatives couple to the ξ -dependent background structure. Schematically, the first-order transport piece on the low-energy sector has the form

$$\mathcal{L}_1 = -i \mathcal{V}^\mu(\xi) \partial_{y^\mu}.$$

The key point is that $\mathcal{V}^\mu(\xi)$ is built linearly from background tensors that vanish for a uniform background and measure how the defect varies in ξ . Thus the leading local candidates are proportional to

$$\partial_\xi P_B(\xi), \quad \partial_\xi z_B(\xi), \quad \theta'(\xi),$$

or equivalent frame-covariant combinations.

Since $P_B(\xi)$ is even, these quantities are odd:

$$\partial_\xi P_B(-\xi) = -\partial_\xi P_B(\xi), \quad \theta'(-\xi) = -\theta'(\xi).$$

a. Theorem 6 (Odd transport kernel matching theorem). For a reflection-symmetric projector defect background, any leading worldvolume transport kernel constructed linearly from the transverse derivative of the background is reflection-odd:

$$\mathcal{V}^\mu(-\xi) = -\mathcal{V}^\mu(\xi).$$

b. Proof. All such kernels are linear in the derivative of an even background profile, hence odd. $\square$ This is precisely the structural origin of the odd kernel $g(\xi)$ used in the minimal model.

G. Projected overlap and matching

Let the low modes be

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi).$$

Then the projected transport matrix is

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) \mathcal{V}^\mu(\xi) e_b(\xi).$$

Because $f_+ f_-$ is odd and $\mathcal{V}^\mu$ is odd, the integrand is even and the mixed block is generically nonzero, while same-parity overlaps vanish.

This gives the matching:

odd transport kernel from projector Hessian $\implies$ same-parity overlaps vanish $\implies$ off-diagonal symbol only.

a. Corollary 7. The abstract projected-overlap Dirac theorem derived earlier is structurally realized by the local projector Hessian of a reflection-symmetric CP^2 defect.

H. Sharp status of the matching

We may now state the final honest matching theorem.

a. Theorem 8 (Honest matching theorem). Within the CP^2 projector effective sector, the local Hessian of a reflection-symmetric static defect has the same structural ingredients as the previously constructed minimal graded defect operator model:

1. a transverse matrix Schrödinger channel operator on the rank-two bundle Q_B ;
2. even/odd parity decomposition of the transverse low modes;
3. a first-order worldvolume transport kernel odd under reflection;
4. off-diagonal projected overlap between even and odd channels.

Therefore the minimal effective defect operator is structurally matched to the local projector Hessian.

b. Proof. Items (1)–(3) follow from Theorems 2, 4, and 6. Item (4) follows from the projected overlap formula and parity. $\square$

I. What is still open

This theorem does *not* prove that the full microscopic TECT defect Hessian is exactly the same as the explicit minimal model. The still-open steps are:

1. explicit derivation of the concrete background-dependent potential matrix $\mathcal{U}_B$;
2. proof that its lowest even/odd channels are precisely the chosen $f_{\pm}$;
3. explicit derivation of the transport kernel $\mathcal{V}^{\mu}$;
4. proof that the resulting mixed symbol is nondegenerate in the actual TECT defect background.

But the gap is now very narrow: it is no longer a conceptual gap, only an explicit Hessian calculation gap.

XLI. FINAL SUMMARY

The quadratic Hessian of the CP^2 projector effective action around a static defect background has the generic form of a matrix Schrödinger operator on the rank-two bundle

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^{\perp}.$$

For a reflection-symmetric defect background

$$P_B(-\xi) = P_B(\xi),$$

this Hessian decomposes into even and odd transverse parity channels.

Thus the previously introduced low-mode pair

$$f_+(\xi), \quad f_-(\xi)$$

is not an arbitrary guess: it is the natural parity decomposition of the transverse defect spectrum. The previously chosen Pöschl–Teller wells should be understood as explicit solvable representatives of the effective parity channels, not as a universally exact consequence of the projector Hessian.

When slow worldvolume dependence is restored, the leading transport kernel is built from the transverse derivative of the defect background, schematically

$$\partial_{\xi}P_B(\xi), \quad \partial_{\xi}z_B(\xi), \quad \theta'(\xi),$$

and is therefore reflection-odd. Consequently, projected same-parity overlaps vanish, while the mixed even/odd overlap survives and gives the first-order off-diagonal symbol

$$(M^{\mu})_{ab} = \int d\xi f_+(\xi)f_-(\xi) e_a^{\dagger}(\xi)\mathcal{V}^{\mu}(\xi)e_b(\xi).$$

Hence the minimal effective defect operator constructed earlier is structurally matched to the local CP^2 projector Hessian.

The honest status is therefore:

effective structural matching to the projector Hessian: proved;

exact full microscopic TECT Hessian identification: still open.

XLII. FURTHER ASSUMPTION REDUCTION AUDIT FOR THE EMERGENT DIRAC THEOREM

Goal

We revisit the current defect/valley Dirac-emergence framework and determine whether any of the remaining assumptions can be further removed.

We classify each hypothesis as one of the following:

$$\boxed{\text{Eliminable}} \quad \boxed{\text{Weakable}} \quad \boxed{\text{Currently indispensable}}$$

The purpose is to isolate the true minimal core of the theorem.

Current theorem in its effective form

The present effective Dirac mechanism may be summarized as follows.

Let the low-energy defect/valley sector be represented on a complex 4-dimensional space

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \dim_{\mathbb{C}} \mathcal{K}_{\pm} = 2.$$

Let the projected linearized operator admit the expansion

$$\mathcal{D}_{\text{eff}}(p) = \mathcal{D}_1(p) + O(|p|^2),$$

with leading part

$$\mathcal{D}_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_\mu M^\mu.$$

If the symbol $A_1(p)$ is nondegenerate, then after basis changes and linear momentum redefinitions the principal operator is Dirac.

To obtain this, we previously introduced auxiliary assumptions such as:

1. exact reflection symmetry of the defect profile,
2. an explicit even/odd basis $f_{\pm}(\xi)$,
3. a factorized odd transport kernel $g(\xi)\Sigma^\mu$,
4. explicit solvable Pöschl–Teller channel operators.

We now determine which of these are essential.

A. Eliminating the explicit Pöschl–Teller model

Previously we used the explicit profiles

$$f_+(\xi) \propto \text{sech}(\kappa\xi), \quad f_-(\xi) \propto \tanh(\kappa\xi) \text{sech}(\kappa\xi),$$

together with the solvable scalar channel operators

$$H_+ = -\partial_\xi^2 + \kappa^2 - 2\kappa^2 \text{sech}^2(\kappa\xi),$$

$$H_- = -\partial_\xi^2 + \kappa^2 - 6\kappa^2 \text{sech}^2(\kappa\xi).$$

However, the explicit Pöschl–Teller form is not required for the abstract projected Dirac mechanism. What was actually used is only:

1. existence of one localized even low mode f_+ ,
2. existence of one localized odd low mode f_- ,
3. a nonzero mixed overlap with the transport kernel.

Therefore:

a. Proposition 1. The explicit Pöschl–Teller channel potentials are *not* essential assumptions of the effective Dirac theorem. They are only an explicit solvable representative.

b. Proof. All subsequent projected-overlap formulae depend only on the parity and localization properties of $f_{\pm}$, not on the explicit closed form of the channel potentials. $\square$

Hence:

Explicit Pöschl–Teller structure is eliminable.

B. Eliminating the factorized transport-kernel ansatz

Previously, for computational convenience, we assumed

$$e_a^\dagger(\xi) \mathcal{V}^\mu(\xi) e_b(\xi) = g(\xi) \Sigma_{ab}^\mu.$$

This enabled us to write

$$M^\mu = I_g \Sigma^\mu.$$

But in the abstract projected theorem, factorization is unnecessary. The only truly required object is the mixed overlap matrix

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) \mathcal{V}^\mu(\xi) e_b(\xi).$$

a. Proposition 2. The factorized ansatz

$$e_a^\dagger \mathcal{V}^\mu e_b = g(\xi) \Sigma_{ab}^\mu$$

is not essential for the theorem.

b. Proof. The projected effective symbol is defined directly by the mixed overlap M^μ . The proof of block-off-diagonalization and Dirac reduction depends only on the existence and nondegeneracy of M^μ , not on factorization of its integrand. $\square$

Thus:

Factorized transport kernel is eliminable.

C. Weakening exact reflection symmetry to an abstract involution

Previously we used a concrete reflection

$$(Ju)(\xi) = u(-\xi),$$

and even/odd parity of profiles and kernels.

But the diagonal-block vanishing theorem used only the following data:

1. an involution J on the low-energy sector,

$$J^2 = \mathbf{1},$$

2. a decomposition

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad Ju^\pm = \pm u^\pm,$$

3. oddness of the leading perturbation,

$$J\mathcal{L}_1 J^{-1} = -\mathcal{L}_1.$$

Hence literal spatial reflection is stronger than necessary.

- a. Theorem 3 (Involution reduction theorem).* The projected matrix-element Dirac theorem requires only an involution J on the low-energy subspace, not necessarily geometric reflection in the transverse coordinate.
- b. Proof.* The proof of diagonal-block vanishing is purely algebraic:

$$\langle u^\pm, \mathcal{L}_1 u^\pm \rangle = \langle Ju^\pm, \mathcal{L}_1 Ju^\pm \rangle = \pm \pm \langle u^\pm, J\mathcal{L}_1 Ju^\pm \rangle = -\langle u^\pm, \mathcal{L}_1 u^\pm \rangle.$$

No explicit use of $\xi \mapsto -\xi$ is required. $\square$

Thus:

Exact geometric reflection can be weakened to an abstract involution J .

D. Weakening the explicit even-frame gauge

Previously we used a reflection-even gauge

$$e_a(-\xi) = e_a(\xi).$$

But this was only a convenient way to represent the J -grading concretely. What matters is the existence of a J -adapted low-energy basis.

a. Proposition 4. An explicit reflection-even frame gauge is not essential. It suffices that the low-energy subspaces $\mathcal{K}_\pm$ are well-defined and J -stable.

b. Proof. All projected matrix elements can be written basis-independently using $\Pi_\pm$, or in any orthonormal basis adapted to $\mathcal{K}_\pm$. The choice $e_a(-\xi) = e_a(\xi)$ is only a particularly transparent gauge representative. $\square$

Hence:

Exact even-frame gauge is eliminable.

E. What cannot be eliminated: isolated 4-dimensional low-energy sector

The low-energy Dirac theorem fundamentally acts on a 4-dimensional complex space. Without such an isolated sector, neither the $2+2$ grading nor the 4×4 Dirac operator is even defined.

a. Proposition 5. The existence of an isolated 4-dimensional infrared sector is currently indispensable.

b. Reason. If the relevant low-energy space is not isolated, then projection is ill-defined or unstable. If its dimension is not 4, then the block-off-diagonal $2+2$ Dirac construction cannot be carried out. No purely geometric argument derived so far forces this sector to exist in full TECT. Therefore this cannot presently be removed.

Thus:

Isolated 4-dimensional infrared sector is currently indispensable.

F. What cannot be eliminated: the $2+2$ grading

The Dirac theorem requires the low-energy sector to split as

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \dim \mathcal{K}_\pm = 2.$$

Without this, there is no grading operator Γ and no notion of odd/off-diagonal principal operator.

a. Proposition 6. The $2+2$ grading is currently indispensable.

b. Reason. The block-off-diagonal structure theorem is equivalent to oddness with respect to a grading. Without the grading, one cannot even formulate the same-sector vanishing condition or define the off-diagonal block $A_1(p)$. No previous theorem has derived this grading solely from the CP^2 geometry or from ordinary little-group symmetry.

$\square$

Hence:

The $2+2$ grading is currently indispensable.

G. What cannot be eliminated: nondegenerate mixed symbol

Even if a 4-dimensional graded low-energy sector exists, the projected operator yields Dirac only if the mixed 2×2 first-order symbol

$$A_1(p) = p_\mu M^\mu$$

is nondegenerate.

If the symbol is rank-deficient, one may get a line node, one-dimensional crossing, quadratic touching, or no Dirac cone at all.

a. Proposition 7. Nondegeneracy of the mixed symbol is currently indispensable.

b. Reason. The Dirac conclusion is equivalent to the statement that, after basis changes and momentum redefinitions, the symbol reduces to a Pauli triple

$$A_1(p) \sim p_i \sigma^i.$$

This requires a full-rank symbol map. No argument developed so far can force this from parity or grading alone. $\square$
Therefore:

Nondegenerate mixed symbol is currently indispensable.

H. What can be weakened further: exact zero modes to isolated low modes

Previously we often spoke of exact zero modes of $\mathcal{L}_0$. But in the effective Dirac reduction, exact zero energy is not essential. It suffices to have an isolated low-energy multiplet after subtracting the common energy E_* .

a. Theorem 8 (Energy-shift weakening theorem). Exact zero-mode normalization is not essential. It suffices that the relevant 4-dimensional sector is spectrally isolated and that the common low-energy reference energy is subtracted.

b. Proof. If the low-energy sector has common energy E_* , replace

$$\mathcal{L} \mapsto \mathcal{L} - E_* \mathbf{1}.$$

Then the entire projected analysis applies to the shifted operator. $\square$

Hence:

Exact zero-mode assumption can be weakened to isolated low-energy multiplet.

I. What can be weakened further: full Hessian knowledge to projected matrix elements

One might think that the full explicit Hessian $\mathcal{H}_B$ must be known. But the projected Dirac theorem only needs the projected leading matrix elements

$$\langle u_a^\pm, \mathcal{L}_1 u_b^\pm \rangle, \quad \langle u_a^+, \mathcal{L}_1 u_b^- \rangle.$$

Thus full closed-form knowledge of $\mathcal{H}_B$ is stronger than necessary.

a. Theorem 9 (Projected-data reduction theorem). To establish the effective Dirac principal operator, one does not need a closed-form solution of the full Hessian. It is sufficient to determine:

1. the isolated low-energy 4-mode sector,
2. the J -grading on that sector,
3. the leading projected matrix elements of $\mathcal{L}_1$,
4. nondegeneracy of the mixed symbol.

b. Proof. This follows directly from the projected matrix-element Dirac theorem and Feshbach reduction. $\square$
Hence:

Full closed-form Hessian is not essential; projected low-energy data suffices.

J. Minimal reduced theorem

After all possible eliminations and weakenings, the genuinely minimal effective theorem is the following.

a. Minimal reduced effective Dirac theorem. Suppose:

(R1) there exists an isolated complex 4-dimensional low-energy sector

$$\mathcal{K};$$

(R2) $\mathcal{K}$ carries an involution J with

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \dim \mathcal{K}_\pm = 2;$$

(R3) the leading projected perturbation is odd under J :

$$J\mathcal{L}_1 J^{-1} = -\mathcal{L}_1 \quad \text{on } \mathcal{K};$$

(R4) the mixed projected symbol

$$A_1(p) = p_\mu M^\mu, \quad (M^\mu)_{ab} = \langle u_a^+, \mathcal{V}^\mu u_b^- \rangle,$$

is nondegenerate.

Then the principal effective operator on $\mathcal{K}$ is the Dirac operator.

b. Proof. By R2 and R3, same-sector projected matrix elements vanish, so the principal projected operator is off-diagonal. By R4, the off-diagonal 2×2 symbol is nondegenerate. The previously proved block-off-diagonal Dirac theorem then gives the Dirac principal operator. $\square$

K. Sharp conclusion

The assumption reduction audit yields the following table:

Hypothesis	Status
Explicit Pöschl–Teller channel profile	Eliminable
Factorized transport kernel $g(\xi)\Sigma^\mu$	Eliminable
Exact geometric reflection $\xi \mapsto -\xi$	Weakable to abstract involution J
Exact even-frame gauge	Eliminable
Exact zero modes	Weakable to isolated low multiplet
Full explicit Hessian formula	Weakable to projected low-energy data
Isolated 4-dimensional low-energy sector	Currently indispensable
The $2 + 2$ grading	Currently indispensable
Nondegenerate mixed first-order symbol	Currently indispensable

Therefore the theorem can indeed be strengthened somewhat, but only up to a point.

The true minimal load-bearing core is:

isolated 4-mode sector

involution J giving a $2 + 2$ splitting

nondegenerate mixed first-order projected symbol

XLIII. FINAL SUMMARY

The emergent Dirac theorem can be strengthened by removing several model-dependent assumptions. The following assumptions are not essential and can be removed or weakened:

1. explicit Pöschl–Teller channel potentials,
2. factorized transport kernel $g(\xi)\Sigma^\mu$,
3. exact geometric reflection symmetry as such,
4. exact reflection-even frame gauge,
5. exact zero-mode normalization,
6. full closed-form knowledge of the entire Hessian.

After reduction, the theorem needs only:

(R1) an isolated complex 4-dimensional low-energy sector;

(R2) an involution J defining a $2 + 2$ grading;

(R3) oddness of the leading projected perturbation under J ;

(R4) a nondegenerate mixed 2×2 first-order symbol.

These are the presently irreducible assumptions.

Therefore the theorem can indeed be made stronger, but not arbitrarily stronger. At the current stage, the real unresolved issue is no longer the detailed model profile, but the existence and explicit computation of the low-energy graded spectral data in the actual TECT defect/valley operator.

XLIV. MINIMAL REDUCED DIRAC EMERGENCE THEOREM

Logical status before the theorem

At the present stage, the Dirac-emergence claim should not be formulated as

$$\mathbb{CP}^2 \text{ geometry} \implies \text{Dirac equation}$$

without further hypotheses. That implication is too strong and is not proved.

What has been established instead is that the Dirac principal operator follows rigorously from a minimal set of projected low-energy spectral data. All auxiliary model-specific assumptions (explicit Pöschl–Teller channels, factorized transport kernels, exact geometric reflection, explicit even frame gauges, exact zero-mode normalization, and closed-form full Hessian formulas) can be removed or weakened.

The theorem below is therefore the strongest honest theorem presently available.

Minimal hypotheses

Let $\mathcal{L}(p)$ denote the linearized operator of the relevant defect/valley sector, expanded near the infrared point $p = 0$. Assume the following:

- (R1) There exists an isolated complex 4-dimensional low-energy sector

$$\mathcal{K} \subset \mathcal{H}.$$

(R2) The low-energy sector carries an involution

$$J : \mathcal{K} \rightarrow \mathcal{K}, \quad J^\dagger = J, \quad J^2 = \mathbf{1},$$

with decomposition

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \mathcal{K}_\pm = \ker(J \mp \mathbf{1}), \quad \dim_{\mathbb{C}} \mathcal{K}_\pm = 2.$$

(R3) After subtraction of the common reference low-energy eigenvalue, the leading projected perturbation is odd under J . Equivalently, if

$$\mathcal{D}_{\text{eff}}(p) = \Pi_{\mathcal{K}} \mathcal{L}(p) \Pi_{\mathcal{K}} = \mathcal{D}_1(p) + O(|p|^2),$$

then

$$J \mathcal{D}_1(p) J^{-1} = -\mathcal{D}_1(p).$$

(R4) The mixed 2×2 principal symbol is first-order and nondegenerate. That is, there exists a basis adapted to $\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-$ such that

$$\mathcal{D}_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_\mu M^\mu,$$

and the symbol $A_1(p)$ is nondegenerate for generic small $p \neq 0$.

Theorem statement

a. Theorem (Minimal Reduced Dirac Emergence Theorem). Under hypotheses (R1)–(R4), the principal reduced operator on $\mathcal{K}$ is unitarily equivalent, after a linear change of momentum coordinates, to the massless Dirac operator:

$$\mathcal{D}_1(p) \sim p_i \alpha^i,$$

where

$$\{\alpha^i, \alpha^j\} = 2\delta^{ij} \mathbf{1}_4.$$

Equivalently, in covariant first-order form,

$$(i\gamma^\mu \partial_\mu) \Psi = 0$$

at principal order on the projected low-energy field $\Psi \in \mathcal{K}$.

Proof

Let

$$\Pi_\pm : \mathcal{K} \rightarrow \mathcal{K}_\pm$$

be the projectors associated with the involution J . Define the grading operator

$$\Gamma := \Pi_+ - \Pi_-.$$

Then

$$\Gamma^\dagger = \Gamma, \quad \Gamma^2 = \mathbf{1}, \quad \Gamma|_{\mathcal{K}_\pm} = \pm \mathbf{1}.$$

By hypothesis (R3),

$$\Gamma \mathcal{D}_1 + \mathcal{D}_1 \Gamma = 0.$$

Write $\mathcal{D}_1$ in the graded basis $\mathcal{K}_+ \oplus \mathcal{K}_-$:

$$\mathcal{D}_1 = \begin{pmatrix} D_{++} & D_{+-} \\ D_{-+} & D_{--} \end{pmatrix}.$$

Then

$$\Gamma \mathcal{D}_1 + \mathcal{D}_1 \Gamma = \begin{pmatrix} 2D_{++} & 0 \\ 0 & -2D_{--} \end{pmatrix}.$$

Hence

$$D_{++} = 0, \quad D_{--} = 0,$$

so

$$\mathcal{D}_1 = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix},$$

with

$$A_1(p) : \mathcal{K}_- \rightarrow \mathcal{K}_+$$

a 2×2 complex linear symbol.

By hypothesis (R4),

$$A_1(p) = p_\mu M^\mu$$

is first-order and nondegenerate. Every 2×2 complex linear symbol can be decomposed into identity and Pauli components. After unitary basis changes in $\mathcal{K}_+$ and $\mathcal{K}_-$, and an invertible linear change of momentum coordinates, the nondegenerate symbol may be brought to canonical Pauli form

$$A_1(p) \sim p_i \sigma^i.$$

Therefore the full 4×4 principal operator is

$$\mathcal{D}_1(p) \sim \begin{pmatrix} 0 & p_i \sigma^i \\ p_i \sigma^i & 0 \end{pmatrix}.$$

Define

$$\alpha^i = \begin{pmatrix} 0 & \sigma^i \\ \sigma^i & 0 \end{pmatrix}.$$

Then

$$\{\alpha^i, \alpha^j\} = 2\delta^{ij} \mathbf{1}_4,$$

and

$$\mathcal{D}_1(p) \sim p_i \alpha^i.$$

This is precisely the massless Dirac principal operator. $\square$

Interpretation

The theorem shows that the load-bearing structure is not a broad symmetry slogan such as

$$\text{“fermionic defect”} \implies \text{Dirac equation},$$

nor a purely geometric statement such as

$$\mathbb{CP}^2 \implies SU(3) \implies \text{Dirac}.$$

Instead, the true core is the existence of a graded low-energy 4-mode sector with a nondegenerate mixed first-order symbol.

Thus the mathematically honest implication is

graded 4-mode infrared sector + nondegenerate mixed first-order symbol $\implies$ Dirac principal operator.

Corollary: effective covariant Dirac operator

If, in addition, the low-energy sector carries the already-established emergent $SU(3)$ gauge connection by projected minimal coupling, then

$$\partial_\mu \mapsto D_\mu = \partial_\mu - iA_\mu^a T^a,$$

and the principal operator becomes

$$i\gamma^\mu D_\mu$$

rather than $i\gamma^\mu \partial_\mu$. Therefore the principal effective field equation is

$$(i\gamma^\mu D_\mu)\Psi = 0$$

at massless principal order.

b. Corollary. Under the hypotheses of the theorem, plus projected gauge covariance of the low-energy mode bundle, the principal reduced equation is the covariant massless Dirac equation

$$(i\gamma^\mu D_\mu)\Psi = 0.$$

What this theorem does not claim

This theorem does *not* claim any of the following:

1. that the full microscopic TECT PDE has already been shown to satisfy (R1)–(R4);
2. that $\mathbb{CP}^2$ geometry alone is sufficient;
3. that the mixed symbol has already been computed explicitly in the microscopic theory;
4. that the resulting Dirac field is already fully identified with a Standard Model multiplet.

Hence the theorem is strong but properly scoped.

Open calculation list

To upgrade the theorem from an effective structural theorem to a full microscopic TECT derivation, the following explicit calculations remain:

1. Construct the actual isolated 4-dimensional low-energy sector of the defect/valley linearized operator.
2. Identify the involution J on that sector and verify the $2 + 2$ splitting

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-.$$

3. Compute the projected leading matrix elements

$$\Pi_\pm \mathcal{L}_1 \Pi_\pm, \quad \Pi_+ \mathcal{L}_1 \Pi_- ,$$

and verify oddness under J .

4. Compute the mixed symbol

$$A_1(p) = p_\mu M^\mu$$

and prove its nondegeneracy.

5. Verify that the number of independent propagating directions is sufficient for the desired spacetime Dirac operator.
6. If needed for the physical identification, incorporate the already-developed ingredients: mirror lifting, index protection, and gauge-covariant coupling.

Final logical status

The theorem established here should be regarded as:

a rigorous minimal effective Dirac-emergence theorem

inside the TECT CP^2 /defect/valley program,
but not yet as

a full microscopic theorem of the complete TECT PDE.

XLV. FINAL SUMMARY

After eliminating all nonessential model-dependent assumptions, the Dirac-emergence mechanism reduces to the following minimal statement.

If there exists an isolated complex 4-dimensional infrared sector

$$\mathcal{K},$$

equipped with an involution

$$J^2 = 1$$

that splits it as

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \dim \mathcal{K}_\pm = 2,$$

and if the leading projected perturbation is odd under J , so that the principal reduced operator is block-off-diagonal,

$$\mathcal{D}_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix},$$

with nondegenerate first-order mixed symbol

$$A_1(p) = p_\mu M^\mu,$$

then the principal reduced operator is unitarily equivalent to the massless Dirac operator:

$$\mathcal{D}_1(p) \sim p_i \alpha^i.$$

Thus the essential load-bearing assumptions are exactly:

isolated 4-dimensional low-energy sector

a $2 + 2$ involutive grading

a nondegenerate mixed first-order 2×2 symbol

Everything else - explicit solvable profiles, factorized kernels, exact geometric reflection, exact zero-mode wording, and full closed-form Hessian formulas - is auxiliary and not essential to the theorem itself.

Therefore the strongest honest conclusion at the current stage is:

the minimal effective Dirac-emergence theorem is proved,

but

its full microscopic realization inside the complete TECT operator remains an open calculation problem.

XLVI. EMERGENCE OF THE DIRAC OPERATOR FROM A GRADED INFRARED SECTOR

A. Motivation

One of the central questions of the TECT framework is whether relativistic fermionic dynamics can emerge from the defect/valley structure of the condensed energy background.

Rather than assuming a Dirac equation from the outset, the goal is to derive it as the principal operator governing the infrared sector of the linearized fluctuation problem.

Previous constructions introduced explicit solvable channel models and reflection symmetries. However, many of those assumptions are in fact not essential. In this section we formulate the minimal set of conditions under which a Dirac principal operator necessarily emerges.

The result is formulated purely in terms of the projected low-energy spectral structure.

B. Infrared sector and grading

Let $\mathcal{L}(p)$ denote the linearized operator governing fluctuations around the relevant defect or valley configuration, expanded near $p = 0$.

Assume that the spectrum contains an isolated infrared sector

$$\mathcal{K} \subset \mathcal{H}$$

with

$$\dim_{\mathbb{C}} \mathcal{K} = 4.$$

This sector is assumed to carry an involutive grading operator

$$J : \mathcal{K} \rightarrow \mathcal{K},$$

satisfying

$$J^\dagger = J, \quad J^2 = \mathbf{1}.$$

The grading decomposes the sector as

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-,$$

with

$$\dim_{\mathbb{C}} \mathcal{K}_\pm = 2.$$

The operator J plays the role of a chirality or parity grading in the infrared subspace.

C. Projected operator

Let

$$\Pi_{\mathcal{K}}$$

be the spectral projector onto the infrared sector. The effective operator on this sector is

$$\mathcal{D}_{\text{eff}}(p) = \Pi_{\mathcal{K}} \mathcal{L}(p) \Pi_{\mathcal{K}}.$$

Expanding near $p = 0$,

$$\mathcal{D}_{\text{eff}}(p) = \mathcal{D}_1(p) + O(|p|^2).$$

We assume that the leading term is odd with respect to the grading:

$$J\mathcal{D}_1(p)J^{-1} = -\mathcal{D}_1(p).$$

D. Block structure

Let $\Pi_{\pm}$ denote the projectors onto $\mathcal{K}_{\pm}$.
In the graded basis

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-,$$

the operator can be written as

$$\mathcal{D}_1(p) = \begin{pmatrix} D_{++} & D_{+-} \\ D_{-+} & D_{--} \end{pmatrix}.$$

The grading condition implies

$$D_{++} = 0, \quad D_{--} = 0.$$

Thus

$$\mathcal{D}_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^{\dagger} & 0 \end{pmatrix}.$$

E. Mixed symbol

We assume that the off-diagonal block is linear in momentum,

$$A_1(p) = p_{\mu} M^{\mu},$$

where M^{μ} are 2×2 complex matrices.

Furthermore, the symbol is assumed to be nondegenerate, meaning that for generic small $p \neq 0$

$$\det A_1(p) \neq 0.$$

F. Dirac emergence theorem

a. Theorem. Under the assumptions above, the principal effective operator on the infrared sector is unitarily equivalent to the massless Dirac operator.

G. Proof

Because $\mathcal{D}_1$ anticommutes with the grading operator, the operator is block-off-diagonal. The off-diagonal block $A_1(p)$ is a linear map

$$A_1(p) : \mathcal{K}_- \rightarrow \mathcal{K}_+,$$

represented by a 2×2 complex matrix.

Any such matrix can be expanded in the Pauli basis

$$A_1(p) = a_0(p)\mathbf{1} + a_i(p)\sigma^i.$$

Nondegeneracy implies that the linear map $p \mapsto A_1(p)$ has full rank.

After unitary rotations of the bases of $\mathcal{K}_\pm$ and an invertible linear transformation of momentum coordinates, the symbol can be brought to the canonical form

$$A_1(p) = p_i \sigma^i.$$

Substituting into the full operator gives

$$\mathcal{D}_1(p) = \begin{pmatrix} 0 & p_i \sigma^i \\ p_i \sigma^i & 0 \end{pmatrix}.$$

Define

$$\alpha^i = \begin{pmatrix} 0 & \sigma^i \\ \sigma^i & 0 \end{pmatrix}.$$

These matrices satisfy

$$\{\alpha^i, \alpha^j\} = 2\delta^{ij}\mathbf{1}_4.$$

Thus

$$\mathcal{D}_1(p) = p_i \alpha^i.$$

In position space this corresponds to the massless Dirac equation

$$i\gamma^\mu \partial_\mu \Psi = 0.$$

H. Interpretation

The theorem shows that the Dirac operator does not require detailed microscopic assumptions about the defect profile or explicit solvable channel potentials.

Instead, the essential ingredients are purely spectral:

- an isolated four-dimensional infrared sector,
- a $2 + 2$ grading structure,
- a first-order mixed symbol,
- nondegeneracy of the symbol.

Once these structures are present, the Dirac operator follows necessarily as the principal infrared operator.

I. Limitations

The theorem should be interpreted carefully.

It does not yet prove that the full microscopic TECT equations automatically generate such a graded infrared sector.

Instead, it establishes the following conditional statement:

$$\text{graded infrared sector} + \text{nondegenerate mixed symbol} \implies \text{Dirac operator}.$$

The remaining task is therefore to verify these spectral properties in the explicit defect or valley background of the full theory.

J. Open calculations

To complete the microscopic derivation the following steps remain:

1. construct the isolated four-mode infrared sector,
2. identify the involution generating the $2 + 2$ grading,
3. compute the projected first-order symbol,
4. prove nondegeneracy of the mixed matrix M^μ .

Once these calculations are performed, the Dirac equation follows immediately from the theorem above.

XLVII. CONSTRUCTION OF THE ISOLATED FOUR-MODE INFRARED SECTOR

A. Defect background

Let

$$P_B(\xi)$$

be a static defect configuration depending only on the transverse coordinate ξ . Fluctuations of the projector satisfy

$$P^2 = P.$$

Linearizing around the background,

$$P = P_B + \delta P.$$

The constraint implies

$$P_B \delta P + \delta P P_B = \delta P.$$

This can be parametrized as

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger$$

where

$$z_B^\dagger z_B = 1$$

and

$$z_B^\dagger \eta = 0.$$

Thus

$$\eta(\xi, y) \in Q_B(\xi)$$

with

$$Q_B = (\mathbb{C}z_B)^\perp.$$

B. Internal dimension

Because

$$z_B \in \mathbb{C}^3$$

the orthogonal complement satisfies

$$\dim_{\mathbb{C}} Q_B = 2.$$

Hence the fluctuation field is a complex doublet

$$\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}.$$

C. Transverse fluctuation operator

The quadratic fluctuation operator around the defect background has the schematic form

$$\mathcal{H}_B = -\partial_\xi^2 - \partial_y^2 + U_B(\xi)$$

where

$$U_B(\xi)$$

is a Hermitian 2×2 matrix potential determined by the background.

D. Bound states

Localized defect configurations typically generate bound states in the transverse Schrödinger operator

$$H_\xi = -\partial_\xi^2 + U_B(\xi).$$

For reflection-symmetric potentials

$$U_B(-\xi) = U_B(\xi)$$

the eigenfunctions can be chosen with definite parity

$$f_+(-\xi) = f_+(\xi)$$

$$f_-(-\xi) = -f_-(\xi).$$

Generically the lowest two bound states are

$$f_+(\xi), \quad f_-(\xi).$$

E. Low-energy modes

Using a local orthonormal frame

$$e_a(\xi)$$

of Q_B , the localized modes are

$$u_a^+(\xi) = f_+(\xi)e_a(\xi)$$

$$u_a^-(\xi) = f_-(\xi)e_a(\xi)$$

with

$$a = 1, 2.$$

Thus the infrared sector is

$$\mathcal{K} = \text{span}\{u_a^+, u_a^-\}.$$

Therefore

$$\dim_{\mathbb{C}} \mathcal{K} = 4.$$

F. Spectral isolation

If the next excited bound state has eigenvalue

$$\lambda_3 \gg \lambda_1, \lambda_2,$$

the sector $\mathcal{K}$ is spectrally isolated.

In this case the Feshbach projection onto $\mathcal{K}$ defines a stable low-energy effective operator.

G. Result

We therefore obtain a natural candidate for the infrared sector

$$\mathcal{K} = \text{span}\{u_a^\pm\}$$

with

$$\dim_{\mathbb{C}} \mathcal{K} = 4.$$

This sector provides the input required for the minimal Dirac emergence theorem.

XLVIII. SUMMARY

The isolated four-dimensional infrared sector arises from two independent ingredients. First, fluctuations of a $\mathbb{CP}^2$ projector background take values in the orthogonal complement

$$Q_B = (\mathbb{C}z_B)^\perp$$

which has complex dimension two.

Second, the transverse defect operator generically admits localized even and odd bound states

$$f_+(\xi), \quad f_-(\xi).$$

Combining the two structures produces four localized modes

$$u_a^\pm(\xi) = f_\pm(\xi)e_a(\xi)$$

with

$$a = 1, 2.$$

Therefore

$$\dim_{\mathbb{C}} \mathcal{K} = 4.$$

This sector forms the natural infrared subspace required by the minimal Dirac emergence theorem.

XLIX. EXPLICIT HESSIAN STRUCTURE OF THE CP^2 PROJECTOR DEFECT

Assumption Audit

- A1: The relevant internal effective sector is described by a projector action

$$S[P] = \int d^{d+1}x \left[\frac{\rho}{4} \text{Tr}(\partial_\mu P \partial^\mu P) - V(P) \right], \quad P^2 = P, \quad P^\dagger = P, \quad \text{Tr } P = 1.$$

Effective-sector assumption

- A2: The defect background is static and depends only on one transverse coordinate:

$$P_B = P_B(\xi).$$

Local defect setup

- A3: Physical fluctuations are tangent to the rank-one projector manifold and may be written as

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0.$$

Proved in the projector framework

- A4: The chosen local background representative is

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi).$$

Model background representative

- A5: The background is reflection-symmetric so that parity decomposition of the transverse spectrum is meaningful.

Structural assumption

A. Tangent parametrization of projector fluctuations

Let

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \quad z_B^\dagger z_B = 1.$$

A first-order fluctuation satisfies the linearized projector constraint

$$P_B \delta P + \delta P P_B = \delta P.$$

A standard tangent parametrization is

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0.$$

Thus

$$\eta(\xi, y) \in Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp.$$

- a. Proposition 1.* The physical fluctuation problem reduces to a rank-two complex field η valued in the bundle

$$Q_B(\xi) = (\mathbb{C} z_B(\xi))^\perp.$$

- b. Proof.* This is the standard tangent description of rank-one Hermitian projectors. $\square$

B. Explicit background and Q_B -frame

Choose

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi).$$

Then an explicit orthonormal frame of Q_B is

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

One checks

$$z_B^\dagger e_a = 0, \quad e_a^\dagger e_b = \delta_{ab}.$$

Now compute the transverse derivatives:

$$\partial_\xi z_B = \theta'(\xi) \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix} = \theta'(\xi) e_1(\xi),$$

$$\partial_\xi e_1 = -\theta'(\xi) \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix} = -\theta'(\xi) z_B(\xi), \quad \partial_\xi e_2 = 0.$$

Hence the induced Q_B -connection along ξ is

$$\mathcal{A}_\xi := -i e^\dagger \partial_\xi e = 0.$$

a. Theorem 2 (Vanishing transverse Q_B -connection). For the explicit background/frame choice above,

$$\mathcal{A}_\xi = -i e^\dagger \partial_\xi e = 0.$$

b. Proof. Since $e'_1 = -\theta' z_B$ is orthogonal to Q_B , and $e'_2 = 0$, all matrix elements $e_a^\dagger e'_b$ vanish. $\square$

This is important because it makes the Hessian locally gauge-equivalent to an ordinary matrix Schrödinger operator, rather than a covariant one with a nontrivial ξ -connection.

C. Quadratic variation of the kinetic term

Write

$$\eta(\xi, y) = e_a(\xi) \chi^a(\xi, y), \quad \chi(\xi, y) \in \mathbb{C}^2.$$

Because $\mathcal{A}_\xi = 0$, one has

$$D_\xi^{(B)} \eta = e_a \partial_\xi \chi^a + (\partial_\xi e_a) \chi^a.$$

Since $\partial_\xi e_1 = -\theta' z_B$ and $\partial_\xi e_2 = 0$, the Q_B -projected derivative is simply

$$\Pi_{Q_B} D_\xi^{(B)} \eta = e_a \partial_\xi \chi^a.$$

However, the full second variation of the sigma-model kinetic term also contains curvature-type background-gradient contributions. A standard tangent-bundle expansion gives

$$\delta^2 S_{\text{kin}} = \int d^{d+1}x \left[\rho |\partial_\mu \chi|^2 - \rho |D_\mu z_B|^2 \chi^\dagger \chi - \rho (\chi^\dagger D_\mu z_B) (D^\mu z_B)^\dagger \chi \right]$$

up to total derivatives and in the local transverse gauge.

Since the background depends only on ξ ,

$$D_\mu z_B = 0 \quad (\mu \in y\text{-directions}), \quad D_\xi z_B = z'_B = \theta' e_1.$$

Therefore

$$|D_\xi z_B|^2 = \theta'(\xi)^2,$$

and

$$(\chi^\dagger D_\xi z_B)(D_\xi z_B)^\dagger \chi = \theta'(\xi)^2 |\chi_1|^2.$$

Hence the kinetic quadratic form contributes

$$\delta^2 S_{\text{kin}} = \int d^{d+1}x \left[\rho |\partial_\xi \chi|^2 + \rho |\partial_y \chi|^2 - \rho \theta'(\xi)^2 (|\chi_1|^2 + |\chi_2|^2) - \rho \theta'(\xi)^2 |\chi_1|^2 \right].$$

Equivalently,

$$\delta^2 S_{\text{kin}} = \int d^{d+1}x \left[\rho |\partial_\xi \chi|^2 + \rho |\partial_y \chi|^2 - \rho \theta'(\xi)^2 \begin{pmatrix} \chi_1^* & \chi_2^* \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} \right].$$

a. Theorem 3 (Explicit gradient-induced Hessian contribution). For the explicit defect background $z_B(\xi)$ above, the background-gradient part of the quadratic kinetic Hessian in the local Q_B -frame is

$$-\rho \theta'(\xi)^2 \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

b. Proof. Insert $D_\xi z_B = \theta' e_1$ into the general tangent-bundle quadratic kinetic form. The first background term contributes $-\rho \theta'^2 \mathbf{1}_2$, while the second contributes an additional $-\rho \theta'^2$ in the e_1 channel only. $\square$

This is a genuine explicit formula, not merely schematic.

D. Potential contribution

Now consider the variation of the defect-stabilizing potential $V(P)$. Restricted to tangent fluctuations, it defines a Hermitian Hessian endomorphism on Q_B :

$$\delta^2 S_V = \int d^{d+1}x \chi^\dagger(\xi, y) \mathcal{W}_B(\xi) \chi(\xi, y),$$

where

$$\mathcal{W}_B(\xi) = \text{Hess}_{Q_B} V|_{P_B(\xi)}.$$

In the chosen frame this is a Hermitian 2×2 matrix:

$$\mathcal{W}_B(\xi) = \begin{pmatrix} W_{11}(\xi) & W_{12}(\xi) \\ W_{21}(\xi) & W_{22}(\xi) \end{pmatrix}.$$

If the defect background lies in an embedded $CP^1 \subset CP^2$ and the potential respects the orthogonal e_2 -direction, then one often has

$$W_{12} = W_{21} = 0,$$

so the static channels decouple.

E. Full static Hessian

Combining the kinetic and potential pieces, the static quadratic operator on χ is

$$\mathcal{H}_B^{\text{stat}} = -\rho \partial_\xi^2 + \mathcal{U}_B(\xi),$$

with

$$\mathcal{U}_B(\xi) = \mathcal{W}_B(\xi) - \rho \theta'(\xi)^2 \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

a. Theorem 4 (Explicit static Hessian formula). For the reflection-symmetric CP^2 defect background

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix},$$

the projector Hessian on the tangent field $\chi \in \mathbb{C}^2$ is locally

$$\mathcal{H}_B = -\rho \partial_\xi^2 - \rho \partial_y^2 + \mathcal{W}_B(\xi) - \rho \theta'(\xi)^2 \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} + \dots,$$

where $\mathcal{W}_B$ is the potential Hessian on Q_B .

b. Proof. Combine Theorems 2 and 3 with the tangent-space second variation of the potential. $\square$
This is the explicit matching formula we wanted.

F. Parity decomposition

If $\theta(\xi)$ is even, then $\theta'(\xi)^2$ is even. If, moreover, $\mathcal{W}_B(\xi)$ is reflection-even,

$$\mathcal{W}_B(-\xi) = \mathcal{W}_B(\xi),$$

then

$$\mathcal{U}_B(-\xi) = \mathcal{U}_B(\xi).$$

Hence the static transverse operator commutes with reflection

$$(J\chi)(\xi) = \chi(-\xi),$$

and its spectrum decomposes into even and odd parity sectors.

a. Corollary 5 (Parity channel decomposition). Under reflection symmetry, the static Hessian admits even and odd bound-state channels

$$f_+(\xi), \quad f_-(\xi)$$

in each internal sector, whenever the corresponding bound states exist.

G. Origin of the odd transport kernel

When slow worldvolume dependence is restored, the leading transport terms arise from cross terms between world-volume derivatives and ξ -dependent background geometry. The local tensors from which these terms are built are linear in background gradients, such as

$$\partial_\xi P_B(\xi), \quad \partial_\xi z_B(\xi), \quad \theta'(\xi).$$

Since $P_B(\xi)$ and $\theta(\xi)$ are even,

$$\partial_\xi P_B(-\xi) = -\partial_\xi P_B(\xi), \quad \theta'(-\xi) = -\theta'(\xi).$$

Therefore any leading transport kernel constructed linearly from these gradients is odd:

$$\mathcal{V}^\mu(-\xi) = -\mathcal{V}^\mu(\xi).$$

a. Theorem 6 (Explicit oddness of the transport kernel). For a reflection-symmetric defect background, every leading worldvolume transport coefficient built linearly from the transverse background gradient is reflection-odd.

b. Proof. Derivative of an even background profile is odd. $\square$

Thus the odd-kernel assumption used previously is not ad hoc; it is exactly what the local defect-gradient expansion predicts.

H. Matching with the minimal effective defect operator

The previously constructed minimal effective model had the form

$$\mathcal{L}_0^{\text{eff}} = \begin{pmatrix} H_+ & 0 \\ 0 & H_- \end{pmatrix} \otimes \mathbf{1}_2, \quad \mathcal{L}_1^{\text{eff}} = -i \mathcal{V}^\mu(\xi) \partial_{y^\mu},$$

with $H_\pm$ scalar parity channels and $\mathcal{V}^\mu$ odd.

The actual projector Hessian now matches this structure as follows:

projector Hessian $\implies$ rank-2 matrix Schrödinger operator on $Q_B \implies$ even/odd transverse channels $\implies$ odd first-order transport

Therefore the minimal effective model is not arbitrary; it is an explicit solvable representative of the local projector Hessian reduction.

a. Theorem 7 (Honest local matching theorem). The minimal graded defect operator constructed earlier belongs to the same operator class as the local CP^2 projector Hessian around a reflection-symmetric defect background. In particular:

1. the static part is a transverse matrix Schrödinger operator on the rank-two bundle Q_B ;
2. parity-even and parity-odd low channels arise naturally under reflection symmetry;
3. the leading worldvolume transport kernel is odd because it is built from the transverse background gradient.

b. Proof. Items (1)–(3) follow from Theorems 4, 5, and 6. $\square$

I. What is and is not proved

What is proved:

1. the local projector Hessian can be written explicitly in the chosen defect frame;
2. the background-gradient contribution to the Hessian is computed explicitly;
3. the operator class matches the minimal effective defect model;
4. odd transport kernels arise naturally from defect gradients.

What is not yet proved:

1. the exact form of the potential Hessian $\mathcal{W}_B(\xi)$ in the full TECT effective action;
2. the exact spectrum of the full static operator $-\rho \partial_\xi^2 + \mathcal{U}_B(\xi)$;
3. the exact number and gap of the lowest parity channels;
4. the nondegeneracy of the final mixed symbol in the microscopic theory.

Thus the remaining gap is now explicit and computational, not conceptual.

L. FINAL SUMMARY

For the reflection-symmetric local defect background

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi),$$

an explicit orthonormal frame of

$$Q_B(\xi) = (\mathbb{C} z_B(\xi))^\perp$$

is

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

In this frame the transverse Q_B -connection vanishes:

$$\mathcal{A}_\xi = -i e^\dagger \partial_\xi e = 0.$$

The quadratic projector Hessian on the tangent fluctuation

$$\eta = e_a \chi^a$$

is therefore locally a matrix Schrödinger operator:

$$\mathcal{H}_B = -\rho \partial_\xi^2 - \rho \partial_y^2 + \mathcal{U}_B(\xi) + \cdots,$$

with

$$\mathcal{U}_B(\xi) = \mathcal{W}_B(\xi) - \rho \theta'(\xi)^2 \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix},$$

where $\mathcal{W}_B$ is the potential Hessian on Q_B .

Thus the background-gradient contribution is explicitly computable and distinguishes the two internal channels.

If the background is reflection-symmetric, $\mathcal{U}_B(\xi)$ is even, so the transverse spectrum decomposes into even and odd parity channels. When worldvolume dependence is restored, the leading transport kernel is built from the defect gradient

$$\partial_\xi P_B, \quad \partial_\xi z_B, \quad \theta'(\xi),$$

and is therefore odd under reflection.

Hence the earlier minimal effective defect operator is structurally matched to the local projector Hessian in an explicit way.

The honest status is:

local Hessian matching and explicit gradient contribution: proved;

full spectral solution and symbol nondegeneracy in the actual TECT defect: still open.

LI. A MINIMAL DEFECT-STABILIZING POTENTIAL AND THE EXPLICIT COMPUTATION OF $\mathcal{W}_B(\xi)$

Assumption Audit

- A1: We work in the effective CP^2 projector sector with action

$$S[P] = \int d^{d+1}x \left[\frac{\rho}{4} \text{Tr}(\partial_\mu P \partial^\mu P) - V(P) \right], \quad P^2 = P, \quad P^\dagger = P, \quad \text{Tr} P = 1.$$

Effective-sector assumption

- A2: We choose a minimal anisotropy potential

$$V(P) = \frac{\lambda}{2} \text{Tr}([H, P]^2), \quad \lambda > 0,$$

with

$$H = \text{diag}(1, 0, -1).$$

Minimal effective model choice

- A3: The static defect background is represented by

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad P_B = z_B z_B^\dagger, \quad \theta(-\xi) = \theta(\xi).$$

Background ansatz

- A4: The local orthonormal frame of $Q_B = (\mathbb{C}z_B)^\perp$ is

$$e_1 = \begin{pmatrix} -\sin \theta \\ 0 \\ \cos \theta \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Already established

A. The chosen anisotropy potential

We choose

$$V(P) = \frac{\lambda}{2} \text{Tr}([H, P]^2), \quad H = \text{diag}(1, 0, -1).$$

Since

$$[H, P]^\dagger = -[H, P],$$

one has

$$\text{Tr}([H, P]^2) \leq 0,$$

so it is convenient to rewrite the potential in the positive form

$$V(P) = -\frac{\lambda}{2} \text{Tr}([H, P]^2) = \lambda \left(\text{Tr}(H^2 P) - \text{Tr}(H P H P) \right).$$

Because P is rank one, this is minimized when P commutes with H , i.e. when the line selected by P aligns with one of the eigenvectors of H .

Thus $V(P)$ is a standard pinning anisotropy that stabilizes an internal orientation and permits defect profiles interpolating between locally preferred orientations.

B. Tangent fluctuation of the projector

Let

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0.$$

Expand the tangent field in the Q_B -frame:

$$\eta = \chi^1 e_1 + \chi^2 e_2.$$

We compute the second variation of V with respect to χ^1, χ^2 .

A convenient identity for the chosen potential is

$$V(P) = \lambda \|Q_H P\|^2$$

up to an irrelevant constant, but it is simplest to compute directly from the commutator.

Define

$$\delta C := [H, \delta P].$$

Then the quadratic variation is

$$\delta^2 V = -\frac{\lambda}{2} \text{Tr}((\delta C)^2) = \frac{\lambda}{2} \text{Tr}(\delta C^\dagger \delta C),$$

since δC is anti-Hermitian when δP is Hermitian.

Thus the Hessian on tangent fluctuations is determined by the norm of $[H, \delta P]$.

C. Channel e_1

Take first

$$\eta = \chi^1 e_1.$$

Since

$$z_B = \begin{pmatrix} \cos \theta \\ 0 \\ \sin \theta \end{pmatrix}, \quad e_1 = \begin{pmatrix} -\sin \theta \\ 0 \\ \cos \theta \end{pmatrix},$$

we find

$$\delta P_1 = \chi^1 e_1 z_B^\dagger + \chi^{1*} z_B e_1^\dagger.$$

For the Hessian it suffices to compute with real χ^1 and then extend sesquilinearly. Then

$$\delta P_1 = \chi^1 \begin{pmatrix} -2 \sin \theta \cos \theta & 0 & \cos^2 \theta - \sin^2 \theta \\ 0 & 0 & 0 \\ \cos^2 \theta - \sin^2 \theta & 0 & 2 \sin \theta \cos \theta \end{pmatrix}.$$

Using

$$H = \text{diag}(1, 0, -1),$$

we get

$$[H, \delta P_1] = 2\chi^1 \cos 2\theta \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

Hence

$$\frac{1}{2} \text{Tr}([H, \delta P_1]^\dagger [H, \delta P_1]) = 2|\chi^1|^2 \cos^2 2\theta.$$

Therefore the e_1 -channel contribution to the potential Hessian is

$$W_{11}(\xi) = \lambda \cos^2 2\theta(\xi).$$

D. Channel e_2

Now take

$$\eta = \chi^2 e_2, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Again, for real χ^2 ,

$$\delta P_2 = \chi^2 \begin{pmatrix} 0 & \cos \theta & 0 \\ \cos \theta & 0 & \sin \theta \\ 0 & \sin \theta & 0 \end{pmatrix}.$$

Then

$$[H, \delta P_2] = \chi^2 \begin{pmatrix} 0 & \cos \theta & 0 \\ -\cos \theta & 0 & \sin \theta \\ 0 & -\sin \theta & 0 \end{pmatrix}.$$

Therefore

$$\frac{1}{2} \text{Tr}([H, \delta P_2]^\dagger [H, \delta P_2]) = |\chi^2|^2 (\cos^2 \theta + \sin^2 \theta) = |\chi^2|^2.$$

Thus

$$W_{22}(\xi) = \lambda.$$

E. Off-diagonal channel coupling

Finally consider mixed fluctuations

$$\eta = \chi^1 e_1 + \chi^2 e_2.$$

Because the commutator norm is quadratic and the e_1 - and e_2 -channels transform under different symmetry sectors of the chosen background, the mixed term vanishes:

$$W_{12}(\xi) = W_{21}(\xi) = 0.$$

More explicitly, one may compute

$$\text{Tr}([H, \delta P_1]^\dagger [H, \delta P_2]) = 0.$$

a. Theorem 1 (Explicit anisotropy Hessian). For the minimal anisotropy potential

$$V(P) = \frac{\lambda}{2} \text{Tr}([H, P]^2), \quad H = \text{diag}(1, 0, -1),$$

and the defect background

$$z_B = \begin{pmatrix} \cos \theta \\ 0 \\ \sin \theta \end{pmatrix},$$

the tangent-space Hessian on Q_B in the frame (e_1, e_2) is

$$\boxed{\mathcal{W}_B(\xi) = \lambda \begin{pmatrix} \cos^2 2\theta(\xi) & 0 \\ 0 & 1 \end{pmatrix}.}$$

b. Proof. Combine the channel computations above. $\square$

F. Full static channel operator

From the previous Hessian calculation of the kinetic sector, the background-gradient contribution was

$$-\rho \theta'(\xi)^2 \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

Adding the potential Hessian yields

$$\mathcal{U}_B(\xi) = \mathcal{W}_B(\xi) - \rho \theta'(\xi)^2 \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

Therefore

$$\boxed{\mathcal{U}_B(\xi) = \begin{pmatrix} \lambda \cos^2 2\theta(\xi) - 2\rho \theta'(\xi)^2 & 0 \\ 0 & \lambda - \rho \theta'(\xi)^2 \end{pmatrix}.}$$

Thus the full local static Hessian becomes

$$\boxed{\mathcal{H}_B^{\text{stat}} = -\rho \partial_\xi^2 + \begin{pmatrix} \lambda \cos^2 2\theta(\xi) - 2\rho \theta'(\xi)^2 & 0 \\ 0 & \lambda - \rho \theta'(\xi)^2 \end{pmatrix}.}$$

a. Theorem 2 (Explicit channel-decoupled static Hessian). In the minimal anisotropy model, the local static defect Hessian is channel-diagonal and consists of the two scalar Schrödinger operators

$$H_1 = -\rho \partial_\xi^2 + \lambda \cos^2 2\theta(\xi) - 2\rho \theta'(\xi)^2,$$

$$H_2 = -\rho \partial_\xi^2 + \lambda - \rho \theta'(\xi)^2.$$

b. Proof. Immediate from the explicit formula for $\mathcal{U}_B$. $\square$

G. Reflection symmetry and parity channels

If $\theta(\xi)$ is even, then both

$$\cos^2 2\theta(\xi) \quad \text{and} \quad \theta'(\xi)^2$$

are even. Therefore both channel potentials are even, so each channel admits a parity decomposition into even and odd transverse modes.

a. Corollary 3. The explicit minimal anisotropy model produces parity-even and parity-odd transverse channels in both internal sectors.

This is stronger than the earlier abstract statement, because here the static channel operators are fully explicit.

H. Worldvolume transport kernel

When restoring slow worldvolume dependence y^μ , the leading transport kernel is built from terms linear in the transverse background gradient. In the explicit background,

$$z'_B = \theta' e_1, \quad P'_B = z'_B z_B^\dagger + z_B z_B'^\dagger.$$

Since $\theta(\xi)$ is even,

$$\theta'(-\xi) = -\theta'(\xi), \quad P'_B(-\xi) = -P'_B(\xi).$$

Thus the leading transport kernel is reflection-odd:

$$\mathcal{V}^\mu(-\xi) = -\mathcal{V}^\mu(\xi).$$

a. Corollary 4. The explicit minimal anisotropy model satisfies the odd-kernel condition required by the projected overlap theorem.

I. How far this actually goes

At this point, the channel structure is no longer schematic:

1. the tangent fluctuation bundle is explicitly reduced to two scalar channels;
2. the channel potentials are written explicitly in terms of $\theta(\xi)$;
3. parity decomposition is automatic if θ is even;
4. the oddness of the transport kernel is automatic if it is built from the defect gradient.

What still remains is the explicit spectral analysis of the scalar channel operators H_1, H_2 .

In particular, to obtain the isolated four-mode sector one must still show that each of the two channel Hamiltonians has a low even mode and a low odd mode separated by a gap from higher modes.

J. Special solvable matching

If one further chooses $\theta(\xi)$ so that $\theta'(\xi)^2$ and $\cos^2 2\theta(\xi)$ reduce to combinations of $\text{sech}^2(\kappa\xi)$, then the two channel potentials can be brought close to the previously used solvable Pöschl–Teller forms. This is not automatic for a generic $\theta(\xi)$, but it shows how the earlier explicit solvable model sits inside the present more general Hessian framework.

a. Final honest statement. The exact TECT defect spectrum is still open, but the explicit local static Hessian and the odd transport kernel are now derived from a concrete minimal CP^2 projector effective model, not simply postulated.

LII. FINAL SUMMARY

For the minimal anisotropy potential

$$V(P) = \frac{\lambda}{2} \text{Tr}([H, P]^2), \quad H = \text{diag}(1, 0, -1),$$

and the reflection-symmetric defect background

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi),$$

the tangent-space Hessian on the rank-two bundle

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp$$

is explicitly computable.

In the orthonormal frame

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix},$$

the transverse connection vanishes,

$$\mathcal{A}_\xi = -i e^\dagger \partial_\xi e = 0,$$

and the potential Hessian is

$$\mathcal{W}_B(\xi) = \lambda \begin{pmatrix} \cos^2 2\theta(\xi) & 0 \\ 0 & 1 \end{pmatrix}.$$

Combining this with the previously derived gradient-induced kinetic contribution gives the explicit static channel operator

$$\mathcal{H}_B^{\text{stat}} = -\rho \partial_\xi^2 + \begin{pmatrix} \lambda \cos^2 2\theta(\xi) - 2\rho \theta'(\xi)^2 & 0 \\ 0 & \lambda - \rho \theta'(\xi)^2 \end{pmatrix}.$$

Thus the local projector Hessian is now explicitly reduced to two decoupled scalar Schrödinger channels. If $\theta(\xi)$ is even, both channel potentials are even and admit parity decomposition into even and odd transverse modes.

Moreover, any leading worldvolume transport kernel built linearly from the defect gradient

$$\partial_\xi P_B(\xi) \quad \text{or} \quad \theta'(\xi)$$

is automatically reflection-odd.

Therefore the local CP^2 projector Hessian now matches the previously constructed minimal graded defect operator in a fully explicit operator sense.

What remains open is the actual spectral analysis of the two scalar channels and the resulting nondegeneracy of the mixed first-order symbol in the full TECT setting.

LIII. EXPLICIT SPECTRAL ANALYSIS OF THE MINIMAL STATIC CHANNELS FOR

$$\theta(\xi) = \theta_0 \text{sech}(\kappa \xi)$$

Assumption Audit

- A1: We choose the simplest even localized defect profile

$$\theta(\xi) = \theta_0 \text{sech}(\kappa \xi), \quad \theta_0 > 0, \quad \kappa > 0.$$

Model choice

- A2: The static channel Hamiltonians derived previously are

$$H_1 = -\rho \partial_\xi^2 + \lambda \cos^2 2\theta(\xi) - 2\rho \theta'(\xi)^2,$$

$$H_2 = -\rho \partial_\xi^2 + \lambda - \rho \theta'(\xi)^2.$$

Already established

- A3: Bound-state existence may be tested by the variational principle relative to the continuum threshold λ .

Proved

A. Explicit channel potentials

For

$$\theta(\xi) = \theta_0 \operatorname{sech}(\kappa\xi),$$

we have

$$\theta'(\xi) = -\theta_0 \kappa \operatorname{sech}(\kappa\xi) \tanh(\kappa\xi),$$

hence

$$\theta'(\xi)^2 = \theta_0^2 \kappa^2 \operatorname{sech}^2(\kappa\xi) \tanh^2(\kappa\xi) = \theta_0^2 \kappa^2 \left(\operatorname{sech}^2(\kappa\xi) - \operatorname{sech}^4(\kappa\xi) \right).$$

Therefore the two channel potentials are

$$U_1(\xi) = \lambda \cos^2(2\theta_0 \operatorname{sech}(\kappa\xi)) - 2\rho \theta_0^2 \kappa^2 \left(\operatorname{sech}^2(\kappa\xi) - \operatorname{sech}^4(\kappa\xi) \right),$$

$$U_2(\xi) = \lambda - \rho \theta_0^2 \kappa^2 \left(\operatorname{sech}^2(\kappa\xi) - \operatorname{sech}^4(\kappa\xi) \right).$$

Both are even in ξ , so both channels decompose into even and odd parity sectors.

B. Small-amplitude expansion

For $\theta_0^2 \ll 1$,

$$\cos^2(2\theta) = 1 - 4\theta^2 + O(\theta^4),$$

hence

$$\cos^2(2\theta_0 \operatorname{sech}(\kappa\xi)) = 1 - 4\theta_0^2 \operatorname{sech}^2(\kappa\xi) + O(\theta_0^4).$$

Thus

$$U_1(\xi) = \lambda - \left(4\lambda\theta_0^2 + 2\rho\kappa^2\theta_0^2 \right) \operatorname{sech}^2(\kappa\xi) + 2\rho\kappa^2\theta_0^2 \operatorname{sech}^4(\kappa\xi) + O(\theta_0^4),$$

and

$$U_2(\xi) = \lambda - \rho\kappa^2\theta_0^2 \operatorname{sech}^2(\kappa\xi) + \rho\kappa^2\theta_0^2 \operatorname{sech}^4(\kappa\xi).$$

Thus H_1 is the deeper channel, while H_2 is shallower.

C. Continuum threshold

Since

$$\theta(\xi) \rightarrow 0, \quad \theta'(\xi) \rightarrow 0 \quad (\xi \rightarrow \pm\infty),$$

both channel potentials satisfy

$$U_1(\xi) \rightarrow \lambda, \quad U_2(\xi) \rightarrow \lambda.$$

Hence the continuum threshold is

$$E_{\text{cont}} = \lambda.$$

A localized bound state exists whenever one finds a normalized trial function ϕ with

$$\langle \phi, H_j \phi \rangle < \lambda.$$

D. Even trial state

Take the normalized even trial function

$$\phi_0(\xi) = \sqrt{\frac{\kappa}{2}} \operatorname{sech}(\kappa\xi).$$

The needed expectation values are

$$\int d\xi |\phi'_0|^2 = \frac{\kappa^2}{3},$$

$$\int d\xi \phi_0^2 \operatorname{sech}^2(\kappa\xi) = \frac{2}{3}, \quad \int d\xi \phi_0^2 \operatorname{sech}^4(\kappa\xi) = \frac{8}{15}.$$

E. Even variational bound for the first channel

Using the small-amplitude potential for H_1 ,

$$\langle \phi_0, H_1 \phi_0 \rangle \leq \lambda + \rho \frac{\kappa^2}{3} - \left(4\lambda\theta_0^2 + 2\rho\kappa^2\theta_0^2\right) \frac{2}{3} + \left(2\rho\kappa^2\theta_0^2\right) \frac{8}{15} + O(\theta_0^4).$$

Therefore

$$\boxed{\langle \phi_0, H_1 \phi_0 \rangle \leq \lambda + \frac{\rho\kappa^2}{3} - \frac{8}{3}\lambda\theta_0^2 - \frac{4}{15}\rho\kappa^2\theta_0^2 + O(\theta_0^4).}$$

Hence a sufficient condition for an even bound state in the first channel is

$$\frac{\rho\kappa^2}{3} < \frac{8}{3}\lambda\theta_0^2 + \frac{4}{15}\rho\kappa^2\theta_0^2 + O(\theta_0^4).$$

Ignoring higher orders, this becomes

$$\boxed{\theta_0^2 > \frac{5\rho\kappa^2}{4(10\lambda + \rho\kappa^2)}}.$$

a. Theorem 1 (Sufficient condition for an even bound state in H_1). In the small-amplitude regime, if

$$\theta_0^2 > \frac{5\rho\kappa^2}{4(10\lambda + \rho\kappa^2)},$$

then the first static channel H_1 has at least one even bound state.

b. Proof. By the variational principle, it suffices to exhibit a normalized even trial function with energy below the continuum threshold λ . The estimate above provides such a condition. $\square$

F. Even variational bound for the second channel

Similarly,

$$\langle \phi_0, H_2 \phi_0 \rangle \leq \lambda + \frac{\rho \kappa^2}{3} - (\rho \kappa^2 \theta_0^2) \frac{2}{3} + (\rho \kappa^2 \theta_0^2) \frac{8}{15}.$$

Hence

$$\boxed{\langle \phi_0, H_2 \phi_0 \rangle \leq \lambda + \frac{\rho \kappa^2}{3} - \frac{2}{15} \rho \kappa^2 \theta_0^2.}$$

Thus a sufficient condition for an even bound state in H_2 is

$$\theta_0^2 > \frac{5}{2}.$$

a. Corollary 2. The second channel is significantly shallower than the first in the small-amplitude regime. In particular, its even bound state requires a much stronger deformation:

$$\theta_0^2 > \frac{5}{2}$$

at this level of variational estimate.

This condition is not naturally compatible with a strict small-amplitude regime, so no robust general even bound-state conclusion follows for H_2 from this estimate alone.

G. Odd trial state

Take the normalized odd trial function

$$\phi_1(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \operatorname{sech}(\kappa\xi).$$

The required expectation values are

$$\begin{aligned} \int d\xi |\phi_1'|^2 &= \frac{7}{5} \kappa^2, \\ \int d\xi \phi_1^2 \operatorname{sech}^2(\kappa\xi) &= \frac{2}{5}, \quad \int d\xi \phi_1^2 \operatorname{sech}^4(\kappa\xi) = \frac{8}{35}. \end{aligned}$$

H. Odd variational bound for the first channel

Using the same small-amplitude approximation,

$$\langle \phi_1, H_1 \phi_1 \rangle \leq \lambda + \frac{7}{5} \rho \kappa^2 - \left(4\lambda \theta_0^2 + 2\rho \kappa^2 \theta_0^2 \right) \frac{2}{5} + \left(2\rho \kappa^2 \theta_0^2 \right) \frac{8}{35} + O(\theta_0^4).$$

Thus

$$\boxed{\langle \phi_1, H_1 \phi_1 \rangle \leq \lambda + \frac{7}{5} \rho \kappa^2 - \frac{8}{5} \lambda \theta_0^2 - \frac{12}{35} \rho \kappa^2 \theta_0^2 + O(\theta_0^4).}$$

Therefore a sufficient condition for an odd bound state in H_1 is

$$\boxed{\theta_0^2 > \frac{49\rho\kappa^2}{4(14\lambda + 3\rho\kappa^2)}}$$

to leading order.

a. Theorem 3 (Sufficient condition for an odd bound state in H_1). In the small-amplitude regime, if

$$\theta_0^2 > \frac{49\rho\kappa^2}{4(14\lambda + 3\rho\kappa^2)},$$

then the first channel H_1 has at least one odd bound state.

b. Proof. Again this follows from the variational estimate with the odd trial state. $\square$

I. Odd variational bound for the second channel

For the second channel,

$$\langle \phi_1, H_2 \phi_1 \rangle \leq \lambda + \frac{7}{5}\rho\kappa^2 - (\rho\kappa^2\theta_0^2)\frac{2}{5} + (\rho\kappa^2\theta_0^2)\frac{8}{35}.$$

So

$$\boxed{\langle \phi_1, H_2 \phi_1 \rangle \leq \lambda + \frac{7}{5}\rho\kappa^2 - \frac{6}{35}\rho\kappa^2\theta_0^2.}$$

A sufficient condition for an odd bound state would therefore require

$$\theta_0^2 > \frac{49}{6},$$

which is far outside the natural small-amplitude regime.

Thus the minimal model does not generically guarantee an odd bound state in H_2 .

J. Consequences for the four-mode sector

Recall that the desired 4-mode infrared sector requires, at minimum, a robust pair of low modes sufficient to populate the graded $2 + 2$ structure after including the internal doublet.

The variational analysis shows:

1. the first channel H_1 can support an even bound state under a mild depth condition;
2. the first channel can also support an odd bound state under a stronger condition;
3. the second channel is shallower and does not generically support the same parity pair in the small-amplitude regime.

Therefore the explicit minimal anisotropy model demonstrates realizability of localized low modes, but does not by itself prove a universal isolated 4-mode infrared sector.

a. Theorem 4 (Honest spectral status of the minimal model). The minimal anisotropy model with

$$\theta(\xi) = \theta_0 \operatorname{sech}(\kappa\xi)$$

provides explicit sufficient conditions for low bound states in the first channel, but does not generally guarantee a full isolated four-mode sector across both channels.

b. Proof. This follows from Theorems 1 and 3 together with the much weaker bounds in the second channel. $\square$

K. Interpretation

This is an important honest conclusion.

The minimal anisotropy model is already enough to prove that the Dirac mechanism is not empty: localized low modes and a nontrivial projected symbol can occur in an explicit controlled model.

However, it is not yet the strongest model for proving the isolated 4-mode sector.

To prove the latter, one likely needs at least one of the following:

1. a deeper or differently shaped defect profile $\theta(\xi)$;
2. a richer effective potential $V(P)$ that lowers the second channel;
3. a direct microscopic calculation of the projector Hessian beyond the minimal anisotropy model.

LIV. FINAL SUMMARY

For the explicit even defect profile

$$\theta(\xi) = \theta_0 \operatorname{sech}(\kappa\xi),$$

the two static channel Hamiltonians become

$$H_1 = -\rho \partial_\xi^2 + \lambda \cos^2(2\theta_0 \operatorname{sech}(\kappa\xi)) - 2\rho\theta_0^2\kappa^2(\operatorname{sech}^2 - \operatorname{sech}^4),$$

$$H_2 = -\rho \partial_\xi^2 + \lambda - \rho\theta_0^2\kappa^2(\operatorname{sech}^2 - \operatorname{sech}^4).$$

In the small-amplitude regime,

$$H_1$$

is the deeper channel, while

$$H_2$$

is significantly shallower.

Using standard even and odd localized trial functions, one obtains sufficient conditions for bound states in H_1 . In particular:

$$\theta_0^2 > \frac{5\rho\kappa^2}{4(10\lambda + \rho\kappa^2)}$$

is sufficient for an even bound state in H_1 , and

$$\theta_0^2 > \frac{49\rho\kappa^2}{4(14\lambda + 3\rho\kappa^2)}$$

is sufficient for an odd bound state in H_1 , at leading order.

By contrast, the corresponding sufficient conditions in H_2 are much stronger and do not generically hold in the same regime.

Therefore the minimal anisotropy model explicitly demonstrates the existence of localized low modes, but does not yet prove a universal isolated four-mode infrared sector.

The honest conclusion is:

explicit low-mode realizability: proved;

full isolated 4-mode sector in the minimal model: not yet proved.

This sharply identifies what must be improved next: either the defect profile, the effective potential, or the explicit microscopic Hessian.

LIV. IMPROVED EFFECTIVE POTENTIAL DESIGN FOR REALIZING THE FOUR-MODE INFRARED SECTOR

Motivation

The minimal anisotropy model based on a single pinning term

$$V_H(P) = -\frac{\lambda}{2} \operatorname{Tr}([H, P]^2), \quad H = \operatorname{diag}(1, 0, -1),$$

was sufficient to generate explicit low localized modes in the first static channel, but it did not generically provide a comparably deep second channel. As a result, the existence of a full isolated four-mode infrared sector was not guaranteed.

To improve this, one should add a second anisotropy term that lowers the e_2 -channel as well.

Improved model

We consider the effective potential

$$V(P) = V_H(P) + V_K(P),$$

with

$$V_H(P) = -\frac{\lambda}{2} \text{Tr}([H, P]^2), \quad H = \text{diag}(1, 0, -1),$$

and

$$V_K(P) = -\frac{\mu}{2} \text{Tr}([K, P]^2), \quad K = \text{diag}(1, -1, -1), \quad \lambda, \mu > 0.$$

The defect background is again

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi),$$

with tangent frame

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

A. General diagonal-anisotropy Hessian formula

Let

$$A = \text{diag}(a, b, c)$$

be any diagonal Hermitian matrix, and consider

$$V_A(P) = -\frac{\Lambda_A}{2} \text{Tr}([A, P]^2).$$

Then, for tangent fluctuations

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad \eta = \chi^1 e_1 + \chi^2 e_2,$$

the quadratic Hessian on the (e_1, e_2) -channels is diagonal and given by

$$W_{11}^{(A)}(\xi) = \frac{\Lambda_A}{4} (a - c)^2 \cos^2 2\theta(\xi),$$

$$W_{22}^{(A)}(\xi) = \frac{\Lambda_A}{2} \left((a - b)^2 \cos^2 \theta(\xi) + (c - b)^2 \sin^2 \theta(\xi) \right),$$

with

$$W_{12}^{(A)} = W_{21}^{(A)} = 0.$$

a. Proof sketch. The proof is a direct channel computation from

$$\delta^2 V_A = \frac{\Lambda_A}{2} \text{Tr}([A, \delta P]^\dagger [A, \delta P]),$$

using the explicit background and frame. The e_1 -channel fluctuation contains only the $1 \leftrightarrow 3$ transition, whereas the e_2 -channel fluctuation contains the $1 \leftrightarrow 2$ and $2 \leftrightarrow 3$ transitions. Orthogonality kills the mixed term. $\square$

B. Application to the two anisotropies

For

$$H = \text{diag}(1, 0, -1),$$

one recovers the previously obtained result:

$$\mathcal{W}_B^{(H)}(\xi) = \lambda \begin{pmatrix} \cos^2 2\theta(\xi) & 0 \\ 0 & 1 \end{pmatrix}.$$

For

$$K = \text{diag}(1, -1, -1),$$

the same formula gives

$$\mathcal{W}_B^{(K)}(\xi) = \mu \begin{pmatrix} \cos^2 2\theta(\xi) & 0 \\ 0 & 2 \cos^2 \theta(\xi) \end{pmatrix}.$$

Therefore the full potential Hessian is

$$\mathcal{W}_B(\xi) = \begin{pmatrix} (\lambda + \mu) \cos^2 2\theta(\xi) & 0 \\ 0 & \lambda + 2\mu \cos^2 \theta(\xi) \end{pmatrix}.$$

C. Static channel operator

From the previously established kinetic-gradient contribution,

$$-\rho \theta'(\xi)^2 \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix},$$

the full static Hessian becomes

$$\mathcal{H}_B^{\text{stat}} = -\rho \partial_\xi^2 + \begin{pmatrix} (\lambda + \mu) \cos^2 2\theta(\xi) - 2\rho \theta'(\xi)^2 & 0 \\ 0 & \lambda + 2\mu \cos^2 \theta(\xi) - \rho \theta'(\xi)^2 \end{pmatrix}.$$

Hence the two scalar channels are

$$H_1 = -\rho \partial_\xi^2 + (\lambda + \mu) \cos^2 2\theta(\xi) - 2\rho \theta'(\xi)^2,$$

$$H_2 = -\rho \partial_\xi^2 + \lambda + 2\mu \cos^2 \theta(\xi) - \rho \theta'(\xi)^2.$$

a. Interpretation. The new term lowers the second channel relative to its asymptotic threshold, because

$$\cos^2 \theta(\xi) < 1$$

near the defect core whenever $\theta(\xi) \neq 0$.

D. Continuum thresholds

Since

$$\theta(\xi) \rightarrow 0, \quad \theta'(\xi) \rightarrow 0 \quad (|\xi| \rightarrow \infty),$$

the continuum thresholds are now

$$E_{1,\infty} = \lambda + \mu, \quad E_{2,\infty} = \lambda + 2\mu.$$

Thus the relative channel wells are

$$U_1(\xi) - E_{1,\infty} = -(\lambda + \mu) \sin^2 2\theta(\xi) - 2\rho \theta'(\xi)^2,$$

$$U_2(\xi) - E_{2,\infty} = -2\mu \sin^2 \theta(\xi) - \rho \theta'(\xi)^2.$$

This is the key improvement: both channels now have genuine negative wells relative to their own continuum thresholds.

E. Choice of defect profile

Take again the explicit even defect profile

$$\theta(\xi) = \theta_0 \operatorname{sech}(\kappa\xi).$$

Then

$$\theta'(\xi)^2 = \theta_0^2 \kappa^2 \left(\operatorname{sech}^2(\kappa\xi) - \operatorname{sech}^4(\kappa\xi) \right).$$

In the small-amplitude regime $\theta_0^2 \ll 1$,

$$\sin^2 \theta(\xi) = \theta_0^2 \operatorname{sech}^2(\kappa\xi) + O(\theta_0^4),$$

$$\sin^2 2\theta(\xi) = 4\theta_0^2 \operatorname{sech}^2(\kappa\xi) + O(\theta_0^4).$$

Hence

$$H_1 - (\lambda + \mu) = -\rho \partial_\xi^2 - \left(4(\lambda + \mu)\theta_0^2 + 2\rho\kappa^2\theta_0^2 \right) \operatorname{sech}^2 + 2\rho\kappa^2\theta_0^2 \operatorname{sech}^4 + O(\theta_0^4),$$

and

$$H_2 - (\lambda + 2\mu) = -\rho \partial_\xi^2 - \left(2\mu\theta_0^2 + \rho\kappa^2\theta_0^2 \right) \operatorname{sech}^2 + \rho\kappa^2\theta_0^2 \operatorname{sech}^4 + O(\theta_0^4).$$

F. Even bound-state estimate for H_2

Use the normalized even trial function

$$\phi_0(\xi) = \sqrt{\frac{\kappa}{2}} \operatorname{sech}(\kappa\xi).$$

The standard integrals are

$$\int |\phi_0'|^2 d\xi = \frac{\kappa^2}{3}, \quad \int \phi_0^2 \operatorname{sech}^2(\kappa\xi) d\xi = \frac{2}{3}, \quad \int \phi_0^2 \operatorname{sech}^4(\kappa\xi) d\xi = \frac{8}{15}.$$

Therefore

$$\langle \phi_0, H_2 \phi_0 \rangle - (\lambda + 2\mu) \leq \frac{\rho\kappa^2}{3} - \left(2\mu\theta_0^2 + \rho\kappa^2\theta_0^2 \right) \frac{2}{3} + (\rho\kappa^2\theta_0^2) \frac{8}{15} + O(\theta_0^4).$$

Thus

$$\boxed{\langle \phi_0, H_2 \phi_0 \rangle - (\lambda + 2\mu) \leq \frac{\rho\kappa^2}{3} - \frac{2}{3}\mu\theta_0^2 - \frac{2}{15}\rho\kappa^2\theta_0^2 + O(\theta_0^4).}$$

A sufficient condition for an even bound state in H_2 is therefore

$$\boxed{\theta_0^2 > \frac{5\rho\kappa^2}{2(5\mu + \rho\kappa^2)}}$$

to leading order.

a. Theorem 1. In the improved model, the second channel H_2 admits an even bound state whenever

$$\theta_0^2 > \frac{5\rho\kappa^2}{2(5\mu + \rho\kappa^2)}$$

at leading order in the small-amplitude regime.

G. Odd bound-state estimate for H_2

Now use the normalized odd trial function

$$\phi_1(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \operatorname{sech}(\kappa\xi).$$

The needed integrals are

$$\int |\phi_1'|^2 d\xi = \frac{7}{5}\kappa^2, \quad \int \phi_1^2 \operatorname{sech}^2(\kappa\xi) d\xi = \frac{2}{5}, \quad \int \phi_1^2 \operatorname{sech}^4(\kappa\xi) d\xi = \frac{8}{35}.$$

Hence

$$\langle \phi_1, H_2 \phi_1 \rangle - (\lambda + 2\mu) \leq \frac{7}{5}\rho\kappa^2 - \left(2\mu\theta_0^2 + \rho\kappa^2\theta_0^2\right)\frac{2}{5} + (\rho\kappa^2\theta_0^2)\frac{8}{35} + O(\theta_0^4).$$

Therefore

$$\boxed{\langle \phi_1, H_2 \phi_1 \rangle - (\lambda + 2\mu) \leq \frac{7}{5}\rho\kappa^2 - \frac{4}{5}\mu\theta_0^2 - \frac{6}{35}\rho\kappa^2\theta_0^2 + O(\theta_0^4).}$$

A sufficient condition for an odd bound state in H_2 is therefore

$$\boxed{\theta_0^2 > \frac{49\rho\kappa^2}{2(14\mu + 3\rho\kappa^2)}}$$

to leading order.

a. Theorem 2. In the improved model, the second channel H_2 admits an odd bound state whenever

$$\theta_0^2 > \frac{49\rho\kappa^2}{2(14\mu + 3\rho\kappa^2)}$$

at leading order in the small-amplitude regime.

H. Comparison with the first channel

The first channel satisfies

$$H_1 - (\lambda + \mu) = -\rho \partial_\xi^2 - \left(4(\lambda + \mu)\theta_0^2 + 2\rho\kappa^2\theta_0^2\right) \operatorname{sech}^2 + 2\rho\kappa^2\theta_0^2 \operatorname{sech}^4 + O(\theta_0^4).$$

This is deeper than in the minimal one-anisotropy model, so the corresponding bound-state conditions are strengthened relative to the earlier analysis.

In particular, the previous sufficient conditions for even/odd bound states in H_1 remain true after replacing $\lambda \mapsto \lambda + \mu$, so H_1 is at least as favorable as before.

I. Four-mode regime

The desired isolated four-mode infrared sector is naturally supported when both channels possess at least one even and one odd low bound state below the continuum threshold and when the next excited levels are separated by a gap.

The improved model now supplies explicit sufficient conditions for the second channel as well. Hence there exists a parameter regime in which the realization of a four-mode sector is no longer merely plausible, but supported by simultaneous variational bounds.

a. Theorem 3 (Improved realizability of the four-mode sector). In the improved two-anisotropy model, there exists a nonempty parameter regime

$$(\lambda, \mu, \rho, \kappa, \theta_0)$$

for which both static channels H_1 and H_2 admit low even and odd bound states. In this regime, the existence of a four-mode infrared sector is strongly supported at the effective-model level.

b. Proof. Theorem 1 and Theorem 2 provide explicit sufficient conditions for the second channel. The first channel is deeper than in the minimal model and therefore also admits corresponding low bound states in an overlapping region of parameter space. $\square$

J. Honest limitation

The statement above is still an effective-model statement, not yet a full microscopic TECT theorem. What remains to be shown explicitly is:

1. the actual bound-state counting and spectral gap beyond variational sufficiency;
2. the overlap of the two channel conditions in the exact spectrum;
3. the nondegeneracy of the mixed first-order symbol in that same parameter region;
4. the microscopic derivation of the improved anisotropy structure from the full TECT operator.

Nevertheless, the situation is now much stronger than before: the obstruction seen in the minimal single-anisotropy model has been explicitly repaired by a controlled improvement of the effective potential.

LVI. FINAL SUMMARY

The failure of the minimal single-anisotropy model to robustly produce the full four-mode infrared sector came from the fact that the second static channel H_2 was too shallow.

This can be repaired by adding a second anisotropy term

$$V_K(P) = -\frac{\mu}{2} \text{Tr}([K, P]^2), \quad K = \text{diag}(1, -1, -1),$$

to the original pinning potential.

The resulting potential Hessian becomes

$$\mathcal{W}_B(\xi) = \begin{pmatrix} (\lambda + \mu) \cos^2 2\theta(\xi) & 0 \\ 0 & \lambda + 2\mu \cos^2 \theta(\xi) \end{pmatrix},$$

and the static channel operators are

$$H_1 = -\rho \partial_\xi^2 + (\lambda + \mu) \cos^2 2\theta(\xi) - 2\rho \theta'(\xi)^2,$$

$$H_2 = -\rho \partial_\xi^2 + \lambda + 2\mu \cos^2 \theta(\xi) - \rho \theta'(\xi)^2.$$

Relative to the continuum thresholds, both channels now contain genuine attractive wells:

$$H_1 - (\lambda + \mu) \sim -(\lambda + \mu) \sin^2 2\theta - 2\rho \theta'^2,$$

$$H_2 - (\lambda + 2\mu) \sim -2\mu \sin^2 \theta - \rho \theta'^2.$$

For the explicit defect profile

$$\theta(\xi) = \theta_0 \text{sech}(\kappa \xi),$$

variational estimates give explicit sufficient conditions for even and odd bound states in the second channel:

$$\theta_0^2 > \frac{5\rho\kappa^2}{2(5\mu + \rho\kappa^2)}$$

for an even bound state, and

$$\theta_0^2 > \frac{49\rho\kappa^2}{2(14\mu + 3\rho\kappa^2)}$$

for an odd bound state, at leading order in the small-amplitude regime.

Therefore the improved two-anisotropy effective model supports a genuine parameter regime in which the realization of the four-mode infrared sector is substantially strengthened.

The honest status is:

effective-model realizability of the four-mode sector: strongly supported;

full microscopic TECT spectral proof: still open.

LVII. RANK OF THE MIXED FIRST-ORDER SYMBOL IN THE IMPROVED EFFECTIVE MODEL

Assumption Audit

- A1: The improved static model admits low even/odd modes in both channels,

$$f_{1,+}, f_{1,-}, f_{2,+}, f_{2,-}.$$

Strongly supported at the effective level

- A2: The low-mode basis is

$$u_1^\pm(\xi) = f_{1,\pm}(\xi)e_1(\xi), \quad u_2^\pm(\xi) = f_{2,\pm}(\xi)e_2(\xi).$$

Valid once A1 holds

- A3: The leading transport operator is of first order,

$$\mathcal{L}_1 = -i \mathcal{V}^\mu(\xi) \partial_{y^\mu},$$

with $\mathcal{V}^\mu(-\xi) = -\mathcal{V}^\mu(\xi)$.

Structurally established

- A4: The mixed symbol is defined by projected overlaps.

Proved

- A5: Full-rank Weyl/Dirac reduction is equivalent to nondegeneracy of the traceless part of the mixed 2×2 symbol.

Proved

A. Low-mode basis in the improved model

Assume the two static scalar channels H_1, H_2 admit low even/odd states

$$f_{1,+}(\xi), f_{1,-}(\xi), f_{2,+}(\xi), f_{2,-}(\xi).$$

Using the explicit orthonormal frame of Q_B ,

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix},$$

define the low-energy basis

$$\begin{aligned} u_1^+ &= f_{1,+}e_1, & u_2^+ &= f_{2,+}e_2, \\ u_1^- &= f_{1,-}e_1, & u_2^- &= f_{2,-}e_2. \end{aligned}$$

This gives the $2 + 2$ decomposition

$$\mathcal{K}_+ = \text{span}\{u_1^+, u_2^+\}, \quad \mathcal{K}_- = \text{span}\{u_1^-, u_2^-\}.$$

B. General mixed symbol

Let the leading transport operator be

$$\mathcal{L}_1 = -i \mathcal{V}^\mu(\xi) \partial_{y^\mu}.$$

The projected mixed block is

$$(M^\mu)_{ab} = \int d\xi (u_a^+)^\dagger \mathcal{V}^\mu(\xi) u_b^-, \quad a, b = 1, 2.$$

In the chosen basis this becomes

$$M^\mu = \begin{pmatrix} I_1^\mu & J^\mu \\ \frac{1}{J^\mu} & I_2^\mu \end{pmatrix},$$

where

$$I_1^\mu = \int d\xi f_{1,+}(\xi) f_{1,-}(\xi) e_1^\dagger(\xi) \mathcal{V}^\mu(\xi) e_1(\xi),$$

$$I_2^\mu = \int d\xi f_{2,+}(\xi) f_{2,-}(\xi) e_2^\dagger(\xi) \mathcal{V}^\mu(\xi) e_2(\xi),$$

and

$$J^\mu = \int d\xi f_{1,+}(\xi) f_{2,-}(\xi) e_1^\dagger(\xi) \mathcal{V}^\mu(\xi) e_2(\xi).$$

a. Theorem 1 (General form of the mixed symbol). The mixed first-order symbol in the improved model is always of the Hermitian form

$$M^\mu = a^\mu \mathbf{1} + b_i^\mu \sigma^i,$$

with

$$a^\mu = \frac{I_1^\mu + I_2^\mu}{2}, \quad b_3^\mu = \frac{I_1^\mu - I_2^\mu}{2}, \quad b_1^\mu = \Re J^\mu, \quad b_2^\mu = -\Im J^\mu.$$

b. Proof. Every Hermitian 2×2 matrix admits a unique decomposition in the basis

$$\mathbf{1}, \sigma^1, \sigma^2, \sigma^3.$$

Substituting the entries gives the stated coefficients. $\square$

C. Criterion for nondegeneracy

The full mixed symbol is

$$A_1(p) = p_\mu M^\mu.$$

Its scalar part $a^\mu \mathbf{1}$ is irrelevant for chirality and may be removed by energy shift or interpreted as cone tilt. The decisive object is the traceless part

$$\tilde{A}_1(p) = p_\mu b_i^\mu \sigma^i.$$

Define the real $3 \times d$ matrix

$$B_{i\mu} := b_i^\mu.$$

a. Theorem 2 (Rank criterion for the mixed symbol). Assume there are three independent propagating directions p_i , $i = 1, 2, 3$. Then the traceless mixed symbol is nondegenerate if and only if the three vectors

$$b^1, b^2, b^3 \in \mathbb{R}^3$$

span $\mathbb{R}^3$, equivalently

$$\det(b_i^j) \neq 0$$

after choosing three directions.

In that case

$$\tilde{A}_1(p)$$

is unitarily equivalent, after linear momentum redefinition, to

$$v_i p_i \sigma^i,$$

and the full projected principal operator is Dirac.

b. Proof. This is exactly the previously established classification of nondegenerate 2×2 first-order symbols. Full rank of the coefficient map $p \mapsto b_i^\mu p_\mu$ is equivalent to Weyl form of the traceless part. $\square$

D. Explicit effective realization of full rank

We now show that the improved model admits an explicit realization with full-rank mixed symbol.

Assume, for simplicity, a near-symmetric low-mode regime in which the overlap magnitudes in the two channels are comparable:

$$f_{1,+} \approx f_{2,+} \approx \phi_0, \quad f_{1,-} \approx f_{2,-} \approx \phi_1,$$

where ϕ_0 is even and ϕ_1 is odd. This assumption is natural in the regime where both channels are simultaneously shallow bound-state channels of similar width.

Choose three odd transport kernels

$$\mathcal{V}^1(\xi) = g(\xi)\sigma^1, \quad \mathcal{V}^2(\xi) = g(\xi)\sigma^2, \quad \mathcal{V}^3(\xi) = g(\xi)\sigma^3,$$

acting in the channel basis (e_1, e_2) , with

$$g(-\xi) = -g(\xi).$$

Then

$$M^i = \int d\xi \phi_0(\xi) \phi_1(\xi) g(\xi) \sigma^i = I_g \sigma^i,$$

where

$$I_g := \int d\xi \phi_0(\xi) \phi_1(\xi) g(\xi).$$

a. Theorem 3 (Explicit Pauli-triplet realization). If

$$I_g \neq 0,$$

then the mixed symbol takes the form

$$A_1(p) = p_i M^i = I_g p_i \sigma^i,$$

which is nondegenerate for generic $p \neq 0$.

b. Proof. Direct substitution gives

$$\det A_1(p) = -I_g^2(p_1^2 + p_2^2 + p_3^2),$$

which is nonzero for generic $p \neq 0$. $\square$

E. Explicit overlap value

Using the previously employed trial profiles

$$\phi_0(\xi) = \sqrt{\frac{\kappa}{2}} \operatorname{sech}(\kappa\xi), \quad \phi_1(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \operatorname{sech}(\kappa\xi),$$

and taking

$$g(\xi) = g_0 \tanh(\kappa\xi),$$

one finds

$$I_g = \int d\xi \phi_0(\xi) \phi_1(\xi) g(\xi) = \frac{g_0}{\sqrt{3}}.$$

Therefore

$$M^i = \frac{g_0}{\sqrt{3}} \sigma^i, \quad A_1(p) = \frac{g_0}{\sqrt{3}} p_i \sigma^i.$$

a. Corollary 4. The improved effective model admits an explicit nondegenerate mixed symbol

$$A_1(p) = \frac{g_0}{\sqrt{3}} p_i \sigma^i$$

whenever $g_0 \neq 0$.

F. General non-factorized effective criterion

The explicit Pauli-triplet construction above is stronger than necessary. In general, one only needs

$$b^1, b^2, b^3$$

to be linearly independent. Off-diagonal transport kernels are therefore not required to be exactly the Pauli matrices; it suffices that their projected overlaps generate the full traceless Hermitian algebra.

a. Theorem 5 (General effective full-rank criterion). The improved effective model yields a full-rank mixed symbol whenever the projected odd transport kernels generate three linearly independent traceless Hermitian matrices on the channel doublet.

b. Proof. This is an immediate consequence of Theorem 2. $\square$

G. What is actually proved

At the effective-model level, the situation is now strong:

1. the improved static potential supports a parameter regime with low bound states in both channels;
2. the mixed symbol always has the general Hermitian 2×2 form;
3. there exists an explicit odd-kernel realization in which the symbol is a full Pauli triple;
4. therefore the Dirac principal operator is explicitly realizable in the improved effective model.

H. What is still open

The remaining unresolved issues are now sharply localized to the microscopic theory:

1. proving the exact isolated four-mode sector in the full TECT operator rather than only in the improved effective model;
2. deriving the actual transport kernel $\mathcal{V}^\mu(\xi)$ from the microscopic TECT linearization;
3. proving that its projected overlap matrices are full rank in the actual theory.

LVIII. FINAL SUMMARY

In the improved two-anisotropy effective model, the low-energy basis may be chosen as

$$u_1^\pm = f_{1,\pm} e_1, \quad u_2^\pm = f_{2,\pm} e_2.$$

The mixed first-order projected symbol then has the general Hermitian form

$$M^\mu = \begin{pmatrix} I_1^\mu & J^\mu \\ \bar{J}^\mu & I_2^\mu \end{pmatrix} = a^\mu \mathbf{1} + b_i^\mu \sigma^i.$$

The scalar part $a^\mu \mathbf{1}$ is irrelevant for chirality, while the traceless part

$$p_\mu b_i^\mu \sigma^i$$

determines whether the symbol is Weyl/Dirac.

The symbol is nondegenerate precisely when the coefficient vectors $b^\mu \in \mathbb{R}^3$ span $\mathbb{R}^3$.

Moreover, the improved effective model admits an explicit realization in which

$$M^1 = I_g \sigma^1, \quad M^2 = I_g \sigma^2, \quad M^3 = I_g \sigma^3,$$

so that

$$A_1(p) = I_g p_i \sigma^i$$

is manifestly full rank.

With the concrete odd overlap profile

$$g(\xi) = g_0 \tanh(\kappa \xi),$$

and standard even/odd trial profiles, one obtains

$$I_g = \frac{g_0}{\sqrt{3}} \neq 0,$$

hence

$$A_1(p) = \frac{g_0}{\sqrt{3}} p_i \sigma^i.$$

Therefore the honest effective-level conclusion is:

the improved effective model supports not only the four-mode sector, but also an explicit full-rank mixed symbol realizing the

What remains open is the microscopic derivation of the same structure from the full TECT linearized operator.

LIX. EMERGENCE OF THE DIRAC OPERATOR: MINIMAL THEOREM AND IMPROVED EFFECTIVE REALIZATION

A. Scope and logical status

The goal of this section is to isolate the strongest mathematically honest statement currently available for the emergence of the Dirac operator in the TECT defect/valley framework.

At the present stage, one should *not* claim that the $\mathbb{CP}^2$ geometry alone automatically implies a Dirac equation. That implication is too strong and is not proved.

What can be established rigorously is the following two-layer structure:

1. a minimal reduced theorem, showing that a graded 4-mode infrared sector with a nondegenerate mixed first-order symbol necessarily yields the Dirac principal operator;
2. an improved effective realization, showing that these hypotheses are not empty but are explicitly realizable in a concrete CP^2 -projector defect model with improved anisotropy.

B. Minimal reduced Dirac-emergence theorem

Let $\mathcal{L}(p)$ denote the linearized operator near the infrared point $p = 0$, and let $\Pi_{\mathcal{K}}$ denote the projector onto an isolated infrared sector $\mathcal{K} \subset \mathcal{H}$.

Assume:

(R1) there exists an isolated complex 4-dimensional infrared sector

$$\mathcal{K}, \quad \dim_{\mathbb{C}} \mathcal{K} = 4;$$

(R2) $\mathcal{K}$ carries an involution

$$J : \mathcal{K} \rightarrow \mathcal{K}, \quad J^\dagger = J, \quad J^2 = \mathbf{1},$$

with splitting

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \mathcal{K}_\pm = \ker(J \mp \mathbf{1}), \quad \dim_{\mathbb{C}} \mathcal{K}_\pm = 2;$$

(R3) after subtraction of the common reference low-energy eigenvalue, the leading projected perturbation is odd under J :

$$\mathcal{D}_{\text{eff}}(p) = \Pi_{\mathcal{K}} \mathcal{L}(p) \Pi_{\mathcal{K}} = \mathcal{D}_1(p) + O(|p|^2), \quad J \mathcal{D}_1(p) J^{-1} = -\mathcal{D}_1(p);$$

(R4) in a basis adapted to $\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-$, the leading operator takes the form

$$\mathcal{D}_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_\mu M^\mu,$$

and the symbol $A_1(p)$ is nondegenerate for generic small $p \neq 0$.

a. Theorem 1 (Minimal Reduced Dirac Emergence Theorem). Under assumptions (R1)–(R4), the principal infrared operator is unitarily equivalent, after a linear change of momentum coordinates, to the massless Dirac operator:

$$\mathcal{D}_1(p) \sim p_i \alpha^i, \quad \{\alpha^i, \alpha^j\} = 2\delta^{ij} \mathbf{1}_4.$$

b. Proof. Let

$$\Gamma := \Pi_+ - \Pi_-.$$

Then

$$\Gamma^\dagger = \Gamma, \quad \Gamma^2 = \mathbf{1},$$

and the oddness condition implies

$$\Gamma \mathcal{D}_1 + \mathcal{D}_1 \Gamma = 0.$$

In the graded basis,

$$\mathcal{D}_1 = \begin{pmatrix} D_{++} & D_{+-} \\ D_{-+} & D_{--} \end{pmatrix},$$

and therefore

$$\Gamma \mathcal{D}_1 + \mathcal{D}_1 \Gamma = \begin{pmatrix} 2D_{++} & 0 \\ 0 & -2D_{--} \end{pmatrix}.$$

Hence

$$D_{++} = D_{--} = 0,$$

so

$$\mathcal{D}_1 = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}.$$

Every 2×2 complex linear symbol admits a Pauli decomposition

$$A_1(p) = a_0(p)\mathbf{1} + a_i(p)\sigma^i.$$

Nondegeneracy implies that, after unitary basis changes on the two graded subspaces and an invertible linear momentum redefinition, the symbol can be brought to canonical Weyl form

$$A_1(p) \sim p_i \sigma^i.$$

Substituting into the full 4×4 operator gives

$$\mathcal{D}_1(p) \sim \begin{pmatrix} 0 & p_i \sigma^i \\ p_i \sigma^i & 0 \end{pmatrix} = p_i \alpha^i.$$

This is precisely the massless Dirac principal operator. $\square$

C. Projector-Hessian origin of the low-energy channels

We now explain how the structure above is realized in the effective $\mathbb{CP}^2$ -projector defect sector. Let

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger$$

be a static reflection-symmetric defect background depending on one transverse coordinate ξ , with

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi).$$

Projector fluctuations may be written as

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0,$$

so the physical fluctuation field takes values in

$$Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp, \quad \dim_{\mathbb{C}} Q_B = 2.$$

An explicit orthonormal frame is

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

In this frame the transverse Q_B -connection vanishes:

$$\mathcal{A}_\xi = -i e^\dagger \partial_\xi e = 0.$$

The local quadratic projector Hessian then reduces to a matrix Schrödinger operator on the channel coefficients:

$$\mathcal{H}_B^{\text{stat}} = -\rho \partial_\xi^2 + \mathcal{U}_B(\xi).$$

D. Improved effective anisotropy model

To support low modes in both channels, we use the improved effective potential

$$V(P) = V_H(P) + V_K(P),$$

with

$$V_H(P) = -\frac{\lambda}{2} \text{Tr}([H, P]^2), \quad H = \text{diag}(1, 0, -1),$$

and

$$V_K(P) = -\frac{\mu}{2} \text{Tr}([K, P]^2), \quad K = \text{diag}(1, -1, -1), \quad \lambda, \mu > 0.$$

For the background above, the tangent-space potential Hessian in the (e_1, e_2) channel basis is

$$\mathcal{W}_B(\xi) = \begin{pmatrix} (\lambda + \mu) \cos^2 2\theta(\xi) & 0 \\ 0 & \lambda + 2\mu \cos^2 \theta(\xi) \end{pmatrix}.$$

Combining this with the background-gradient contribution from the kinetic Hessian,

$$-\rho \theta'(\xi)^2 \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix},$$

one obtains the explicit static channel operator

$$\mathcal{H}_B^{\text{stat}} = -\rho \partial_\xi^2 + \begin{pmatrix} (\lambda + \mu) \cos^2 2\theta(\xi) - 2\rho \theta'(\xi)^2 & 0 \\ 0 & \lambda + 2\mu \cos^2 \theta(\xi) - \rho \theta'(\xi)^2 \end{pmatrix}.$$

Thus the two scalar channels are

$$H_1 = -\rho \partial_\xi^2 + (\lambda + \mu) \cos^2 2\theta(\xi) - 2\rho \theta'(\xi)^2,$$

$$H_2 = -\rho \partial_\xi^2 + \lambda + 2\mu \cos^2 \theta(\xi) - \rho \theta'(\xi)^2.$$

E. Improved four-mode support

Take the even defect profile

$$\theta(\xi) = \theta_0 \text{sech}(\kappa \xi).$$

Then both channel potentials are even, hence admit parity decomposition into even and odd sectors.

Relative to the continuum thresholds

$$E_{1,\infty} = \lambda + \mu, \quad E_{2,\infty} = \lambda + 2\mu,$$

the effective wells are

$$H_1 - (\lambda + \mu) = -\rho \partial_\xi^2 - (\lambda + \mu) \sin^2 2\theta - 2\rho \theta'^2,$$

$$H_2 - (\lambda + 2\mu) = -\rho \partial_\xi^2 - 2\mu \sin^2 \theta - \rho \theta'^2.$$

Therefore both channels are genuinely attractive relative to their own thresholds.

Using standard even and odd trial functions,

$$\phi_0(\xi) = \sqrt{\frac{\kappa}{2}} \text{sech}(\kappa \xi), \quad \phi_1(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa \xi) \text{sech}(\kappa \xi),$$

one obtains explicit sufficient conditions for low even/odd bound states in the second channel:

$$\theta_0^2 > \frac{5\rho\kappa^2}{2(5\mu + \rho\kappa^2)} \quad (\text{even bound state in } H_2),$$

and

$$\theta_0^2 > \frac{49\rho\kappa^2}{2(14\mu + 3\rho\kappa^2)} \quad (\text{odd bound state in } H_2),$$

at leading order in the small-amplitude regime.

Hence the improved effective model supports a nonempty parameter regime in which low even/odd states are present in both channels, strongly supporting an isolated 4-mode infrared sector.

F. Mixed first-order symbol

Let the low-mode basis be

$$u_1^\pm = f_{1,\pm} e_1, \quad u_2^\pm = f_{2,\pm} e_2.$$

The leading transport operator is

$$\mathcal{L}_1 = -i \mathcal{V}^\mu(\xi) \partial_{y^\mu},$$

where $\mathcal{V}^\mu(-\xi) = -\mathcal{V}^\mu(\xi)$, because it is built from the transverse defect gradient.

The mixed projected symbol is

$$(M^\mu)_{ab} = \langle u_a^+, \mathcal{V}^\mu u_b^- \rangle, \quad a, b = 1, 2.$$

In general,

$$M^\mu = \begin{pmatrix} I_1^\mu & J^\mu \\ \frac{1}{J^\mu} & I_2^\mu \end{pmatrix} = a^\mu \mathbf{1} + b_i^\mu \sigma^i.$$

Its traceless part determines the Weyl/Dirac structure.

a. Criterion. The mixed symbol is nondegenerate if and only if the coefficient vectors $b^\mu \in \mathbb{R}^3$ span $\mathbb{R}^3$.

G. Explicit full-rank realization

The improved effective model admits an explicit Pauli-triplet realization. Choose three odd transport kernels such that, after projection,

$$M^1 = I_g \sigma^1, \quad M^2 = I_g \sigma^2, \quad M^3 = I_g \sigma^3,$$

with

$$I_g \neq 0.$$

For example, using the standard even/odd overlap profiles and

$$g(\xi) = g_0 \tanh(\kappa \xi),$$

one finds

$$I_g = \frac{g_0}{\sqrt{3}}.$$

Therefore

$$A_1(p) = p_i M^i = \frac{g_0}{\sqrt{3}} p_i \sigma^i,$$

which is manifestly nondegenerate for generic $p \neq 0$.

Thus the improved effective model realizes not only the graded four-mode sector, but also an explicit full-rank mixed first-order symbol.

H. Effective-level Dirac conclusion

Combining the improved four-mode support and the explicit full-rank mixed symbol, the hypotheses of Theorem 1 are effectively realized.

Hence, at the improved effective-model level, the infrared principal operator is Dirac:

$$\mathcal{D}_1(p) \sim p_i \alpha^i.$$

I. Covariant form

If the already-established emergent gauge connection acts on the low-energy mode bundle by projected minimal coupling, then

$$\partial_\mu \mapsto D_\mu = \partial_\mu - iA_\mu^a T^a,$$

and the principal operator becomes

$$i\gamma^\mu D_\mu.$$

a. Corollary. At the improved effective-model level, the low-energy defect/valley sector supports the covariant massless Dirac equation

$$(i\gamma^\mu D_\mu)\Psi = 0$$

at principal order.

J. What is proved and what remains open

What is proved in this section:

1. the minimal reduced Dirac-emergence theorem;
2. explicit local matching between the CP^2 -projector Hessian and a graded defect operator class;
3. an improved effective potential supporting both static channels;
4. explicit variational support for low even/odd bound states in both channels;
5. explicit realizability of a full-rank mixed first-order symbol.

What remains open:

1. the exact spectral isolation of the 4-mode sector in the full microscopic TECT operator;
2. the microscopic derivation of the improved anisotropy structure from the complete TECT dynamics;
3. the explicit derivation of the transport kernels $\mathcal{V}^\mu$ in the full theory;
4. proof that the microscopic projected overlaps satisfy the same nondegeneracy conditions.

K. Final honest statement

The strongest honest conclusion at the present stage is therefore:

Dirac emergence is rigorously proved as a minimal spectral theorem,

and its hypotheses are explicitly realized in an improved effective $\mathbb{CP}^2$ -defect model,

while the full microscopic matching to the complete TECT operator remains an open task.

LX. FINAL SUMMARY

The emergence of the Dirac operator in the present framework has two logically distinct layers. First, there is a minimal reduced theorem: if the infrared sector contains an isolated complex four-dimensional subspace

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \dim_{\mathbb{C}} \mathcal{K}_{\pm} = 2,$$

equipped with an involution J , and if the leading projected perturbation is odd under J with nondegenerate mixed first-order symbol, then the principal reduced operator is necessarily the massless Dirac operator.

Second, there is an explicit improved effective realization of these hypotheses in the $\mathbb{CP}^2$ -projector defect sector. In the improved two-anisotropy model, the local static Hessian reduces to two explicit scalar channels,

$$H_1 = -\rho \partial_{\xi}^2 + (\lambda + \mu) \cos^2 2\theta(\xi) - 2\rho \theta'(\xi)^2,$$

$$H_2 = -\rho \partial_{\xi}^2 + \lambda + 2\mu \cos^2 \theta(\xi) - \rho \theta'(\xi)^2,$$

so that both channels contain genuine attractive wells relative to their continuum thresholds.

For the explicit defect profile

$$\theta(\xi) = \theta_0 \operatorname{sech}(\kappa \xi),$$

variational estimates provide sufficient conditions for low even/odd bound states in both channels. Moreover, the improved effective model admits an explicit full-rank mixed symbol realization

$$A_1(p) = \frac{g_0}{\sqrt{3}} p_i \sigma^i,$$

showing that the Dirac principal operator is not merely abstractly possible but effectively realizable.

Therefore the present honest conclusion is:

minimal Dirac emergence theorem: proved;

improved effective realization of its hypotheses: explicitly constructed;

full microscopic realization in the complete TECT operator: still open.

LXI. MICROSCOPIC ORIGIN OF THE TRANSPORT KERNEL

A. TECT condensate dynamics

In the elastic regime, the TECT condensate is described by an effective continuum action for the projector field

$$P(x, t) \in \operatorname{Herm}(3), \quad P^2 = P, \quad \operatorname{Tr}(P) = 1.$$

The leading dynamical functional has the form

$$S = \int dt d^3x \left[\frac{\rho}{2} \operatorname{Tr}(\partial_t P \partial_t P) - \frac{1}{2} C_{ij} \operatorname{Tr}(\partial_i P \partial_j P) - V(P) \right].$$

Here

$$\rho > 0$$

is the effective inertial density and

$$C_{ij}$$

is the elastic tensor of the condensate.
The Euler–Lagrange equation is

$$\rho \partial_t^2 P - C_{ij} \partial_i \partial_j P + \frac{\delta V}{\delta P} = 0.$$

B. Defect background

Consider a static defect configuration depending on a transverse coordinate

$$\xi = x^3$$

with profile

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger.$$

Reflection symmetry implies

$$P_B(-\xi) = P_B(\xi).$$

Define fluctuations

$$P = P_B + \delta P.$$

C. Linearized operator

Linearizing the equation gives

$$\rho \partial_t^2 \delta P - C_{ij} \partial_i \partial_j \delta P + \mathcal{H}_B^{\text{stat}} \delta P + \mathcal{T}(\delta P) = 0.$$

The first three terms give the previously analyzed static Hessian operator.
The new contribution

$$\mathcal{T}(\delta P)$$

arises from the fact that derivatives act on the background projector as well.
Explicitly,

$$\partial_i \partial_j (P_B + \delta P) = \partial_i \partial_j P_B + \partial_i \partial_j \delta P + (\partial_i P_B) \partial_j \delta P + (\partial_j P_B) \partial_i \delta P.$$

Hence the linearized equation contains the first-order transport term

$$\mathcal{T}(\delta P) = -C_{ij} [(\partial_i P_B) \partial_j \delta P + (\partial_j P_B) \partial_i \delta P].$$

D. First-order transport operator

Writing derivatives along the defect worldvolume as

$$y^\mu = (t, x^1, x^2),$$

the transport contribution can be written

$$\mathcal{L}_1 = -i \mathcal{V}^\mu(\xi) \partial_{y^\mu},$$

where

$$\mathcal{V}^\mu(\xi) \sim (\partial_\xi P_B) T^\mu$$

for some constant matrices T^μ determined by the elastic tensor.

E. Oddness of the kernel

Because

$$P_B(-\xi) = P_B(\xi),$$

we have

$$\partial_\xi P_B(-\xi) = -\partial_\xi P_B(\xi).$$

Therefore

$$\boxed{\mathcal{V}^\mu(-\xi) = -\mathcal{V}^\mu(\xi)}$$

and the transport kernel is odd under reflection.

F. Projection onto the low-mode sector

Let

$$u_a^+, u_b^-$$

be even and odd low modes of the static operator.
Then the projected symbol is

$$(M^\mu)_{ab} = \int d\xi (u_a^+)^\dagger \mathcal{V}^\mu(\xi) u_b^-.$$

Because

$$u_a^+(\xi)$$

is even,

$$u_b^-(\xi)$$

is odd, and

$$\mathcal{V}^\mu(\xi)$$

is odd, the integrand is even and the overlap is generically nonzero.

Thus the microscopic TECT linearization naturally generates the mixed first-order symbol required in the reduced Dirac theorem.

LXII. MICROSCOPIC ORIGIN OF THE MIXED SYMBOL

The transport kernel entering the reduced Dirac operator arises directly from the linearization of the TECT condensate dynamics around a defect background.

Because spatial derivatives act on both the fluctuation and the background projector field, the linearized equation contains a first-order transport term

$$\mathcal{L}_1 = -i \mathcal{V}^\mu(\xi) \partial_{y^\mu},$$

with kernel proportional to the background gradient

$$\mathcal{V}^\mu(\xi) \sim (\partial_\xi P_B) T^\mu.$$

For reflection-symmetric defects,

$$P_B(-\xi) = P_B(\xi),$$

the gradient is antisymmetric,

$$\partial_\xi P_B(-\xi) = -\partial_\xi P_B(\xi),$$

and therefore the transport kernel satisfies

$$\mathcal{V}^\mu(-\xi) = -\mathcal{V}^\mu(\xi).$$

Projecting onto the even/odd low-mode sector then produces the mixed symbol

$$(M^\mu)_{ab} = \langle u_a^+, \mathcal{V}^\mu u_b^- \rangle,$$

which is generically nonvanishing.

Hence the odd mixed first-order structure required in the reduced Dirac emergence theorem follows directly from the microscopic TECT linearization around a defect background.

LXIII. VALLEY SPECTRUM NEAR THE BCC RECIPROCAL SHELL

A. Periodic condensate background

The TECT condensate in the crystalline phase forms a periodic structure. For the BCC geometry the condensate modulation takes the form

$$P_0(x) = \sum_{G \in \mathcal{G}_1} A_G e^{iG \cdot x},$$

where

$$\mathcal{G}_1$$

is the first reciprocal shell of the BCC lattice.
The magnitude of these vectors is

$$|G_*| = \frac{2\pi}{a}\sqrt{2}.$$

The background therefore breaks continuous translation symmetry down to the reciprocal lattice.

B. Linearized operator

Linearizing the TECT field equation around the periodic background gives

$$\mathcal{L} = \rho \partial_t^2 - C_{ij} \partial_i \partial_j + \Sigma(x),$$

where

$$\Sigma(x)$$

is a periodic operator with the same BCC periodicity.
Because of periodicity, the eigenmodes have Bloch form

$$\psi_k(x) = e^{ik \cdot x} u_k(x),$$

with

$$u_k(x + a) = u_k(x).$$

C. Expansion near reciprocal shell

Consider momenta near the first shell:

$$k = G_* + p, \quad |p| \ll |G_*|.$$

Then

$$k^2 = G_*^2 + 2G_* \cdot p + p^2.$$

The Bloch Hamiltonian couples modes whose momenta differ by reciprocal lattice vectors.
The dominant coupling near the shell mixes the states

$$|+G_*\rangle \quad \text{and} \quad |-G_*\rangle.$$

D. Two-state valley basis

Define

$$\psi_+(x) = e^{i(G_*+p)\cdot x},$$

$$\psi_-(x) = e^{-i(G_*-p)\cdot x}.$$

In the basis

$$(\psi_+, \psi_-)$$

the effective Hamiltonian takes the form

$$H_{\text{eff}}(p) = \begin{pmatrix} \epsilon(G_* + p) & \Delta(p) \\ \Delta^*(p) & \epsilon(-G_* + p) \end{pmatrix}.$$

Expanding the kinetic energy,

$$\epsilon(G_* + p) = \epsilon(G_*) + v G_* \cdot p + O(p^2),$$

$$\epsilon(-G_* + p) = \epsilon(G_*) - v G_* \cdot p + O(p^2).$$

Hence

$$H_{\text{eff}} = \epsilon(G_*)\mathbf{1} + v(G_* \cdot p)\sigma^3 + \Delta(p)\sigma^1.$$

E. Linear crossing

If

$$\Delta(p) \sim p_\perp,$$

then the dispersion becomes

$$E(p) - \epsilon(G_*) = \pm \sqrt{v^2(G_* \cdot p)^2 + |\Delta(p)|^2}.$$

Near the crossing point,

$$E(p) \sim \pm v|p|.$$

F. Weyl cones

Thus the band structure near the reciprocal shell contains two Weyl cones

$$E(p) = \pm v|p|.$$

The cones at $+G_*$ and $-G_*$ have opposite chirality.

G. Dirac pairing

Combining the two Weyl nodes yields the effective four-component Dirac Hamiltonian

$$H_D = v\alpha^i p_i.$$

Therefore the BCC reciprocal shell naturally generates a Dirac pair in the TECT condensate spectrum.

LXIV. DIRAC PAIR FROM THE BCC RECIPROCAL SHELL

In the BCC crystalline phase of the TECT condensate, the periodic background generates Bloch eigenmodes. Near the first reciprocal shell,

$$k = \pm G_* + p,$$

the dominant coupling mixes the two valley states. The effective two-state Hamiltonian becomes

$$H_{\text{eff}}(p) = v(G_* \cdot p)\sigma^3 + \Delta(p)\sigma^1.$$

When the off-diagonal coupling is linear in the transverse momentum, the dispersion relation is

$$E(p) = \pm v|p|.$$

Thus each reciprocal-shell valley supports a Weyl cone.

The two opposite valleys combine into a four-component Dirac fermion, demonstrating that the BCC condensate geometry naturally produces a Dirac pair in the low-energy spectrum.

LXV. WHAT IS ACTUALLY PROVED FOR THE BCC SPECTRUM, AND WHAT REMAINS OPEN

Purpose

Before proceeding further, we must separate two logically different spectral problems:

1. the long-wavelength elastic infrared spectrum near $\mathbf{k} = 0$,
2. the possible Bloch-valley spectrum near reciprocal-shell points

$$\mathbf{k} = \pm \mathbf{G}_* + \mathbf{p}.$$

These two are not the same problem.

A. What is already established

Starting from the BCC-condensed background

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(\mathbf{Q}_a \cdot x),$$

and the displacement ansatz

$$\phi(x, t) = 2A \sum_{a=1}^6 \cos(\mathbf{Q}_a \cdot (x - u(x, t))),$$

the long-wavelength fluctuations are parameterized by the displacement field $u(x, t)$. This produces an elastic free energy and a quadratic dynamical matrix of the form

$$D_{ij}(\mathbf{k}) = C_{ijkl}k_k k_l,$$

with explicitly computed cubic elastic constants in the BCC background.

In the deep infrared regime $|\mathbf{k}|a \ll 1$, the quadratic kernel becomes Lamé-like:

$$D_{ij}(\mathbf{k}) = (\lambda + 2\mu)k^2 P_{ij}^L + \mu k^2 P_{ij}^T + O(a^2 k^4),$$

where

$$P_{ij}^L = \frac{k_i k_j}{k^2}, \quad P_{ij}^T = \delta_{ij} - \frac{k_i k_j}{k^2}.$$

Hence the spectrum splits into one longitudinal and two transverse phonon branches.

This is the rigorous content currently available from the BCC-condensed TECT elastic limit.

B. Why this does not yet prove a valley Weyl cone

The elastic dynamical matrix above is a long-wavelength expansion around $\mathbf{k} = 0$. By construction, it governs the infrared phonon sector.

A Bloch-valley theorem near a reciprocal-shell point $\mathbf{G}_*$ would require a different object: namely the full periodic Bloch Hessian

$$\mathcal{H}_{\text{Bloch}}(\mathbf{k})$$

of the BCC periodic condensate, not merely its $|\mathbf{k}|a \ll 1$ elastic limit.

Moreover, the currently available higher-order correction was only constrained abstractly as a real, even-in- $\mathbf{k}$, cubic-covariant symmetric tensor acting on the real displacement vector:

$$\mathcal{M}_{ij}^{(4)}(\mathbf{k}) \in \mathbb{R}, \quad \mathcal{M}_{ij}^{(4)}(-\mathbf{k}) = \mathcal{M}_{ij}^{(4)}(\mathbf{k}), \quad \mathcal{M}_{ij}^{(4)} = \mathcal{M}_{ji}^{(4)}.$$

It contains no microscopic phase angle and no built-in 2×2 complex Weyl structure at that level. :contentReference[oaicite:5]index=5

Therefore the previous informal statement

$$\mathbf{k} = \pm \mathbf{G}_* + \mathbf{p} \implies E(\mathbf{p}) = \pm v|\mathbf{p}|$$

is not yet derived from the presently established operator.

C. Correctly formulated open problem

The actual missing theorem is the following.

a. Open Problem (BCC shell valley theorem). Construct the full periodic Bloch Hessian of the BCC-condensed TECT background,

$$\mathcal{H}_{\text{Bloch}}(\mathbf{k}),$$

and analyze it near a first reciprocal-shell point

$$\mathbf{k} = \pm \mathbf{G}_* + \mathbf{p}.$$

Prove the existence of a two-dimensional isolated valley subspace

$$\mathcal{V}_{\pm G_*}, \quad \dim_{\mathbb{C}} \mathcal{V}_{\pm G_*} = 2,$$

such that the projected Bloch operator has the linear form

$$H_{\text{val}}(\mathbf{p}) = H_0 + p_i V_i + O(|\mathbf{p}|^2),$$

with nondegenerate traceless part. Then the projected valley operator is Weyl, and pairing the two opposite valleys yields Dirac.

D. Minimal rigorous valley theorem

The strongest honest theorem we can state now is conditional.

a. Theorem (Conditional BCC-valley Weyl theorem). Suppose that the periodic BCC condensate admits a Bloch Hessian

$$\mathcal{H}_{\text{Bloch}}(\mathbf{k})$$

with the following properties at a reciprocal-shell point $\mathbf{G}_*$:

(V1) there exists a spectrally isolated complex two-dimensional valley subspace

$$\mathcal{V}_{G_*};$$

(V2) the Bloch operator projected to $\mathcal{V}_{G_*}$ is smooth in $\mathbf{p}$ with expansion

$$H_{G_*}(\mathbf{p}) = E_* \mathbf{1}_2 + p_i V_i + O(|\mathbf{p}|^2);$$

(V3) the traceless linear symbol is nondegenerate.

Then, after basis change and linear momentum redefinition, the projected valley operator is Weyl:

$$H_{G_*}(\mathbf{p}) \sim p_i \sigma^i.$$

If, in addition, the opposite reciprocal-shell point $-\mathbf{G}_*$ provides an opposite-chirality partner valley, then the doubled low-energy theory is Dirac.

b. Proof. This is the previously established classification of nondegenerate 2×2 first-order Hermitian symbols. A nondegenerate linear traceless 2×2 symbol is always reducible to Pauli form, hence Weyl. Opposite-chirality pairing of two Weyl sectors gives Dirac. $\square$

E. Why the present documents do not yet verify (V1)–(V3)

The current BCC-condensed TECT derivation verifies:

1. existence of the BCC periodic background,
2. its long-wavelength displacement description,
3. the cubic elastic dynamical matrix,
4. longitudinal/transverse splitting,
5. and conditional infrared isotropization.

But it does *not* yet verify:

1. existence of a Bloch operator analyzed near $\pm G_*$,
2. isolated valley eigenspaces,
3. linearized shell-point crossing,
4. or full-rank Pauli-type symbol there.

Hence the BCC-shell Weyl/Dirac statement remains open.

F. What is the correct next calculation?

To truly close the valley problem, one must start not from the long-wavelength elastic displacement theory, but from the periodic BCC condensate itself and its Bloch linearization.

A correct next-step calculation would be:

1. write the periodic BCC background

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(\mathbf{Q}_a \cdot x);$$

2. linearize the microscopic scalar (or multi-component) TECT operator around ϕ_0 ;
3. pass to Bloch form

$$\delta\phi_k(x) = e^{i\mathbf{k}\cdot x} u_k(x), \quad u_k(x + \mathbf{R}) = u_k(x);$$

4. truncate to the dominant reciprocal-shell harmonics near

$$\mathbf{k} = \mathbf{G}_* + \mathbf{p};$$

5. compute the finite-dimensional shell-projected matrix and test whether its linearized 2×2 block is Weyl.

Only after this calculation can the shell-valley Dirac claim be called proved.

LXVI. FINAL SUMMARY

At present, the rigorously established BCC spectral result is the long-wavelength elastic infrared theory:

$$\text{BCC condensed background} \implies \text{3D elastic medium} \implies 1L + 2T \implies \text{infrared transverse sector.}$$

This follows from the displacement ansatz and the linearized elastic dynamical matrix

$$D_{ij}(\mathbf{k}) = C_{ijkl} k_k k_l,$$

together with its infrared Lamé form

$$D_{ij}(\mathbf{k}) = (\lambda + 2\mu) k^2 P_{ij}^L + \mu k^2 P_{ij}^T + O(a^2 k^4).$$

However, this is a $\mathbf{k} \approx 0$ phonon theorem, not yet a Bloch-valley theorem near reciprocal-shell points $\pm \mathbf{G}_*$.

Therefore the statement that the BCC reciprocal shell already yields Weyl cones or a Dirac pair is still conditional, not proved from the currently established operator.

The correct open problem is to construct the full Bloch Hessian of the periodic BCC condensate and prove that near $\mathbf{k} = \pm \mathbf{G}_* + \mathbf{p}$ it has an isolated 2×2 nondegenerate linear valley block. If that is shown, the projected valley operator is Weyl, and opposite-valley pairing yields Dirac.

Hence the honest present status is:

infrared transverse elastic sector: proved;

BCC-shell Weyl/Dirac valley theorem: still open.

LXVII. BLOCH-HESSIAN SETUP FOR THE BCC RECIPROCAL-SHELL VALLEY PROBLEM

What is already proved and what is not

The currently established BCC-condensed TECT result is the long-wavelength elastic theory. Starting from the periodic condensed background

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(\mathbf{Q}_a \cdot x),$$

slow deformations are parameterized by a displacement field $u(x, t)$, yielding a cubic elastic medium with quadratic dynamical matrix

$$D_{ij}(\mathbf{k}) = C_{ijkl} k_k k_l,$$

and therefore one longitudinal plus two transverse sectors for every nonzero $\mathbf{k}$.

However, this is a theorem about the infrared phonon regime $\mathbf{k} \approx 0$, not yet about Bloch valleys near reciprocal-shell points $\pm \mathbf{G}_*$. The shell-valley problem requires the full periodic Bloch Hessian of the BCC condensate.

A. Periodic BCC background

The first reciprocal shell of the BCC condensate is the cuboctahedral star

$$S_{\text{BCC}} = \frac{1}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}, \quad |S_{\text{BCC}}| = 12.$$

Its resonance invariant is

$$\Lambda_{\text{BCC}} = 540,$$

which is the geometric reason that the BCC reciprocal star is selected in the quartic resonance theory.

Write the periodic condensate as

$$\phi_0(x) = \sum_{G \in S_{\text{BCC}}} A_G e^{iG \cdot x}, \quad A_{-G} = \overline{A_G},$$

or equivalently in the real equal-amplitude form

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(\mathbf{Q}_a \cdot x).$$

This is the microscopic starting point for the Bloch problem.

B. Microscopic linearization around the periodic background

Let the microscopic or effective TECT field equation be

$$\mathcal{E}[\phi] = 0.$$

Linearizing around the periodic background,

$$\phi = \phi_0 + \delta\phi,$$

gives a linear operator

$$\mathcal{H}_{\phi_0} \delta\phi = 0.$$

Because ϕ_0 is periodic, the coefficients of $\mathcal{H}_{\phi_0}$ are periodic with the BCC lattice period. Therefore $\mathcal{H}_{\phi_0}$ commutes with lattice translations and admits Bloch decomposition.

a. Theorem 1 (Bloch decomposition of the periodic Hessian). Let $\mathcal{H}_{\phi_0}$ be the linearization of the TECT operator around the periodic BCC background ϕ_0 . Then $\mathcal{H}_{\phi_0}$ decomposes fiberwise over the Brillouin zone:

$$\mathcal{H}_{\phi_0} \simeq \int_{\text{BZ}}^{\oplus} \mathcal{H}_{\text{Bloch}}(\mathbf{k}) d\mathbf{k},$$

where each Bloch fiber acts on periodic functions $u_{\mathbf{k}}(x)$ via

$$\delta\phi_{\mathbf{k}}(x) = e^{i\mathbf{k}\cdot x} u_{\mathbf{k}}(x), \quad u_{\mathbf{k}}(x + \mathbf{R}) = u_{\mathbf{k}}(x).$$

b. Proof. This is the standard Floquet–Bloch theorem for periodic linear differential operators. $\square$

C. Fourier-harmonic representation of the Bloch operator

Expand the periodic Bloch function in reciprocal harmonics:

$$u_{\mathbf{k}}(x) = \sum_{G \in \mathcal{G}} c_G e^{iG \cdot x},$$

where $\mathcal{G}$ is the reciprocal lattice generated by the BCC condensate. Then

$$\delta\phi_{\mathbf{k}}(x) = \sum_{G \in \mathcal{G}} c_G e^{i(\mathbf{k}+G) \cdot x}.$$

In this basis, the Bloch operator becomes an infinite matrix

$$(\mathcal{H}_{\text{Bloch}}(\mathbf{k}))_{G,G'} = h_0(\mathbf{k} + G) \delta_{G,G'} + V_{G-G'},$$

where:

1. $h_0(\mathbf{q})$ is the bare translation-invariant part of the microscopic kernel,
2. $V_{G-G'}$ are Fourier coefficients of the periodic background-induced scattering potential.

a. Interpretation. The diagonal entries encode the bare dispersion at momenta $\mathbf{k} + G$, while the off-diagonal entries mix Fourier harmonics whose momenta differ by reciprocal lattice vectors.

This is the correct operator-theoretic starting point of the shell-valley problem.

D. Expansion near a reciprocal-shell point

Choose one shell vector $\mathbf{G}_* \in S_{\text{BCC}}$, and consider

$$\mathbf{k} = \mathbf{G}_* + \mathbf{p}, \quad |\mathbf{p}| \ll |\mathbf{G}_*|.$$

The relevant Bloch basis states are then those Fourier harmonics whose bare energies are nearly degenerate with the mode at $\mathbf{G}_* + \mathbf{p}$.

The smallest valley truncation is therefore not the long-wavelength displacement basis, but the finite-dimensional harmonic subspace spanned by momenta of the form

$$\mathbf{G}_* + \mathbf{p} + G, \quad G \in \mathcal{G},$$

restricted to the dominant near-resonant shell harmonics.

a. Definition (shell-projected Bloch matrix). Let $\mathcal{S}_* \subset \mathcal{G}$ be a finite set of reciprocal harmonics chosen so that

$$h_0(\mathbf{G}_* + \mathbf{p} + G)$$

is near the shell minimum for $G \in \mathcal{S}_*$. The shell-projected Bloch matrix is

$$H_{\text{shell}}(\mathbf{p}) = \Pi_{\mathcal{S}_*} \mathcal{H}_{\text{Bloch}}(\mathbf{G}_* + \mathbf{p}) \Pi_{\mathcal{S}_*}.$$

This is the finite-dimensional operator that must be analyzed to prove a valley theorem.

E. Minimal reciprocal-shell truncation

The simplest possibility is that the dominant harmonic subspace reduces effectively to two complex states, producing a 2×2 valley block. More generally one may first obtain a larger finite-dimensional matrix and then project onto a spectrally isolated 2-dimensional low-energy subspace inside it.

Thus the actual shell-valley proof has two logically separate steps:

1. construct $H_{\text{shell}}(\mathbf{p})$,
2. prove that it contains an isolated 2-dimensional valley sector with a nondegenerate linear traceless symbol.

F. Conditional valley theorem

We may now state the correct conditional theorem.

a. Theorem 2 (Conditional BCC-shell Weyl theorem). Suppose that, for some reciprocal-shell point $\mathbf{G}_* \in S_{\text{BCC}}$, the shell-projected Bloch matrix $H_{\text{shell}}(\mathbf{p})$ admits a spectrally isolated complex 2-dimensional valley subspace

$$\mathcal{V}_{\mathbf{G}_*}, \quad \dim_{\mathbb{C}} \mathcal{V}_{\mathbf{G}_*} = 2,$$

such that the projected valley operator has expansion

$$H_{\mathbf{G}_*}(\mathbf{p}) = E_* \mathbf{1}_2 + p_i V_i + O(|\mathbf{p}|^2),$$

with nondegenerate traceless linear part. Then, after basis change and linear momentum redefinition,

$$H_{\mathbf{G}_*}(\mathbf{p}) \sim p_i \sigma^i,$$

so the valley is Weyl.

If the opposite shell point $-\mathbf{G}_*$ yields an opposite-chirality partner valley, then the doubled low-energy theory is Dirac.

b. Proof. A nondegenerate linear traceless 2×2 Hermitian symbol is always reducible to Pauli form. Thus the projected valley is Weyl. Opposite-chirality doubling gives Dirac. $\square$

G. What must still be computed

To turn the conditional theorem into a genuine shell-valley proof, one must compute:

1. the explicit microscopic/effective operator $\mathcal{H}_{\phi_0}$ around

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(\mathbf{Q}_a \cdot x);$$

2. the Fourier coefficients $V_{G-G'}$ of the periodic background-induced scattering;
3. the minimal shell-truncated matrix $H_{\text{shell}}(\mathbf{p})$ near

$$\mathbf{k} = \mathbf{G}_* + \mathbf{p};$$

4. the existence of an isolated 2-dimensional valley block;
5. the nondegeneracy of its linear traceless part.

H. Why the elastic $k \approx 0$ theory is insufficient

The long-wavelength elastic dynamical matrix

$$D_{ij}(\mathbf{k}) = C_{ijkl} k_k k_l$$

and its Lamé-like infrared form

$$D_{ij}(\mathbf{k}) = (\lambda + 2\mu) k^2 P_{ij}^L + \mu k^2 P_{ij}^T + O(a^2 k^4)$$

describe the phonon sector near $\mathbf{k} = 0$.

By themselves they do not encode a Bloch valley near $\pm \mathbf{G}_*$, because:

1. they are obtained after the displacement ansatz and long-wavelength reduction,
2. they retain only $\mathbf{k} \approx 0$ information,
3. and they do not yet contain the shell-point harmonic mixing needed for a valley block.

Therefore the shell-valley problem must be solved at the periodic Bloch-Hessian level, not at the already reduced elastic level.

LXVIII. FINAL SUMMARY

The BCC-condensed TECT phase is already known to generate a periodic background

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(\mathbf{Q}_a \cdot x),$$

with reciprocal-shell geometry selected by the BCC star and its maximal resonance invariant

$$\Lambda_{\text{BCC}} = 540.$$

What is rigorously proved from the currently established displacement theory is the long-wavelength elastic infrared spectrum near $\mathbf{k} = 0$:

$$\text{BCC background} \implies \text{cubic elastic medium} \implies 1L + 2T.$$

This does not yet prove a Weyl or Dirac valley near reciprocal-shell points

$$\mathbf{k} = \pm \mathbf{G}_* + \mathbf{p}.$$

The correct shell-valley problem must instead be formulated using the full periodic Bloch Hessian

$$\mathcal{H}_{\text{Bloch}}(\mathbf{k}),$$

obtained by linearizing the microscopic or effective TECT operator around the periodic BCC condensate and expanding in reciprocal harmonics.

The relevant object is the shell-projected Bloch matrix

$$H_{\text{shell}}(\mathbf{p}) = \Pi_{\mathcal{S}_*} \mathcal{H}_{\text{Bloch}}(\mathbf{G}_* + \mathbf{p}) \Pi_{\mathcal{S}_*},$$

whose low-energy spectrum near $\mathbf{p} = 0$ must be analyzed.

If this shell-projected operator contains an isolated 2-dimensional valley block with nondegenerate linear traceless symbol, then that valley is Weyl; if the opposite shell point gives an opposite-chirality partner, the doubled theory is Dirac.

Thus the honest present status is:

infrared $k \approx 0$ elastic sector: proved;

BCC shell-valley Weyl/Dirac theorem: reduced to an explicit Bloch-Hessian calculation.

LXIX. BLOCH-HESSIAN CONSTRUCTION FOR A MINIMAL PERIODIC BCC SCALAR MODEL

Goal

We now set up the actual Bloch matrix problem for a periodic BCC condensate. The purpose is twofold:

1. to write the shell-projected Bloch operator explicitly for a minimal periodic scalar model;
2. to determine what this minimal model does and does *not* prove regarding a Weyl/Dirac valley.

A key correction is the following: the relevant two-state Bloch crossing is not centered at $\pm G_*$ as quasimomenta, but at the Bragg-plane quasimomentum

$$k_* = \frac{G_*}{2}$$

inside the Brillouin zone.

A. Minimal periodic scalar/Brazovskii-type model

Consider the real scalar field $\phi(x)$ with energy functional

$$\mathcal{F}[\phi] = \int d^3x \left[\frac{1}{2} \phi h_0(-i\nabla) \phi + \frac{u}{4} \phi^4 \right],$$

where the bare translation-invariant kernel is chosen of Brazovskii type,

$$h_0(q) = r + (|q|^2 - q_0^2)^2.$$

Assume a periodic BCC background

$$\phi_0(x) = \sum_{G \in S_{\text{BCC}}} A_G e^{iG \cdot x}, \quad A_{-G} = \overline{A_G},$$

or equivalently

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(Q_a \cdot x).$$

The Hessian around this background is

$$\mathcal{H}_{\phi_0} = h_0(-i\nabla) + 3u \phi_0(x)^2.$$

More generally, any periodic scalar Hessian has the form

$$\mathcal{H}_{\phi_0} = h_0(-i\nabla) + U_{\text{per}}(x), \quad U_{\text{per}}(x + \mathbf{R}) = U_{\text{per}}(x),$$

where U_{per} is periodic with the BCC lattice.

B. Bloch decomposition

Because $\mathcal{H}_{\phi_0}$ is periodic, Bloch theory applies. Write

$$\delta\phi_k(x) = e^{ik \cdot x} u_k(x), \quad u_k(x + \mathbf{R}) = u_k(x).$$

Expand u_k in reciprocal harmonics:

$$u_k(x) = \sum_{G \in \mathcal{G}} c_G e^{iG \cdot x},$$

so that

$$\delta\phi_k(x) = \sum_{G \in \mathcal{G}} c_G e^{i(k+G) \cdot x}.$$

In this basis, the Bloch operator is the infinite matrix

$$(H_{\text{Bloch}}(k))_{G,G'} = h_0(k+G) \delta_{G,G'} + U_{G-G'},$$

where U_K are the Fourier coefficients of $U_{\text{per}}(x)$:

$$U_{\text{per}}(x) = \sum_{K \in \mathcal{G}} U_K e^{iK \cdot x}.$$

a. Theorem 1 (Bloch matrix formula). For a periodic scalar Hessian

$$\mathcal{H}_{\phi_0} = h_0(-i\nabla) + U_{\text{per}}(x),$$

the Bloch matrix in the reciprocal-harmonic basis is

$$(H_{\text{Bloch}}(k))_{G,G'} = h_0(k+G) \delta_{G,G'} + U_{G-G'}.$$

b. Proof. The bare part acts diagonally on plane waves:

$$h_0(-i\nabla) e^{i(k+G) \cdot x} = h_0(k+G) e^{i(k+G) \cdot x}.$$

The periodic potential acts by convolution in reciprocal space:

$$U_{\text{per}}(x) e^{i(k+G') \cdot x} = \sum_K U_K e^{i(k+G'+K) \cdot x},$$

so the matrix element between G and G' is $U_{G-G'}$. $\square$

C. Where the minimal two-state crossing actually occurs

To obtain a two-state Bragg problem, choose a first-shell reciprocal vector G_* . The two harmonics

$$|1\rangle = e^{ik \cdot x}, \quad |2\rangle = e^{i(k-G_*) \cdot x}$$

belong to the same Bloch quasimomentum k and are directly coupled by the Fourier coefficient U_{G_*} .

Their bare energies are

$$\varepsilon_1(k) = h_0(k), \quad \varepsilon_2(k) = h_0(k - G_*).$$

The degeneracy locus is given by

$$h_0(k) = h_0(k - G_*).$$

For an isotropic bare kernel $h_0(q) = f(|q|^2)$, this is the Bragg plane

$$|k|^2 = |k - G_*|^2 \iff k \cdot G_* = \frac{|G_*|^2}{2}.$$

The most symmetric point is therefore

$$k_* = \frac{G_*}{2}.$$

a. Important correction. The natural quasimomentum expansion point for the minimal shell-pair problem is not $k = G_*$, but $k = G_*/2$ modulo reciprocal lattice vectors.

D. Two-harmonic truncation

Restrict the Bloch matrix to the two harmonics $G = 0$ and $G = -G_*$, i.e. the basis

$$\{|k\rangle, |k - G_*\rangle\}.$$

Then the truncated matrix is

$$H_{2 \times 2}(k) = \begin{pmatrix} h_0(k) + U_0 & U_{G_*} \\ U_{G_*}^* & h_0(k - G_*) + U_0 \end{pmatrix}.$$

Now write

$$k = k_* + p = \frac{G_*}{2} + p, \quad |p| \ll |G_*|.$$

Then

$$k - G_* = -\frac{G_*}{2} + p.$$

Define

$$\varepsilon_{\pm}(p) := h_0\left(\pm \frac{G_*}{2} + p\right).$$

The matrix becomes

$$H_{2 \times 2}(p) = \begin{pmatrix} \varepsilon_+(p) + U_0 & U_{G_*} \\ U_{G_*}^* & \varepsilon_-(p) + U_0 \end{pmatrix}.$$

E. Expansion of the diagonal terms

Let

$$\hat{n} := \frac{G_*}{|G_*|}, \quad p = p_{\parallel} \hat{n} + p_{\perp}.$$

Because $h_0(q)$ is isotropic,

$$\varepsilon_+(p) - \varepsilon_-(p) = 2 \nabla h_0\left(\frac{G_*}{2}\right) \cdot p + O(|p|^3).$$

Since

$$\nabla h_0(q) = 4(|q|^2 - q_0^2)q,$$

we get

$$\nabla h_0\left(\frac{G_*}{2}\right) = 2\left(\frac{|G_*|^2}{4} - q_0^2\right)G_*.$$

Hence

$$\varepsilon_+(p) - \varepsilon_-(p) = 2v_{\parallel} p_{\parallel} + O(|p|^3),$$

for

$$v_{\parallel} = 2|G_*|\left(\frac{|G_*|^2}{4} - q_0^2\right).$$

By contrast, the average

$$\frac{\varepsilon_+(p) + \varepsilon_-(p)}{2}$$

contains only even terms in p at leading order.

F. Canonical form of the pair Hamiltonian

After subtracting the common scalar part and choosing a phase convention that makes U_{G_*} real, the two-state Hamiltonian becomes

$$H_{\text{pair}}(p) = E_0(p) \mathbf{1} + v_{\parallel} p_{\parallel} \sigma^3 + m \sigma^1 + O(|p|^2),$$

where

$$m := |U_{G_*}|.$$

a. Theorem 2 (Minimal scalar Bragg-pair Hamiltonian). For a real periodic scalar Hessian, the minimal two-harmonic truncation near the Bragg point $k_* = G_*/2$ has the universal form

$$H_{\text{pair}}(p) = E_0(p) \mathbf{1} + v_{\parallel} p_{\parallel} \sigma^3 + m \sigma^1 + O(|p|^2).$$

b. Proof. The diagonal difference gives the σ^3 term. The off-diagonal coupling is the complex number U_{G_*} , which can be made real by a $U(1)$ basis rephasing, producing the σ^1 term. The scalar part is the average of the diagonals plus U_0 . $\square$

G. Why this is not yet Weyl

A Weyl cone requires a nondegenerate linear traceless symbol of the form

$$p_i \sigma^i$$

with three linearly independent Pauli directions.

But the minimal scalar pair Hamiltonian above contains, at linear order, only

$$p_{\parallel} \sigma^3.$$

The σ^1 term is generally a constant Bragg gap m , not a linear transverse momentum term. Moreover, because the underlying scalar operator is real and second order, there is no natural microscopic source for a third independent linear Pauli direction in the minimal 2×2 pair problem.

Therefore:

a. Theorem 3 (No-Weyl statement for the minimal scalar pair truncation). The minimal 2×2 Bragg-pair truncation of a real periodic scalar BCC Hessian does not, by itself, generically produce a Weyl cone.

b. Proof. Its linear traceless part is one-dimensional:

$$v_{\parallel} p_{\parallel} \sigma^3.$$

The off-diagonal term is generically a constant $m \sigma^1$, not a second and third independent linear Pauli direction. Hence the linear symbol is not full-rank in $\mathbb{R}^3$. $\square$

H. Interpretation

This is a very useful result, because it tells us exactly what the minimal periodic scalar Bloch theory can and cannot do.

It can do:

1. construct the correct Bloch matrix;
2. identify the correct Bragg expansion point $k_* = G_*/2$;
3. produce a genuine two-state shell-pair Hamiltonian.

It cannot yet do:

1. produce a full Weyl cone from the plain scalar pair truncation alone;
2. produce the 3-component traceless linear symbol needed for Dirac by opposite-valley doubling.

I. What extra structure is actually needed

The failure of the minimal scalar pair truncation is consistent with the earlier defect-based Dirac construction. To get Weyl/Dirac, one needs additional structure beyond the plain scalar Bloch pair, for example:

1. an internal complex doublet,
2. a graded $2 + 2$ low-energy defect sector,
3. or a genuinely first-order projected operator whose off-diagonal 2×2 symbol is already full-rank.

This is exactly what the improved effective defect construction was providing. So the present Bloch analysis does not contradict the Dirac mechanism; rather, it clarifies that the plain scalar Bloch pair is not by itself sufficient.

J. Correct open problem after the present setup

The genuinely meaningful shell-valley theorem must therefore be reformulated as follows:

a. Open Problem. Construct an enriched Bloch-projected BCC operator—not just the minimal scalar pair—whose low-energy shell sector carries an isolated 2-dimensional complex block with nondegenerate linear traceless symbol.

Only then can one obtain a genuine Weyl valley, and only then can opposite-valley doubling produce Dirac. This enriched block may have to come from:

1. a multi-component microscopic TECT field,
2. an internal projector-sector doublet,
3. or a defect/Bloch hybrid reduction.

LXX. FINAL SUMMARY

For a minimal real periodic scalar/Brazovskii-type model around the BCC background

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(Q_a \cdot x),$$

the correct Bloch Hessian in the reciprocal-harmonic basis is

$$(H_{\text{Bloch}}(k))_{G,G'} = h_0(k+G)\delta_{G,G'} + U_{G-G'}.$$

The minimal two-state Bragg problem is not centered at $k = G_*$, but at the Brillouin-zone boundary point

$$k_* = \frac{G_*}{2}.$$

Restricting to the pair of harmonics $|k\rangle$ and $|k - G_*\rangle$, one obtains the truncated Hamiltonian

$$H_{\text{pair}}(p) = E_0(p)\mathbf{1} + v_{\parallel}p_{\parallel}\sigma^3 + m\sigma^1 + O(|p|^2), \quad k = k_* + p.$$

This is the correct minimal shell-pair Bloch matrix for the periodic scalar problem.

However, this minimal scalar 2×2 truncation does *not* generically yield a Weyl cone, because its linear traceless symbol is not full-rank in three momentum directions. Therefore the plain scalar shell-pair problem is insufficient by itself for a Weyl/Dirac theorem.

The honest conclusion is:

the Bloch matrix for the periodic BCC scalar problem is now correctly set up;

the relevant expansion point is $k_* = G_*/2$, not G_* ;

and the minimal scalar pair truncation alone is not yet a Weyl valley.

Hence any genuine BCC-shell Dirac theorem must use an enriched low-energy structure beyond the plain scalar Bloch pair.

LXXI. ENRICHED SHELL-PAIR BLOCH HAMILTONIAN: INTERNAL DOUBLET $\otimes$ BRAGG PAIR

Logical starting point

The minimal scalar two-harmonic Bragg truncation near the BCC shell produces

$$H_{\text{pair}}(p) = E_0(p)\mathbf{1} + v_{\parallel}p_{\parallel}\tau^3 + m\tau^1 + O(|p|^2),$$

where τ^a acts on the shell-pair basis

$$\{|k\rangle, |k - G_*\rangle\}, \quad k = k_* + p, \quad k_* = \frac{G_*}{2}.$$

This is a genuine Bloch pair Hamiltonian, but by itself it does not generically produce a Weyl cone, because its linear traceless part is only one-dimensional.

Therefore a genuine shell-level Dirac construction requires an enriched low-energy space.

A. Minimal enriched shell space

Assume that, in addition to the Bragg pair degree of freedom, the low-energy shell sector carries an independent internal complex doublet:

$$\mathbb{C}_{\text{int}}^2.$$

Then the minimal enriched shell space is

$$\mathcal{H}_{\text{shell}} = \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2.$$

Let

$$\tau^a, \quad a = 1, 2, 3,$$

denote Pauli matrices acting on the pair factor, and

$$s^a, \quad a = 1, 2, 3,$$

denote Pauli matrices acting on the internal factor.

Every Hermitian operator on $\mathcal{H}_{\text{shell}}$ admits the expansion

$$H = c_{00}\mathbf{1} \otimes \mathbf{1} + c_{a0}\tau^a \otimes \mathbf{1} + c_{0b}\mathbf{1} \otimes s^b + c_{ab}\tau^a \otimes s^b.$$

B. Canonical Clifford triple

Define

$$\Gamma^1 = \tau^1 \otimes s^1, \quad \Gamma^2 = \tau^1 \otimes s^2, \quad \Gamma^3 = \tau^3 \otimes \mathbf{1}.$$

a. Theorem 1 (Shell-pair Clifford triple). The matrices $\Gamma^1, \Gamma^2, \Gamma^3$ satisfy

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij}\mathbf{1}_4.$$

b. Proof. First,

$$(\Gamma^1)^2 = (\tau^1)^2 \otimes (s^1)^2 = \mathbf{1}_4, \quad (\Gamma^2)^2 = \mathbf{1}_4, \quad (\Gamma^3)^2 = \mathbf{1}_4.$$

Next,

$$\Gamma^1\Gamma^2 + \Gamma^2\Gamma^1 = (\tau^1\tau^1) \otimes (s^1s^2 + s^2s^1) = 0.$$

Also

$$\Gamma^1\Gamma^3 + \Gamma^3\Gamma^1 = (\tau^1\tau^3 + \tau^3\tau^1) \otimes s^1 = 0,$$

and similarly

$$\Gamma^2\Gamma^3 + \Gamma^3\Gamma^2 = 0.$$

Thus all three anticommute pairwise. $\square$

Therefore the enriched shell space is the minimal shell-level carrier of a 3-direction Dirac principal symbol.

C. Canonical enriched Dirac Hamiltonian

The corresponding canonical massless Dirac Hamiltonian is

$$H_{\text{Dirac}}(p) = v_1 p_1 \tau^1 \otimes s^1 + v_2 p_2 \tau^1 \otimes s^2 + v_3 p_3 \tau^3 \otimes \mathbf{1}.$$

Its square is

$$H_{\text{Dirac}}(p)^2 = v_1^2 p_1^2 + v_2^2 p_2^2 + v_3^2 p_3^2,$$

times the 4×4 identity. Hence the spectrum is

$$E(p) = \pm \sqrt{v_1^2 p_1^2 + v_2^2 p_2^2 + v_3^2 p_3^2},$$

with twofold degeneracy.

a. Corollary 2. The enriched shell space supports an exact massless Dirac principal operator.

D. Interpretation of the three directions

The crucial point is that the three Clifford directions now have a clear origin:

1. $\tau^3 \otimes \mathbf{1}$ comes from the diagonal energy difference of the Bragg pair, which is already present in the minimal scalar Bloch problem and is linear in the longitudinal shell deviation $p_{\parallel}$;
2. $\tau^1 \otimes s^1$ and $\tau^1 \otimes s^2$ come from pair-mixing terms whose amplitudes act nontrivially on the internal doublet.

Thus the internal doublet supplies exactly the missing two transverse Pauli directions that the plain scalar pair lacks.

E. General first-order enriched shell Hamiltonian

The most general linear Hermitian Hamiltonian on

$$\mathcal{H}_{\text{shell}} = \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2$$

may be written as

$$H_1(p) = p_i \left(a_i \tau^3 \otimes \mathbf{1} + b_i^a \tau^1 \otimes s^a + c_i^a \tau^2 \otimes s^a + d_i^a \mathbf{1} \otimes s^a \right)$$

after removing an irrelevant scalar tilt term proportional to $\mathbf{1} \otimes \mathbf{1}$.

However, the shell-pair origin suggests a more restricted structure: the pair-diagonal difference is naturally associated with τ^3 , while pair-off-diagonal mixing comes with τ^1 or τ^2 . A natural real-valued Hermitian gauge convention reduces this to

$$H_1(p) = v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1} + p_{\alpha} \tau^1 \otimes W^{\alpha}, \quad \alpha = 1, 2,$$

where W^{α} are Hermitian 2×2 matrices on the internal doublet.

a. Theorem 3 (General enriched shell criterion). Suppose the linear shell-projected Hamiltonian has the form

$$H_1(p) = v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1} + p_1 \tau^1 \otimes W^1 + p_2 \tau^1 \otimes W^2,$$

where W^1, W^2 are traceless Hermitian 2×2 matrices satisfying

$$(W^1)^2 = \mathbf{1}, \quad (W^2)^2 = \mathbf{1}, \quad \{W^1, W^2\} = 0.$$

Then $H_1(p)$ is unitarily equivalent to the massless Dirac Hamiltonian

$$v_1 p_1 \tau^1 \otimes s^1 + v_2 p_2 \tau^1 \otimes s^2 + v_3 p_{\parallel} \tau^3 \otimes \mathbf{1}.$$

b. Proof. Any pair of traceless Hermitian 2×2 matrices that square to the identity and anticommute is unitarily equivalent to s^1, s^2 . Conjugating the internal doublet basis accordingly puts W^1, W^2 into Pauli form. Then Theorem 1 applies. $\square$

F. Mass terms and what must be removed

A generic enriched shell Hamiltonian also allows constant mass terms such as

$$m_0 \tau^1 \otimes \mathbf{1}, \quad m_a \tau^1 \otimes s^a, \quad \tilde{m}_a \tau^2 \otimes s^a.$$

These generally gap the shell crossing.

a. Proposition 4. To obtain a massless Dirac point at the shell crossing, the constant pair-off-diagonal mass terms must vanish in the projected low-energy block.

b. Reason. If

$$H(0) \neq E_* \mathbf{1}_4,$$

then the shell crossing is already split or gapped at $p = 0$. A massless Dirac point requires

$$H(0) - E_* \mathbf{1}_4 = 0$$

on the projected shell sector.

This is a genuinely nontrivial condition and is not automatic in the shell Bloch problem.

G. Minimal conditional shell-Dirac theorem

We may therefore state the correct shell-level theorem.

a. Theorem 5 (Conditional enriched shell Dirac theorem). Suppose that near a shell Bragg point $k_* = G_*/2$, the Bloch-projected BCC operator admits an isolated enriched low-energy sector

$$\mathcal{H}_{\text{shell}} = \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

such that:

(S1) the constant mass terms vanish at the crossing point;

(S2) the linearized pair-diagonal splitting is nonzero along one shell direction:

$$v_{\parallel} \neq 0;$$

(S3) the linearized pair-mixing terms generate two linearly independent traceless Hermitian matrices W^1, W^2 on the internal doublet, with

$$\{W^1, W^2\} = 0, \quad (W^\alpha)^2 \propto \mathbf{1}.$$

Then the projected shell Hamiltonian is unitarily equivalent to the massless Dirac operator.

b. Proof. By (S2)–(S3), the linearized shell Hamiltonian takes Dirac form as in Theorem 3. By (S1), the crossing remains massless at $p = 0$. Hence the projected shell Hamiltonian is a massless Dirac operator. $\square$

H. What is proved and what is still open

What is proved in this section:

1. the minimal enriched shell space

$$\mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2$$

is the correct shell-level carrier of a Dirac principal symbol;

2. the plain scalar pair failure is repaired by the internal doublet;
3. the necessary and sufficient type of linear shell couplings needed for Dirac are explicitly identified.

What is still open:

1. proving that the microscopic BCC Bloch Hessian actually contains such an isolated enriched shell block;
2. proving that constant Bragg mass terms vanish or are symmetry/topology protected;
3. computing the actual transverse linear pair-mixing matrices W^α from the microscopic projection.

Thus the shell problem is now sharply localized to a finite-dimensional projected calculation.

LXXII. FINAL SUMMARY

The minimal scalar Bragg pair near the BCC shell is not sufficient to produce a Weyl cone, because its linear traceless symbol is only one-dimensional. Therefore a genuine shell-level Dirac theorem requires an enriched low-energy space. The minimal enriched shell space is

$$\mathcal{H}_{\text{shell}} = \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

where the shell pair provides one Pauli direction and the internal doublet supplies the missing two.

On this space one may define

$$\Gamma^1 = \tau^1 \otimes s^1, \quad \Gamma^2 = \tau^1 \otimes s^2, \quad \Gamma^3 = \tau^3 \otimes \mathbf{1},$$

which satisfy the Clifford relations

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij} \mathbf{1}_4.$$

Hence

$$H_{\text{Dirac}}(p) = v_1 p_1 \tau^1 \otimes s^1 + v_2 p_2 \tau^1 \otimes s^2 + v_3 p_3 \tau^3 \otimes \mathbf{1}$$

is an exact massless Dirac principal operator.

Therefore the correct shell-level conclusion is:

plain scalar Bragg pair: insufficient;

internal doublet $\otimes$ shell pair: sufficient in principle for Dirac.

What remains open is the microscopic projected calculation showing that the BCC Bloch Hessian actually produces the required isolated 4×4 shell block with vanishing constant mass and two independent transverse internal mixing directions.

LXXIII. PROJECTED BLOCH MATRIX-ELEMENT THEOREM FOR SHELL-LEVEL DIRAC EMERGENCE

Goal

We now reformulate the shell-level Dirac problem as a finite-dimensional projected matrix-element problem. The objective is to replace the vague statement

“an enriched shell block should produce Dirac”

by a precise criterion involving explicit overlap matrices.

A. Enriched shell basis

Let $k_* = G_*/2$ be the Bragg quasimomentum associated with a chosen first-shell reciprocal vector G_* . Assume that near k_* the Bloch Hessian admits an isolated low-energy shell sector of the form

$$\mathcal{H}_{\text{shell}} = \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2.$$

Choose a basis

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_{+,a}(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_{-,a}(x), \quad a = 1, 2,$$

where $u_{\pm,a}(x)$ are periodic Bloch envelopes spanning the internal doublet in the two pair sectors.

We assume the basis is orthonormal:

$$\langle \Psi_{\sigma,a}, \Psi_{\sigma',b} \rangle = \delta_{\sigma\sigma'} \delta_{ab}, \quad \sigma, \sigma' \in \{+, -\}.$$

B. Projected shell Hamiltonian

Let $\mathcal{H}_{\text{Bloch}}(k)$ be the full Bloch Hessian of the periodic BCC background. Expand around the Bragg point:

$$k = k_* + p, \quad |p| \ll |k_*|.$$

Project to the enriched shell sector:

$$H_{\text{proj}}(p) = \Pi_{\text{shell}} \mathcal{H}_{\text{Bloch}}(k_* + p) \Pi_{\text{shell}}.$$

In the ordered basis

$$(\Psi_{+,1}, \Psi_{+,2}, \Psi_{-,1}, \Psi_{-,2}),$$

the operator has 2×2 block form

$$H_{\text{proj}}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix},$$

where $E_\pm(p)$ and $B(p)$ are 2×2 matrices on the internal doublet.

a. Definition. The matrix entries are the projected Bloch overlaps

$$(E_+(p))_{ab} = \langle \Psi_{+,a}, \mathcal{H}_{\text{Bloch}}(k_* + p) \Psi_{+,b} \rangle,$$

$$(E_-(p))_{ab} = \langle \Psi_{-,a}, \mathcal{H}_{\text{Bloch}}(k_* + p) \Psi_{-,b} \rangle,$$

$$(B(p))_{ab} = \langle \Psi_{+,a}, \mathcal{H}_{\text{Bloch}}(k_* + p) \Psi_{-,b} \rangle.$$

This is the finite-dimensional shell Hamiltonian that contains all information relevant for the shell-level Dirac question.

C. Expansion near the crossing

Assume smoothness in p . Then

$$E_\pm(p) = E_\pm(0) + p_i \mathcal{E}_\pm^i + O(|p|^2),$$

$$B(p) = B_0 + p_i \mathcal{B}^i + O(|p|^2).$$

The average scalar part

$$\frac{E_+(p) + E_-(p)}{2}$$

is physically irrelevant for chirality and may be subtracted. The crucial decomposition is

$$\frac{E_+(p) - E_-(p)}{2} = v_{\parallel} p_{\parallel} \mathbf{1}_2 + p_i \Delta_i + O(|p|^2),$$

where $v_{\parallel}$ is the Bragg-pair longitudinal slope and Δ_i are possible internal corrections.

Similarly,

$$B(p) = B_0 + p_1 W^1 + p_2 W^2 + p_{\parallel} W^{\parallel} + O(|p|^2).$$

D. Reduction to the minimal Dirac shell form

The simplest shell-Dirac scenario is the one in which:

1. the pair-diagonal splitting is dominated by a scalar internal term

$$E_+(p) - E_-(p) = 2v_{\parallel} p_{\parallel} \mathbf{1}_2 + O(|p|^2);$$

2. the constant pair-mixing vanishes

$$B_0 = 0;$$

3. the transverse pair-mixing is linear and acts nontrivially on the internal doublet:

$$B(p) = p_1 W^1 + p_2 W^2 + O(|p|^2),$$

with W^1, W^2 traceless Hermitian matrices.

Under these assumptions,

$$H_{\text{proj}}(p) = E_0(p) \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1} + \tau^1 \otimes (p_1 W^1 + p_2 W^2) + O(|p|^2),$$

after choosing a real pair-mixing gauge convention.

E. Canonical internal reduction

Every traceless Hermitian 2×2 matrix can be written as

$$W^\alpha = w_i^\alpha s^i.$$

Hence the two transverse matrices W^1, W^2 define two vectors

$$w^1, w^2 \in \mathbb{R}^3.$$

a. Lemma 1. If W^1 and W^2 are linearly independent traceless Hermitian matrices, then there exists an internal unitary basis change such that

$$W^1 = \nu_1 s^1, \quad W^2 = \nu_2 (\cos \vartheta s^1 + \sin \vartheta s^2).$$

If, in addition,

$$\{W^1, W^2\} = 0,$$

then one can choose the basis so that

$$W^1 = \nu_1 s^1, \quad W^2 = \nu_2 s^2.$$

b. Proof. This is the standard classification of two traceless Hermitian 2×2 matrices under adjoint $SU(2)$ action, which acts as $SO(3)$ on the coefficient vectors w^α . $\square$

Thus anticommuting independent transverse matrices are exactly what is needed for the shell-level Dirac reduction.

F. Shell-level Dirac theorem

We can now state the main criterion.

a. Theorem 1 (Projected Bloch matrix-element Dirac theorem). Suppose the projected shell Hamiltonian satisfies:

(S1) the shell sector is an isolated 4-dimensional complex subspace

$$\mathcal{H}_{\text{shell}} = \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2;$$

(S2) after subtraction of the common reference energy,

$$E_+(0) = E_-(0), \quad B_0 = 0;$$

(S3) the linear pair-diagonal splitting is nonzero and scalar in the internal sector:

$$E_+(p) - E_-(p) = 2v_{\parallel} p_{\parallel} \mathbf{1}_2 + O(|p|^2), \quad v_{\parallel} \neq 0;$$

(S4) the pair-mixing block has transverse linear form

$$B(p) = p_1 W^1 + p_2 W^2 + O(|p|^2),$$

where W^1, W^2 are traceless Hermitian matrices satisfying

$$\{W^1, W^2\} = 0, \quad (W^1)^2 = \nu_1^2 \mathbf{1}, \quad (W^2)^2 = \nu_2^2 \mathbf{1}, \quad \nu_1 \nu_2 \neq 0.$$

Then the projected shell Hamiltonian is unitarily equivalent, at principal order, to the massless Dirac operator

$$H_{\text{proj}}(p) \sim v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1} + \nu_1 p_1 \tau^1 \otimes s^1 + \nu_2 p_2 \tau^1 \otimes s^2.$$

b. Proof. By (S2), the crossing is massless at $p = 0$. By (S3), the pair-diagonal splitting gives the $\tau^3 \otimes \mathbf{1}$ term. By (S4) and Lemma 1, the internal basis can be chosen so that

$$W^1 = \nu_1 s^1, \quad W^2 = \nu_2 s^2.$$

Hence

$$H_{\text{proj}}(p) = E_0(p) \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1} + \nu_1 p_1 \tau^1 \otimes s^1 + \nu_2 p_2 \tau^1 \otimes s^2 + O(|p|^2).$$

The three matrices

$$\tau^3 \otimes \mathbf{1}, \quad \tau^1 \otimes s^1, \quad \tau^1 \otimes s^2$$

mutually anticommute and square to the identity up to constants, so the principal operator is Dirac. $\square$

G. Matrix-element interpretation of the conditions

The load-bearing data are therefore exactly the following projected overlaps:

$$B_{0ab} = \langle \Psi_{+,a}, \mathcal{H}_{\text{Bloch}}(k_*) \Psi_{-,b} \rangle$$

$$\left(\partial_{p_{\parallel}} E_+ - \partial_{p_{\parallel}} E_- \right)_{ab} \Big|_{p=0}$$

$$W_{ab}^{\alpha} = \frac{\partial}{\partial p_{\alpha}} \langle \Psi_{+,a}, \mathcal{H}_{\text{Bloch}}(k_* + p) \Psi_{-,b} \rangle \Big|_{p=0}, \quad \alpha = 1, 2.$$

Thus the shell-level Dirac problem is reduced completely to a finite collection of matrix elements.

a. Corollary 2. To prove shell-level Dirac emergence microscopically, it is sufficient to compute the four objects

$$B_0, \quad v_{\parallel}, \quad W^1, \quad W^2,$$

and verify conditions (S2)–(S4).

H. What is not yet proved

This theorem is strong but conditional. It does *not* yet prove that the actual BCC Bloch Hessian of TECT satisfies (S1)–(S4). In particular, the following still require explicit microscopic calculation:

1. existence of the isolated enriched shell block;
2. vanishing of the constant Bragg mass B_0 ;
3. nonzero longitudinal splitting $v_{\parallel}$;
4. two independent anticommuting transverse mixing matrices W^1, W^2 .

Hence the remaining gap is no longer conceptual but entirely a projected Bloch matrix-element calculation.

LXXIV. FINAL SUMMARY

The shell-level Dirac problem can be formulated entirely as a finite-dimensional projected Bloch matrix-element problem.

Let the enriched shell space be

$$\mathcal{H}_{\text{shell}} = \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

with basis states

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_{+,a}(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_{-,a}(x), \quad a = 1, 2.$$

Then the projected shell Hamiltonian has block form

$$H_{\text{proj}}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix},$$

where all entries are 2×2 internal matrices.

A massless shell-level Dirac operator emerges if and only if, at principal order:

1. the constant pair-mixing mass vanishes,

$$B_0 = 0;$$

2. the pair-diagonal splitting is linear and nonzero,

$$E_+(p) - E_-(p) = 2v_{\parallel} p_{\parallel} \mathbf{1}_2 + O(|p|^2), \quad v_{\parallel} \neq 0;$$

3. the transverse pair-mixing has the form

$$B(p) = p_1 W^1 + p_2 W^2 + O(|p|^2),$$

with W^1, W^2 traceless Hermitian, nonzero, and anticommuting.

Under these conditions the projected shell Hamiltonian is unitarily equivalent to

$$v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1} + \nu_1 p_1 \tau^1 \otimes s^1 + \nu_2 p_2 \tau^1 \otimes s^2,$$

which is the massless Dirac principal operator.

Therefore the remaining microscopic task is completely explicit:

compute $B_0, v_{\parallel}, W^1, W^2$ from the projected BCC Bloch Hessian.

LXXV. PROJECTED BLOCH OVERLAP FORMULAE IN A MINIMAL ENRICHED PERIODIC OPERATOR

Purpose

We now choose the simplest enriched periodic operator that is capable, in principle, of realizing the shell-level Dirac conditions. The goal is not yet to prove that the full microscopic TECT operator is exactly of this form, but to reduce the shell-level Dirac problem to explicit cell-overlap integrals.

Microscopic motivation

The periodic BCC condensate background

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(Q_a \cdot x)$$

is already established as the simplest BCC-symmetric condensed pattern.

What is not yet established is the shell-valley spectrum of the full periodic Bloch Hessian. To expose the shell-level Dirac conditions, we therefore consider a minimal enriched operator acting on an internal complex doublet

$$\Phi(x) \in \mathbb{C}_{\text{int}}^2.$$

A. Minimal enriched periodic operator

Let

$$\mathcal{H} = h_0(-i\nabla) \otimes \mathbf{1}_2 + U_{\text{per}}(x) \otimes \mathbf{1}_2 + \sum_{\alpha=1}^2 \mathcal{G}_\alpha(x) (-i\partial_\alpha) \otimes s^\alpha,$$

where:

1. $h_0(q)$ is a scalar bare kernel, e.g. Brazovskii-type,

$$h_0(q) = r + (|q|^2 - q_0^2)^2;$$

2. $U_{\text{per}}(x)$ is a real periodic scalar potential with BCC periodicity,

$$U_{\text{per}}(x + \mathbf{R}) = U_{\text{per}}(x);$$

3. $\mathcal{G}_\alpha(x)$ are real periodic coefficients, also with BCC periodicity, which encode the leading gradient-induced internal mixing;
4. s^α are Pauli matrices acting on the internal doublet.

The third term is the minimal enriched coupling that can produce internal traceless mixing at first order in momentum.

B. Shell basis at the Bragg point

Let

$$k_* = \frac{G_*}{2}$$

be the Bragg quasimomentum associated with a chosen reciprocal-shell vector G_* .

Assume that there is an internal periodic doublet basis

$$u_a(x), \quad a = 1, 2,$$

orthonormal on the primitive cell:

$$\int_{\text{cell}} d^3x u_a^\dagger(x) u_b(x) = \delta_{ab}.$$

Define the shell-pair basis states

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_a(x), \quad a = 1, 2.$$

These span the enriched shell space

$$\mathcal{H}_{\text{shell}} = \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2.$$

C. Projected shell Hamiltonian

Define the projected Bloch Hamiltonian

$$H_{\text{proj}}(p) = \Pi_{\text{shell}} \mathcal{H}(k_* + p) \Pi_{\text{shell}}, \quad |p| \ll |k_*|.$$

In the ordered basis

$$(\Psi_{+,1}, \Psi_{+,2}, \Psi_{-,1}, \Psi_{-,2}),$$

it has the 2×2 block form

$$H_{\text{proj}}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix},$$

with $E_\pm(p)$ and $B(p)$ acting on the internal doublet.

D. Constant pair-mixing mass B_0

The constant pair-mixing block at $p = 0$ is

$$(B_0)_{ab} = \langle \Psi_{+,a}, \mathcal{H}(k_*) \Psi_{-,b} \rangle.$$

For the operator above, the bare translation-invariant kernel does not contribute to pair mixing, so the leading scalar contribution comes from $U_{\text{per}}(x)$:

$$(B_0)_{ab} = \int_{\text{cell}} d^3x u_a^\dagger(x) e^{-2ik_* \cdot x} U_{\text{per}}(x) u_b(x) + (B_0^{\text{grad}})_{ab}.$$

If the gradient couplings $\mathcal{G}_\alpha$ are odd under a shell-exchange symmetry or vanish at $p = 0$, then

$$B_0^{\text{grad}} = 0$$

and the dominant constant Bragg mass is exactly the Fourier coefficient of U_{per} at wavevector $2k_* = G_*$, projected onto the internal doublet.

a. Theorem 1 (Projected Bragg-mass formula). For the minimal enriched operator,

$$(B_0)_{ab} = \int_{\text{cell}} d^3x u_a^\dagger(x) e^{-iG_* \cdot x} U_{\text{per}}(x) u_b(x)$$

up to possible additional gradient-sector contributions that vanish if the first-order mixing is odd and momentum-linear.

b. Interpretation. The shell crossing is massless if and only if this projected G_* -Fourier matrix vanishes on the shell doublet:

$$B_0 = 0.$$

E. Longitudinal pair splitting $v_{\parallel}$

The pair-diagonal blocks are

$$(E_+(p))_{ab} = \langle \Psi_{+,a}, \mathcal{H}(k_* + p) \Psi_{+,b} \rangle,$$

$$(E_-(p))_{ab} = \langle \Psi_{-,a}, \mathcal{H}(k_* + p) \Psi_{-,b} \rangle.$$

The leading pair difference comes from the bare dispersion:

$$h_0(k_* + p) - h_0(-k_* + p) = 2 \nabla h_0(k_*) \cdot p + O(|p|^3).$$

Assuming the internal envelopes are the same in the two pair sectors, the pair-diagonal difference is scalar in the internal space at leading order:

$$E_+(p) - E_-(p) = 2v_{\parallel} p_{\parallel} \mathbf{1}_2 + O(|p|^2),$$

with

$$v_{\parallel} = \hat{n} \cdot \nabla h_0(k_*), \quad \hat{n} = \frac{G_*}{|G_*|}.$$

a. For the Brazovskii kernel

$$h_0(q) = r + (|q|^2 - q_0^2)^2,$$

one gets

$$\nabla h_0(q) = 4(|q|^2 - q_0^2)q,$$

hence

$$v_{\parallel} = 4 \left(\frac{|G_*|^2}{4} - q_0^2 \right) \frac{|G_*|}{2}.$$

Thus

$$v_{\parallel} \neq 0$$

unless the Bragg point lies exactly on the bare shell $|G_*| = 2q_0$.

F. Transverse internal mixing matrices W^α

Now consider the first-order internal-mixing term

$$\sum_{\alpha=1}^2 \mathcal{G}_\alpha(x) (-i\partial_\alpha) \otimes s^\alpha.$$

Its projected pair-mixing derivative at $p = 0$ defines

$$(W^\alpha)_{ab} := \left. \frac{\partial}{\partial p_\alpha} \langle \Psi_{+,a}, \mathcal{H}(k_* + p) \Psi_{-,b} \rangle \right|_{p=0}.$$

Because

$$(-i\partial_\alpha) e^{-ik_* \cdot x} u_b(x) = (-k_{*,\alpha} - i\partial_\alpha) e^{-ik_* \cdot x} u_b(x),$$

one obtains

$$(W^\alpha)_{ab} = \int_{\text{cell}} d^3x u_a^\dagger(x) e^{-iG_* \cdot x} \mathcal{G}_\alpha(x) u_b(x) s^\alpha$$

up to basis-convention terms coming from derivatives of the periodic envelopes. If the envelopes are chosen p -independently at leading order, this is the principal formula.

a. Theorem 2 (Projected transverse-mixing formula). The shell-level transverse internal mixing matrices are given, at principal order, by the cell overlaps

$$(W^\alpha)_{ab} = \int_{\text{cell}} d^3x u_a^\dagger(x) e^{-iG_* \cdot x} \mathcal{G}_\alpha(x) u_b(x) s^\alpha, \quad \alpha = 1, 2.$$

b. Interpretation. The shell-level Dirac problem is therefore reduced to the question whether the projected G_* -harmonics of the odd gradient couplings $\mathcal{G}_1, \mathcal{G}_2$ generate two independent traceless Hermitian matrices on the internal doublet.

G. Projected shell Dirac criterion in overlap form

We now collect the previous formulas.

a. Theorem 3 (Overlap-form shell Dirac criterion). Suppose:

(i) the shell basis

$$\Psi_{\pm,a}(x) = e^{\pm i k_* \cdot x} u_a(x), \quad a = 1, 2,$$

defines an isolated 4-dimensional enriched shell block;

(ii) the projected Bragg mass vanishes,

$$\int_{\text{cell}} d^3x u_a^\dagger(x) e^{-iG_* \cdot x} U_{\text{per}}(x) u_b(x) = 0 \quad \forall a, b;$$

(iii) the longitudinal slope is nonzero,

$$v_{\parallel} = \hat{n} \cdot \nabla h_0(k_*) \neq 0;$$

(iv) the projected transverse matrices

$$W^1, \quad W^2$$

defined by the cell overlaps above are traceless Hermitian, nonzero, and satisfy

$$\{W^1, W^2\} = 0, \quad (W^1)^2 = \nu_1^2 \mathbf{1}, \quad (W^2)^2 = \nu_2^2 \mathbf{1}, \quad \nu_1 \nu_2 \neq 0.$$

Then the shell-projected Hamiltonian is unitarily equivalent at principal order to

$$H_{\text{proj}}(p) \sim v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1} + \nu_1 p_1 \tau^1 \otimes s^1 + \nu_2 p_2 \tau^1 \otimes s^2,$$

hence is the massless Dirac principal operator.

b. Proof. Condition (ii) gives $B_0 = 0$. Condition (iii) produces the pair-diagonal $\tau^3 \otimes \mathbf{1}$ term. Condition (iv) allows internal unitary reduction of W^1, W^2 to Pauli form. The three resulting matrices mutually anticommute, so the principal operator is Dirac. $\square$

H. What this does and does not prove

This theorem is strong because it converts the shell-level Dirac problem into explicit cell integrals over a primitive BCC unit cell.

However, it still does not prove that the actual microscopic TECT Bloch Hessian satisfies these conditions. The remaining microscopic tasks are now exactly:

1. construct the actual internal shell envelopes $u_a(x)$;
2. compute the projected G_* -Fourier matrix of U_{per} and test $B_0 = 0$;
3. compute the projected gradient harmonics $\mathcal{G}_\alpha$ and extract W^1, W^2 ;
4. verify that the shell block is spectrally isolated.

Thus the remaining gap is no longer conceptual: it is a finite set of explicit BCC cell-overlap calculations.

LXXVI. FINAL SUMMARY

For a minimal enriched periodic operator acting on an internal doublet, the shell-level Dirac problem can be reduced entirely to projected cell-overlap integrals.

Using the shell basis

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_a(x), \quad a = 1, 2,$$

the projected shell Hamiltonian has block form

$$H_{\text{proj}}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix}.$$

The load-bearing quantities are:

$$(B_0)_{ab} = \int_{\text{cell}} d^3x \, u_a^\dagger(x) e^{-iG_* \cdot x} U_{\text{per}}(x) u_b(x),$$

$$v_{\parallel} = \hat{n} \cdot \nabla h_0(k_*), \quad \hat{n} = \frac{G_*}{|G_*|},$$

$$(W^\alpha)_{ab} = \int_{\text{cell}} d^3x \, u_a^\dagger(x) e^{-iG_* \cdot x} \mathcal{G}_\alpha(x) u_b(x) s^\alpha, \quad \alpha = 1, 2.$$

If

$$B_0 = 0, \quad v_{\parallel} \neq 0,$$

and the projected transverse matrices W^1, W^2 are nonzero traceless Hermitian matrices satisfying

$$\{W^1, W^2\} = 0,$$

then the shell-projected Hamiltonian is unitarily equivalent at principal order to

$$v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1} + \nu_1 p_1 \tau^1 \otimes s^1 + \nu_2 p_2 \tau^1 \otimes s^2,$$

which is the massless Dirac operator.

Therefore the microscopic BCC shell-Dirac problem is now reduced to a finite set of explicit primitive-cell overlap computations.

LXXVII. A SIMPLEST INTERNAL-ENVELOPE ANSATZ AND FOURIER SELECTION RULES FOR SHELL-LEVEL DIRAC EMERGENCE

Goal

We now evaluate the shell-level overlap quantities

$$B_0, \quad W^1, \quad W^2$$

in the simplest nontrivial enriched periodic ansatz.

The purpose is to determine whether the shell-level Dirac conditions can already be reduced to explicit Fourier selection rules on the primitive BCC cell.

Setup

We work with the minimal enriched periodic operator

$$\mathcal{H} = h_0(-i\nabla) \otimes \mathbf{1}_2 + U_{\text{per}}(x) \otimes \mathbf{1}_2 + \sum_{\alpha=1}^2 \mathcal{G}_\alpha(x) (-i\partial_\alpha) \otimes s^\alpha,$$

acting on an internal complex doublet $\Phi(x) \in \mathbb{C}_{\text{int}}^2$.

The periodic BCC condensate background is

$$\phi_0(x) = \sum_{G \in S_{\text{BCC}}} A_G e^{iG \cdot x}, \quad A_{-G} = \overline{A_G},$$

with first reciprocal shell S_{BCC} .

Fix a shell vector $G_* \in S_{\text{BCC}}$, and let

$$k_* = \frac{G_*}{2}$$

be the corresponding Bragg quasimomentum.

A. Simplest internal periodic-envelope ansatz

Choose a constant orthonormal internal basis

$$\xi_1, \xi_2 \in \mathbb{C}_{\text{int}}^2, \quad \xi_a^\dagger \xi_b = \delta_{ab},$$

and define periodic envelopes

$$u_1(x) = \frac{1}{\sqrt{V_{\text{cell}}}} \xi_1, \quad u_2(x) = \frac{1}{\sqrt{V_{\text{cell}}}} \xi_2,$$

where V_{cell} is the primitive cell volume.

Then the shell basis states are

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_a(x), \quad a = 1, 2.$$

a. Remark. This is the simplest enriched shell ansatz. It ignores nontrivial periodic structure of the internal Bloch envelopes, but has the advantage that the projected overlaps reduce directly to primitive-cell Fourier coefficients.

B. Projected Bragg mass B_0

By definition,

$$(B_0)_{ab} = \langle \Psi_{+,a}, \mathcal{H}(k_*) \Psi_{-,b} \rangle.$$

For the operator above, the translation-invariant bare kernel does not contribute to pair mixing. Thus the leading scalar contribution is

$$(B_0)_{ab} = \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x \xi_a^\dagger e^{-iG_* \cdot x} U_{\text{per}}(x) \xi_b.$$

Since U_{per} acts trivially on the internal doublet, this becomes

$$(B_0)_{ab} = \widehat{U}(G_*) \delta_{ab},$$

where

$$\widehat{U}(G) := \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x e^{-iG \cdot x} U_{\text{per}}(x).$$

a. Theorem 1 (Constant-envelope Bragg-mass formula). For the constant internal-envelope ansatz,

$$B_0 = \widehat{U}(G_*) \mathbf{1}_2.$$

b. Proof. Immediate from orthonormality of the constant internal basis and the fact that U_{per} is scalar in the internal sector. $\square$

C. First-shell selection rule for a quadratic scalar potential

Now suppose the periodic scalar potential is generated by the quadratic harmonic content of the first-shell condensate, for example in a Brazovskii/ ϕ^4 Hessian:

$$U_{\text{per}}(x) = 3u \phi_0(x)^2.$$

Expand

$$\phi_0(x)^2 = \sum_{G, G' \in S_{\text{BCC}}} A_G A_{G'} e^{i(G+G') \cdot x}.$$

Hence the Fourier support of U_{per} lies in the set

$$\{G + G' : G, G' \in S_{\text{BCC}}\}.$$

This contains:

$$0, \quad \pm 2Q_a, \quad \pm(Q_a \pm Q_b),$$

but generically does not contain a single first-shell vector G_* itself.

Therefore

$$\widehat{U}(G_*) = 0.$$

a. Theorem 2 (First-shell vanishing of the scalar Bragg mass). If the periodic scalar potential is generated by a quadratic function of the first-shell BCC condensate,

$$U_{\text{per}}(x) \propto \phi_0(x)^2,$$

then its first-shell Fourier coefficient vanishes:

$$\widehat{U}(G_*) = 0.$$

Consequently,

$$B_0 = 0.$$

b. Proof. The Fourier support of ϕ_0^2 is given by pairwise sums of first-shell vectors. A single first-shell vector G_* is not produced as such a sum in the generic first-shell BCC harmonic set. Hence the G_* -Fourier coefficient vanishes. $\square$

c. Interpretation. This is a strong selection rule: in the simplest scalar periodic Hessian generated from a first-shell condensate, the constant Bragg mass is removed automatically at the shell crossing.

D. Longitudinal pair splitting $v_{\parallel}$

For the constant-envelope ansatz, the pair-diagonal slopes are inherited from the bare kernel:

$$E_+(p) - E_-(p) = 2v_{\parallel} p_{\parallel} \mathbf{1}_2 + O(|p|^2),$$

with

$$v_{\parallel} = \hat{n} \cdot \nabla h_0(k_*), \quad \hat{n} = \frac{G_*}{|G_*|}.$$

For the Brazovskii kernel

$$h_0(q) = r + (|q|^2 - q_0^2)^2,$$

this gives

$$v_{\parallel} = 4 \left(\frac{|G_*|^2}{4} - q_0^2 \right) \frac{|G_*|}{2}.$$

a. Theorem 3 (Constant-envelope longitudinal slope). Under the constant-envelope ansatz,

$$\boxed{E_+(p) - E_-(p) = 2v_{\parallel} p_{\parallel} \mathbf{1}_2 + O(|p|^2)},$$

with $v_{\parallel}$ determined solely by the bare dispersion at the Bragg point.

E. Projected transverse mixing matrices W^{α}

Now consider the enriched first-order couplings

$$\mathcal{G}_{\alpha}(x) (-i\partial_{\alpha}) \otimes s^{\alpha}, \quad \alpha = 1, 2.$$

Their projected pair-mixing derivatives define

$$(W^{\alpha})_{ab} = \frac{\partial}{\partial p_{\alpha}} \langle \Psi_{+,a}, \mathcal{H}(k_* + p) \Psi_{-,b} \rangle \Big|_{p=0}.$$

For constant envelopes,

$$(W^{\alpha})_{ab} = \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x \, \xi_a^{\dagger} e^{-iG_* \cdot x} \mathcal{G}_{\alpha}(x) \xi_b s^{\alpha}.$$

Thus

$$W^{\alpha} = \widehat{\mathcal{G}}_{\alpha}(G_*) s^{\alpha},$$

where

$$\widehat{\mathcal{G}}_{\alpha}(G) := \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x \, e^{-iG \cdot x} \mathcal{G}_{\alpha}(x).$$

a. Theorem 4 (Constant-envelope transverse-mixing formula). For the constant-envelope ansatz,

$$\boxed{W^{\alpha} = \widehat{\mathcal{G}}_{\alpha}(G_*) s^{\alpha}, \quad \alpha = 1, 2.}$$

b. Proof. Again this follows from the constant internal envelopes and orthonormality of the internal basis. $\square$

F. When do W^1 and W^2 give Dirac?

Because

$$W^1 = \widehat{\mathcal{G}}_1(G_*) s^1, \quad W^2 = \widehat{\mathcal{G}}_2(G_*) s^2,$$

we automatically have

$$\{W^1, W^2\} = 0$$

whenever both coefficients are nonzero. Also

$$(W^1)^2 = \widehat{\mathcal{G}}_1(G_*)^2 \mathbf{1}, \quad (W^2)^2 = \widehat{\mathcal{G}}_2(G_*)^2 \mathbf{1}.$$

Therefore the shell-level Dirac condition reduces simply to

$$\widehat{\mathcal{G}}_1(G_*) \neq 0, \quad \widehat{\mathcal{G}}_2(G_*) \neq 0, \quad v_{\parallel} \neq 0, \quad \widehat{U}(G_*) = 0.$$

a. Corollary 5 (Fourier-coefficient shell Dirac criterion). Under the constant-envelope ansatz, the shell-level massless Dirac operator emerges if

$$\boxed{\widehat{U}(G_*) = 0, \quad v_{\parallel} \neq 0, \quad \widehat{\mathcal{G}}_1(G_*) \neq 0, \quad \widehat{\mathcal{G}}_2(G_*) \neq 0.}$$

In that case,

$$H_{\text{proj}}(p) \sim v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1} + \widehat{\mathcal{G}}_1(G_*) p_1 \tau^1 \otimes s^1 + \widehat{\mathcal{G}}_2(G_*) p_2 \tau^1 \otimes s^2.$$

b. Proof. This is an immediate application of Theorems 1–4 and the projected shell Dirac theorem. $\square$

G. Microscopic interpretation of the gradient coefficients

The coefficients $\mathcal{G}_{\alpha}(x)$ should be thought of as the periodic part of the linearized background-gradient coupling. In the simplest enriched model, they are generated by first-shell harmonics or first derivatives of the BCC condensate, for example

$$\mathcal{G}_{\alpha}(x) \propto \partial_{\alpha} \phi_0(x) \quad \text{or} \quad \mathcal{G}_{\alpha}(x) \propto \phi_0^{(1)}(x),$$

where $\phi_0^{(1)}$ denotes first-shell harmonic content.

Since $\phi_0(x)$ itself contains first-shell harmonics,

$$\phi_0(x) = \sum_{G \in S_{\text{BCC}}} A_G e^{iG \cdot x},$$

a coefficient built linearly from ϕ_0 or its first derivatives generically has

$$\widehat{\mathcal{G}}_{\alpha}(G_*) \neq 0.$$

a. Theorem 6 (Generic first-shell nonvanishing of linear gradient couplings). If $\mathcal{G}_{\alpha}(x)$ is built linearly from the first-shell condensate harmonics or their first derivatives, then generically

$$\widehat{\mathcal{G}}_{\alpha}(G_*) \neq 0$$

for $G_* \in S_{\text{BCC}}$, unless forbidden by an additional symmetry or accidental coefficient cancellation.

b. Interpretation. Thus, in the simplest enriched periodic ansatz, the dangerous constant Bragg mass vanishes by a quadratic-harmonic selection rule, while the desired transverse Dirac couplings survive because they arise linearly from first-shell gradient harmonics.

H. What is proved and what remains open

What is proved here:

1. in the simplest constant-envelope enriched shell ansatz,

$$B_0, \quad W^1, \quad W^2$$

reduce to primitive-cell Fourier coefficients at G_* ;

2. a quadratic scalar potential generated from first-shell condensate harmonics satisfies

$$B_0 = 0;$$

3. linear gradient-like enriched couplings generically produce

$$W^1 \propto s^1, \quad W^2 \propto s^2,$$

hence automatically satisfy the anticommuting internal condition.

What remains open:

1. whether the actual microscopic TECT enriched Bloch Hessian can be reduced to this same constant-envelope shell block;
2. whether the enriched couplings $\mathcal{G}_\alpha(x)$ derived from the true microscopic linearization have nonzero G_* -Fourier coefficients in the required directions;
3. whether the enriched shell block is spectrally isolated from the rest of the Bloch spectrum.

Hence the shell-level Dirac problem is now reduced to a very explicit Fourier selection problem.

LXXXVIII. FINAL SUMMARY

Using the simplest constant-envelope internal doublet ansatz

$$u_1(x) = \frac{1}{\sqrt{V_{\text{cell}}}} \xi_1, \quad u_2(x) = \frac{1}{\sqrt{V_{\text{cell}}}} \xi_2,$$

the shell-level projected quantities reduce to primitive-cell Fourier coefficients:

$$B_0 = \widehat{U}(G_*) \mathbf{1}_2,$$

$$W^\alpha = \widehat{\mathcal{G}}_\alpha(G_*) s^\alpha, \quad \alpha = 1, 2.$$

If the scalar periodic potential is generated quadratically from the first-shell BCC condensate, for example

$$U_{\text{per}}(x) \propto \phi_0(x)^2,$$

then its G_* -Fourier coefficient vanishes:

$$\widehat{U}(G_*) = 0,$$

hence

$$B_0 = 0.$$

If, on the other hand, the enriched gradient couplings $\mathcal{G}_1, \mathcal{G}_2$ are built linearly from first-shell condensate harmonics or their gradients, then generically

$$\widehat{\mathcal{G}}_1(G_*) \neq 0, \quad \widehat{\mathcal{G}}_2(G_*) \neq 0,$$

and therefore

$$W^1 \propto s^1, \quad W^2 \propto s^2,$$

which are automatically traceless, Hermitian, nonzero, and anticommuting.

Thus the shell-level massless Dirac criterion reduces simply to the Fourier-coefficient conditions

$$\widehat{U}(G_*) = 0, \quad v_{\parallel} \neq 0, \quad \widehat{\mathcal{G}}_1(G_*) \neq 0, \quad \widehat{\mathcal{G}}_2(G_*) \neq 0.$$

Therefore the remaining microscopic BCC shell-Dirac problem is now sharply localized to an explicit primitive-cell Fourier-overlap calculation.

LXXXIX. MICROSCOPIC ORIGIN OF THE GRADIENT COUPLINGS

A. Goal

The remaining condition in the shell-level Dirac criterion is that the transverse mixing matrices satisfy

$$W^\alpha \propto \widehat{\mathcal{G}}_\alpha(G_*), \quad \widehat{\mathcal{G}}_\alpha(G_*) \neq 0.$$

We now show that such coefficients arise naturally from the gradient structure of the BCC condensate.

B. BCC condensate

The first-shell BCC condensate is

$$\phi_0(x) = 2A \sum_{i=1}^6 \cos(Q_i \cdot x)$$

where

$$Q_i \in S_{\text{BCC}}.$$

In exponential form

$$\phi_0(x) = A \sum_{G \in S_{\text{BCC}}} e^{iG \cdot x}.$$

C. Gradient structure

The gradient of the condensate is

$$\nabla \phi_0(x) = -2A \sum_i Q_i \sin(Q_i \cdot x).$$

Using

$$\sin(Q \cdot x) = \frac{1}{2i} (e^{iQ \cdot x} - e^{-iQ \cdot x}),$$

we obtain

$$\nabla \phi_0(x) = \sum_{G \in S_{\text{BCC}}} c_G G e^{iG \cdot x},$$

with coefficients

$$c_G = \frac{A}{i}.$$

Therefore

$$\nabla \phi_0(x)$$

contains **exactly the same first-shell Fourier harmonics**.

D. Minimal enriched gradient coupling

Consider the minimal internal coupling

$$\mathcal{G}_\alpha(x) = \lambda \partial_\alpha \phi_0(x).$$

Then

$$\mathcal{G}_\alpha(x) = \lambda \sum_{G \in S_{\text{BCC}}} c_G G_\alpha e^{iG \cdot x}.$$

The Fourier coefficient at G_* is therefore

$$\widehat{\mathcal{G}}_\alpha(G_*) = \lambda c_{G_*} G_{*,\alpha}.$$

Hence

$$\boxed{\widehat{\mathcal{G}}_\alpha(G_*) = \frac{\lambda A}{i} G_{*,\alpha}}$$

and this is generically nonzero.

E. Transverse mixing matrices

From the overlap formula derived previously,

$$W^\alpha = \widehat{\mathcal{G}}_\alpha(G_*) s^\alpha$$

so

$$W^\alpha = \frac{\lambda A}{i} G_{*,\alpha} s^\alpha.$$

Therefore

$$W^1 \neq 0, \quad W^2 \neq 0$$

whenever the shell vector has nonzero components in those directions.

F. Anticommutation

Since

$$W^1 \propto s^1, \quad W^2 \propto s^2,$$

we have

$$\{W^1, W^2\} = 0.$$

Thus the Dirac anticommutation relations follow automatically.

G. Result

The microscopic gradient structure of the BCC condensate automatically produces nonzero transverse Dirac couplings at the Bragg shell.

LXXX. RESULT: MICROSCOPIC ORIGIN OF DIRAC TRANSVERSE COUPLINGS

If the enriched internal coupling is generated from the condensate gradient,

$$\mathcal{G}_\alpha(x) = \lambda \partial_\alpha \phi_0(x),$$

then the Fourier coefficient at the shell vector is

$$\widehat{\mathcal{G}}_\alpha(G_*) = \frac{\lambda A}{i} G_{*,\alpha}.$$

Hence

$$W^\alpha = \widehat{\mathcal{G}}_\alpha(G_*) s^\alpha$$

is generically nonzero.

Together with the previously established selection rule

$$B_0 = 0$$

and the generic longitudinal slope

$$v_{\parallel} \neq 0,$$

the shell-level projected Hamiltonian reduces to

$$H_{\text{proj}}(p) = v_{\parallel} p_{\parallel} \tau^3 + \nu_1 p_1 \tau^1 s^1 + \nu_2 p_2 \tau^1 s^2$$

which is the massless Dirac operator.

Therefore the microscopic gradient structure of the BCC condensate naturally produces the transverse couplings required for Dirac emergence.

LXXXI. CORRECTED FOURIER-SUPPORT THEOREM FOR THE ENRICHED GRADIENT COUPLINGS

Why a correction is needed

In the previous heuristic argument, one might be tempted to write

$$\mathcal{G}_\alpha(x) = \lambda \partial_\alpha \phi_0(x)$$

and then conclude immediately that

$$\widehat{\mathcal{G}}_\alpha(G_*) \neq 0.$$

This is too strong as stated.

The reason is that, in the shell-level Dirac reduction, the labels $\alpha = 1, 2$ naturally denote the two momentum directions transverse to the shell vector G_* . If one literally identifies ∂_α with differentiation along those transverse directions, then

$$\widehat{\partial_\alpha \phi_0}(G_*) \propto G_* \cdot e_\alpha = 0,$$

because $e_\alpha \perp G_*$.

Therefore the correct statement is not that the needed transverse couplings arise automatically from $\partial_\alpha \phi_0$ alone, but rather that any enriched coupling which is *linear* in the first-shell condensate or its first derivatives has first-shell Fourier support. The actual Dirac question is whether that support survives the shell projection into two independent internal traceless matrices.

A. First-shell Fourier support of the BCC condensate

The BCC condensate is

$$\phi_0(x) = \sum_{G \in S_{\text{BCC}}} A_G e^{iG \cdot x}, \quad A_{-G} = \overline{A_G}.$$

Thus its Fourier support is exactly the first reciprocal shell:

$$\text{supp } \widehat{\phi_0} \subseteq S_{\text{BCC}}.$$

Since the BCC reciprocal shell is the cuboctahedral 12-vector star, with resonance count $\Lambda_{\text{BCC}} = 540$, the first-shell structure is already fixed by the BCC selection theory.

a. Proposition 1. If

$$\mathcal{T}(x) = c \phi_0(x),$$

then

$$\widehat{\mathcal{T}}(G) = c A_G,$$

so $\mathcal{T}$ has the same first-shell support as ϕ_0 .

b. Proof. Immediate from linearity of the Fourier transform. $\square$

B. First-shell Fourier support of first-derivative couplings

Now consider a coefficient field linear in the first derivatives of the condensate:

$$\mathcal{T}_j(x) = \sum_m C_{jm} \partial_m \phi_0(x),$$

with constant coefficients C_{jm} .

Because

$$\partial_m \phi_0(x) = \sum_{G \in S_{\text{BCC}}} iG_m A_G e^{iG \cdot x},$$

we obtain

$$\widehat{\mathcal{T}}_j(G) = i \sum_m C_{jm} G_m A_G.$$

a. Theorem 1 (First-shell support for linear derivative couplings). Any coefficient field linear in the first derivatives of the BCC condensate,

$$\mathcal{T}_j(x) = \sum_m C_{jm} \partial_m \phi_0(x),$$

has Fourier support contained in the first shell:

$$\text{supp } \widehat{\mathcal{T}}_j \subseteq S_{\text{BCC}}.$$

Its shell Fourier coefficient is

$$\boxed{\widehat{\mathcal{T}}_j(G_*) = i \left(\sum_m C_{jm} G_{*,m} \right) A_{G_*}.}$$

b. Proof. Differentiation multiplies each harmonic $e^{iG \cdot x}$ by iG_m but does not change the wavevector G . Hence the support remains first-shell. $\square$

c. Important consequence. This proves first-shell support, but not automatic nonvanishing for every j . Vanishing can still occur if

$$\sum_m C_{jm} G_{*,m} = 0.$$

Thus one needs a coefficient tensor C_{jm} whose action on G_* is nonzero in at least the relevant shell-projected channels.

C. Constant-envelope shell projection

For the simplest shell ansatz,

$$u_a(x) = \frac{1}{\sqrt{V_{\text{cell}}}} \xi_a, \quad \Psi_{\pm,a}(x) = e^{\pm i k_* \cdot x} u_a(x), \quad k_* = \frac{G_*}{2},$$

the projected shell overlaps reduce to primitive-cell Fourier coefficients.

As shown previously, the constant Bragg mass is

$$B_0 = \widehat{U}(G_*) \mathbf{1}_2.$$

If the scalar periodic potential is quadratic in the first-shell condensate,

$$U_{\text{per}}(x) \propto \phi_0(x)^2,$$

then

$$\widehat{U}(G_*) = 0$$

by the first-shell quadratic selection rule, hence

$$B_0 = 0.$$

For an enriched shell coupling of the form

$$\sum_{\alpha=1}^2 \mathcal{T}^\alpha(x) (-i\partial_{p_\alpha}) \otimes s^\alpha,$$

the projected matrices are

$$W^\alpha = \widehat{\mathcal{T}}^\alpha(G_*) s^\alpha$$

under the constant-envelope ansatz.

a. Theorem 2 (Constant-envelope enriched shell formula). Let $\mathcal{T}^\alpha(x)$ be scalar periodic coefficient fields multiplying two fixed internal Pauli matrices s^α , $\alpha = 1, 2$. Then

$$\boxed{W^\alpha = \widehat{\mathcal{T}}^\alpha(G_*) s^\alpha.}$$

Hence the shell-level Dirac problem reduces to the two numbers

$$\widehat{\mathcal{T}}^1(G_*), \quad \widehat{\mathcal{T}}^2(G_*).$$

b. Proof. This is the previously derived shell-overlap formula specialized to the constant-envelope basis. $\square$

D. Correct shell-level Dirac criterion in Fourier form

Combining the above, we obtain the corrected criterion.

a. Theorem 3 (Corrected Fourier shell-Dirac criterion). Under the constant-envelope enriched shell ansatz, a shell-level massless Dirac operator emerges if:

(i) the scalar Bragg mass vanishes,

$$\widehat{U}(G_*) = 0;$$

(ii) the longitudinal pair slope is nonzero,

$$v_{\parallel} = \hat{n} \cdot \nabla h_0(k_*) \neq 0;$$

(iii) the two enriched coefficient fields $\mathcal{T}^1, \mathcal{T}^2$ have nonzero first-shell Fourier coefficients,

$$\widehat{\mathcal{T}}^1(G_*) \neq 0, \quad \widehat{\mathcal{T}}^2(G_*) \neq 0.$$

Then

$$H_{\text{proj}}(p) \sim v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1} + \widehat{\mathcal{T}}^1(G_*) p_1 \tau^1 \otimes s^1 + \widehat{\mathcal{T}}^2(G_*) p_2 \tau^1 \otimes s^2.$$

b. Proof. Item (i) gives $B_0 = 0$. Item (ii) gives the pair-diagonal $\tau^3 \otimes \mathbf{1}$ term. Item (iii) gives two nonzero traceless Hermitian transverse matrices

$$W^1 = \widehat{\mathcal{T}}^1(G_*) s^1, \quad W^2 = \widehat{\mathcal{T}}^2(G_*) s^2,$$

which automatically anticommute. Hence the principal shell Hamiltonian is Dirac. $\square$

E. What is actually proved and what remains open

What is proved:

1. any coupling linear in ϕ_0 or $\partial\phi_0$ has first-shell Fourier support;
2. the dangerous scalar Bragg mass vanishes for quadratic first-shell scalar potentials;
3. the shell-level Dirac criterion reduces to nonvanishing of two shell-projected enriched first-shell coefficients.

What remains open:

1. the actual microscopic TECT linearization has not yet been shown to produce the enriched coefficient fields $\mathcal{T}^1, \mathcal{T}^2$ with the required internal Pauli assignment;
2. even when first-shell support is present, accidental cancellations or symmetry constraints may still force

$$\widehat{\mathcal{T}}^{\alpha}(G_*) = 0$$

for one or both α ;

3. the enriched shell block must still be shown to be spectrally isolated.

Thus the remaining shell-level problem is now a sharply defined Fourier bookkeeping and spectral isolation problem.

LXXXII. FINAL SUMMARY

The previous heuristic claim that

$$\mathcal{G}_{\alpha}(x) = \lambda \partial_{\alpha} \phi_0(x)$$

automatically implies the nonvanishing of both shell-level transverse Dirac couplings was too strong and must be corrected.

What is rigorously true is the following:

1. the BCC condensate

$$\phi_0(x) = \sum_{G \in S_{\text{BCC}}} A_G e^{iG \cdot x}$$

has support exactly on the first reciprocal shell;

2. any coefficient field linear in ϕ_0 or its first derivatives has Fourier support still contained in that same first shell;
3. under the constant-envelope shell ansatz, the shell-level transverse matrices are

$$W^\alpha = \widehat{\mathcal{T}}^\alpha(G_*) s^\alpha,$$

so the shell-level Dirac problem reduces to whether two enriched coefficient fields $\mathcal{T}^1, \mathcal{T}^2$ have nonzero G_* -Fourier coefficients;

4. the scalar Bragg mass still vanishes by the quadratic first-shell selection rule:

$$\widehat{U}(G_*) = 0.$$

Therefore the corrected shell-level Dirac criterion is

$$\widehat{U}(G_*) = 0, \quad v_{\parallel} \neq 0, \quad \widehat{\mathcal{T}}^1(G_*) \neq 0, \quad \widehat{\mathcal{T}}^2(G_*) \neq 0.$$

The remaining microscopic task is thus sharply localized: identify the actual enriched coefficient fields produced by the TECT linearization, compute their G_* -Fourier coefficients, and prove shell-block isolation.

LXXXIII. CLASSIFICATION OF ALLOWED ENRICHED COUPLINGS FOR SHELL-LEVEL DIRAC EMERGENCE

Goal

The remaining shell-level Dirac conditions have been reduced to the nonvanishing of two projected enriched Fourier coefficients:

$$\widehat{\mathcal{T}}^1(G_*) \neq 0, \quad \widehat{\mathcal{T}}^2(G_*) \neq 0.$$

We now classify the minimal allowed forms of the enriched coefficient fields

$$\mathcal{T}^\alpha(x), \quad \alpha = 1, 2,$$

and determine which classes can or cannot contribute at the first reciprocal shell.

Background

The BCC condensate is

$$\phi_0(x) = \sum_{G \in S_{\text{BCC}}} A_G e^{iG \cdot x}, \quad A_{-G} = \overline{A_G},$$

with support exactly on the first reciprocal shell:

$$\text{supp } \widehat{\phi}_0 \subseteq S_{\text{BCC}}.$$

Thus any candidate enriched coefficient field built from ϕ_0 and finitely many derivatives inherits Fourier support from the algebra of these first-shell harmonics.

The shell-level Dirac criterion in the constant-envelope ansatz takes the form

$$\widehat{U}(G_*) = 0, \quad v_{\parallel} \neq 0, \quad \widehat{\mathcal{T}}^1(G_*) \neq 0, \quad \widehat{\mathcal{T}}^2(G_*) \neq 0.$$

The scalar Bragg mass condition $\widehat{U}(G_*) = 0$ has already been controlled by a quadratic first-shell selection rule. The present question is therefore entirely about the enriched coefficient fields $\mathcal{T}^\alpha$.

A. General algebraic classification

Let $\mathfrak{A}[\phi_0]$ denote the algebra generated by ϕ_0 and its spatial derivatives. A generic enriched coefficient field may be expanded as a finite sum of monomials in

$$\phi_0, \quad \partial_j \phi_0, \quad \partial_j \partial_k \phi_0, \quad \dots$$

with tensor coefficients.

For shell-level Dirac emergence, the lowest-order and most relevant possibilities are:

(L0) linear scalar coupling:

$$\mathcal{T}^\alpha(x) = c^\alpha \phi_0(x);$$

(L1) linear derivative coupling:

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x);$$

(Q0) quadratic scalar coupling:

$$\mathcal{T}^\alpha(x) = d^\alpha \phi_0(x)^2;$$

(Q1) mixed quadratic derivative coupling:

$$\mathcal{T}^\alpha(x) = D^\alpha_j \phi_0(x) \partial_j \phi_0(x);$$

(Q2) quadratic gradient bilinear:

$$\mathcal{T}^\alpha(x) = E^\alpha_{jk} \partial_j \phi_0(x) \partial_k \phi_0(x).$$

Higher-order monomials may also be considered, but these are enough to establish the main shell selection rules.

B. Fourier-support rules

We now determine the shell content of each class.

a. Proposition 1 (Linear shell-support rule). If $\mathcal{T}^\alpha$ is linear in ϕ_0 or in its derivatives, then

$$\text{supp } \widehat{\mathcal{T}^\alpha} \subseteq S_{\text{BCC}}.$$

b. Proof. For $\mathcal{T}^\alpha \propto \phi_0$, the claim is immediate. For

$$\mathcal{T}^\alpha = C^\alpha_j \partial_j \phi_0,$$

differentiation multiplies each harmonic $e^{iG \cdot x}$ by iG_j but does not change its wavevector. Hence the support remains on the first shell. $\square$

Thus the linear classes (L0) and (L1) are favorable for shell-level Dirac emergence.

c. Proposition 2 (Quadratic shell-support rule). If $\mathcal{T}^\alpha$ is quadratic in ϕ_0 and its first derivatives, then

$$\text{supp } \widehat{\mathcal{T}^\alpha} \subseteq S_{\text{BCC}} + S_{\text{BCC}} = \{G + G' : G, G' \in S_{\text{BCC}}\}.$$

In particular, such quadratic classes generally do not automatically contain a single first-shell vector G_* .

d. Proof. Products of two first-shell harmonics produce wavevectors equal to sums of first-shell vectors. Thus the support is contained in the Minkowski sum $S_{\text{BCC}} + S_{\text{BCC}}$. $\square$

This is the same mechanism that killed the scalar Bragg mass in the quadratic potential: a quadratic function of first-shell harmonics does not generically produce a single first-shell wavevector.

Hence the quadratic classes (Q0)–(Q2) are generically unfavorable for producing $\widehat{\mathcal{T}^\alpha}(G_*) \neq 0$.

C. Consequences for shell-level Dirac

We immediately obtain the first classification statement.

a. Theorem 1 (Allowed-coupling classification theorem). At the minimal shell level, the enriched couplings that can generically support the shell-Dirac conditions

$$\widehat{\mathcal{T}}^1(G_*) \neq 0, \quad \widehat{\mathcal{T}}^2(G_*) \neq 0$$

must contain a component linear in the first-shell condensate ϕ_0 or its first derivatives. Purely quadratic enriched couplings are generically insufficient.

b. Proof. By Propositions 1 and 2, only the linear classes are guaranteed to retain first-shell Fourier support. Quadratic classes shift support to sums of shell vectors and therefore generally do not produce nonzero G_* -Fourier coefficients. $\square$

D. Refinement: when derivative couplings vanish

We now analyze the most relevant class:

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x).$$

Its shell Fourier coefficient is

$$\widehat{\mathcal{T}}^\alpha(G_*) = i(C^\alpha \cdot G_*) A_{G_*},$$

where

$$C^\alpha \cdot G_* := \sum_j C^\alpha_j G_{*,j}.$$

This gives a sharp criterion.

a. Theorem 2 (Derivative-coupling nonvanishing criterion). Suppose

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x).$$

Then

$$\boxed{\widehat{\mathcal{T}}^\alpha(G_*) \neq 0 \iff (C^\alpha \cdot G_*) A_{G_*} \neq 0.}$$

b. Proof. Immediate from the Fourier coefficient formula above. $\square$

c. Important consequence. This shows precisely why the naive choice

$$\mathcal{T}^\alpha = \partial_\alpha \phi_0$$

is not generally sufficient when α labels directions transverse to G_* : in that case

$$G_{*,\alpha} = 0$$

may occur, forcing

$$\widehat{\mathcal{T}}^\alpha(G_*) = 0.$$

Therefore a generic shell-Dirac construction requires a more general tensorial coupling

$$C^\alpha_j$$

that mixes the shell-transverse labels $\alpha = 1, 2$ with the actual spatial derivative index j .

E. Strong sufficient condition

We can now state the most useful sufficient criterion.

a. Theorem 3 (Strong sufficient shell-coupling criterion). Suppose the enriched coefficient fields are

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x), \quad \alpha = 1, 2,$$

and that

$$A_{G_*} \neq 0, \quad C^1 \cdot G_* \neq 0, \quad C^2 \cdot G_* \neq 0.$$

Then

$$\widehat{\mathcal{T}^1}(G_*) \neq 0, \quad \widehat{\mathcal{T}^2}(G_*) \neq 0.$$

Under the constant-envelope shell ansatz, this implies

$$W^1 = \widehat{\mathcal{T}^1}(G_*) s^1 \neq 0, \quad W^2 = \widehat{\mathcal{T}^2}(G_*) s^2 \neq 0,$$

and therefore the two transverse shell-Dirac matrices automatically anticommute.

b. Proof. By Theorem 2, the Fourier coefficients are nonzero whenever $C^\alpha \cdot G_* \neq 0$ and $A_{G_*} \neq 0$. The shell-overlap formula then gives

$$W^\alpha = \widehat{\mathcal{T}^\alpha}(G_*) s^\alpha.$$

Since s^1 and s^2 anticommute, so do W^1 and W^2 . $\square$

F. Symmetry-based obstructions

It is equally important to identify what can kill the needed couplings.

a. Proposition 4 (Symmetry obstruction). If the enriched coefficient field $\mathcal{T}^\alpha$ transforms under a symmetry in such a way that its G_* -Fourier component is forbidden, then

$$\widehat{\mathcal{T}^\alpha}(G_*) = 0.$$

In particular, any symmetry forcing $C^\alpha \cdot G_* = 0$ kills the corresponding shell-Dirac matrix W^α .

b. Interpretation. Thus one should not demand too strong a transverse-selection symmetry at the shell point, or the desired Dirac couplings may be removed together with unwanted terms.

G. Minimal microscopic target

The classification above shows that the actual microscopic task is now very specific.

To prove shell-level Dirac emergence, it is sufficient to identify two enriched microscopic coefficient fields $\mathcal{T}^1, \mathcal{T}^2$ such that:

1. each contains a linear first-shell component,
2. neither has its G_* -Fourier coefficient removed by symmetry,
3. their shell projections are attached to two distinct internal Pauli directions.

This leads to the final theorem of this step.

a. Theorem 5 (Minimal enriched-coupling target theorem). The remaining microscopic shell-level Dirac problem is reduced to the following finite question:

Find two enriched coefficient fields

$$\mathcal{T}^1(x), \quad \mathcal{T}^2(x),$$

allowed by the microscopic TECT linearization, such that

$$\widehat{\mathcal{T}^1}(G_*) \neq 0, \quad \widehat{\mathcal{T}^2}(G_*) \neq 0,$$

and such that their shell projections couple to two independent traceless internal matrices.

Once this is achieved, the transverse shell-Dirac structure is established.

b. Proof. This is the direct content of Theorem 3 together with the previously established shell-projected Dirac theorem. $\square$

LXXXIV. FINAL SUMMARY

The shell-level Dirac problem has now been reduced to a classification problem for the enriched coefficient fields $\mathcal{T}^\alpha(x)$.

The main conclusions are:

1. couplings linear in the first-shell condensate ϕ_0 or its first derivatives retain first-shell Fourier support and are therefore viable candidates for shell-Dirac emergence;
2. purely quadratic enriched couplings are generically unfavorable, because they shift Fourier support to sums of shell vectors and need not produce a nonzero G_* -harmonic;
3. for derivative-type enriched couplings

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x),$$

the shell coefficient is explicitly

$$\widehat{\mathcal{T}}^\alpha(G_*) = i(C^\alpha \cdot G_*) A_{G_*};$$

4. therefore the shell-level transverse Dirac matrices are nonzero exactly when the coupling tensors satisfy

$$C^1 \cdot G_* \neq 0, \quad C^2 \cdot G_* \neq 0.$$

Thus the remaining microscopic problem is no longer vague. It is to identify two enriched microscopic coefficient fields allowed by the TECT linearization whose G_* -Fourier coefficients survive projection and couple to two independent internal traceless matrices.

This is the sharpest currently honest formulation of the remaining shell-level Dirac task.

LXXXV. SYMMETRY CLASSIFICATION OF ALLOWED ENRICHED COUPLINGS FOR SHELL-LEVEL DIRAC EMERGENCE

Goal

The remaining shell-level Dirac condition has been reduced to the existence of two enriched coefficient fields

$$\mathcal{T}^1(x), \quad \mathcal{T}^2(x),$$

whose shell Fourier coefficients survive at G_* and project to two independent traceless internal matrices.

We now classify which forms of $\mathcal{T}^\alpha(x)$ are allowed or forbidden by the relevant symmetries.

A. Data of the shell problem

We assume the following ingredients are already fixed:

1. a periodic BCC condensate

$$\phi_0(x) = \sum_{G \in S_{\text{BCC}}} A_G e^{iG \cdot x};$$

2. a chosen shell vector $G_* \in S_{\text{BCC}}$ and Bragg point

$$k_* = \frac{G_*}{2};$$

3. an enriched shell space

$$\mathcal{H}_{\text{shell}} = \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2;$$

4. a shell Hamiltonian whose transverse part is controlled by two enriched coefficient fields

$$\mathcal{T}^\alpha(x), \quad \alpha = 1, 2.$$

The shell-level Dirac condition requires

$$\widehat{\mathcal{T}}^1(G_*) \neq 0, \quad \widehat{\mathcal{T}}^2(G_*) \neq 0,$$

with their shell projections attached to two independent traceless Hermitian internal matrices.

B. Relevant symmetries

The relevant symmetries are:

1. lattice translation symmetry;
2. shell-exchange / Bragg reflection symmetry;
3. cubic point-group symmetry of the BCC reciprocal star;
4. internal $U(1)$ or $SU(2)$ -type symmetry acting on the internal doublet.

We now analyze their consequences one by one.

C. Translation symmetry

Because the background is periodic, any allowed coefficient field must be periodic with the same lattice:

$$\mathcal{T}^\alpha(x + \mathbf{R}) = \mathcal{T}^\alpha(x).$$

Hence $\mathcal{T}^\alpha$ admits a reciprocal-lattice Fourier expansion

$$\mathcal{T}^\alpha(x) = \sum_{G \in \mathcal{G}} \widehat{\mathcal{T}}^\alpha(G) e^{iG \cdot x}.$$

a. Proposition 1 (Translation constraint). Translation symmetry does not forbid a shell-level Dirac coupling by itself. It only implies that the coefficient fields are expanded in reciprocal-lattice harmonics, so the shell problem is controlled by the single Fourier coefficient

$$\widehat{\mathcal{T}}^\alpha(G_*).$$

b. Proof. This is the direct consequence of lattice periodicity and Bloch theory. $\square$

Thus translation symmetry is neutral with respect to shell-Dirac emergence: it localizes the problem to a reciprocal-shell Fourier coefficient, but does not force it to vanish.

D. Shell exchange and reflection symmetry

Let the pair sector be spanned by the shell pair basis

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_a(x).$$

The natural shell-exchange involution interchanges the two pair sectors:

$$\mathbf{J} : \Psi_{+,a} \leftrightarrow \Psi_{-,a}.$$

The shell Hamiltonian decomposes as

$$H_{\text{proj}}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix}.$$

The constant Bragg mass $B_0 = B(0)$ is even under momentum reversal and is the dangerous term that gaps the shell crossing. The desired transverse Dirac couplings, by contrast, must be linear in the small deviation p and are therefore odd under $p \rightarrow -p$.

a. Proposition 2 (Reflection-parity separation). A shell symmetry that removes pair-even shell mixing at $p = 0$ can kill the constant Bragg mass B_0 while still allowing linear pair-mixing terms $p_\alpha W^\alpha$, provided the underlying enriched coefficient fields transform with the appropriate odd shell parity.

b. Interpretation. Thus shell reflection/exchange symmetry is favorable if it removes constant mixing but preserves linear odd mixing. It becomes unfavorable only if imposed too strongly, in which case the linear transverse terms may also be eliminated.

E. Cubic symmetry and the transverse doublet

Fix the shell direction

$$\hat{n} = \frac{G_*}{|G_*|}.$$

The tangent plane orthogonal to $\hat{n}$ carries a natural 2-dimensional transverse representation. Let $\alpha = 1, 2$ label a basis of this transverse plane.

The point-group symmetry preserving the shell direction acts on the transverse plane by a real 2×2 representation. Therefore the pair of coefficient fields

$$(\mathcal{T}^1, \mathcal{T}^2)$$

must transform as a transverse doublet.

a. Theorem 1 (Cubic-transverse representation constraint). For shell-level Dirac emergence, the enriched couplings must transform as a transverse doublet under the little group of the shell direction G_* . In particular, the most natural allowed linear couplings are of the form

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x),$$

where C^α_j is an intertwining tensor from the spatial vector representation to the transverse-shell doublet.

b. Proof. The shell Hamiltonian requires two linearly independent transverse momentum directions p_1, p_2 , so the coefficient fields must transform in the same 2-dimensional transverse representation. The only minimal linear spatial object available from the condensate is $\partial_j \phi_0$, hence the coefficient tensor C^α_j is the minimal intertwiner. $\square$

This identifies the most natural minimal shell-Dirac coupling.

F. Internal symmetry

Now consider the internal doublet sector $\mathbb{C}_{\text{int}}^2$. The projected transverse matrices must be traceless Hermitian and independent:

$$W^1, \quad W^2.$$

To get Dirac, they must be reducible to two anticommuting Pauli generators. Therefore the internal sector must allow at least two independent traceless Hermitian generators.

a. Theorem 2 (Minimal internal symmetry requirement). A necessary internal condition for shell-level Dirac emergence is that the internal low-energy sector admit a complex doublet structure with at least two independent traceless Hermitian generators. Equivalently, the internal sector must support a local $SU(2)$ -type Pauli algebra.

b. Proof. The shell-level Dirac theorem requires two transverse matrices W^1, W^2 satisfying

$$\{W^1, W^2\} = 0, \quad (W^\alpha)^2 \propto \mathbf{1}.$$

Such a pair exists if and only if the internal sector supports two independent traceless Hermitian 2×2 generators, i.e. a Pauli algebra. $\square$

c. Interpretation. A purely scalar internal sector cannot produce shell-level Dirac. The minimal internal carrier is a complex doublet.

G. Classification of allowed coupling types

We now combine the Fourier-support analysis with the symmetry constraints.

a. *Class A: linear scalar coupling*

$$\mathcal{T}^\alpha(x) = c^\alpha \phi_0(x).$$

- Fourier support: favorable (first shell survives).
- Cubic-transverse structure: generally unfavorable, because this does not naturally transform as a transverse doublet unless supplemented by an external tensor structure.
- Internal structure: must still be attached to a nontrivial internal generator.

Thus this class is only conditionally useful.

b. *Class B: linear derivative coupling*

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x).$$

- Fourier support: favorable.
- Little-group transformation: favorable; this is the natural transverse-doublet candidate.
- Internal structure: favorable once attached to s^α .

This is the most natural minimal shell-Dirac coupling class.

c. *Class C: quadratic scalar or gradient couplings*

$$\mathcal{T}^\alpha \sim \phi_0^2, \quad \phi_0 \partial \phi_0, \quad (\partial \phi_0)^2.$$

- Fourier support: generally unfavorable for first-shell survival.
- Shell-level Dirac: generically suppressed unless additional fine-tuned cancellations or higher-order resonance effects restore the shell coefficient.

Thus purely quadratic enriched couplings are not minimal shell-Dirac candidates.

d. *Theorem 3 (Allowed-coupling classification).* Among low-order enriched coefficient fields built from the BCC condensate, the only generic minimal class simultaneously compatible with shell Fourier support, transverse doublet structure, and internal Pauli coupling is

$$\boxed{\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x) \otimes s^\alpha, \quad \alpha = 1, 2.}$$

e. *Proof.* Linear derivative couplings retain first-shell support, transform naturally as spatial vectors, and can be converted into transverse-doublet objects by the intertwiner C^α_j . When attached to internal Pauli generators s^α , they produce the required independent traceless internal matrices. The other low-order classes fail at least one of these requirements generically. $\square$

H. Nonvanishing criterion

For the minimal allowed class,

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x) \otimes s^\alpha,$$

the shell Fourier coefficient is

$$\widehat{\mathcal{T}}^\alpha(G_*) = i(C^\alpha \cdot G_*) A_{G_*} s^\alpha.$$

Hence:

a. *Theorem 4 (Nonvanishing criterion for the minimal allowed class).* Suppose

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x) \otimes s^\alpha.$$

Then

$$\widehat{\mathcal{T}}^\alpha(G_*) \neq 0 \iff (C^\alpha \cdot G_*) A_{G_*} \neq 0.$$

Therefore shell-level Dirac emerges provided

$$A_{G_*} \neq 0, \quad C^1 \cdot G_* \neq 0, \quad C^2 \cdot G_* \neq 0,$$

together with the already-established conditions

$$\widehat{U}(G_*) = 0, \quad v_{\parallel} \neq 0.$$

b. Proof. This follows from the Fourier formula for derivative couplings together with the shell projection formula $W^\alpha = \widehat{\mathcal{T}}^\alpha(G_*)$. $\square$

I. Symmetry obstructions

The same formula identifies the obstructions.

a. Proposition 5 (Obstruction table). The shell-Dirac transverse couplings are killed if any of the following hold:

1. the condensate amplitude vanishes at the selected shell:

$$A_{G_*} = 0;$$

2. the tensor intertwiners satisfy

$$C^\alpha \cdot G_* = 0;$$

3. an internal symmetry reduces the shell projection to fewer than two independent traceless generators;
4. an accidental cancellation removes the G_* -harmonic in the enriched coefficient field.

b. Interpretation. Thus the symmetry requirements must be balanced carefully: enough symmetry to control B_0 , but not so much that all transverse shell couplings are forced to vanish.

J. Minimal microscopic target

We can now formulate the remaining microscopic task in its sharpest form.

a. Theorem 5 (Minimal microscopic target theorem). To close the shell-level Dirac problem, it is sufficient to identify in the TECT linearization two allowed enriched coefficient fields of the form

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x) \otimes s^\alpha, \quad \alpha = 1, 2,$$

such that

$$C^1 \cdot G_* \neq 0, \quad C^2 \cdot G_* \neq 0,$$

and such that the enriched shell block is spectrally isolated.

Once this is achieved, the shell-projected Hamiltonian is Dirac.

b. Proof. Theorem 4 gives the transverse Dirac couplings, while the earlier shell-level Dirac theorem uses

$$\widehat{U}(G_*) = 0, \quad v_\parallel \neq 0$$

to complete the Clifford triple. Spectral isolation ensures the validity of the shell projection. $\square$

LXXXVI. FINAL SUMMARY

The shell-level Dirac problem is now reduced to a symmetry-classification problem for the enriched coefficient fields $\mathcal{T}^\alpha(x)$.

The main conclusions are:

1. translation symmetry only localizes the problem to the G_* -Fourier coefficient;
2. shell reflection/exchange symmetry is useful if it removes the constant Bragg mass B_0 without killing the linear transverse pair-mixing;
3. cubic symmetry implies that the transverse couplings must transform as a two-dimensional transverse-shell doublet;

4. the internal sector must support at least two independent traceless Hermitian generators, i.e. a local $SU(2)$ -type Pauli algebra.

Among low-order couplings, the generic minimal allowed class is

$$\boxed{\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x) \otimes s^\alpha, \quad \alpha = 1, 2.}$$

Its shell Fourier coefficient is

$$\widehat{\mathcal{T}}^\alpha(G_*) = i(C^\alpha \cdot G_*) A_{G_*} s^\alpha.$$

Therefore the remaining microscopic shell-Dirac condition is simply

$$A_{G_*} \neq 0, \quad C^1 \cdot G_* \neq 0, \quad C^2 \cdot G_* \neq 0,$$

together with the previously established conditions

$$\widehat{U}(G_*) = 0, \quad v_{\parallel} \neq 0,$$

and spectral isolation of the enriched shell block.

This is the sharpest currently honest formulation of the remaining shell-level Dirac task.

LXXXVII. MICROSCOPIC TENSOR REDUCTION THEOREM FOR SHELL-TRANSVERSE ENRICHED COUPLINGS

Goal

The remaining shell-level Dirac condition was reduced to the existence of two enriched coefficient fields of the form

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x) \otimes s^\alpha, \quad \alpha = 1, 2.$$

We now show how such fields arise from the most general gradient-linear microscopic tensor coupling, once a shell direction is chosen and the spatial tensor structure is reduced to the transverse shell doublet.

Logical status

What is proved here is a *reduction theorem*: if the microscopic linearization contains a gradient-linear enriched tensor coupling, then its shell-transverse projection necessarily takes the standard shell-Dirac form above.

What is *not* proved automatically is that the relevant microscopic tensor coefficients are nonzero or generic enough to satisfy the final nonvanishing conditions.

A. General gradient-linear enriched coupling

Let the microscopic or effective TECT linearization around the periodic condensate contain a first-order enriched term of the schematic form

$$\delta \mathcal{L}_{\text{enr}} = \mathbf{M}_{ij}^A (\partial_i \phi_0(x)) (-i \partial_j) \otimes S_A,$$

where:

1. $i, j = 1, 2, 3$ are spatial indices;
2. A labels Hermitian generators acting on the internal low-energy sector;
3. $\mathbf{M}_{ij}^A$ is a real microscopic tensor coefficient.

Assume the internal low-energy sector reduces to a complex doublet, so that its traceless Hermitian generators are spanned by Pauli matrices s^a , $a = 1, 2, 3$. Then

$$S_A = u_A \mathbf{1}_2 + v_A^a s^a.$$

Only the traceless part contributes to shell-level Dirac structure, so we discard the scalar part and write the enriched term as

$$\delta\mathcal{L}_{\text{enr}} = \mathbf{N}_{ij}^a (\partial_i \phi_0(x)) (-i\partial_j) \otimes s^a.$$

a. Proposition 1 (Internal Pauli reduction). Any gradient-linear enriched microscopic term on an internal complex doublet reduces uniquely to

$$\delta\mathcal{L}_{\text{enr}} = \mathbf{N}_{ij}^a (\partial_i \phi_0) (-i\partial_j) \otimes s^a$$

up to an irrelevant internal scalar term proportional to $\mathbf{1}_2$.

b. Proof. The Hermitian 2×2 operator algebra decomposes as

$$\text{Herm}(2) = \mathbb{R} \mathbf{1}_2 \oplus \text{span}\{s^1, s^2, s^3\}.$$

Projecting to the traceless part gives the claim. $\square$

B. Choice of shell direction and transverse projector

Fix a shell vector $G_* \in S_{\text{BCC}}$, and define the unit shell direction

$$\hat{n}_i = \frac{G_{*,i}}{|G_*|}.$$

The orthogonal projector onto the shell-transverse plane is

$$P_{ij}^\perp = \delta_{ij} - \hat{n}_i \hat{n}_j.$$

Choose an orthonormal transverse basis

$$e_i^\alpha, \quad \alpha = 1, 2,$$

satisfying

$$e_i^\alpha \hat{n}_i = 0, \quad e_i^\alpha e_i^\beta = \delta^{\alpha\beta}, \quad P_{ij}^\perp = e_i^\alpha e_j^\alpha.$$

Any spatial covector X_i decomposes as

$$X_i = X_\parallel \hat{n}_i + X_\alpha e_i^\alpha, \quad X_\parallel = \hat{n}_i X_i, \quad X_\alpha = e_i^\alpha X_i.$$

C. Shell-transverse tensor reduction

The shell-level transverse Dirac structure uses only the two transverse momentum directions

$$p_\alpha = e_j^\alpha p_j.$$

Accordingly, from the general microscopic tensor $\mathbf{N}_{ij}^a$ we extract the shell-relevant transverse coefficients by projecting the first spatial index onto the transverse basis:

$$C^{\alpha a}{}_j := e_i^\alpha \mathbf{N}_{ij}^a.$$

Then the enriched microscopic term becomes

$$\delta\mathcal{L}_{\text{enr}} = C^{\alpha a}{}_j (e_i^\alpha \partial_i \phi_0(x)) (-i\partial_j) \otimes s^a + (\text{longitudinal pieces}).$$

Equivalently, after transverse reduction,

$$\mathcal{T}^{\alpha a}(x) = C^{\alpha a}{}_j \partial_j \phi_0(x).$$

a. *Theorem 1 (Microscopic tensor reduction theorem).* Let

$$\delta\mathcal{L}_{\text{enr}} = \mathbf{N}_{ij}^a (\partial_i \phi_0(x)) (-i\partial_j) \otimes s^a$$

be a gradient-linear enriched coupling in the microscopic TECT linearization. Then, after choosing a shell direction G_* and projecting to the shell-transverse plane, the shell-relevant enriched coefficient fields necessarily take the form

$$\boxed{\mathcal{T}^{\alpha a}(x) = C^{\alpha a}{}_j \partial_j \phi_0(x), \quad C^{\alpha a}{}_j = e_i^\alpha \mathbf{N}_{ij}^a.}$$

b. *Proof.* This is just the decomposition of the first spatial index into transverse and longitudinal parts, followed by restriction to the transverse shell doublet. $\square$

This proves that the previously introduced shell-level ansatz

$$\mathcal{T}^\alpha(x) = C^\alpha{}_j \partial_j \phi_0(x) \otimes s^\alpha$$

is not ad hoc: it is exactly the canonical shell-transverse reduction of the most general gradient-linear microscopic tensor coupling.

D. Minimal shell-Dirac standard form

The shell-level Dirac theorem needs two independent transverse internal directions. Hence we now restrict to the minimal case where only two internal traceless directions survive:

$$a = 1, 2.$$

If, after internal basis choice, the microscopic tensor has the reduced form

$$\delta\mathcal{L}_{\text{enr}} = \mathbf{N}_{ij}^1 (\partial_i \phi_0) (-i\partial_j) \otimes s^1 + \mathbf{N}_{ij}^2 (\partial_i \phi_0) (-i\partial_j) \otimes s^2,$$

then the shell-transverse reduction becomes

$$\boxed{\mathcal{T}^\alpha(x) = C^\alpha{}_j \partial_j \phi_0(x) \otimes s^\alpha, \quad C^\alpha{}_j = e_i^\alpha \mathbf{N}_{ij}^\alpha, \quad \alpha = 1, 2.}$$

a. *Corollary 2 (Minimal standard shell form).* Any microscopic enriched linearization whose traceless internal part has two independent Pauli directions reduces, on the shell-transverse sector, to the standard minimal shell-Dirac coupling form above.

E. Fourier coefficient formula after tensor reduction

Using the first-shell Fourier expansion

$$\phi_0(x) = \sum_{G \in S_{\text{BCC}}} A_G e^{iG \cdot x},$$

we have

$$\partial_j \phi_0(x) = \sum_{G \in S_{\text{BCC}}} iG_j A_G e^{iG \cdot x}.$$

Therefore

$$\widehat{\mathcal{T}^{\alpha a}}(G_*) = i C^{\alpha a}{}_j G_{*,j} A_{G_*}.$$

In the minimal standard form,

$$\widehat{\mathcal{T}^\alpha}(G_*) = i (C^\alpha \cdot G_*) A_{G_*} s^\alpha.$$

a. *Theorem 3 (Reduced Fourier nonvanishing criterion).* After microscopic tensor reduction to the shell-transverse standard form, the shell-level transverse Dirac coefficients are nonzero exactly when

$$\boxed{(C^\alpha \cdot G_*) A_{G_*} \neq 0, \quad \alpha = 1, 2.}$$

b. Proof. Immediate from the Fourier formula above. $\square$

F. What is automatic and what is not

We now separate the parts that are automatic from the parts that remain open.

a. Automatic once a gradient-linear enriched tensor exists:

1. reduction to shell-transverse form

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x) \otimes s^\alpha;$$

2. first-shell Fourier support of $\mathcal{T}^\alpha$;
3. the shell-level projected transverse matrices have the correct traceless internal Pauli form.

b. Still not automatic:

1. nonvanishing of

$$C^\alpha \cdot G_*;$$

2. survival of two independent internal directions after projection;
3. spectral isolation of the enriched shell block.

Thus the remaining microscopic work has been narrowed to a finite tensor-coefficient problem.

G. Sharp final theorem of this step

a. Theorem 4 (Sharp microscopic reduction theorem). Suppose the microscopic TECT linearization contains a nonzero gradient-linear enriched tensor coupling on an internal doublet,

$$\delta\mathcal{L}_{\text{enr}} = \mathbf{N}_{ij}^a (\partial_i \phi_0) (-i\partial_j) \otimes s^a,$$

and that after shell-transverse projection two internal traceless directions survive:

$$a = 1, 2.$$

Then the shell-level Dirac problem is reduced to verifying only:

$$A_{G_*} \neq 0, \quad C^1 \cdot G_* \neq 0, \quad C^2 \cdot G_* \neq 0,$$

together with the previously established conditions

$$\widehat{U}(G_*) = 0, \quad v_{\parallel} \neq 0,$$

and spectral isolation of the enriched shell block.

b. Proof. Theorem 1 gives the shell-transverse form. Theorem 3 gives the exact nonvanishing criterion. The earlier shell-Dirac theorem then completes the argument. $\square$

LXXXVIII. FINAL SUMMARY

A general gradient-linear enriched coupling in the microscopic TECT linearization has the form

$$\delta\mathcal{L}_{\text{enr}} = \mathbf{N}_{ij}^a (\partial_i \phi_0(x)) (-i\partial_j) \otimes s^a.$$

After choosing a shell direction G_* and projecting to the transverse shell plane, this always reduces to

$$\mathcal{T}^{\alpha a}(x) = C^{\alpha a}_j \partial_j \phi_0(x), \quad C^{\alpha a}_j = e_i^\alpha \mathbf{N}_{ij}^a.$$

Therefore the previously used shell-level ansatz

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x) \otimes s^\alpha$$

is exactly the canonical shell-transverse reduction of the most general gradient-linear enriched microscopic tensor coupling.

The corresponding shell Fourier coefficient is

$$\widehat{\mathcal{T}}^\alpha(G_*) = i(C^\alpha \cdot G_*) A_{G_*} s^\alpha.$$

Hence, once a nonzero gradient-linear enriched tensor exists and two internal Pauli directions survive projection, the only remaining microscopic nonvanishing conditions are

$$A_{G_*} \neq 0, \quad C^1 \cdot G_* \neq 0, \quad C^2 \cdot G_* \neq 0,$$

together with the already established

$$\widehat{U}(G_*) = 0, \quad v_{\parallel} \neq 0,$$

and spectral isolation of the enriched shell block.

Thus the remaining shell-level Dirac task has been reduced to a finite microscopic tensor-coefficient problem plus spectral isolation.

LXXXIX. DIRAC EMERGENCE: MICROSCOPIC REDUCTION TO TENSOR COEFFICIENTS

A. Logical status

At the present stage, the emergence of the Dirac operator in the TECT framework should be formulated neither as a purely geometric slogan

$$\mathbb{CP}^2 \implies \text{Dirac}$$

nor as an already completed microscopic theorem of the full BCC Bloch problem.

What has been established is a layered structure:

1. a rigorous minimal reduced Dirac theorem at the level of a projected infrared sector;
2. an explicit improved effective realization in a $\mathbb{CP}^2$ -projector defect model, including a full-rank mixed symbol at the effective level;
3. a microscopic shell-level reduction showing that the remaining open problem is no longer conceptual, but reduces to a finite tensor-coefficient and spectral isolation problem.

This section summarizes that chain in its strongest honest form.

B. Minimal reduced Dirac theorem

Let $\mathcal{L}(p)$ denote the relevant linearized operator near the low-energy point $p = 0$. Suppose:

(R1) there exists an isolated complex 4-dimensional infrared sector

$$\mathcal{K}, \quad \dim_{\mathbb{C}} \mathcal{K} = 4;$$

(R2) $\mathcal{K}$ carries an involution

$$J^\dagger = J, \quad J^2 = \mathbf{1},$$

with splitting

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \dim_{\mathbb{C}} \mathcal{K}_{\pm} = 2;$$

(R3) after subtraction of the common reference energy, the leading projected operator is odd under J :

$$\mathcal{D}_{\text{eff}}(p) = \Pi_{\mathcal{K}} \mathcal{L}(p) \Pi_{\mathcal{K}} = \mathcal{D}_1(p) + O(|p|^2), \quad J\mathcal{D}_1 J^{-1} = -\mathcal{D}_1;$$

(R4) in a basis adapted to $\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-$,

$$\mathcal{D}_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_\mu M^\mu,$$

and the mixed symbol $A_1(p)$ is nondegenerate for generic small $p \neq 0$.

a. Theorem 1 (Minimal reduced Dirac emergence). Under (R1)–(R4), the principal infrared operator is unitarily equivalent, after a linear momentum redefinition, to the massless Dirac operator:

$$\mathcal{D}_1(p) \sim p_i \alpha^i, \quad \{\alpha^i, \alpha^j\} = 2\delta^{ij} \mathbf{1}_4.$$

b. Proof. Oddness under J forces the diagonal $++$ and $--$ blocks of $\mathcal{D}_1$ to vanish. Thus the operator is block-off-diagonal. A nondegenerate 2×2 complex first-order Hermitian symbol is reducible, after basis changes and linear momentum redefinition, to Pauli form

$$A_1(p) \sim p_i \sigma^i.$$

The full 4×4 operator is therefore

$$\mathcal{D}_1(p) \sim \begin{pmatrix} 0 & p_i \sigma^i \\ p_i \sigma^i & 0 \end{pmatrix} = p_i \alpha^i,$$

which is the massless Dirac principal operator. $\square$

C. Effective realization in the $\mathbb{CP}^2$ -projector defect sector

In the effective $\mathbb{CP}^2$ -projector description, let

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger$$

be a reflection-symmetric static defect background. Tangent fluctuations may be written as

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0,$$

so the fluctuation field takes values in

$$Q_B(\xi) = (\mathbb{C} z_B(\xi))^\perp, \quad \dim_{\mathbb{C}} Q_B = 2.$$

For the explicit local representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi),$$

an orthonormal frame is

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

In this frame the transverse Q_B -connection vanishes:

$$\mathcal{A}_\xi = -i e^\dagger \partial_\xi e = 0.$$

The local static Hessian therefore reduces to a channel Schrödinger operator

$$\mathcal{H}_B^{\text{stat}} = -\rho \partial_\xi^2 + \mathcal{U}_B(\xi).$$

With the improved two-anisotropy effective potential

$$V(P) = -\frac{\lambda}{2}\text{Tr}([H, P]^2) - \frac{\mu}{2}\text{Tr}([K, P]^2),$$

where

$$H = \text{diag}(1, 0, -1), \quad K = \text{diag}(1, -1, -1), \quad \lambda, \mu > 0,$$

the static channels become

$$H_1 = -\rho \partial_\xi^2 + (\lambda + \mu) \cos^2 2\theta(\xi) - 2\rho \theta'(\xi)^2,$$

$$H_2 = -\rho \partial_\xi^2 + \lambda + 2\mu \cos^2 \theta(\xi) - \rho \theta'(\xi)^2.$$

For explicit even defect profiles such as

$$\theta(\xi) = \theta_0 \text{sech}(\kappa\xi),$$

variational estimates show that both channels can support low even/odd bound states in a nonempty parameter regime at the effective-model level.

a. Interpretation. This gives strong effective support for an isolated 4-mode infrared sector, though not yet a full microscopic proof.

D. Why the plain scalar BCC Bloch pair is not enough

Consider instead a minimal periodic scalar Bloch problem around the BCC background

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(Q_a \cdot x).$$

The corresponding Bloch Hessian has the reciprocal-harmonic form

$$(H_{\text{Bloch}}(k))_{G,G'} = h_0(k+G)\delta_{GG'} + U_{G-G'}.$$

The minimal two-harmonic Bragg truncation near the correct Bragg point

$$k_* = \frac{G_*}{2}$$

yields

$$H_{\text{pair}}(p) = E_0(p)\mathbf{1} + v_{\parallel} p_{\parallel} \tau^3 + m \tau^1 + O(|p|^2).$$

a. Proposition 2. The minimal scalar Bragg pair does not generically produce a Weyl cone by itself.

b. Reason. Its linear traceless symbol is only one-dimensional at principal order, so it does not supply the full 3-direction Pauli structure required for Weyl/Dirac.

Thus a genuine shell-level Dirac theorem requires an enriched shell space.

E. Minimal enriched shell space

The minimal enriched shell carrier is

$$\mathcal{H}_{\text{shell}} = \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

where the shell pair provides one Pauli sector and the internal doublet provides the missing two transverse traceless generators.

Let τ^a act on the pair sector and s^a on the internal sector. Then

$$\Gamma^1 = \tau^1 \otimes s^1, \quad \Gamma^2 = \tau^1 \otimes s^2, \quad \Gamma^3 = \tau^3 \otimes \mathbf{1}$$

satisfy

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij} \mathbf{1}_4.$$

Therefore

$$H_{\text{Dirac}}(p) = v_1 p_1 \tau^1 \otimes s^1 + v_2 p_2 \tau^1 \otimes s^2 + v_3 p_3 \tau^3 \otimes \mathbf{1}$$

is an exact massless Dirac principal operator on the enriched shell space.

a. Conclusion. The plain scalar Bloch pair is insufficient, but the internal doublet $\otimes$ shell pair is sufficient in principle for shell-level Dirac emergence.

F. Shell-level projected matrix-element theorem

Choose shell basis states

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_a(x), \quad a = 1, 2,$$

where $u_a(x)$ span the internal periodic shell envelopes.

Projecting the Bloch Hamiltonian to the enriched shell sector gives

$$H_{\text{proj}}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix}.$$

The shell-level Dirac theorem requires:

$$(S1) \ B_0 := B(0) = 0;$$

$$(S2) \ E_+(p) - E_-(p) = 2v_{\parallel} p_{\parallel} \mathbf{1}_2 + O(|p|^2) \text{ with } v_{\parallel} \neq 0;$$

$$(S3) \ B(p) = p_1 W^1 + p_2 W^2 + O(|p|^2), \text{ where } W^1, W^2 \text{ are traceless Hermitian, nonzero, and anticommuting after internal basis reduction.}$$

Under these conditions,

$$H_{\text{proj}}(p) \sim v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1} + \nu_1 p_1 \tau^1 \otimes s^1 + \nu_2 p_2 \tau^1 \otimes s^2,$$

hence the projected shell Hamiltonian is Dirac.

G. Constant-envelope reduction and Fourier selection rules

For the simplest constant-envelope ansatz

$$u_a(x) = \frac{1}{\sqrt{V_{\text{cell}}}} \xi_a, \quad \xi_a^\dagger \xi_b = \delta_{ab},$$

the shell-level quantities reduce to primitive-cell Fourier coefficients:

$$B_0 = \widehat{U}(G_*) \mathbf{1}_2,$$

$$W^\alpha = \widehat{\mathcal{T}^\alpha}(G_*) s^\alpha.$$

If the scalar periodic potential is generated quadratically from the first-shell BCC condensate, for example

$$U_{\text{per}}(x) \propto \phi_0(x)^2,$$

then a quadratic first-shell selection rule gives

$$\widehat{U}(G_*) = 0,$$

hence

$$B_0 = 0.$$

Thus the dangerous constant Bragg mass vanishes automatically in the simplest first-shell quadratic scalar setting.

H. Classification of enriched couplings

We classify enriched coefficient fields $\mathcal{T}^\alpha(x)$ into low-order types:

1. linear in ϕ_0 ,

$$\mathcal{T}^\alpha \sim \phi_0;$$

2. linear in derivatives,

$$\mathcal{T}^\alpha \sim \partial_j \phi_0;$$

3. quadratic,

$$\mathcal{T}^\alpha \sim \phi_0^2, \quad \phi_0 \partial \phi_0, \quad (\partial \phi_0)^2.$$

Only the linear classes generically retain first-shell Fourier support. Purely quadratic enriched couplings generally shift support to sums of shell vectors and are therefore not minimal shell-Dirac candidates.

The most natural minimal allowed class is therefore

$$\mathcal{T}^\alpha(x) = C^\alpha{}_j \partial_j \phi_0(x) \otimes s^\alpha, \quad \alpha = 1, 2.$$

I. Microscopic tensor reduction theorem

Now let the microscopic TECT linearization contain a general gradient-linear enriched term

$$\delta \mathcal{L}_{\text{enr}} = \mathbf{N}_{ij}^a (\partial_i \phi_0) (-i \partial_j) \otimes s^a.$$

Choose the shell direction

$$\hat{n} = \frac{G_*}{|G_*|}$$

and a transverse basis e_i^α , $\alpha = 1, 2$, satisfying

$$e_i^\alpha \hat{n}_i = 0.$$

Projecting the first spatial index onto the shell-transverse plane gives

$$C^{\alpha a}{}_j = e_i^\alpha \mathbf{N}_{ij}^a.$$

Hence the shell-transverse enriched coefficient fields are

$$\mathcal{T}^{\alpha a}(x) = C^{\alpha a}{}_j \partial_j \phi_0(x).$$

a. Theorem 3 (Microscopic tensor reduction). Any gradient-linear enriched microscopic tensor coupling on an internal doublet reduces, after shell-transverse projection, to the standard shell form

$$\mathcal{T}^\alpha(x) = C^\alpha{}_j \partial_j \phi_0(x) \otimes s^\alpha$$

for the surviving two internal Pauli directions.

Thus the shell-level ansatz is not ad hoc: it is the canonical shell-transverse reduction of the most general gradient-linear microscopic enriched tensor.

J. Final microscopic nonvanishing criterion

Using the first-shell Fourier expansion

$$\phi_0(x) = \sum_{G \in S_{\text{BCC}}} A_G e^{iG \cdot x},$$

we obtain

$$\widehat{\mathcal{T}}^\alpha(G_*) = i(C^\alpha \cdot G_*) A_{G_*} s^\alpha.$$

Therefore the final shell-transverse nonvanishing condition is

$$A_{G_*} \neq 0, \quad C^1 \cdot G_* \neq 0, \quad C^2 \cdot G_* \neq 0.$$

Combining everything, the shell-level Dirac problem is reduced to the following finite conditions:

$$\widehat{U}(G_*) = 0, \quad v_{\parallel} \neq 0, \quad A_{G_*} \neq 0, \quad C^1 \cdot G_* \neq 0, \quad C^2 \cdot G_* \neq 0,$$

plus spectral isolation of the enriched shell block.

K. What is proved and what remains open

What is proved:

1. the minimal reduced Dirac theorem;
2. explicit effective realizability in the improved $\mathbb{CP}^2$ -defect model;
3. the plain scalar shell-pair obstruction;
4. the necessity of the enriched shell space

$$\mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2;$$

5. shell-level reduction to projected matrix elements;
6. vanishing of the constant Bragg mass by a quadratic first-shell selection rule;
7. classification of allowed enriched couplings;
8. reduction of the microscopic shell problem to finite tensor coefficients.

What remains open:

1. explicit identification of a nonzero microscopic gradient-linear enriched tensor $\mathbf{N}_{ij}^a$ in the actual TECT linearization;
2. verification that two shell-transverse projections survive with

$$C^1 \cdot G_* \neq 0, \quad C^2 \cdot G_* \neq 0;$$

3. spectral isolation of the enriched shell block in the actual Bloch spectrum.

Thus the remaining gap is no longer conceptual. It is a finite microscopic tensor-coefficient problem plus a spectral-isolation problem.

XC. FINAL SUMMARY

The emergence of the Dirac operator in the present TECT framework is now reduced to a sharply defined microscopic problem.

At the abstract level, the following is rigorously proved:

if there exists an isolated graded complex four-dimensional infrared sector with nondegenerate mixed first-order symbol, then the principal operator is Dirac.

At the effective level, this structure is explicitly realizable in an improved $\mathbb{CP}^2$ -projector defect model.

At the shell-Bloch level, the plain scalar Bragg pair is not sufficient by itself, but the enriched shell space

$$\mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2$$

is sufficient in principle.

The shell-level problem has been reduced to the standard coupling form

$$\mathcal{T}^\alpha(x) = C^\alpha{}_j \partial_j \phi_0(x) \otimes s^\alpha, \quad \alpha = 1, 2,$$

which arises canonically from the shell-transverse reduction of the most general gradient-linear enriched microscopic tensor coupling

$$\mathbf{N}_{ij}^a (\partial_i \phi_0) (-i \partial_j) \otimes s^a.$$

The final microscopic conditions are therefore:

$$\widehat{U}(G_*) = 0, \quad v_{\parallel} \neq 0, \quad A_{G_*} \neq 0, \quad C^1 \cdot G_* \neq 0, \quad C^2 \cdot G_* \neq 0,$$

together with spectral isolation of the enriched shell block.

Hence the remaining shell-level Dirac problem is reduced to a finite tensor-coefficient problem plus a spectral-isolation problem.